

FINITE FAILURE SCHEMES AND THE RECONSTRUCTION OF QUADRATIC PENCILS

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ABSTRACT. We prove that an unmarked, ungraded finite local algebra associated with a complex quadratic pencil determines the pencil up to congruence. The truncation order $n^2 + 2n - 4$ is sharp, and no regularity assumption is imposed. Multiplication recovers the first relation, its determinant rulings, and the orientation that identifies the pencil coefficient line. The construction extends to families over arbitrary complex bases. On the effectively rigidified failure stack, the recovered source carries a scalar-normalized apolar algebra bundle. The associated multiple-incidence boundary admits explicit multi-Rees equations and relative lifting conditions. The sharp inverse is proved independently of this boundary application; its relative, covering, recognition, and spectral extensions are retained in the appendices. The paired boundary theory identifies the full multiple-reduced-incidence modification and computes arbitrary-order Jordan contact slices, thick Hilbert tails, transverse graph singularities and covered stable-map tails. This master preserves both mathematical bodies and is not a third submission.

1. INTRODUCTION

The first paper proves the sharp unmarked inverse for every complex quadratic pencil. The second identifies the full normalized Hilbert modification over the reduced corank-two incidence open and computes nonreduced-contact slices of every order. The respective principal results are Theorems 8.3, 36.3, and 37.2. The singularity and stable-map comparison are Proposition 37.3 and Theorem 37.4. Their effective stack and relative lifting interpretation is given in Theorem 38.2 and Proposition 38.3.

This master contains all earlier statements and proofs. The focused papers place their core arguments before supplementary extensions; no extension is silently used as a premise of the sharp inverse. Classical arrangement blow-ups, complete quadrics, hypersurface singularities and symmetric canonical forms are credited at their points of use. The source-normalized envelope is not a subquotient of the finite failure algebra, and a closed algebra does not choose a normal deformation direction. The Ballico 1993 comparison remains documentary-limited [1]; no priority inference is drawn from unavailable text.

2. FIRST RELATIONS, UNGRADED ISOMORPHISMS, AND INVERSE SYSTEMS

The passage from a finite local algebra to its first relation space is independent of pencils. We isolate it before undertaking the geometric inverse. In this section E denotes a space of degree-one *generators*; thus it is a cotangent space, rather than the dual normal space used later. Let $e = \dim E$, $d \geq 2$, and $K \subset \mathrm{Sym}^d E$. Set

$$A(K) = \mathrm{Sym} E / ((K) + \mathfrak{m}^{d+1}), \quad \mathfrak{m} = (E), \quad \mathfrak{n} = \mathfrak{m}A(K). \quad (2.1)$$

The grading in this presentation is not supplied to an isomorphism.

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Proposition 2.1 (The first-relation orbit principle). *The abstract algebra $A(K)$ recovers E and K , up to linear isomorphism, by*

$$E = \mathfrak{n}/\mathfrak{n}^2, \quad K = \ker[\mathrm{Sym}^d(\mathfrak{n}/\mathfrak{n}^2) \longrightarrow \mathfrak{n}^d]. \quad (2.2)$$

Consequently $A(K) \simeq A(K')$ as ungraded complex algebras if and only if some $a : E \xrightarrow{\sim} E'$ satisfies $(\mathrm{Sym}^d a)K = K'$.

More precisely, after fixing the homogeneous inclusions of generators, all isomorphisms with tangent map a are exactly the substitutions

$$x \longmapsto a(x) + u(x), \quad u \in \mathrm{Hom}_{\mathbb{C}}(E, \mathfrak{n}^2).$$

These assertions about substitutions remain valid after scalar extension to any commutative complex algebra.

Proof. The positive ideal is the unique maximal ideal, and $\mathfrak{n}^{d+1} = 0$. Multiplication of d lifts of cotangent vectors is independent of their lifts: replacing any one by an element of $\mathfrak{n}^2$ changes the product in $\mathfrak{n}^{d+1}$. There are no relations of smaller degree, and the degree- d quotient is $\mathrm{Sym}^d E/K$. This proves (2.2). In particular every isomorphism induces the stated linear equivalence.

Conversely substitute $a(x) + u(x)$ for each generator. A relation of degree d changes, beyond its linear substitution, only in degrees at least $d + 1$. Those terms vanish. The relations of degree $d + 1$ also vanish. The induced map on associated graded algebras is the isomorphism induced by a ; a finite filtration then makes the algebra map invertible. Every map with this tangent map has precisely such generator images. These polynomial identities hold over any commutative complex coefficient algebra. The intrinsic preservation of the augmentation under that scalar extension is verified in the next proof, so no preservation of a chosen origin has tacitly been added to the isomorphism problem. $\square$

Proposition 2.2 (The full automorphism group scheme). *Write $\ell = \dim_{\mathbb{C}} A(K) = \binom{e+d}{d} - \dim K$ and $H_K = \mathrm{Stab}_{\mathrm{GL}(E)}(K)$, with its closed subgroup scheme structure. The functor $D \mapsto \mathrm{Aut}_D(A(K) \otimes_{\mathbb{C}} D)$ is an affine group scheme of finite type and fits into a split exact sequence of affine group schemes over $\mathbb{C}$:*

$$1 \longrightarrow U_K \longrightarrow \mathrm{Aut}(A(K)) \longrightarrow H_K \longrightarrow 1. \quad (2.3)$$

The kernel is unipotent. Choosing the homogeneous lifting of E identifies its underlying scheme, but not generally its group law, with the affine space associated to $\mathrm{Hom}_{\mathbb{C}}(E, \mathfrak{n}^2)$. Its dimension is $e(\ell - 1 - e)$. Intrinsically, U_K acts simply transitively on the affine space of linear splittings of $\mathfrak{n} \twoheadrightarrow \mathfrak{n}/\mathfrak{n}^2$. The splitting of (2.3) uses a choice of such a lifting; it is not an additional intrinsic grading.

Proof. Choose a basis of the finite-dimensional vector space $A = A(K)$. Inside $\mathrm{GL}(A)$, the equations $f(1) = 1$ and $f(xy) = f(x)f(y)$, imposed on basis elements, are polynomial equations. They represent the automorphism functor, including its values on nonreduced complex algebras.

The augmentation can be recovered without mistaking the absolute nilradical after scalar extension for the augmentation ideal. If M_z denotes multiplication by z , the homogeneous filtration gives

$$\mathrm{tr}(M_z) = \ell \epsilon(z). \quad (2.4)$$

Indeed multiplication by a positive-degree element is strictly triangular, whereas multiplication by a scalar has that scalar on every diagonal entry. Identity (2.4) holds over every commutative complex algebra D . Algebra automorphisms conjugate multiplication matrices, so they preserve $\epsilon = \ell^{-1} \mathrm{tr}(M_{(-)})$, its kernel $\mathfrak{n}_D$, and all its powers. Thus the tangent map is a morphism of group schemes into H_K , not merely a map on complex points.

Linear substitutions give a morphism $H_K \rightarrow \text{Aut}(A)$ sectioning this tangent map. By Proposition 2.1, its kernel has, functorially in D , the unique parametrization $x \mapsto x + u(x)$ with $u : E_D \rightarrow \mathfrak{n}_D^2$. Reading generator images and making these substitutions are inverse polynomial maps. On a basis ordered by the positive filtration all kernel matrices are unitriangular, proving unipotence. Multiplication is substitution, which need not be addition of corrections. The dimension follows from $\dim \mathfrak{n}^2 = \ell - 1 - e$.

Finally two splittings of $\mathfrak{n} \twoheadrightarrow E$ differ by exactly a map $E \rightarrow \mathfrak{n}^2$. Any splitting generates A , and its degree- d products impose exactly K ; the preceding substitution argument therefore gives the unique element of U_K relating any two splittings. This is the asserted coordinate-free torsor description. $\square$

2.1. The inverse-system description and its precise role. Put $E^\vee = \text{Hom}_{\mathbb{C}}(E, \mathbb{C})$. Use the apolar pairing in which dual divided monomials satisfy $\langle x^\alpha, y^\beta / \beta! \rangle = \delta_{\alpha\beta}$. Here the pairing means evaluation at the origin after applying the constant-coefficient differential operator. For $I = (K) + \mathfrak{m}^{d+1}$, its Macaulay inverse system is

$$I^\perp = \bigoplus_{j < d} \text{Sym}^j E^\vee \oplus K^\perp, \quad K^\perp \subset \text{Sym}^d E^\vee. \quad (2.5)$$

To check this equality, the relations of degree $d + 1$ bound the degree of every inverse-system element by d . The degree- d relations pair only with its top component. All lower components are unrestricted. Conversely that direct sum is stable under every partial derivative: differentiating its top part lands in the full lower-degree space. It is therefore annihilated by I , proving the displayed equality. Thus the orbit problem for K is equivalently the contragredient orbit problem for $K^\perp$. This uses the classical inverse-system correspondence; it does not assert a new canonical-grading theorem.

Elias and Rossi prove that Gorenstein local algebras with Hilbert function $(1, e, e, 1)$ are canonically graded [7, Theorem 3.3 and Corollaries 3.4–3.5]. Their proof removes a lower-degree term of the inverse-system generator by a nonlinear coordinate change. In their later work, compressed Gorenstein algebras of socle degree at most four are canonically graded, with counterexamples beyond the stated range [8, Theorem 3.1 and Examples 3.3–3.5]. In contrast, (2.1) is homogeneous by construction, and its truncation makes all higher substitutions invisible to its first relation. We do not deduce that an arbitrary filtered deformation with the same Hilbert function is graded.

For the pencil algebras, $e = n^2$, $d = n^2 + 2n - 4 \geq 11$, and $\dim K = \binom{N}{2}$. They are not Gorenstein: their socles contain $\mathfrak{n}^d$, whose dimension is $\binom{e+d-1}{d} - \binom{N}{2} > 1$. For example, the first binomial is at least $\binom{e+1}{2}$, while $e = n^2 > N$. The issue left by Proposition 2.1 is accordingly not the removal of a nonhomogeneous inverse-system term. It is the recovery of the matrix factors and defining coefficient data from the *unrestricted* $\text{GL}(E)$ -orbit of K . The next sections establish that recovery and its global descent.

3. INTRINSIC TENSOR RULINGS

The first step of the inverse problem is to recognize the two tensor factors of a determinant cone without an ambient marking. The following scheme-theoretic ruling calculation and descent lemma supply precisely this step. They apply to the pencil argument independently of the web and exterior-contraction constructions in the separate research archive.

Lemma 3.1 (The relative scheme of maximal rank-one spaces). *Let B be any complex scheme and E, F vector bundles of rank $n \geq 2$. In $\text{Gr}_B(n, E \otimes F)$, let $Z(E, F)$ be the zero scheme of the restriction of the quadratic rank-one equations to the universal subbundle $\mathcal{T}$. Equivalently it is*

the zero scheme of

$$\mathrm{Sym}^2 \mathcal{T} \longrightarrow \bigwedge^2 E \otimes \bigwedge^2 F.$$

The natural morphism

$$\mathbb{P}_B(E) \sqcup \mathbb{P}_B(F) \longrightarrow Z(E, F), \quad \ell \mapsto \ell \otimes F, \quad m \mapsto E \otimes m, \quad (3.1)$$

is an isomorphism. It commutes with arbitrary change of complex base, including change to a nonreduced base.

Proof. We first work over $\mathbb{C}$. Each displayed map to the Grassmannian is a closed immersion. Its composition with the Pluecker embedding is the degree- n Veronese embedding on $\ell^{\otimes n} \otimes \det F$, respectively $\det E \otimes m^{\otimes n}$, followed by its natural linear inclusion into $\bigwedge^n(E \otimes F)$. This linear inclusion is injective: polarization of the nonzero pure-power map gives the corresponding irreducible symmetric-power summand. The rank-one equations vanish identically on both universal families, so the closed immersions factor through $Z(E, F)$.

Every geometric point is in one of the two images. Indeed the sum of two rank-one tensors is rank at most one only if their first factors or their second factors are proportional. Starting with two independent tensors, this observation forces a linear space consisting of rank-one tensors to fix the same factor throughout. A space of dimension n is then exactly one of the displayed spaces. The two images are disjoint: a space fixing both factors has dimension one, not n .

We verify that these pointwise statements omit no infinitesimal structure. At $\ell \otimes F$, choose a generator of ℓ and a complement in E . A Grassmannian tangent vector is a map $\phi : F \rightarrow (E/\ell) \otimes F$. For each coordinate of E/ℓ , write its component as $A : F \rightarrow F$. Linearizing the restricted minors gives

$$u \wedge A(u) = 0 \quad \text{for every } u \in F.$$

Polarizing this diagonal identity gives

$$u \wedge A(v) + v \wedge A(u) = 0 \quad (u, v \in F).$$

In characteristic zero the diagonal and polarized identities are equivalent. These are precisely all coefficients of the bilinear linearization of the restricted quadratic map on $\mathrm{Sym}^2 F$, so no infinitesimal equation is omitted by testing the diagonal. Taking the basis vectors shows that A is diagonal; taking their pairwise sums makes all its diagonal entries equal. Conversely a scalar A satisfies all the linearized equations. Thus the Zariski tangent space to $Z(E, F)$ is precisely $\mathrm{Hom}(\ell, E/\ell)$, of dimension $n - 1$. The other image has the symmetric calculation.

At every closed point the closed immersed projective space gives local dimension at least $n - 1$, whereas the tangent calculation gives embedding dimension exactly $n - 1$. The local ring is therefore regular. A finite-type complex scheme whose closed-point local rings are regular is reduced: a nonempty support of its nilradical would contain a closed point. In particular there is no nilpotent thickening or embedded associated structure at either image. The disjoint union of the two closed immersions is itself a closed immersion, surjective on geometric points. Its ideal in the now reduced target vanishes at every closed point, so it is zero. This proves the scheme isomorphism over $\mathbb{C}$.

For general B , trivialize E, F on a common open cover. The Grassmannian and the zero scheme of the displayed universal quadratic map are, on such an open, the base change of the constant construction just proved. Thus (3.1) is an isomorphism there even if the base has nilpotents. The maps are natural under changes of frame and hence glue. This argument is an identification by equations and base change, not an inference of relative flatness from a fibrewise tangent calculation. $\square$

Lemma 3.2 (Oriented relative Segre descent). *Let B be a connected projective complex variety with $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and let V, V' have the same dimension $n \geq 2$. Let $\mathcal{U}, \mathcal{U}'$ be vector bundles of rank n on B . Suppose a linear bundle isomorphism*

$$\alpha : V \otimes \mathcal{U}^* \xrightarrow{\sim} V' \otimes \mathcal{U}'^*$$

preserves the rank-one cones and the ruling whose parameter bundle is $\mathbb{P}(V) \times B$. The induced isomorphism $\mathbb{P}(V) \times B \simeq \mathbb{P}(V') \times B$ is regular and is induced by one constant projective isomorphism $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$. Locally $\alpha(T) = gTh^{-1}$, with the same $[g]$ on every chart. The possible line-bundle twists of vector-bundle lifts do not change the projective left coefficient lines of any homogeneous functorial coefficient construction.

Proof. Lemma 3.1, applied to $E = V \otimes \mathcal{O}_B$ and $F = \mathcal{U}^*$, identifies the closed relative Fano scheme with $(\mathbb{P}(V) \times B) \sqcup \mathbb{P}_B(\mathcal{U}^*)$. In particular its two components carry their displayed scheme structures, with no additional nilpotent or embedded components. The isomorphism α induces a regular map of these relative parameter schemes, and the orientation hypothesis preserves the first component.

For completeness, an isomorphism of two trivial $\mathbb{P}^{n-1}$ -bundles over any base gives a regular projective-matrix-valued map, not merely a fibrewise collection. Its pullback of the relative $\mathcal{O}(1)$ has the form $\mathcal{O}(1) \otimes p^*L$ for a line bundle L on the base. To see this, tensor by $\mathcal{O}(-1)$; the resulting line bundle is trivial on every projective-space fibre, and its pushforward and evaluation identify it with a pullback from the base. Pushing forward $\mathcal{O}(1)$ now gives an isomorphism of the corresponding trivial rank- n bundles, with that line-bundle twist. On charts trivializing L , it is an invertible matrix g_i of regular functions. On overlaps, $g_j = u_{ij}g_i$ for an invertible regular function u_{ij} . The classes $[g_i]$ thus glue to a regular morphism to the projective-isomorphism scheme, identified after choosing fixed bases with PGL_n . This accounts explicitly for the possible absence of a global linear lift.

The group PGL_n is affine: it is the determinant-nonzero principal open in the projective space of matrices. A morphism from B to an affine scheme factors through $\mathrm{Spec} H^0(B, \mathcal{O}_B) = \mathrm{Spec} \mathbb{C}$, so the projective map is constant. Choose a constant linear lift g . Each local g_i then differs from g by a scalar unit. After removing its two projective-factor actions, a linear map preserving every decomposable line is scalar: apply it to sums of two tensors with one common factor. Consequently the remaining action is a right bundle map, and the scalar units may be absorbed in its local lifts h . This proves the local tensor formula.

Finally a scalar multiplying a lift acts on a homogeneous functor of degree a by its scalar to the power a (or $-a$ on its dual). It cannot alter a projective coefficient line. The conclusion applies to every homogeneous degree. In particular, no right change of frame can replace the single projective left map by separate transformations on different coefficient spaces. $\square$

4. ACTUAL FACTOR DESCENT AND MOVING COEFFICIENT SPACES

We now separate the linear-preserver input, the descent of vector bundles, and the recovery of coefficients. The resulting principle allows several varying coefficient spaces, of arbitrary prescribed ranks, rather than only the line associated with a pencil.

Lemma 4.1 (Linear preservers of the determinant cone). *Let U, V, U', V' be complex vector spaces of dimension $n \geq 2$. An invertible linear map $\Phi : \mathrm{Hom}(U, V) \rightarrow \mathrm{Hom}(U', V')$ carrying the determinant hypersurface onto the determinant hypersurface has one of the forms*

$$T \mapsto gTh^{-1}, \quad T \mapsto gT^{\mathrm{t}}h^{-1},$$

where the second formula is written after choosing identifications for transposition. The two cases respectively preserve and exchange the two rank-one rulings. In the first case the pair (g, h) is unique up to a common nonzero scalar.

Proof. At a matrix of rank r , an invertible r -minor reduces the local rank condition to the rank condition on a generic $(n - r)$ -square matrix. The singular locus of the rank-at-most s variety is therefore the rank-at-most $s - 1$ variety for $1 \leq s < n$: on the rank- s stratum the Schur-complement entries give independent equations, whereas on a lower stratum the linear terms of its defining $(s + 1)$ -minors vanish. Starting with the determinant cone and iterating reduced singular loci hence recovers the rank-one cone. For $n = 2$ that cone is already the determinant cone.

If $v \otimes u$ and $w \otimes z$ have rank one, their sum has rank at most one only if v, w or u, z are dependent. It follows that each maximal linear space of rank-one tensors has one factor fixed. The two families are $\mathbb{P}(V)$ and $\mathbb{P}(U^*)$. The induced Segre automorphism must preserve these families or exchange them. In the preserving case its restriction to a ruling gives projective linear maps on each factor. Remove linear lifts of both maps. The remaining linear transformation fixes each decomposable line. In tensor bases its eigenvalues at $v_i \otimes u_j$ agree along each row and each column by applying it to $(v_i + v_k) \otimes u_j$ and $v_i \otimes (u_j + u_l)$. They are all equal, so the remaining map is scalar. This proves the first form and its uniqueness. Composing with transposition reduces the exchanging case to the first.

This is the square invertible case of the classical rank-one preserver theorem of Marcus and Moyls [9, Theorem 1, pp. 1218–1219]; the argument above specifies the reduction from the determinant hypersurface that is needed here. $\square$

Lemma 4.2 (An actual tensor map removes the line twist). *Let B be a reduced complex variety, V a complex vector space with $\dim V \geq 2$, and $\mathcal{A}_1, \mathcal{A}_2$ vector bundles. Suppose an actual bundle isomorphism*

$$\Psi : V \otimes \mathcal{A}_1 \xrightarrow{\sim} V \otimes \mathcal{A}_2$$

preserves the rank-one cones and induces the identity on the $\mathbb{P}(V)$ -parameter space of rulings. There is a unique bundle isomorphism $a : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ with $\Psi = 1_V \otimes a$.

Equivalently, the intrinsic operation of taking the commutant of the constant matrix algebra gives

$$\mathcal{H}om_{\text{End}(V) \otimes \mathcal{O}_B}(V \otimes \mathcal{A}_1, V \otimes \mathcal{A}_2) \simeq \mathcal{H}om_{\mathcal{O}_B}(\mathcal{A}_1, \mathcal{A}_2). \quad (4.1)$$

Thus after a constant left projective map has been lifted linearly, no residual line-bundle twist is left in the recovery of the right factor from Ψ .

Proof. For every constant line $\mathbb{C}v \subset V$, the corresponding ruling is fixed, so on every geometric fibre Ψ maps $v \otimes \mathcal{A}_1$ onto $v \otimes \mathcal{A}_2$. Choose a basis $v_1, \dots, v_n$ for the purpose of checking this assertion as an equality of bundle maps. The rulings for v_i force all off-diagonal blocks of Ψ to vanish. The rulings for $v_i + v_j$ force its diagonal blocks to agree. Since B is reduced, fibrewise zero morphisms of vector bundles are zero. Therefore Ψ commutes with every constant matrix unit, hence with $\text{End}(V) \otimes \mathcal{O}_B$.

For clarity, commutant recovery itself is canonical and does not use these chosen matrix units as extra data. Tensoring a bundle map a with 1_V gives the forward map in (4.1). Commutation with matrix units shows locally that it is bijective, with inverse reading any diagonal block. This proves that it is an isomorphism of sheaves, independent of the checking basis. Its inverse applied to the global section Ψ produces a global section a , not projective local sections requiring arbitrary scalar lifts. Applying the same operation to Ψ^{-1} proves that a is invertible. Uniqueness follows from injectivity of $a \mapsto 1_V \otimes a$.

Projective-factor data alone would permit replacing a bundle by a line twist. The additional datum used here is the actual linear map of the tensor bundles. It is this datum, recovered from the dual of the first nilradical quotient, that makes the commutant argument applicable. $\square$

4.1. A general finite-order coefficient inverse. Fix $n \geq 2$, $q \geq 1$, and integers $0 \leq r_\lambda \leq m_\lambda = \dim \mathbb{S}_\lambda \mathbb{C}^n$ for partitions λ of q of length at most n . Assume that for at least one partition

$$0 < r_\lambda < m_\lambda. \quad (4.2)$$

Let B be a smooth connected projective complex variety, V a constant n -dimensional space, and $\mathcal{U}$ a rank- n bundle. For each λ , choose a subbundle with locally free quotient

$$\mathcal{L}_\lambda \subset \mathbb{S}_\lambda V^* \otimes \mathcal{O}_B, \quad \text{rank } \mathcal{L}_\lambda = r_\lambda.$$

Zero and full subbundles are permitted. In the matrix bundle $\mathcal{E} = \text{Hom}(\mathcal{U}, V \otimes \mathcal{O}_B)$, the multiplicity-one Cauchy decomposition supplies the subbundle

$$\mathcal{J}_q = \bigoplus_{\lambda} \mathcal{L}_\lambda \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^q \mathcal{E}^*. \quad (4.3)$$

Let $\mathcal{D}_n \subset \text{Sym}^n \mathcal{E}^*$ be the determinant line, put $D = n + q$, and define

$$\mathcal{K} = \mathcal{D}_n \mathcal{J}_q, \quad \mathcal{A} = \text{Sym } \mathcal{E}^* / ((\mathcal{K}) + \mathfrak{a}^{D+1}), \quad Y = \text{Spec}_B \mathcal{A}, \quad (4.4)$$

where $\mathfrak{a}$ is the positive homogeneous ideal. The notation D in this subsection is an integer, not a failure locus. The scheme Y is finite over B , and need not be zero-dimensional over $\mathbb{C}$.

Theorem 4.3 (Unmarked descent for moving coefficient spaces). *For fixed $n, q, (r_\lambda)$ satisfying (4.2), the abstract complex scheme Y recovers its coefficient data as follows. Two such schemes Y_1, Y_2 are isomorphic if and only if there are an isomorphism $\sigma : B_1 \rightarrow B_2$, a constant linear isomorphism $g : V_1 \rightarrow V_2$, and an actual bundle isomorphism $h : \mathcal{U}_1 \rightarrow \sigma^* \mathcal{U}_2$ such that, for every λ ,*

$$\mathcal{L}_{1,\lambda} = (\mathbb{S}_\lambda g)^* \sigma^* \mathcal{L}_{2,\lambda}. \quad (4.5)$$

Every unmarked isomorphism induces a normal-bundle map $T \mapsto gTh^{-1}$; for that map the pair (g, h) is unique up to a common nonzero constant scalar. Conversely every such triple lifts homogeneously to an isomorphism. The scheme is finite locally free over its reduced base of rank

$$\binom{n^2 + D}{D} - \sum_{\lambda} r_\lambda m_\lambda.$$

No nontriviality hypothesis on $\mathbb{P}_B(\mathcal{U}^)$ is required.*

Proof. Recovering the intrinsic linear data. Multiplication by a determinant polynomial is an injective linear map of polynomial spaces. In local bundle frames its matrix is constant over $\mathbb{C}$ and its cokernel is free. The Cauchy summands and the $\mathcal{L}_\lambda$ are subbundles, so $\mathcal{K}$ is a subbundle of rank $\sum r_\lambda m_\lambda$. The graded pieces of $\mathcal{A}$ below D are the full symmetric powers; its top piece is $\text{Sym}^D \mathcal{E}^* / \mathcal{K}$. This proves local freeness and the rank formula.

The nilradical $\mathfrak{n}$ of $\mathcal{O}_Y$ is its positive ideal, since B is reduced. Thus $Y_{\text{red}} = B$ and $\mathfrak{n}/\mathfrak{n}^2 = \mathcal{E}^*$ intrinsically. The multiplication kernel

$$\ker[\text{Sym}^D(\mathfrak{n}/\mathfrak{n}^2) \rightarrow \mathfrak{n}^D]$$

is precisely $\mathcal{K}$, by the proof of Proposition 2.1 in local frames. An abstract isomorphism consequently gives the base isomorphism σ and an actual linear normal-bundle isomorphism carrying $\mathcal{K}_1$ to $\mathcal{K}_2$ contravariantly. The normal map is linear over the reduced base even when the original scheme map does not preserve the chosen graph projection. Indeed the action on $\mathfrak{n}/\mathfrak{n}^2$ is linear over $\mathcal{O}_Y/\mathfrak{n}$.

Recovering and orienting the factors. Generate the homogeneous ideal $(\mathcal{K})$ in the recovered symmetric algebra; do not confuse its cone with the finite truncation. Its radical is the determinant ideal. To verify this, at an invertible matrix T , the map $\mathbb{S}_\lambda T$ is invertible. For a partition with $r_\lambda > 0$, a nonzero covector in $\mathcal{L}_\lambda$ has a nonzero matrix coefficient there. Thus the residual

coefficients cannot all vanish off the determinant. All generators vanish on the determinant. The relative determinant hypersurface is reduced over the smooth base; the equality of radicals follows on affine charts from this equality of closed points. Successive reduced singular loci and Lemma 3.1 recover its two unordered tensor rulings with their relative scheme structures.

Cancellation of the determinant line is intrinsic and recovers the ideal generated by $\mathcal{J}_q$, and hence its degree- q piece. In a Cauchy summand the two contraction supports of $\mathcal{L}_\lambda \otimes \mathbb{S}_\lambda \mathcal{U}$ have ranks (r_λ, m_λ) . Separate factor changes preserve these ranks. By Lemma 4.1, an exchange of the two matrix factors interchanges the ranks in each summand. At a partition satisfying (4.2), the result (m_λ, r_λ) cannot equal (r_λ, m_λ) for the other datum. Thus no fibre permits an exchange. Twisting local factor lifts by lines changes no support rank. The V -ruling is thereby oriented intrinsically, including when both projective ruling bundles are trivial.

Descending the actual map. By Lemma 3.2, the induced morphism $B_1 \rightarrow \text{Isom}(\mathbb{P}(V_1), \mathbb{P}(V_2))$ is constant: the target is affine and $H^0(B_1, \mathcal{O}_{B_1}) = \mathbb{C}$. Choose a constant linear lift g . Remove it from the actual normal map. Lemma 4.2, applied to $\mathcal{A}_1 = \mathcal{U}_1^*$ and $\mathcal{A}_2 = (\sigma^* \mathcal{U}_2)^*$, supplies the unique global right factor. Dualizing its inverse gives the actual isomorphism h . This step uses more than an isomorphism of projective bundles; it is exactly what excludes an undetected line twist. Replacing g by cg replaces h by ch . These are all ambiguities because the left lift is constant.

Recovering every coefficient subbundle. The oriented Cauchy decomposition is now functorial for the recovered left-right map. Project the residual degree- q module onto each summand and contract against the full right dual factor. The image is precisely $\mathcal{L}_\lambda$, as a subbundle of the constant left module, not just a collection of fibrewise projective subspaces. The right map acts invertibly on the full Schur factor and hence does not change this image. Pullback by the normal map therefore gives (4.5) for the same constant g in all summands. Scalar choices in the determinant line do not change these subbundles. This proves necessity.

Conversely $T \mapsto gTh^{-1}$ carries the determinant line, all coefficient subbundles, and every power of the positive ideal to the corresponding objects. It induces the required homogeneous scheme isomorphism. The stated uniqueness refers to the induced normal map; it does not eliminate the higher-order automorphisms of that scheme. $\square$

4.2. Identical fibres and graded bundles do not determine a family. The global conclusion has content even when neither fibre isomorphism classes nor the underlying graded vector bundles vary. The following example uses the general coefficient theorem, not a further spectral readout of a reconstructed pencil.

Example 4.4 (Two isotrivial thickenings separated by coefficient descent). Take $B = \mathbb{P}^1$, $V = \mathbb{C}^3$, $\mathcal{U} = \mathcal{O}_B^3$, $q = 1$, and $r_{(1)} = 1$. The two everywhere nonzero columns

$$(s^3, t^3, 0)^t, \quad (s^3 + st^2, t^3, 0)^t$$

define subbundles $\mathcal{L}_0, \mathcal{L}_1 \subset V^* \otimes \mathcal{O}_B$, both isomorphic to $\mathcal{O}_B(-3)$. Let Y_0, Y_1 be their order-4 thickenings (4.4). They are finite flat of rank 712 over B and Zariski locally trivial with the same Artin algebra as fibre. Their nilradical graded vector bundles are isomorphic in every degree, although Y_0 and Y_1 are not isomorphic as abstract complex schemes.

Proof. Neither column vanishes, since its second entry is nonzero when $t \neq 0$, and its first is nonzero at $t = 0$. The left coefficient map to its image line $\mathbb{P}^1 \subset \mathbb{P}(V^*)$ is, in the coordinate $z = s/t$, respectively

$$f_0(z) = z^3, \quad f_1(z) = z^3 + z.$$

A local frame completing a generator of $\mathcal{L}_i$ to a basis of $V^* \otimes \mathcal{O}$ changes the coefficient line to a fixed line. This gives Zariski local product presentations of both thickenings. Alternatively,

equality of the fibre isomorphism classes follows from transitivity of $\mathrm{GL}(V)$ on nonzero coefficient lines. The rank formula is $\binom{13}{4} - 3 = 712$.

Let $E = V^* \otimes \mathbb{C}^3$, of dimension nine. For $j < 4$, the degree- j nilradical quotient is $\mathrm{Sym}^j E \otimes \mathcal{O}_B$. Multiplication by the fixed determinant embeds E as a fixed nine-dimensional subspace of $\mathrm{Sym}^4 E$. Choosing a constant linear complement gives, for the top quotient,

$$\mathfrak{n}_i^4 \simeq \mathcal{O}_B^{486} \oplus ((V^* \otimes \mathcal{O}_B)/\mathcal{L}_i)^{\oplus 3}.$$

Each column has only its first two entries nonzero, and these two entries have no common zero. The rank-one quotient of $\mathcal{O}_B^2$ by $\mathcal{O}_B(-3)$ is consequently $\mathcal{O}_B(3)$, by determinants. Thus the displayed top bundle is $\mathcal{O}_B^{489} \oplus \mathcal{O}_B(3)^{\oplus 3}$ for both i . The assertions about every graded vector bundle follow. They are not assertions that the graded *algebra* structures agree.

If an abstract isomorphism $Y_0 \simeq Y_1$ existed, Theorem 4.3 would identify their coefficient maps by one automorphism of B and one constant projective transformation of V^* . Both maps have as image the same projective line, so the latter transformation would restrict to a projective automorphism of that line. Hence f_0 and f_1 would be equivalent by independent projective changes in source and target. This is impossible. The map f_0 has two ramification points, each of index three. The map f_1 has the two simple zeros of $3z^2 + 1$, each of index two, and infinity, of index three. Ramification indices are invariant under those projective changes. Thus the unmarked algebra scheme distinguishes these two globally different coefficient maps even though its fibre algebras, conormal bundle, abstract first-relation bundle, and all graded vector bundles agree. $\square$

4.3. Pencils and the category of their automorphisms. For pencils, take $q = 2p$, $p = N - 2$, and $F(H) = \bigwedge^{N-2} \mathrm{Sym}^2 H$. By Lemma 6.2 this is a single Schur module of dimension $M = \binom{N}{2}$, and Lemma D.1 identifies its coefficient subbundle with $\mathcal{L}_{\mathcal{R}} = \bigwedge^p (\mathrm{Sym}^2 V/\mathcal{R})^*$. Its rank is one, so the strict-support hypothesis holds for every pencil. Exterior duality recovers the subbundle $\mathcal{R}$ from that line. Theorem 4.3 therefore proves the unrestricted moving-pencil inverse, including its actual source-bundle conclusion. This is a simultaneous reconstruction theorem for a map into a coefficient Grassmannian, not just a classification of its individual fibres.

Corollary 4.5 (Algebraic automorphism sequence for a pencil). *Let $\mathfrak{A}_R^{[d]}$ be the unmarked local algebra of Theorem 8.3, and let $G_R = \{g \in \mathrm{GL}(V) : (\mathrm{Sym}^2 g)R = R\}$. There is a split exact sequence of complex affine algebraic groups*

$$1 \longrightarrow U_R \longrightarrow \mathrm{Aut}(\mathfrak{A}_R^{[d]}) \longrightarrow (G_R \times \mathrm{GL}(U))/\mathbb{G}_m \longrightarrow 1, \quad (4.6)$$

where $\mathbb{G}_m$ is the diagonal scalar subgroup. Its intrinsic quotient is the closed stabilizer of K_R in the general linear group of the cotangent space. In that form the sequence represents automorphisms after extension to every commutative complex algebra. Its splitting is homogeneous after choosing generators, and U_R has the intrinsic description and dimension in Proposition 2.2.

Proof. Proposition 2.2 gives the represented sequence with quotient H_{K_R} . The coefficient inverse and Lemma 4.1 identify its complex points with the image of the left-right action; transposition is excluded by the support ranks $(1, M)$. Use the contragredient of the normal action to obtain a homomorphism on the cotangent space. Its kernel is exactly the diagonal scalar subgroup: a tensor product acting identically has both factors scalar, with reciprocal actions on the dual factors. The image is a closed algebraic subgroup and all the groups in question are smooth in characteristic zero [10, Lemma 39.8.2]. Thus the stabilizer and this image agree as closed subgroup schemes, not only on complex points. The induced quotient identification is an isomorphism of algebraic groups: after division by the kernel it is a bijective separable homomorphism between smooth groups. The splitting and kernel are those already constructed by substitutions. $\square$

For a positive-dimensional projective reduction, Theorem D.4 is used as an exact sequence of groups of complex scheme automorphisms, with its stated filtered kernel. We do not identify that sequence with a finite-type affine group scheme or a moduli-stack equivalence. Likewise, the marked construction over nonreduced parameter bases later in the paper retains its specified relative socle: it does not replace that ideal by the absolute nilradical. These distinctions preserve the full unmarked geometric assertion while specifying its category.

5. AN INTRINSIC COEFFICIENT PRINCIPLE

Extracting coefficient data from an abstract scheme and reconstructing a defining relation space from those data are distinct operations. The criterion below isolates the first. Its hypotheses will be verified directly for the multiplication schemes and then on curves.

5.1. The structural criterion. Let X be a finite-type complex scheme with a closed subvariety B specified by a construction preserved by every abstract scheme isomorphism. Write $\mathfrak{b} \subset \mathcal{O}_X$ for its ideal and

$$\mathcal{G} = \bigoplus_{r \geq 0} \mathfrak{b}^r / \mathfrak{b}^{r+1}, \quad \mathcal{A} = \text{Sym}_{\mathcal{O}_B}(\mathcal{G}_1), \quad \mathcal{I} = \ker(\mathcal{A} \twoheadrightarrow \mathcal{G}).$$

Assume the following conditions, for fixed integers $n \geq 2, m \geq 1$.

- (i) B is connected and projective, $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and $\mathcal{G}_1$ is locally free. Its dual admits a presentation $V \otimes \mathcal{U}^*$, where $\dim V = \text{rank } \mathcal{U} = n$.
- (ii) The reduction of $\text{Spec}_B \mathcal{G}$ in its degree-one ambient bundle is the generic determinant cone. The bundle $\mathbb{P}_B(\mathcal{U}^*)$ is not projectively trivial. Thus precisely one of the two ruling parameter bundles of its rank-one locus is projectively trivial.
- (iii) If $\mathcal{D} \subset \mathcal{A}$ is the ideal of that reduced cone and $\mathcal{L} = \mathcal{D}_n$, then $\mathcal{I} = \mathcal{D}\mathcal{J}$, where $\mathcal{J}$ is generated in degree m . Under the Cauchy decomposition in any local frame of $\mathcal{U}$,

$$\mathcal{J}_m = \bigoplus_{\lambda \in \Lambda} L_\lambda \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^m(V^* \otimes \mathcal{U}), \quad L_\lambda \subset \mathbb{S}_\lambda V^*, \quad (5.1)$$

where Λ is a specified collection of partitions of m of length at most n , and the left subspaces are constant over B . Zero subspaces are permitted. The Cauchy inclusions, not just their representation labels, are the coefficient realizations $\beta \otimes \xi \mapsto [T \mapsto \beta(\mathbb{S}_\lambda(T)\xi)]$.

These conditions are hypotheses on the normal cone and its residual coefficient module. They do not assume that an abstract isomorphism extends to an ambient space, preserves a chosen frame, or already acts on all the left subspaces through a common projective transformation.

Theorem 5.1 (Intrinsic coefficient reconstruction). *Suppose X, B satisfy (i)–(iii). The abstract scheme X determines the collection $(L_\lambda)_{\lambda \in \Lambda}$ up to one common projective coordinate transformation on V . More precisely, an isomorphism between two such schemes induces a constant projective isomorphism $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$ such that*

$$L_\lambda = \mathbb{S}_\lambda(g)^* L'_\lambda \quad (\lambda \in \Lambda).$$

The equality means equality of subspaces; scalar changes of a linear lift of $[g]$ do not affect it.

Proof. An isomorphism transports B , every power of $\mathfrak{b}$, $\mathcal{G}_1$, and the surjection $\mathcal{A} \twoheadrightarrow \mathcal{G}$. In particular it transports $\mathcal{I}$ in the intrinsic degree-one ambient bundle. Iterated reduced singular loci of the determinant cone give its rank-one locus. Lemma 3.1 reconstructs its two ruling parameter schemes, including their scheme structure. Condition (ii) distinguishes the trivial ruling intrinsically. Lemma 3.2 then gives one constant projective map $[g]$. After choosing a constant linear lift and local frames on both bases, the ambient bundle map has the form

$T \mapsto gTh^{-1}$. These local choices are made only after constructing the graded algebra, its degree-one part, and the ruling.

Locally $\mathcal{D} = (d)$ with $d = \det T$. Its leading coefficient is a unit, so it is a nonzerodivisor over every coefficient ring. Hence $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$, and

$$\mathcal{J}_m \simeq \mathcal{L}^* \otimes \mathcal{I}_{n+m}.$$

This recovers the residual module without choosing a scalar determinant equation. Under $T \mapsto gTh^{-1}$, the Cauchy coefficient map transports a tensor by $\mathbb{S}_\lambda(g)^* \otimes \mathbb{S}_\lambda(h^{-1})$. Projection to the specified Cauchy summand, followed by contraction against its whole right dual, therefore transports exactly L'_λ to L_λ . The right map is invertible and has no effect on that left image. Both determinant-line transition functions and changes of a lift of $[g]$ act by scalar units. Since every partition has size m , this ambiguity is common in degree and disappears on subspaces. This proves the assertion. $\square$

6. QUADRATIC RELATION SPACES AND INTRINSIC RECONSTRUCTION OF PENCILS

The multiplication construction itself supplies a wider natural class for the coefficient criterion. In this section the number of quadratic relations is allowed to vary independently of the embedding dimension. For two relations the resulting theorem recovers every pencil of quadrics, including singular pencils. No extra block is added to the multiplication map.

6.1. Automatic coefficient extraction for quadratic multiplication. Let $\dim V = n \geq 3$, $W = \text{Sym}^2 V$, $N = n(n+1)/2$, and $R \in \text{Gr}(r, W)$, where $1 \leq r < N - n$. Put $p = N - r > n$, $S = W/R$, and

$$A_R = \mathbb{C} \oplus V \oplus S, \quad X_R = \text{Gr}(n, V \oplus S).$$

The only nonzero product in the maximal ideal is the quotient map $\gamma : \text{Sym}^2 V \rightarrow S$. Write $\mathcal{K}$ for the tautological bundle on X_R , and use the same definition (A.1) of $\hat{D}_R$, with A_R in place of B_R . Thus $\hat{D}_R$ is the failure scheme of generating A_R by a varying n -plane and the unit. The notation in this section agrees with the web construction when $(n, r) = (4, 4)$.

Let $\Lambda_{n,p}$ denote the partitions occurring in the classical multiplicity-free decomposition

$$F(V) := \bigwedge^p \text{Sym}^2 V = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{S}_\lambda V. \quad (6.1)$$

Every partition has size $2p$; terms of length greater than n are omitted. Choose a nonzero volume form on S and let $\beta_R \in F(V)^*$ be its pullback by $\bigwedge^p \gamma$. Its component in $\mathbb{S}_\lambda V^*$ is denoted by $\beta_{R,\lambda}$. Only the lines of the nonzero components will be used, so the volume choice is immaterial.

Theorem 6.1 (Automatic intrinsic coefficient extraction). *For every $R \in \text{Gr}(r, \text{Sym}^2 V)$ in the stated range, the abstract scheme $\hat{D}_R$ determines*

$$(\mathbb{C}\beta_{R,\lambda})_{\lambda \in \Lambda_{n,p}}$$

up to one common projective transformation of V , with zero components recorded as zero subspaces. More precisely, the intrinsic construction recovers the base $B = \text{Gr}(n, S)$, its degree-one normal-cone bundle, and the residual ideal generated in degree $2p$ with coefficient module

$$\mathcal{J}_{2p} = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{C}\beta_{R,\lambda} \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^{2p}(V^* \otimes \mathcal{U}), \quad (6.2)$$

where $\mathcal{U}$ is the tautological rank- n bundle on B . Consequently all geometric hypotheses of Theorem 5.1 hold for this entire Grassmannian family, not just for smooth webs.

Proof. On a frame chart of $\mathcal{K}$, write its inclusion as $\begin{pmatrix} M \\ A_0 \end{pmatrix}$, with M an $n \times n$ matrix. Since A_R is cube-zero, columns of degree greater than two vanish. After retaining the scalar, linear, and quadratic columns, multiplication has the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \operatorname{Sym}^2 M \end{pmatrix}. \quad (6.3)$$

A nonzero maximal minor must use the scalar column and all n linear columns: these are the only columns meeting the first $1+n$ rows. Expanding along those rows gives the identity of ideals

$$I_{\widehat{D}_R} = (\det M) I_p(\gamma \operatorname{Sym}^2 M). \quad (6.4)$$

The same expansion works for every $m \geq 2$. Where M is invertible, the quadratic map is surjective. Hence the reduction of $\widehat{D}_R$ is exactly the Schubert divisor where M is singular, independent of R after identifying the vector spaces S .

Successive reduced singular loci of that divisor are the projection rank loci. Their last nonempty member is $B = \operatorname{Gr}(n, S)$. For clarity, the usual determinantal assertion here is local: at a projection-rank- k point, an invertible k -minor splits away, leaving a generic square $(n-k)$ -matrix and smooth coordinates. Every remaining entry can vary independently by moving the kernel of the projection in V modulo its image. Thus its reduced singular locus is the next rank locus. Since $p > n$, the final base is nonempty, connected and projective, with $H^0(B, \mathcal{O}_B) = \mathbb{C}$.

Near B , a Grassmannian chart consists of a varying plane in S and its graph $T : \mathcal{U} \rightarrow V$. Formula (6.4) is independent of the other graph coordinates and homogeneous in T . It therefore gives its actual normal-cone ideal, not merely an initial reduced support:

$$C_B \widehat{D}_R \subset \operatorname{Tot} \operatorname{Hom}(\mathcal{U}, V), \quad \mathcal{I} = (\det T) \mathcal{J}, \quad \mathcal{J} = I_p(\gamma \operatorname{Sym}^2 T). \quad (6.5)$$

There are no degree-one equations. The ambient bundle in (6.5) is consequently the dual of the intrinsically recovered degree-one conormal. Its reduced ideal is the determinant ideal $\mathcal{D}$. The determinant polynomial is monic for a suitable monomial order, so cancellation gives $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$. This is an intrinsic ideal quotient; choosing a generator of the determinant line is not part of the recovered datum.

The rank-one locus of the reduced cone determines the two ruling parameter schemes by Lemma 3.1. They are $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial. Indeed a Schubert line in $\operatorname{Gr}(n, S)$ restricts $\mathcal{U}^*$ to $\mathcal{O}(1) \oplus \mathcal{O}^{n-1}$. A trivial projective bundle on that line would make this bundle a line-bundle twist of $\mathcal{O}^n$; its determinant would then have degree divisible by n , rather than degree one. Thus the trivial ruling is intrinsically oriented. Lemma 3.2 applies: an abstract isomorphism gives a morphism $B \rightarrow \operatorname{PGL}(V)$, constant because B is connected projective and $\operatorname{PGL}(V)$ is affine. Local linear lifts and right changes of frame give $T \mapsto gTh^{-1}$ for this one fixed $[g]$.

It remains to prove (6.2), rather than assume it as a condition on the example. The minors are the image, for fixed β_R , of the universal coefficient map

$$F(V)^* \otimes F(U) \longrightarrow \operatorname{Sym}^{2p}(V^* \otimes U), \quad \beta \otimes \xi \longmapsto [T \mapsto \beta(F(T)\xi)]. \quad (6.6)$$

In characteristic zero, the polynomial functor F has the multiplicity-free decomposition (6.1); see [3, Theorem 3.5]. The Cauchy decomposition of the target contains each diagonal $\mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U$ once, and no off-diagonal pair. Equivariance kills every off-diagonal pair. On each diagonal pair the coefficient map is nonzero: identify U with V , evaluate at $T = 1$, and choose β, ξ in that summand with $\beta(\xi) \neq 0$. Schur's lemma then identifies its image with the specified Cauchy summand. This is the actual matrix-coefficient inclusion, not merely a dimension count. Keeping β_R fixed proves (6.2). All generators have degree $2p$, and at least one minor is nonzero since γ is surjective.

Under $T \mapsto gTh^{-1}$, the right map acts within the full factor $\mathbb{S}_\lambda U$. Contracting against its dual therefore leaves exactly the left coefficient line transformed by $\mathbb{S}_\lambda(g)^*$. Determinant-line scalars and local lift ambiguities do not change these lines. This proves the theorem. $\square$

The theorem does not assert that separately projectivized Schur components always determine their relative scalars. For webs this is the recombination problem solved in the separate research archive. For pencils that issue disappears altogether.

6.2. Every pencil, without a regularity hypothesis.

Lemma 6.2 (The exterior square of the quadratic representation). *For $n \geq 2$, $\bigwedge^2 \text{Sym}^2 V = \mathbb{S}_{(3,1)} V$ is irreducible. Consequently $\bigwedge^{N-2} \text{Sym}^2 V$ is irreducible as well.*

Proof. The vector $x_1^2 \wedge x_1 x_2$ is a highest-weight vector of weight $(3, 1)$. By complete reducibility it generates a copy of $\mathbb{S}_{(3,1)} V$. The hook formula gives

$$\dim \mathbb{S}_{(3,1)} V = \frac{n(n-1)(n+1)(n+2)}{8} = \binom{N}{2}.$$

This exhausts the exterior square. The last assertion follows by exterior duality. These are classical representation identities. $\square$

Theorem 6.3 (Intrinsic reconstruction of all quadratic pencils). *Let $n \geq 3$ and let R, R' be two-dimensional subspaces of $\text{Sym}^2 V$. For the natural multiplication-failure schemes of their cube-zero algebras of Hilbert function $(1, n, N-2)$,*

$$\widehat{D}_R \simeq \widehat{D}_{R'} \iff R' = gR \text{ for some } g \in \text{PGL}(V). \quad (6.7)$$

No pencil, base locus, or discriminant is required to be smooth or regular. The isomorphism on the left is abstract and carries no supplied ambient embedding, base, or tensor-factor marking. All the reduced schemes are isomorphic to the same Schubert divisor.

Proof. Here $p = N - 2 > n$. By Theorem 6.1 and Lemma 6.2, the normal cone has only one left coefficient line to recover: the entire line $\mathbb{C}\beta_R$. The canonical duality

$$\bigwedge^{N-2} W^* \simeq \bigwedge^2 W \otimes (\det W)^* = \bigwedge^2 W \otimes (\det V)^{-(n+1)} \quad (6.8)$$

sends that line to the Pluecker line of R , with the same projective transformation g . Indeed the pullback of a quotient volume is the annihilator-volume dual of the kernel R . The common determinant character is a scalar on projective space. A nonzero decomposable two-vector recovers its plane as $\{w \in W : w \wedge w_R = 0\}$. Thus an abstract scheme isomorphism implies $R' = gR$.

Conversely such a linear coordinate change induces the quotient isomorphism $S_R \rightarrow S_{R'}$, the algebra isomorphism $A_R \rightarrow A_{R'}$, and the corresponding Grassmannian isomorphism. It intertwines multiplication and hence preserves its zeroth Fitting ideal. This proves the reverse implication. The assertion about reductions was proved in (6.4). $\square$

6.3. Discriminants, elementary divisors, and hyperelliptic moduli. A regular pencil means one containing an invertible symmetric matrix; it need not have a reduced discriminant. The classical invariant of such a pencil consists of its discriminant points together with their Jordan-block partitions, usually called its Weierstrass–Segre data. We use the root-labelled partitions, not a pairing of an unlabelled list with an independently unlabelled divisor. Their classification is classical; compare [4, Theorem 1.1, Corollary 2.1 and §2]. The following consequence identifies precisely what the intrinsic failure scheme remembers beyond the discriminant.

Corollary 6.4 (Intrinsic recovery of elementary divisors). *For a regular pencil in any dimension $n \geq 3$, the abstract scheme $\widehat{D}_R$ determines all root-labelled elementary divisors of a matrix pencil, up to projective reparametrization of its parameter line. Conversely these data determine its abstract failure scheme. In particular the invariant distinguishes different Jordan partitions over a fixed multiple discriminant root.*

Proof. The first implication follows by reconstructing R , then choosing an invertible member A and a complementary member B . For completeness, the relevant classical sufficiency can be seen directly. Put $C = A^{-1}B$; it is self-adjoint for A . If $hC = C'h$ identifies the Jordan forms for another pair (A', B') , set

$$H = A^{-1}h^t A'h.$$

Then H is invertible, A -self-adjoint, and commutes with C . There is a polynomial k in H with $k^2 = H$: on each nonzero eigenvalue choose a square root and truncate its Taylor series to the size of the largest Jordan block, then combine by the Chinese remainder theorem. Thus k is invertible, self-adjoint, and commutes with C . The map hk^{-1} is an isometry from A to A' and intertwines C, C' , so it simultaneously identifies A, B with A', B' .

Changing the basis of the pencil replaces C by a fractional linear expression with invertible denominator. At each eigenvalue its derivative is nonzero; hence each Jordan-block size is preserved while its root moves by the same projective transformation. A parameter change can avoid infinity for both finite spectra. Root-labelled elementary divisors therefore suffice, and (6.7) proves the converse for the failure schemes. This argument recalls the classical classification; the new input is recovery from an unmarked abstract failure scheme. $\square$

Example 6.5 (Identical discriminant, distinct intrinsic failure schemes). Choose distinct nonzero $\mu_3, \dots, \mu_n$, and put

$$S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Consider the symmetric pairs

$$\begin{aligned} A_s &= I_n, & B_s &= \text{diag}(0, 0, \mu_3, \dots, \mu_n), \\ A_j &= S_2 \oplus I_{n-2}, & B_j &= S_2 J_2 \oplus \text{diag}(\mu_3, \dots, \mu_n). \end{aligned}$$

Their determinant binary forms agree up to a nonzero scalar:

$$\det(sA_s + tB_s) = -\det(sA_j + tB_j) = s^2 \prod_{i=3}^n (s + \mu_i t). \quad (6.9)$$

At the unique double root the ranks are respectively $n-2$ and $n-1$, so no simultaneous projective equivalence exists. The two root partitions are $(1, 1)$ and (2) . Their abstract failure schemes are therefore nonisomorphic although both their reduced failure schemes and their discriminant divisors, including multiplicities, agree. This realizes inside the same multiplication construction the classical loss of elementary-divisor information under passage to the determinant.

Corollary 6.6 (Hyperelliptic isomorphism classes). *Fix $g \geq 2$ and $n = 2g + 2$. On the locus of pencils with simple discriminant, $\widehat{D}_R$ gives an injective invariant of the isomorphism classes of smooth hyperelliptic curves of genus g . All these failure schemes have the same reduction; their nonreduced structures distinguish a $(2g - 1)$ -dimensional family of curve classes. The pencil invariant also distinguishes the two regular boundary pencils in Example 6.5, where their singular discriminant double covers alone do not.*

Proof. The classical curve associated with a simple-discriminant pencil is the double cover of its parameter line branched at $\det(sA + tB) = 0$; see [6, §4.2, equation (4.3)]. Choose an invertible member. Distinct eigenvalues of $A^{-1}B$ have mutually A -orthogonal eigenlines, each nonisotropic,

so simultaneous diagonalization identifies the pencil with $\langle \sum x_i^2, \sum \lambda_i x_i^2 \rangle$. The unordered branch divisor modulo PGL_2 therefore determines the pencil, and every reduced divisor of degree $2g + 2$ arises. A smooth curve of genus at least two has a unique degree-two map to a rational curve up to automorphism of that curve: two distinct such maps would give a birational map to a curve of bidegree $(2, 2)$ in $\mathbb{P}^1 \times \mathbb{P}^1$, whose arithmetic genus is one. Thus curve isomorphism is exactly branch-divisor equivalence. Applying Theorem 6.3 gives the asserted injectivity and well-definedness.

Ordered distinct branch points have dimension $2g + 2$. Normalizing three of them removes the three-dimensional projective action; taking the finite permutation quotient leaves dimension $2g - 1$. Equality of all reductions follows from (6.4). Finally the two binary forms in (6.9) differ only by a scalar, a square over $\mathbb{C}$, and define isomorphic double covers, whereas their failure schemes have already been separated. $\square$

These are statements about complex isomorphism classes. They do not identify moduli stacks, assert a universal family over a coarse moduli space, or eliminate the different automorphism groups of pencils and curves. Nor is the classical pencil–hyperelliptic correspondence a new Torelli theorem. The conclusion here is reconstruction of those objects and of the elementary-divisor information on their boundary from the abstract nonreduced multiplication-failure scheme, while all reduced failure data are fixed.

7. THE EXACT INFINITESIMAL ORDER OF PENCIL RECONSTRUCTION

The all-pencil theorem admits a finite-order strengthening. It identifies the first infinitesimal neighbourhood that retains the pencil, and shows that every preceding neighbourhood is independent of it. We use ideal-adic neighbourhoods of the deepest stratum; these are not arc spaces or jet schemes.

Throughout this section let $n \geq 3$, $W = \mathrm{Sym}^2 V$, and

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n + 2p = n^2 + 2n - 4. \quad (7.1)$$

For a pencil $R \subset W$, put $S_R = W/R$ and $B = \mathrm{Gr}(n, S_R) \subset \widehat{D}_R$. This notation for the socle Grassmannian is local to this section; the cube-zero algebra is denoted by A_R , as in Section 6. Write $\mathfrak{b} \subset \mathcal{O}_{\widehat{D}_R}$ for the ideal of B , and define

$$Z_R^{[k]} = (B, \mathcal{O}_{\widehat{D}_R} / \mathfrak{b}^{k+1}) \quad (k \geq 0). \quad (7.2)$$

Thus $Z_R^{[0]} = B$. No embedding or identification of its reduction is supplied with the abstract scheme $Z_R^{[k]}$.

Theorem 7.1 (Sharp finite-neighbourhood reconstruction). *For any two pencils $R, R' \subset \mathrm{Sym}^2 V$, the following conditions are equivalent:*

$$\begin{aligned} Z_R^{[d]} \simeq Z_{R'}^{[d]} &\iff \widehat{D}_R \simeq \widehat{D}_{R'} \\ &\iff R' = gR \text{ for some } g \in \mathrm{PGL}(V). \end{aligned} \quad (7.3)$$

In contrast, for every $0 \leq k < d$, all the schemes $Z_R^{[k]}$ are isomorphic. Consequently d is the smallest uniform order at which these neighbourhoods classify all pencils up to projective equivalence. Singular pencils are included.

7.1. The first relation in the conormal algebra. Let $\mathcal{U}$ be tautological on B . The open set of X_R where projection to S_R is injective is canonically $\text{Tot } E$, where $E = \text{Hom}(\mathcal{U}, V)$: a plane is the graph of a unique map from its image in S_R to V . In particular this is one open neighbourhood of all of B , not merely a collection of unrelated formal charts. Let $\mathfrak{a}$ be its zero-section ideal. The exact multiplication calculation (6.4) gives

$$\mathcal{I}_R = (\det T)I_p(\gamma_R \text{Sym}^2 T) \subset \text{Sym } E^*. \quad (7.4)$$

It is generated in degree d , with a nonzero degree- d piece K_R . Indeed each quadratic minor has degree $2p$, and an invertible T makes $\gamma_R \text{Sym}^2 T$ surjective. There are no terms of lower degree, even on special pencils. Homogeneity is an equality of ideals in the actual graph neighbourhood, not just an equality of initial supports.

Lemma 7.2 (Intrinsic recovery from the first relation). *The abstract scheme $Z_R^{[d]}$ determines, as a graded $\mathcal{O}_B$ -algebra, the complete normal-cone algebra $\text{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R}$.*

Proof. The reduction of $Z_R^{[d]}$ is B , since its zero-section ideal is nilpotent and B is reduced. Hence its nilradical $\mathfrak{n}$ is intrinsic. Put $L = \mathfrak{n}/\mathfrak{n}^2$. Formula (7.4) and $d > 1$ identify $L = E^*$. Multiplication in the given finite scheme recovers

$$K_R = \ker[\text{Sym}^d L \longrightarrow \mathfrak{n}^d/\mathfrak{n}^{d+1}]. \quad (7.5)$$

The last denominator is zero, but retaining it displays the grading. No splitting of the reduction map is needed to form this kernel. Because $\mathcal{I}_R$ is generated in this single degree,

$$\text{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R} = \text{Sym}_{\mathcal{O}_B} L/(K_R). \quad (7.6)$$

Here (K_R) denotes the homogeneous ideal generated by K_R , not its radical. The right side is intrinsic to the finite scheme, proving the assertion. $\square$

Proof of Theorem 7.1. For $k < d$, equations (7.4) and (7.2) give

$$\mathcal{O}_{Z_R^{[k]}} = \text{Sym } E^*/\mathfrak{a}^{k+1}.$$

A linear identification $S_R \simeq S_{R'}$ identifies B , its tautological bundle, and E . It therefore identifies every one of these lower-order neighbourhoods.

An abstract isomorphism at order d induces an isomorphism of reductions, of conormal bundles, and, by Lemma 7.2, of complete graded normal cones. We spell out why the reconstruction in Theorem 6.1 requires no information outside this cone. Its reduced support is the square-matrix determinant cone; its iterated singular loci give the rank-one cone. The relative scheme of maximal linear spaces in that cone has the two rulings $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial, as is seen by restriction to a Schubert line. Thus it is intrinsically oriented. The induced projective transformation of the trivial factor is constant on the connected projective base. Local lifts have the form $T \mapsto gTh^{-1}$ for this one $[g]$. The determinant ideal line is intrinsic, so its ideal quotient in (7.4) recovers the residual ideal. Finally Lemma 6.2 identifies the entire left coefficient line $\mathbb{C}\beta_R$, and exterior duality (6.8) recovers the Pluecker line of R . These are operations on the recovered normal cone; we do not assume that its isomorphism extends to X_R or $\widehat{D}_R$. They imply $R' = gR$.

Conversely such a projective transformation induces the algebra and failure-scheme isomorphisms, carrying the socle Grassmannian to the socle Grassmannian and preserving all its ideal powers. This proves (7.3), also using Theorem 6.3. For sharpness, there are inequivalent pencils in every dimension under consideration: $\langle x_1^2, x_1x_2 \rangle$ has a common linear factor, whereas $\langle x_1^2, x_2^2 \rangle$ does not. Their neighbourhoods are nevertheless isomorphic at every order below d . Thus no smaller order classifies all pencils. $\square$

7.2. A closed immersion of the full parameter scheme. The finite-order result has a scheme-theoretic parameter version, which records infinitesimal variation as well as closed points. Here tensor coordinates are fixed. This additional qualification is important: we are not asserting an equivalence of unmarked moduli stacks.

Fix an auxiliary n -dimensional vector space U , and put

$$F(V) = \bigwedge^p \mathrm{Sym}^2 V, \quad F(U) = \bigwedge^p \mathrm{Sym}^2 U, \quad M = \dim F(U) = \binom{N}{2}, \quad H = \mathrm{Hom}(U, V).$$

Choose a nonzero scalar representative δ of the determinant polynomial on H ; its scalar has no effect on the subspaces below. Let $K_R \subset \mathrm{Sym}^d H^*$ be the degree- d space of $(\det T)I_p(\gamma_R \mathrm{Sym}^2 T)$ in these fixed coordinates.

Theorem 7.3 (Faithful finite-order parameterization). *The assignment*

$$\Phi : \mathrm{Gr}(2, \mathrm{Sym}^2 V) \longrightarrow \mathrm{Gr}(M, \mathrm{Sym}^d H^*), \quad R \longmapsto K_R, \quad (7.7)$$

is a closed immersion. It is defined on the whole Grassmannian, including singular pencils. Its construction and its inverse on the image commute with arbitrary base change over $\mathbb{C}$. In particular it separates families over nonreduced bases and has injective differential at every pencil.

Lemma 7.4 (Tensoring a line with a fixed space). *For finite-dimensional vector spaces A, B_0 , with $b = \dim B_0 > 0$, the map $\mathbb{P}(A) \rightarrow \mathrm{Gr}(b, A \otimes B_0)$, $\ell \mapsto \ell \otimes B_0$, is a closed immersion.*

Proof. Choose bases $a_0, \dots, a_t$ and a basis of B_0 . Above the projective chart $\ell = \mathbb{C}(a_0 + \sum x_i a_i)$, the subspace is the graph of the block matrices $x_i I_b$. The corresponding Grassmannian chart has arbitrary $b \times b$ blocks. The image is cut out by their off-diagonal entries and the differences between their diagonal entries; an inverse reads one diagonal entry in each block. These are scheme-theoretic equations over any $\mathbb{C}$ -algebra, not just pointwise conditions. The preimage of this Grassmannian chart is exactly the stated projective chart, since the projection has determinant a_0^b . Such charts cover the image. The map is proper because its source is projective, so these chartwise closed immersions give a global closed immersion. $\square$

Proof of Theorem 7.3. The coefficient map (6.6) becomes

$$\mu : F(V)^* \otimes F(U) \longrightarrow \mathrm{Sym}^{2p} H^*.$$

It is injective: by Lemma 6.2 its source is irreducible for $\mathrm{GL}(V) \times \mathrm{GL}(U)$, and evaluation at an isomorphism $U \simeq V$ shows that the equivariant map is nonzero. Multiplication by the nonzero polynomial δ is injective. Thus $\nu = \delta\mu$ is a fixed linear inclusion and

$$K_R = \nu(\mathbb{C}\beta_R \otimes F(U)). \quad (7.8)$$

In particular $\dim K_R = M$, also on every singular pencil. By exterior duality the quotient-volume map $R \mapsto \mathbb{C}\beta_R$ is the Pluecker closed immersion $\mathrm{Gr}(2, W) \hookrightarrow \mathbb{P}(F(V)^*)$. Factor Φ through this map, the closed immersion of Lemma 7.4, and the Grassmannian closed immersion induced by ν . This proves the theorem. All are morphisms of schemes defined by the displayed polynomial and linear equations. A closed immersion is a monomorphism and remains one after arbitrary base change. This proves the family assertion, not merely separation of complex-valued points. $\square$

Corollary 7.5 (A flat family of finite normal algebras). *Over $\mathrm{Gr}(2, W)$, the algebras*

$$Q_R = \mathrm{Sym} H^* / ((K_R) + \mathfrak{m}^{d+1}), \quad \mathfrak{m} = \bigoplus_{j>0} \mathrm{Sym}^j H^*, \quad (7.9)$$

form a finite locally free family of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}.$$

In fixed coordinates this family determines every base-changed family of pencils. All its truncations below degree d are constant families.

Proof. Theorem 7.3 makes the spaces K_R a rank- M subbundle of the constant degree- d bundle, with locally free quotient. In degree $j < d$ the algebra is $\text{Sym}^j H^*$, and its degree- d piece is $\text{Sym}^d H^*/K_R$; higher degrees vanish. This proves local freeness and base-change compatibility. Summing the ranks gives the displayed binomial identity. The kernel of the coordinate quotient in degree d recovers K_R , and Φ recovers the pencil family. This is a statement about coordinate quotient algebras; it does not discard their coordinate maps and then infer a family reconstruction theorem. $\square$

7.3. Classical moduli at a sharp order.

Corollary 7.6 (Hyperelliptic classes in finite neighbourhoods). *Let $g \geq 2$, $n = 2g + 2$, and $d = 4g^2 + 12g + 4$. For pencils with simple discriminant, $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic if and only if their associated hyperelliptic curves are isomorphic. Every lower-order neighbourhood is independent of the curve. Thus the same $(2g - 1)$ -dimensional family of curve classes is invisible to all these lower-order neighbourhoods and is separated at order d .*

Proof. Combine Theorem 7.1 with Corollary 6.6, and substitute $n = 2g + 2$ in (7.1). $\square$

The order here concerns a neighbourhood of the entire projective socle Grassmannian. Its global ruling distinguishes the constant V -factor from the tautological factor. A single unmarked Artinian normal slice does not supply this orientation; the coordinate statement of Corollary 7.5 is deliberately different. The result is therefore not a claim that one unmarked local algebra replaces the projective family. The matrix-pencil classification and the hyperelliptic correspondence remain classical. The assertion proved here is that this classical information first appears at the specified order in the intrinsic multiplication-failure neighbourhoods.

8. COEFFICIENT-SUPPORT ORIENTATION AND A LOCAL ALGEBRAIC INVERSE

For pencils there is a second, entirely local way to distinguish the two tensor rulings. It uses the asymmetry of the residual coefficient module rather than the topology of a projective bundle. Consequently the nontrivial-projectivization hypothesis in the preceding family theorem can be removed, and a single unmarked Artin local algebra already recovers the pencil.

8.1. The coefficient module orients its own tensor factors. If $K \subset A \otimes B$ is a linear subspace, define its two factor supports by

$$s_A(K) = \text{im}(K \otimes B^* \longrightarrow A), \quad s_B(K) = \text{im}(K \otimes A^* \longrightarrow B).$$

The maps here are contractions. Their ranks are invariant under separate changes of basis and are exchanged by interchanging the factors. In particular, if $K = L \otimes B$, then $s_A(K) = L$ and $s_B(K) = B$. The support ranks are therefore $(\dim L, \dim B)$, not the common dimension $\dim K$.

Lemma 8.1 (Asymmetric coefficient support). *Suppose a homogeneous ideal in the matrix space $\text{Hom}(U, V)$, with $\dim U = \dim V = n \geq 3$, has reduced support the determinant cone. Suppose its intrinsic residual ideal after determinant cancellation has, in one multiplicity-one Cauchy summand of degree q , the coefficient subspace*

$$L \otimes \mathbb{S}_\lambda U \subset \mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U, \quad 0 < \dim L < \dim \mathbb{S}_\lambda U.$$

A linear isomorphism carrying this ideal to another ideal with a coefficient subspace of the same displayed type cannot interchange the two matrix rulings. The conclusion is unchanged by line twists of the locally chosen tensor factors. It also applies fibrewise to subbundles of this type over a reduced base.

Proof. A linear isomorphism of determinant cones preserves their successive rank loci and therefore the rank-one cone. Its action on the two ruling components either preserves them or interchanges them. To make the familiar two alternatives explicit, identify the rank-one projectivization with $\mathbb{P}(V) \times \mathbb{P}(U^*)$. An automorphism preserving its two families of maximal linear spaces acts separately on the two projective factors. Each factor map is projective linear, by restricting the ambient linear map to a ruling. Removing their linear lifts leaves a map fixing every decomposable line; sums of tensors with one factor in common show that this remaining map is scalar. The preserving alternative is thus a left-right matrix map. In the interchanging alternative, compose with transposition to obtain the preserving alternative.

The determinant ideal is carried to the determinant ideal, so its ideal quotient is carried to the residual ideal. The Cauchy decomposition is functorial for separate changes of matrix factors; transposition exchanges the factors in its λ -summand. Put $m = \dim \mathbb{S}_\lambda U$, and $r = \dim L < m$. The displayed subspace has support ranks (r, m) . Separate factor maps preserve these ranks, while transposition changes them to (m, r) . An ideal of the prescribed type on the other side has support ranks (r', m) with $r' < m$. It cannot be the transposed image. This excludes interchanging the rulings.

Tensoring a factor by a line changes its Schur module by a line power, not its contraction rank. The argument therefore does not require a specified linear lift of either projective factor. For bundle maps it applies to each geometric fibre; an interchange on any fibre would contradict the same strict rank inequality there. $\square$

For pencils, take $F(H) = \bigwedge^{N-2} \text{Sym}^2 H$. Lemma 6.2 makes this a single irreducible Schur module, of dimension $M = \binom{N}{2} > 1$. The residual module is precisely $\mathcal{L}_R \otimes F(\mathcal{U})$, by Lemma D.1. Its support ranks are $(1, M)$. The relevant strict inequality holds for every pencil, including singular pencils; it is not a generic-rank condition.

Theorem 8.2 (Removal of the projective-bundle restriction). *In Theorem D.2, the hypothesis that $\mathbb{P}_B(\mathcal{U}^*)$ is nontrivial can be omitted. The coefficient-subbundle reconstruction, the smallest uniform order, and the automorphism exact sequence remain valid for every rank- n source bundle on a smooth connected projective complex base. In particular the base is allowed to be a point.*

Proof. The first-relation construction, the recovered homogeneous cone, and cancellation of the determinant use neither projective nontriviality nor positive dimension of the base. They recover both the unordered rulings and the residual coefficient module before a ruling is chosen. At this stage apply Lemma 8.1 with support ranks $(1, M)$. It excludes the transposed alternative and hence orients the left ruling even when both ruling bundles are trivial. This replaces exactly the nontrivial-projectivization step in the proof of Theorem D.2.

The left ruling is the trivial bundle $\mathbb{P}(V) \times B$. Projectivity and connectedness still make its projective-matrix map constant. Choose its constant linear lift; uniqueness of the right bundle map on overlaps gives the same actual isomorphism h , and coefficient contraction recovers the entire pencil subbundle. The converse uses the same homogeneous map. The lower-order algebras and the two inequivalent constant pencils in Corollary D.3 are unchanged. Finally, the proof of Theorem D.4 uses only the resulting classification of graded data, not the earlier way of orienting the rulings, so it too applies without the restriction. $\square$

8.2. Every pencil from one Artin local algebra. Choose n -dimensional vector spaces U, V . Let $T = (t_{ij})$ be the universal matrix, and put $\mathfrak{m} = (t_{ij})$. For a pencil $R \subset \text{Sym}^2 V$, let γ_R be its quotient map and set

$$\mathfrak{A}_R^{[k]} = \mathbb{C}[t_{ij}] / ((\det T) I_p(\gamma_R \text{Sym}^2 T) + \mathfrak{m}^{k+1}). \quad (8.1)$$

An isomorphism below means an isomorphism of complex algebras, without a grading, tangent coordinates, matrix decomposition or chosen generators.

Theorem 8.3 (Unmarked local reconstruction). *For every $n \geq 3$ and every pair of quadratic pencils,*

$$\mathfrak{A}_R^{[d]} \simeq \mathfrak{A}_{R'}^{[d]} \iff R' = (\text{Sym}^2 g)R \text{ for some } g \in \text{GL}(V). \quad (8.2)$$

The algebra has embedding dimension n^2 and length $\binom{n^2+d}{d} - \binom{N}{2}$. Every lower-order algebra $\mathfrak{A}_R^{[k]}$, $k < d$, is independent of R . Thus $d = n^2 + 2n - 4$ is the smallest uniform reconstruction order already among these single-point objects.

Proof. Theorem 8.2 applies with $B = \text{Spec } \mathbb{C}$ and the bundle U . To spell out what its intrinsic construction uses in this case, the unique maximal ideal $\mathfrak{n}$ of the finite algebra determines its tangent space and the multiplication kernel

$$K_R = \ker \left[\text{Sym}^d(\mathfrak{n}/\mathfrak{n}^2) \longrightarrow \mathfrak{n}^d \right].$$

This is precisely the degree- d ideal in the chosen presentation; nonlinear changes of generators contribute only terms above degree d to a degree- d relation. Hence an ungraded algebra isomorphism identifies the homogeneous cones generated by these kernels. Their determinant reductions recover the unordered matrix factors. The residual support ranks $(1, M)$ rule out transposition, and the remaining left-right map carries the left coefficient line to the left coefficient line. Exterior duality then gives $R' = (\text{Sym}^2 g)R$. Conversely such a map, together with any chosen right identification, induces the algebra isomorphism. There are no relations below degree d , so the embedding dimension and the lower-order assertion are immediate. The relation subspace at degree d has dimension M , giving the asserted length. The inequivalent common-factor and no-common-factor pencils prove uniform minimality as before. $\square$

In particular the earlier Schubert-line construction is a sufficient geometric realization, not a lower bound on the dimension of a supporting reduced scheme. The local inverse just proved retains no curve at all. Its additional ingredient is the coefficient-support asymmetry, which was not used in the earlier geometric orientation argument. This is a stronger inverse statement, not a conclusion from knowing the discriminant or the reduced determinant cone.

8.3. The local automorphism kernel is explicit. Put $a = n^2$, $\ell = \text{length } \mathfrak{A}_R^{[d]}$, and $G_R = \{g \in \text{GL}(V) : (\text{Sym}^2 g)R = R\}$. The symbol $\mathfrak{n}^2$ in the next statement is the square of the maximal ideal of the finite algebra.

Proposition 8.4 (Tangent-identity automorphisms). *There is a split exact sequence of complex automorphism groups*

$$1 \longrightarrow U_R \longrightarrow \text{Aut}_{\mathbb{C}}(\mathfrak{A}_R^{[d]}) \longrightarrow (G_R \times \text{GL}(U))/\mathbb{C}^* \longrightarrow 1, \quad (8.3)$$

where the scalar subgroup is diagonal. The kernel U_R is a unipotent algebraic group whose underlying variety is an affine space of dimension

$$a(\ell - 1 - a).$$

More explicitly, after choosing the a degree-one generators $x_1, \dots, x_a$, its elements are exactly the substitutions $x_i \mapsto x_i + u_i$ with arbitrary $u_i \in \mathfrak{n}^2$. The affine-space parametrization is not an assertion that this group law is additive.

Proof. The local inverse and the support-rank argument identify the linear stabilizer of K_R with the indicated left-right quotient; transposition is absent. These linear maps lift homogeneously, giving the quotient and splitting. An automorphism in the kernel is the identity on $\mathfrak{n}/\mathfrak{n}^2$, and consequently has exactly the displayed generator images.

Conversely choose arbitrary $u_i \in \mathfrak{n}^2$. Under this substitution a homogeneous defining relation of degree d changes only by terms in $\mathfrak{n}^{d+1} = 0$, so all defining relations are respected. The induced map is the identity on the associated graded algebra, hence is invertible as a finite filtered linear map and therefore as an algebra map. There are no further conditions on the u_i . Reading the generator images and performing substitutions are polynomial mutually inverse maps between the kernel and the vector space $(\mathfrak{n}^2)^{\oplus a}$. Since $\dim \mathfrak{n}^2 = \ell - 1 - a$, this proves the variety and dimension assertions. On a basis ordered by degree every such automorphism is upper unitriangular; the algebraic group is therefore unipotent. Its law includes higher substitution terms, which accounts for its generally noncommutative structure. $\square$

The positive-dimensional family statement remains useful after the local inverse: it recovers an entire varying subbundle and the actual source bundle up to one global left transformation, whereas separate fibrewise local isomorphisms need not glue. The geometric automorphism statements here still do not identify unmarked moduli stacks or all deformations over nonreduced parameter bases.

9. THE INTRINSIC POLARIZATION AND A PROJECTIVE MODULI COMPLETION

Put $P = \text{Gr}(2, \text{Sym}^2 V)$, $N = \binom{n+1}{2}$, $M = \binom{N}{2}$, and $D = n^2 + 2n - 4$. Fix temporarily an auxiliary n -space U and put $E = \text{Hom}(U, V)^*$. Write $L = \mathcal{O}_P(1)$ for the Pluecker line and $\Gamma = \Phi(P) \subset \text{Gr}(M, \text{Sym}^D E)$ for the closed image of Theorem 7.3. The group in this section is $\text{SL}(V)$, acting by congruence on pencils and by its induced action on E , with U fixed. It is not the full group $\text{GL}(E)$.

Theorem 9.1 (Polarized completion of the effective failure moduli). *There is an $\text{SL}(V)$ -equivariant isomorphism*

$$\Phi^* \mathcal{O}_{\text{Gr}(M, \text{Sym}^D E)}(1) \simeq L^{\otimes M} \otimes C_U, \quad (9.1)$$

where C_U is a constant one-dimensional space on which $\text{SL}(V)$ acts trivially. Consequently the stable and semistable loci of Γ , for the restricted Pluecker linearization, are respectively the images of P^s and P^{ss} . The quotient defined by the complete restricted section ring is canonically

$$\overline{\mathcal{M}}_n^{\text{fail}} = \text{Proj} \left(\bigoplus_{q \geq 0} H^0(P, L^{\otimes q})^{\text{SL}(V)} \right) \simeq \Gamma // \text{SL}(V). \quad (9.2)$$

It is a normal projective variety. It contains the geometric quotient of the simple-discriminant pencils as a dense open subvariety, and has dimension $n - 3$. The complement of that open is the image of the semistable discriminant-zero locus.

The intrinsic inverse identifies the effective semistable failure stack with $[P^{ss} / \text{PGL}(V)]$ and gives a moduli morphism to (9.2). On framed charts the failure-algebra family extends over all P , remains finite locally free, and commutes with arbitrary base change. These assertions do not require, and do not assert, the existence of a universal ordinary algebra on the coarse quotient or properness of the entire quotient stack.

Proof. Let $\mathcal{K}$ be the tautological relation bundle pulled back by Φ . The coefficient factorization (7.8) identifies it with $\mathcal{B} \otimes F(U)$, where $\mathcal{B}$ is the quotient-volume coefficient line. Exterior duality gives $\mathcal{B} \simeq L^{-1}$ up to a fixed determinant character of $\text{GL}(V)$. That character is trivial on $\text{SL}(V)$. Taking the dual determinant gives (9.1), with the harmless constant factor $(\det F(U))^*$ and the corresponding fixed determinant line of V . A trivialization of the constant factor identifies the section rings; a different trivialization rescales each graded piece and has no effect on its projective spectrum. The section ring for L^M is the M th Veronese of the section ring for L . Taking invariants commutes with this operation. The Hilbert–Mumford weights are multiplied by $M > 0$, so stability and semistability are unchanged. This also explains why one uses the

complete section ring on Γ , rather than assuming surjectivity of unspecified ambient invariant restrictions.

Finite generation and the projective good quotient are the reductive GIT construction of [25, Chapter 1]. The complete section ring of the normal projective variety P is integrally closed: it is the intersection of the valuation conditions for rational sections of powers of L at the prime divisors of P . Its invariant subring is integrally closed as well. Indeed an element of its fraction field integral over the invariants is integral over the original ring, belongs to that ring by normality, and is invariant. This proves normality of the projective quotient.

For completeness, the simple-discriminant open is nonempty and stable. In a frame (Q_0, Q_1) of a pencil, take the discriminant of $\det(sQ_0 + tQ_1)$. Under a frame change $h \in \mathrm{GL}_2$ it has weight $(\det h)^{n(n-1)}$, and it is invariant under $\mathrm{SL}(V)$. Thus it is a nonzero invariant section of $L^{n(n-1)}$. It is nonzero at a diagonal pencil with n distinct roots, so its nonvanishing locus is semistable and saturated. A pencil there has a nonsingular member. Relative to that member, the other member is self-adjoint with distinct eigenvalues; orthogonal diagonalization identifies its congruence class with an unordered configuration of n distinct points of $\mathbb{P}^1$. The latter configurations are stable for the usual binary-form linearization: the root-multiplicity criterion is strict since $1 < n/2$ for $n \geq 3$. Their quotient is geometric and separated. It follows that the congruence orbits are closed in the saturated nonvanishing locus. Their stabilizers are finite: the induced projective transformation permutes a set of at least three points, and the kernel consists of diagonal sign changes modulo scalars. They are therefore stable. This gives a geometric dense open of dimension $n - 3$. The complement is the image of the invariant section's zero locus by the good-quotient property. This reasoning uses classical pencil classification and GIT, not a new classification of binary forms; compare [4, 25].

Finally apply Theorem E.2 and the rigidification results of Section F. The Pluecker polarization is taken on the recovered pencil, not on an arbitrary first-relation orbit. A sufficiently divisible power of L , for example L^n , has trivial central character and descends from $\mathrm{SL}(V)$ to $\mathrm{PGL}(V)$. A change of auxiliary frame changes the constant factor in (9.1), not the resulting projective quotient. On the framed Grassmannian the extended algebras are exactly those of Corollary 7.5. Their equivariance yields the stated classifying morphisms. Removing ineffective inertia does not turn these into a universal ordinary algebra on the coarse space, and no such descent is used. $\square$

Remark 9.2 (What is compactified). The completion is the classical polarized pencil quotient recovered from the intrinsic failure invariant, with its polarization verified by (9.1). It is not a new construction of classical GIT. Its role is to supply a proper coarse target for the effective invariant and a specified class of semistable limiting families. Strictly semistable points represent closed-orbit representatives, not every orbit in their fibres. In particular Segre data on the stack need not descend to an invariant separating all points of a coarse-quotient fibre. For $n = 3$ the quotient is a point and its discriminant boundary can be empty. The full $\mathrm{GL}(E)$ common limit of Theorem 42.1 is a different construction.

10. SPECTRAL DIVISOR FIBRES AND THE RECOVERY OF DEGENERATION TYPE

The preceding completion concerns the native congruence action. To retain degeneration type before passage to a coarse quotient, we associate a tower of binary divisor incidences to a regular pencil. This construction is intrinsic after the unmarked inverse and is unchanged by a simultaneous change of the pencil and source frames.

Let $\Lambda = \mathbb{P}(R)$ be the pencil line and let $\mathcal{T}_R$ be the cokernel of the universal symmetric map on Λ . A regular pencil means that its determinant is not identically zero. Thus $\mathcal{T}_R$ is a torsion sheaf. Define $D_j(R)$ by

$$\mathcal{I}_{D_j(R)} = \mathrm{Fitt}_j(\mathcal{T}_R), \quad 0 \leq j \leq n, \quad (10.1)$$

where $D_n = \emptyset$. These are effective divisors on the smooth curve Λ . Write $s_j = \deg D_j$. If $s_j > 0$, let G_j be its binary equation and put

$$J_j = G_j H^0(\Lambda, \mathcal{O}_\Lambda(1)) \subset H^0(\Lambda, \mathcal{O}_\Lambda(s_j + 1)).$$

Changing the scalar representative of G_j does not change J_j . Let $\mathcal{F}_j(R)$ be the whole scheme fibre at J_j of the degree-one divisor normalization ν_1 , with $r = 2$ and $D = s_j + 1$. For $s_j = 0$ put $\mathcal{F}_j = \emptyset$.

Theorem 10.1 (A fibre-theoretic Segre invariant). *There is a natural identification of finite schemes over Λ*

$$\mathcal{F}_j(R) \simeq D_j(R). \quad (10.2)$$

At a spectral point x , write the positive elementary divisors of $\mathcal{T}_R$ as $\lambda_{x,1} \geq \dots \geq \lambda_{x,r_x} > 0$, and extend this sequence by zeroes. Then

$$(\mathcal{F}_j(R))_x = \text{Spec } \mathbb{C}[\epsilon]/(\epsilon^{b_{x,j}}), \quad b_{x,j} = \sum_{i>j} \lambda_{x,i}, \quad (10.3)$$

with the empty-scheme convention when $b_{x,j} = 0$. Consequently

$$\lambda_{x,j+1} = b_{x,j} - b_{x,j+1}. \quad (10.4)$$

The tower, with its maps to the same intrinsic spectral line, therefore determines the full Segre symbol. Conversely, the elementary-divisor partitions together with their spectral support on that line determine the tower. The tower retains the projective configuration of the support as well as the Segre symbol. It is an invariant of the unmarked finite failure algebra.

The construction is compatible with base change on every family stratum where the Fitting ideals in (10.1) define relative effective Cartier divisors of fixed degrees. On such a stratum $\mathcal{F}_j$ is finite locally free of degree s_j . These hypotheses are part of the relative statement; no assertion of flatness across a jump of s_j is intended.

Proof. Over the discrete valuation ring $\mathcal{O}_{\Lambda,x}$, Smith normal form gives diagonal entries with exponents $\lambda_{x,r_x}, \dots, \lambda_{x,1}$ and $n - r_x$ zero exponents. The minors of size $n - j$ have greatest common divisor of order $\sum_{i>j} \lambda_{x,i}$. This proves the formula for D_j , including its nonreduced structure. It is the usual Fitting-ideal description of elementary divisors, used here on the intrinsic spectral line.

The residual two-space $H^0(\mathcal{O}_\Lambda(1))$ is gcd-free. Hence Theorem 14.1 identifies the degree-one normalization fibre at J_j with the scheme of linear factors of G_j . On an affine chart write a monic linear factor as $z - a$. Monic division gives $G_j = (z - a)H + G_j(a)$, so the entire factor scheme is $G_j(a) = 0$, not its reduction. These descriptions agree on overlaps of the projective line. They give (10.2) and the local algebra (10.3). Taking successive differences proves (10.4). The common map to Λ is essential: forgetting it separately in each member of the tower would discard the alignment of the spectral supports.

Pencil frame changes act on Λ by projective transformations; source congruence changes give isomorphic cokernels and preserve all Fitting ideals. The intrinsic inverse of Theorem 8.3 therefore defines this invariant from the unmarked algebra. The equivalence with Segre data uses the classical elementary-divisor convention of [4, §5].

For a family with the stated Cartier and degree conditions, the same monic division argument works over any base algebra, including one with nilpotents. It represents the divisor's functor of points, so it commutes with base change. A relative effective divisor of fixed degree on a projective-line bundle is finite locally free of that degree. Local choices of G_j differ by units and glue. Formation of Fitting ideals itself commutes with base change; requiring them to remain flat Cartier divisors is precisely what permits the additional finite-flat conclusion. $\square$

Example 10.2 (Equal discriminant, different fibre towers). At a root of multiplicity four, the partitions $(3, 1)$ and $(2, 2)$ have the same zeroth fibre $\mathbb{C}[\epsilon]/(\epsilon^4)$. Their first fibres are respectively $\mathbb{C}$ and $\mathbb{C}[\epsilon]/(\epsilon^2)$; all higher fibres are empty there. Thus the discriminant alone does not detect the distinction, whereas the divisor-fibre tower does. Both are realized by regular symmetric pencils: for a block of size l , take the reversal matrix H_l and the symmetric pair $(H_l, H_l J_l(\alpha))$, where $J_l(\alpha)$ is a Jordan block. Direct sums realize every indicated partition. Additional simple blocks may be appended. The example is a statement on the regular-pencil stack, not a claim that distinct Segre strata always remain distinct after strictly semistable GIT identification.

Remark 10.3 (The two divisor constructions). The primitive first relation of an actual pencil has exact common factor $(\det T)^{n-1}$, by Lemma E.1. Its common-factor degree does not jump when the pencil changes Segre type. The divisors D_j above instead lie on the recovered spectral line and define auxiliary binary first-relation subspaces J_j . Thus (10.4) relates pencil data to intrinsically induced divisor ramification, not to a nonexistent change in the conductor of the original exact-gcd locus. The identification supplies a moduli consequence of the all-binary-fibre theorem while respecting this distinction.

Part II. Divisor incidence and homological fibres

11. COMMON-DIVISOR STRATA AND INTRINSIC FAMILIES

The coefficient condition in Theorem B.1 is not imposed in this section. We instead retain the common divisor of a space of first relations. This geometric condition gives a larger smooth parameter space, includes relations without either factor symmetry, and has a scheme-theoretic inverse. The distinction between a stratum and the condition on its geometric points is essential when the parameter base is nonreduced.

Write $S_j(E) = \text{Sym}^j E$, where E is a complex vector space. For $r \geq 2$, denote by

$$X_{h,r}(E) \subset \text{Gr}(r, S_h(E))$$

the locus of subspaces with greatest common divisor equal to 1, up to a scalar. Empty Grassmannians are omitted throughout. When $h = 0$ this notation can also be used for $r = 1$.

Lemma 11.1 (Universal common divisor). *Let $D = g + h$. The spaces of r degree- D forms whose exact greatest common divisor has degree g form a locally closed reduced subscheme Z_g of $\text{Gr}(r, S_D(E))$. Multiplication induces an isomorphism*

$$\mathbb{P}(S_g(E)) \times X_{h,r}(E) \xrightarrow{\sim} Z_g, \quad ([f], J) \mapsto fJ. \quad (11.1)$$

In particular, the common-divisor line and the residual subbundle of a family with classifying map in Z_g are recovered uniquely and commute with arbitrary complex base change.

Proof. For each a , multiplication from $\mathbb{P}(S_a(E)) \times \text{Gr}(r, S_{D-a}(E))$ has closed image: its domain is projective. The union for $a > g$ is closed. Remove this union from the image for $a = g$, and give the resulting locally closed set its reduced structure. Its points are exactly those in the statement. The inverse image of this set under multiplication is $\mathbb{P}(S_g(E)) \times X_{h,r}(E)$. The same closed-image argument with $D = h$ shows that $X_{h,r}(E)$ is open.

The restricted multiplication map is proper, by base change. It is injective on geometric points over every algebraically closed extension of $\mathbb{C}$, since a greatest common divisor is unique up to scalar in a polynomial ring. Its differential is injective as well. Indeed, a tangent vector in its kernel satisfies

$$\dot{f}j + f\dot{j} \in fJ \quad (j \in J).$$

Consequently $f \mid \dot{f}j$ for every $j \in J$. For each irreducible factor of f , some element of J is not divisible by that factor. Comparison of its valuation gives $f \mid \dot{f}$. Equality of degrees makes $\dot{f}$ a scalar multiple of f ; then $\dot{J} = 0$ in $\text{Hom}(J, S_h(E)/J)$.

In characteristic zero the geometric-point injectivity and this differential calculation imply that the map is universally injective and unramified. A proper, universally injective, unramified morphism is a closed immersion [12, Lemma 41.7.2]. Here its image is all of the reduced target. The defining ideal of a closed immersion with this support is zero in a reduced scheme. This proves the isomorphism, including its assertion for nonreduced test schemes. Equivalently, one can check properness on the base change of the original projective multiplication map; that base change has the same underlying open preimage. Its source is reduced because it is this open subscheme of the original smooth product. $\square$

Remark 11.2. A map from a nonreduced scheme is required to factor through the locally closed scheme Z_g , not merely to have geometric points in its underlying set. These are different requirements. For example, over $\mathbb{C}[\epsilon]/(\epsilon^2)$ the span of $x^2 + \epsilon y^2, xy$ specializes to a pair with common divisor x , but has no factor $x + \epsilon(ax + by)$. Comparison of the y^2 coefficient in the first form proves this. The stratum in Lemma 11.1 excludes this transverse infinitesimal family. No use of a fibrewise radical can replace its scheme structure.

Now let $\dim V = \dim U = n \geq 3$, $E = V^* \otimes U$, and $\delta = \det T$. Denote by $G \subset \text{GL}(E)$ the stabilizer of the line $\mathbb{C}\delta$. By Lemma 4.1,

$$G^\circ = (\text{GL}(V) \times \text{GL}(U))/\mathbb{G}_m, \quad G/G^\circ = \mathbf{Z}/2\mathbf{Z}. \quad (11.2)$$

The second component exchanges the two tensor factors; the diagonal scalars in the displayed product act trivially on the matrix space. Fix $s \geq 1$ and $D = sn + h$. Inside Z_{sn} , impose the condition that the common-divisor line is the s -th power of a determinant form after a linear change of the n^2 variables. Call the resulting locally closed scheme $Z_{s,h,r}^{\det}$.

Theorem 11.3 (Intrinsic determinant-divisor stratum). *There are canonical isomorphisms of schemes and algebraic stacks*

$$Z_{s,h,r}^{\det} \simeq \text{GL}(E) \times^G X_{h,r}(E), \quad [Z_{s,h,r}^{\det} / \text{GL}(E)] \simeq [X_{h,r}(E)/G]. \quad (11.3)$$

They hold over arbitrary complex parameter schemes. The stack is smooth. At a residual relation J , its tangent complex, in degrees $-1, 0$, is

$$[\text{Lie}(G) \longrightarrow \text{Hom}(J, S_h(E)/J)], \quad X \longmapsto (j \mapsto Xj \bmod J). \quad (11.4)$$

No invariance of J under either full factor group is assumed. The equivalence identifies stabilizers, infinitesimal families, and specializations whose classifying maps remain in this stratum.

Proof. The power map $\mathbb{P}(S_n(E)) \rightarrow \mathbb{P}(S_{sn}(E))$, $[f] \mapsto [f^s]$, is a closed immersion in characteristic zero. It is proper and geometrically injective, and its differential has kernel given by $sf^{s-1}\dot{f} \in \mathbb{C}f^s$, hence by $\dot{f} \in \mathbb{C}f$. The same closed-immersion criterion as above applies. The orbit of $[\delta]$ is the smooth locally closed homogeneous space $\text{GL}(E)/G$. The preimage of its power orbit under the first projection in (11.1) is therefore a locally closed subscheme. It is exactly $Z_{s,h,r}^{\det}$.

Pulling back along the G -torsor $\text{GL}(E) \rightarrow \text{GL}(E)/G$ trivializes the common-divisor form, leaving the unrestricted residual relation $J \in X_{h,r}(E)$. Descent gives the first isomorphism in (11.3); descent of equivariant maps and morphisms gives the second. This proof uses morphisms of schemes, not a comparison only on complex points. It also supplies local lifts of all isomorphisms over an fppf cover.

The space $X_{h,r}(E)$ is open in a Grassmannian and G is smooth. The quotient is algebraic and smooth [13, Theorem 97.17.2]; see also [14, Lemma 101.33.7]. For completeness, write a first-order residual subspace as the graph of $\epsilon\phi : J \rightarrow S_h(E)/J$. An infinitesimal change

$1 + \epsilon X$ replaces ϕ by $\phi + X|_J \bmod J$. This proves the tangent complex. Its kernel records infinitesimal automorphisms and its cokernel records first-order isomorphism classes. The same torsor description proves compatibility with specialization. No assertion is made about limits outside the specified common-divisor stratum. $\square$

The next step connects these parameter spaces with ungraded finite algebras. It is useful to state the family condition intrinsically, since an absolute nilradical is unsuitable over a nonreduced base. Put $e = \dim E$ and $\ell = \binom{e+D}{D} - r$. Consider finite locally free commutative unital algebras $\mathcal{A}$ of rank ℓ on a complex scheme T_0 , subject to the following conditions. The map $\epsilon(a) = \ell^{-1} \operatorname{tr}(M_a)$ is multiplicative. For $\mathcal{N} = \ker \epsilon$, one has $\mathcal{N}^{D+1} = 0$; all quotients of its power filtration are locally free; $\mathcal{E} = \mathcal{N}/\mathcal{N}^2$ has rank e ; and multiplication gives isomorphisms $\operatorname{Sym}^j \mathcal{E} \simeq \mathcal{N}^j/\mathcal{N}^{j+1}$ for $j < D$. In degree D its kernel has rank r . These conditions specify the maximal-lower-Hilbert-function stratum; they do not assert that arbitrary smoothings of a local algebra remain in it.

Proposition 11.4 (Relative first-relation algebra). *Every such algebra is locally isomorphic, over its base, to*

$$\mathcal{A}(\mathcal{K}) = \operatorname{Sym} \mathcal{E} / ((\mathcal{K}) + \mathfrak{m}^{D+1}), \quad \mathcal{K} = \ker(\operatorname{Sym}^D \mathcal{E} \longrightarrow \mathcal{N}^D).$$

Its trace augmentation, $\mathcal{E}$, and $\mathcal{K}$ are intrinsic and compatible with arbitrary base change in this stratum. For two such algebras, taking the linear part gives a locally surjective map of isomorphism sheaves to the sheaf of isomorphisms of $(\mathcal{E}, \mathcal{K})$. Its kernel is the smooth substitution group whose underlying scheme in a local presentation is $\operatorname{Hom}(\mathcal{E}, \mathcal{N}^2)$.

After quotienting morphisms by this tangent-identity kernel and fppf stackifying, the algebras with first-relation map in $Z_{s,h,r}^{\det}$ have stack $[X_{h,r}(E)/G]$.

Proof. Locally lift a basis of $\mathcal{E}$ to $\mathcal{N}$. Nilpotence and multiplication on the successive quotients make the induced map from the truncated symmetric algebra surjective. A relation with a nonzero term of least degree $j < D$ would give a nonzero kernel in degree j of the associated graded algebra, which is impossible. Its entire kernel therefore lies in degree D and is precisely $\mathcal{K}$. Changing a lift by an element of $\mathcal{N}^2$ does not alter a product of D generators, since the difference has degree at least $D + 1$. This also proves the asserted uniqueness of the first relation.

In this presentation multiplication by a positive-degree element is strictly filtration-increasing. Its trace is zero over any base ring. Multiplication by a scalar has trace ℓ times that scalar, proving that the presentation's augmentation is the specified intrinsic trace augmentation. All filtration sequences are locally split sequences of vector bundles, so their kernels and quotients commute with base change.

An isomorphism must preserve trace and induces a linear map carrying $\mathcal{K}$ to $\mathcal{K}'$. Conversely such a map lifts locally to an isomorphism by the homogeneous presentations. All other lifts are obtained by substituting generators plus arbitrary elements of $\mathcal{N}^2$; the inverse is constructed degree by degree in a finite filtration. This gives the stated kernel and its smoothness, without replacing its substitution law by addition. After identifying morphisms with the same linear part, frames of $\mathcal{E}$ give the quotient $[Z_{s,h,r}^{\det}/\operatorname{GL}(E)]$. Theorem 11.3 completes the proof. $\square$

Remark 11.5. The original algebra stack is not this quotient stack: it has the additional nonlinear inertia just computed. The phrase *effective first-relation stack* will refer to the specified quotient on morphisms followed by stackification. The construction retains the full automorphism group before taking that quotient. It does not identify algebras with the same geometric fibres, nor does it discard nilpotent parameters. In particular, its relation to a stratum of first relations is stronger than a bijection of isomorphism classes over $\mathbb{C}$.

12. THE COMMON-DIVISOR BOUNDARY AND ITS NORMALIZATION

The exact-common-divisor stratum has a canonical completion which need not be normal. Its normalization records a divisor, and its failure to be an isomorphism is controlled by two different phenomena: choices of that divisor and overlap with the residual common factor. We establish these assertions for arbitrary spaces of forms. No determinant, Cauchy, or Pluecker equation is imposed in this section.

Let E have dimension $e \geq 2$, put $S_j = \text{Sym}^j E$ and $s_j = \dim S_j = \binom{e+j-1}{j}$, and fix

$$g, h \geq 1, \quad D = g + h, \quad 2 \leq r \leq s_h.$$

Write $\mathcal{G} = \text{Gr}(r, S_D)$. A *degree- g divisor incidence* over a complex scheme T_0 consists of a line subbundle $\mathcal{L} \subset S_g \otimes \mathcal{O}_{T_0}$ and a rank- r subbundle $\mathcal{K} \subset S_D \otimes \mathcal{O}_{T_0}$ such that

$$\mathcal{K} \longrightarrow (S_D \otimes \mathcal{O}_{T_0})/(\mathcal{L} \cdot S_h) = 0. \quad (12.1)$$

Subbundle always means that the quotient is locally free. In particular, the form generating $\mathcal{L}$ is nonzero on every geometric fibre. Condition (12.1) is the vanishing of a bundle map, not a condition imposed only on closed points.

Proposition 12.1 (Canonical divisor incidence). *The incidence functor is represented, with its full scheme structure, by*

$$\mathcal{X}_g = \mathbb{P}(S_g) \times \text{Gr}(r, S_h).$$

Its forgetful morphism is multiplication

$$\nu_g : \mathcal{X}_g \longrightarrow \mathcal{G}, \quad ([f], J) \longmapsto fJ. \quad (12.2)$$

This morphism is finite. Its scheme-theoretic image Y_g is an integral projective scheme. If K is a geometric point of Y_g and $F_K = \gcd(K)$, the underlying set of $\nu_g^{-1}(K)$ is

$$\{[f] : f \mid F_K, \deg f = g\}, \quad (12.3)$$

with residual space $J = K/f$. The incidence construction and its scheme-theoretic fibres commute with arbitrary complex base change.

Proof. On $\mathbb{P}(S_g)$, multiplication from the tautological line tensored with S_h to S_D has constant rank s_h . Indeed, multiplication by a nonzero polynomial over a field is injective. A nonzero maximal minor locally splits this map, so its image and cokernel are vector bundles, also after a nonreduced base change. The closed incidence (12.1) is therefore the relative Grassmannian of rank- r subbundles of $\mathcal{L} \otimes S_h$. Tensoring by $\mathcal{L}^{-1}$ identifies it canonically with the stated product. In more elementary terms, $\mathcal{K}$ has a unique inverse image in $\mathcal{L} \otimes S_h$; its quotient there is locally free because the quotient of $S_D/\mathcal{K}$ onto $S_D/(\mathcal{L} \cdot S_h)$ is a surjection of vector bundles. This proves the functorial assertion, including its scheme structure.

The source of ν_g is projective. Unique factorization in the polynomial ring over an algebraically closed field gives (12.3). A fixed polynomial has only finitely many divisors of a fixed degree, up to scalar. The morphism is thus proper with finite fibres and is finite [15, Tag 02LS]. Define Y_g by the kernel of $\mathcal{O}_{\mathcal{G}} \rightarrow (\nu_g)_* \mathcal{O}_{\mathcal{X}_g}$, as in [16, Tag 01R5]. Its integrality follows from the integrality of $\mathcal{X}_g$: on an affine open meeting the image, the coordinate ring of the image injects into the domain which is the coordinate ring of its finite inverse image. Reducedness is therefore a consequence of this incidence construction, not an extra choice. Representability of incidence gives the base-change assertion. $\square$

Remark 12.2. The condition defining Y_g is the existence of a divisor of *degree* g . In several variables it is not the condition $\deg \gcd(K) \geq g$: an irreducible polynomial of larger degree need not have a degree- g factor. Nor do we assert that formation of the scheme-theoretic image, or normalization itself, commutes with arbitrary nonflat base change. The assertion for arbitrary

bases in Proposition 12.1 concerns the incidence functor and the base change of its representing morphism.

We next retain the nilpotents of a fibre. This will distinguish a ramified divisor choice from several distinct divisor choices.

Lemma 12.3 (Fibre equations and the overlap space). *Let $K = fJ$, set $H = \gcd(J)$, and put $c = \gcd(f, H)$ and $k = \deg c$. Then the tangent space of the scheme-theoretic fibre at $([f], J)$ is naturally, after the displayed representatives have been chosen,*

$$\ker(d\nu_g)_{([f], J)} \simeq S_k/\mathbb{C}c, \quad \dim \ker(d\nu_g) = s_k - 1. \quad (12.4)$$

More explicitly, choose a projective coefficient chart $f(u) = f + \sum u_\alpha m_\alpha$, with one coefficient normalized to its value in f . Let $M(f(u))$ be the matrix of multiplication $S_h \rightarrow S_D$. Choose a set A_0 of s_h rows for which $M_{A_0}(f)$ is invertible, and let B_0 denote the remaining rows. For a matrix C whose columns form a basis of K , the full fibre on this chart is defined by

$$C_{B_0} - M_{B_0}(f(u))M_{A_0}(f(u))^{-1}C_{A_0} = 0, \quad \det M_{A_0}(f(u)) \neq 0. \quad (12.5)$$

In particular its completed local ring is the power-series ring in the $s_g - 1$ variables u_α , modulo the entries in (12.5); no radical is taken.

Proof. For each column of C , its possible quotient by $f(u)$ is forced by the A_0 rows to be $M_{A_0}^{-1}C_{A_0}$. The remaining rows are exactly the condition that this quotient works. These quotients span a subbundle wherever the equations hold, by the incidence argument above. Thus elimination has introduced neither additional solutions nor discarded nilpotents, proving the fibre presentation over arbitrary coefficient rings in the chart.

A tangent vector is a pair $(\dot{f} \bmod \mathbb{C}f, \dot{J})$. It lies in the fibre precisely when

$$\dot{f}j + f\dot{j} \in fJ \quad \text{for all } j \in J.$$

It is necessary and sufficient that $f \mid \dot{f}j$ for every $j \in J$; the class of $\dot{j}$ modulo J is then forced. Taking the minimum valuation over a basis of J , for each irreducible polynomial, makes this equivalent to $f \mid \dot{f}H$. Writing $f = cf_1$, $H = cH_1$, with $\gcd(f_1, H_1) = 1$, gives $\dot{f} = f_1w$ with $w \in S_k$. The scalar tangent direction $\dot{f} \in \mathbb{C}f$ is exactly $w \in \mathbb{C}c$. This proves (12.4) and its dimension assertion. $\square$

Theorem 12.4 (Normalization and the exact normal locus). *The finite map*

$$\nu_g : \mathbb{P}(S_g) \times \text{Gr}(r, S_h) \longrightarrow Y_g$$

is the normalization of the natural degree- g divisor locus. Its dimension is

$$\dim Y_g = s_g - 1 + r(s_h - r). \quad (12.6)$$

For a geometric point K , the scheme Y_g is normal at K if and only if F_K has exactly one degree- g divisor $[f]$ and

$$\gcd(f, F_K/f) = 1. \quad (12.7)$$

Normality at a point is also equivalent to smoothness there. Thus the two precise obstructions are more than one divisor choice, or overlap between the selected divisor and the residual common factor. The latter gives the fibre tangent dimension in (12.4), and (12.5) determines its higher nilpotents.

Proof. The locus of gcd-free r -planes in S_h is nonempty: a space containing x_1^h, x_2^h is one example, and it can be extended to every allowed rank r . It is open, since the loci possessing a positive-degree divisor are the finitely many closed images of projective multiplication maps. It is therefore dense in the irreducible Grassmannian. Its product with $\mathbb{P}(S_g)$ maps to the locus where the exact gcd has degree g .

This last locus is open in Y_g : remove the finitely many closed images for divisor degrees $a > g$. Call it U_g . Every fibre over a geometric point of U_g has one point and, by Lemma 12.3 with $H = 1$, zero tangent space. A finite local algebra over an algebraically closed field with zero cotangent space is the field itself, by Nakayama's lemma. The fibre consequently has length one, not merely one point.

Here is the local argument which also settles the scheme structure. At a closed point of U_g , write A for the local ring of Y_g , and B for its finite inverse-image algebra. The map $A \rightarrow B$ is injective, and $B/\mathfrak{m}_A B = \mathbb{C}$, generated by 1. Nakayama's lemma, applied to the finite module B/A , gives $B = A$. The non-isomorphism locus of a finite morphism is closed, so the same conclusion holds throughout U_g . Hence ν_g is birational. The source is smooth and integral, thus normal. On affine opens, a finite birational extension $A \subset B$ with B integrally closed is the integral closure of A in the common fraction field: every element integral over A is integral over B and hence belongs to B . This proves the normalization assertion; compare [17, Tag 035E]. The dimension follows from the product description.

If Y_g is normal at K , its normalization is an isomorphism there, so its fibre is one reduced point. Formula (12.3) gives uniqueness of $[f]$, and (12.4) gives $s_k - 1 = 0$, equivalent to $k = 0$ because $e \geq 2$. Conversely uniqueness and $k = 0$ again make the entire fibre a single reduced point. The same finite local Nakayama argument shows that the normalization is an isomorphism near K . Its source is normal, so Y_g is normal at K . The source is smooth, so every normal point of the target is smooth; the converse holds for every complex variety. This proves both directions without replacing a nonreduced fibre by its set of points. $\square$

Corollary 12.5 (Universality of the exact stratum). *The open subscheme $U_g \subset Y_g$ represents the unmarked relation families in the divisor incidence whose residual space is gcd-free on every geometric fibre. Over it the divisor line and residual subbundle are unique and commute with arbitrary base change. Moreover U_g , as a locally closed subscheme of $\mathcal{G}$, is the reduced exact-gcd stratum of Lemma 11.1.*

Proof. The inverse image of U_g is $\mathbb{P}(S_g) \times X_{h,r}(E)$, and the proof above shows that its finite map to U_g is an isomorphism. Its source is reduced, so U_g has the reduced induced structure on its locally closed support. The incidence functor supplies the asserted universal property. A family in the ambient Grassmannian which only has the right geometric gcd need not factor through Y_g . In particular, the transverse dual-number family of Remark 11.2 is not admitted by this corollary. The corollary is a scheme-theoretic identification, not a claim that fibrewise gcd tests detect nilpotent equations. $\square$

Example 12.6 (A nonreduced fibre at a divisor collision). Take $E = \langle x, y, z \rangle$, $g = 1, h = 2, r = 2$, and $K = \langle x^2 y, x^2 z \rangle$. Its gcd is x^2 , so the only point of the normalization fibre has divisor x . In the chart $f = x + ay + bz$, division by this monic linear form amounts to substitution $x = -ay - bz$. The two remainders are

$$(ay + bz)^2 y, \quad (ay + bz)^2 z.$$

Their coefficient ideal is (a^2, ab, b^2) . Thus the full fibre is

$$\text{Spec } \mathbb{C}[a, b]/(a, b)^2, \tag{12.8}$$

of length three and tangent dimension two. It is not a reduced point, despite uniqueness of the geometric divisor. The family

$$K_t = x \langle xy + ty^2, xz + tz^2 \rangle$$

has independent generators for every t , exact gcd x when $t \neq 0$, and the displayed collision at $t = 0$. Consequently this nonreduced fibre occurs on the boundary of the exact stratum, not in an unrelated example.

Example 12.7 (Several branches and a normal gcd jump). For the same g, h, r, E , let $K = xy\langle x, z \rangle$. The divisors x and y give two points of the normalization fibre. At both points the overlap is one, so the fibre consists of two reduced points. The point of Y_1 is nonnormal. By contrast, put $q = x^2 + yz$ and $K = xq\langle y, z \rangle$, now with $h = 3$. The polynomial q is irreducible and is not divisible by x . There is exactly one linear divisor of $\gcd K = xq$, and its overlap with the residual $\gcd q$ is one. This point of Y_1 is normal although its gcd degree is three. These two examples show why neither gcd degree nor the number of geometric lifts alone is a sufficient normality criterion.

Corollary 12.8 (An intrinsic boundary for first-relation algebras). *On the maximal-lower-Hilbert-function stratum of Proposition 11.4, the algebras whose intrinsic degree- D kernel has classifying map in Y_g have, after removing only tangent-identity substitutions, stack $[Y_g/\mathrm{GL}(E)]$. Its normalization is the finite representable morphism*

$$[\mathcal{X}_g/\mathrm{GL}(E)] \longrightarrow [Y_g/\mathrm{GL}(E)]. \quad (12.9)$$

All members of the universal relation family give finite locally free algebras of rank $\binom{e+D}{D} - r$. This construction imposes no pencil image equations and permits the divisor and residual common factor to collide.

Proof. For every subbundle $\mathcal{K}$ classified by the Grassmannian, $\mathrm{Sym} E/((\mathcal{K}) + \mathfrak{m}^{D+1})$ is, as a module, the sum of S_j for $j < D$ and $S_D/\mathcal{K}$. This proves finite local freeness and the rank. The first-relation proposition identifies the quotient after the specified nonlinear inertia is removed. Incidence and multiplication are $\mathrm{GL}(E)$ -equivariant. Pulling (12.9) back along the smooth frame atlas $Y_g \rightarrow [Y_g/\mathrm{GL}(E)]$ gives precisely ν_g . Finiteness, representability, and the normalization identification follow by descent; equivalently normality and integral closure may be checked on these smooth atlases. This quotient is not claimed to classify smoothings outside the specified Hilbert stratum. $\square$

For the pencil first relations one can take $e = n^2$, $g = n(n-1)$, $h = 3n-4$, and $r = \binom{n+1}{2}$. Every pencil then lies in the exact-gcd open of this canonical boundary. The boundary theorem concerns the much larger space of all degree- g divisorial first relations, rather than a closure defined by recovering a pencil and reimposing its Pluecker equations. It determines its normalization and normal locus, but does not assert a classification of all its linear orbits or of all finite-flat algebra deformations.

13. FORMAL IMAGES AND MARKED DIVISOR GERMS

All rings and schemes in the following sections are over $\mathbb{C}$. Write $S_j = \mathrm{Sym}^j E$, and let $Y_g \subset \mathrm{Gr}(r, S_D)$ be the scheme-theoretic image of

$$\nu_g : \mathbb{P}(S_g) \times \mathrm{Gr}(r, S_{D-g}) \longrightarrow \mathrm{Gr}(r, S_D), \quad (f, J) \longmapsto fJ,$$

where $\dim E \geq 2$, $g \geq 1$, and $2 \leq r \leq \dim S_{D-g}$. The source is smooth, and ν_g is its finite normalization: a fibre records the degree- g divisors of the common divisor, and on the exact-degree- g locus there is one reduced lift. We shall retain the scheme structures of these fibres and of the image.

Lemma 13.1 (Completed finite images). *Let $X \rightarrow M$ be a finite morphism of complex finite-type schemes, with X reduced and M smooth, and let Y be its scheme-theoretic image. Fix a closed point $y \in Y$, put $R = \hat{\mathcal{O}}_{M,y}$, and write $B_i = \hat{\mathcal{O}}_{X,x_i}$ for the points over y . Then*

$$\hat{\mathcal{O}}_{Y,y} = R / \bigcap_i \ker(R \rightarrow B_i).$$

This completed image is reduced. If the cotangent map from R to B_i is surjective, then $R \rightarrow B_i$ is surjective. If there are two such points, writing $I_i = \ker(R \rightarrow B_i)$ gives

$$\widehat{\mathcal{O}}_{Y,y} = (R/I_1) \times_{R/(I_1+I_2)} (R/I_2).$$

The formal schemes at the different x_i are disjoint. A local Artin point of the finite inverse image belongs to exactly one of them.

Proof. On an affine neighbourhood the image ring is the image of the ambient ring in a finite algebra. Exactness of completion on finite modules preserves this kernel. The completed finite algebra is the product of the B_i : the orthogonal idempotents of its Artin closed fibre lift uniquely through the powers of the maximal ideal. Reduced finite-type complex rings are excellent and analytically unramified, so their completions are reduced; alternatively this applies directly to the reduced image ring. For cotangent surjectivity, lift a basis of the maximal ideal of B_i modulo its square to R . The resulting lifts generate every successive associated-graded piece. Subtracting successive homogeneous lifts and taking their convergent sum expresses any element of B_i as the image of an element of R . The finite map factors through the completed image, so this is also a closed immersion into that image, not only into M . Finally the elementary exact sequence

$$0 \longrightarrow R/(I_1 \cap I_2) \longrightarrow R/I_1 \oplus R/I_2 \longrightarrow R/(I_1 + I_2) \longrightarrow 0$$

proves the fibre-product assertion. The idempotent decomposition gives the last assertion, including over nonreduced Artin bases. $\square$

Lemma 13.2 (Coprime lifts and the split codimension). *If homogeneous forms a, b have coprime reductions over a local Artin base A , then $(a) \cap (b) = (ab)$ in $A[E]$. The relevant graded quotients and Koszul kernels are A -flat. In the split-divisor model, with $s_j = \binom{e+j-1}{j}$, $e \geq 2$, $h > g \geq 1$, and $r \geq 2$,*

$$c = r(s_h - s_{h-g}) - (s_g - 1) \geq (r-1)(s_g - 1) \geq 1.$$

Proof. Over the residue field the two forms form a regular sequence. In each degree their Koszul complex consists of finite free modules and has its prescribed ranks. The acyclic-complex criterion over an Artin local ring, proved by lifting splittings successively along its maximal ideal, gives exactness and flatness of the lifted cokernels. Thus b is a nonzerodivisor modulo a , proving the intersection identity. For the inequality, express $s_h - s_{h-g}$ as the sum of g consecutive first differences of the sequence s_j . These differences are nondecreasing for $e \geq 2$. Their sum is at least $s_g - s_0 = s_g - 1$. $\square$

In particular the two formal marked lifts in a coprime split model are separated before their images are compared. Their intersection functor is precisely $K_A = a_A b_A L_A$. Division gives a subbundle: multiplication by a fibrewise nonzero form is locally split, and the quotient of the ambient bundle by K_A is locally free. Combining this Artin-functor identification with Lemma 13.1 justifies the completed-image fibre product without any assumption that a tangent-space comparison determines an intersection scheme.

14. A GLOBAL FACTORIZATION THEOREM FOR BINARY NORMALIZATION FIBRES

For this section E has dimension two. Set $N_0 = r(D+1-r)$ and $m = r-1$. Let

$$\mathcal{T}_s = Y_s \setminus Y_{s+1}, \quad 1 \leq s \leq D+1-r,$$

with the reduced locally closed structure; an inadmissible Y_j is empty. Binary forms split into linear factors, so the Y_j are nested. The multiplication map identifies

$$\mathcal{T}_s \simeq \mathbb{P}(S_s) \times X_{D-s,r},$$

where $X_{D-s,r}$ is the open Grassmannian of gcd-free residual spaces. The first projection is the algebraic common-divisor morphism, not a choice made independently on individual fibres.

For nonnegative a, b define $H_{a,b} = \mathbb{C}$ when $ab = 0$. Otherwise, put $k = \min(a, b)$ and $l = \max(a, b)$, and set

$$H_{a,b} = \mathbb{C}[q_1, \dots, q_k]/(c_{l+1}, \dots, c_{l+k}), \quad \sum_{j \geq 0} c_j z^j = (1 + q_1 z + \dots + q_k z^k)^{-1}. \quad (14.1)$$

The weight of q_i is i . This grading is not the maximal-ideal adic grading when $k > 1$.

Theorem 14.1 (Stratified finite flatness and all binary fibres). *For $g \leq s$, the restriction of ν_g over $\mathcal{T}_s$ is the base change, along the common-divisor morphism, of*

$$\mathbb{P}(S_g) \times \mathbb{P}(S_{s-g}) \longrightarrow \mathbb{P}(S_s), \quad (f, H) \mapsto fH.$$

It is finite locally free of degree $\binom{s}{g}$. If $\gcd(K) = \prod_{i=1}^b \ell_i^{s_i}$, with distinct ℓ_i and $\sum_i s_i = s$, its entire normalization fibre is

$$\prod_{\substack{0 \leq a_i \leq s_i \\ \sum_i a_i = g}} \bigotimes_{i=1}^b H_{a_i, s_i - a_i}. \quad (14.2)$$

Every local factor is an Artin complete intersection and is Gorenstein. Its length, weighted Hilbert series, and socle weight are, respectively,

$$\prod_i \binom{s_i}{a_i}, \quad \prod_i \prod_{j=1}^{\min(a_i, s_i - a_i)} \frac{1 - z^{\max(a_i, s_i - a_i) + j}}{1 - z^j}, \quad \sum_i a_i (s_i - a_i).$$

The total length is $\binom{s}{g}$, including at arbitrary simultaneous root collisions. Over fixed s , the closure order of common-root partition strata is the order generated by merging parts.

Proof. Over $\mathcal{T}_s$ the common divisor G and residual subbundle L are defined scheme-theoretically by the inverse to the exact-divisor incidence. Locally on any base change, choose a member of L coprime to G on each relevant geometric fibre. There is such a member over $\mathbb{C}$, since only finitely many roots must be avoided. Regular-sequence cancellation, or monic division and its flat quotient, gives $f \mid GL$ if and only if $f \mid G$. This equivalence is an equality of incidence functors over nonreduced bases. The residual subspace is then forced. This proves the base-change description.

The displayed multiplication map is finite by properness and the finiteness of the divisors of a fixed binary form. Its source and target are smooth of dimension s . A finite map between these smooth spaces is flat: a regular system of parameters in the target is a system of parameters in each source local ring, hence a regular sequence; the local flatness criterion applies. Its degree can be computed at a squarefree form, where the $\binom{s}{g}$ divisors are reduced and distinct.

Coprime root clusters split uniquely over Artin bases by Hensel factorization. On the cluster with polynomial x^{s_i} , choosing a degree- a_i factor is the equation $f_i H_i = x^{s_i}$ with both factors monic. Eliminating one factor by reciprocal expansion gives (14.1). This proves the full product decomposition, not only its closed points. Its equations have their sole common zero at the origin: a monic factor of x^{a+b} over $\mathbb{C}$ is a power of x . Thus its k equations in k variables are a homogeneous system of parameters and a regular sequence. The Hilbert series is the quotient of the corresponding weighted polynomial Hilbert series. Taking its value at one gives $\binom{a+b}{a}$; its top weight is ab . A zero-dimensional complete intersection has a one-dimensional socle. Equivalently q_k^l , up to a nonzero scalar, represents the rectangular Schur class. The identification with the usual Grassmannian cohomology presentation is classical; a purely algebraic Schur-basis statement is [24, Theorem 2.7]. Tensoring the local factors proves all three formulas. The total length identity

is the coefficient of z^g in $\prod_i (1+z)^{s_i}$. Finally the exact-gcd stratum is the product with $X_{D-s,r}$ above. In the symmetric power of the projective line, root clusters specialize by coalescence, and every merging is realized by moving the corresponding roots together. This gives precisely the stated closure order within fixed gcd degree. $\square$

Remark 14.2. The finite flat statement is on the exact-gcd stratum with its indicated scheme structure. It does not assert that normalization or formation of scheme-theoretic images commutes with an arbitrary nonflat base change of the ambient Grassmannian. It identifies a specific base change of the already constructed incidence morphism.

15. CONDUCTORS AND ALL BRANCH INTERSECTIONS ON THE SQUAREFREE LOCUS

Let $\mathcal{U} \subset \text{Gr}(r, S_D)$ be the open set on which the common divisor is squarefree. Its complement is the proper image of the incidence $K = \ell^2 L$. This condition concerns the common divisor, not every member of the subspace.

Theorem 15.1 (Squarefree conductor stratification). *Let $K \in Y_g \cap \mathcal{U}$, let $s = \deg \gcd(K)$, and set*

$$t = s - g, \quad m = r - 1, \quad q = N_0 - ms.$$

Introduce s disjoint blocks $u_i = (u_{i1}, \dots, u_{im})$, and let J_j be the ideal generated by products of one variable from each of j distinct blocks. Put $J_0 = (1)$ and $J_{s+1} = 0$. Then

$$\hat{\mathcal{O}}_{Y_g, K} \simeq \mathbb{C}[[z_1, \dots, z_q, u_{11}, \dots, u_{sm}]]/J_{t+1}. \quad (15.1)$$

The branches are indexed by the subsets $A \subset \{1, \dots, s\}$ of cardinality g . The branch A has ideal $I_A = (u_{ij} : i \in A)$ in the ambient power-series ring. For any collection of branches, their full scheme-theoretic intersection has ideal $I_{\cup A}$; it is smooth of dimension $q + m(s - |\cup A|)$.

The conductor in (15.1) is J_t/J_{t+1} . Its restriction to branch A is $\prod_{i \notin A} (u_{i1}, \dots, u_{im})$. Consequently the global conductor sheaf on $Y_g \cap \mathcal{U}$ is exactly

$$\mathfrak{c}_{Y_g|U} = \mathcal{I}_{Y_{g+1}/Y_g}|U. \quad (15.2)$$

In particular it is radical there. These local rings are seminormal. Their dimension, depth and multiplicity are

$$q + mt, \quad q + t, \quad \binom{s}{g}.$$

They are Cohen–Macaulay exactly when $s = g$ or $r = 2$. The Hilbert series of the nonsmooth factor in its standard grading is

$$\sum_{j=0}^t \binom{s}{j} ((1-z)^{-m} - 1)^j. \quad (15.3)$$

The whole collection of branch intersections and conductor strata is invariant under permutation of the roots and descends from any ordered-root etale chart.

Proof. Choose $F_0 \in K$ with a simple zero at each of the s common roots, and complete it to a frame $F_0, \dots, F_m$. Hensel factorization gives moving roots $x_1, \dots, x_s$ in the completion. Use as coordinates the s root parameters and the ms evaluations $u_{ij} = F_j(x_i)$, with harmless invertible normalizing factors. These are part of a regular parameter system: the evaluation map $S_D/K \rightarrow \mathbb{C}^s$ is surjective because K vanishes on the common-root scheme, and $D \geq s - 1$. Thus $\text{Hom}(K, S_D/K)$ allows independent variation of all rs jets. The passage to the displayed coordinates is triangular with invertible diagonal, since F_0 has simple zeros. The other $N_0 - rs$ parameters and the s root parameters give exactly $q = N_0 - ms$ smooth parameters.

A marked degree- g divisor near this fibre chooses a subset A of these roots. Its incidence equations are exactly $u_{ij} = 0$ for $i \in A$, with no nilpotents suppressed. There are no other divisor lifts. Lemma 13.1 therefore identifies the completed image ideal with $\bigcap_{|A|=g} I_A$. A monomial belongs to this intersection exactly when its block support meets every g -element subset, or equivalently has at least $s - g + 1 = t + 1$ blocks. This proves (15.1). Sums of the coordinate ideals give all intersections, including intersections of three or more branches.

For completeness compute the conductor inside the product of the branch rings. Its component on branch A consists of functions whose extension by zero on every other branch belongs to the image ring. The image of $\bigcap_{B \neq A} I_B$ modulo I_A is precisely $\prod_{i \notin A} (u_i)$. Indeed a surviving monomial has support outside A , and must meet every g -subset other than A . Replacing one element of A by each $i \notin A$ forces all those blocks to occur. Conversely their product meets every such subset. Hence a conductor monomial has exactly t surviving blocks; monomials on more blocks are zero. This is J_t/J_{t+1} . The same coordinates identify Y_{g+1} with J_t , proving the sheaf equality by faithful flatness of completion. The case $t = 0$ means that the conductor is the unit ideal and the next stratum is absent locally.

The ring is the ring of tuples on its coordinate branches that agree on every pairwise intersection. To verify this directly, compare the coefficient of each monomial on all branches where it survives; the pairwise agreements give a unique common coefficient. If an element b of the total fraction ring has b^2, b^3 in the image, it first belongs to the normalization, by integrality. Its two restrictions on any intersection have equal squares and cubes, hence are equal in that reduced ring. Thus b lies in the image. This proves seminormality by the square-and-cube criterion.

Formula (15.3) follows by sorting nonzero monomials by their block support. The top-dimensional contributions come from the $\binom{s}{t}$ supports of size t , giving dimension mt and multiplicity $\binom{s}{t}$. We supply the homological step for the depth, since branch counting alone does not give it. The simplicial complex of the squarefree monomial ideal J_{t+1} consists of sets using at most t blocks. Any induced subcomplex meeting b blocks is homotopy equivalent to the $(t-1)$ -skeleton of the simplex on those b blocks: collapse each block to a representative vertex. The two composites are contiguous to the identity, because adjoining the representatives never increases the number of blocks. If $b \leq t$ the complex is a simplex; otherwise its only reduced homology is in degree $t-1$, with rank $\binom{b-1}{t}$. This last assertion follows also by the elementary induction that adds the last vertex to a simplex skeleton.

The squarefree multidegree- W part of the Koszul complex on all sm variables is the augmented simplicial cochain complex of this induced subcomplex, with degree shifted by $|W| - 1$. Non-squarefree multidegrees are contractible in the relevant summands, by pairing faces using a variable whose exponent exceeds one. Consequently a nonzero positive Betti number has homological degree $|W| - t$. The full vertex set gives the nonzero degree $sm - t$, since $s > t$ when $g > 0$. The projective dimension is therefore $sm - t$, and the Auslander–Buchsbaum formula gives depth t . When $t = 0$ the quotient is simply $\mathbb{C}$. Adjoining q power-series parameters gives depth $q + t$ in every case, and comparison with $q + mt$ proves the Cohen–Macaulay criterion. All the constructions commute with relabelling roots, proving descent. $\square$

Corollary 15.2 (Binary divisor-image intersections). *For admissible binary divisor degrees $a_1, \dots, a_v$, their scheme-theoretic intersection is $Y_{\max a_i}$. On the squarefree locus all multiple intersections of normalization branches, rather than only the degree images, are given by the union-of-subsets rule in Theorem 15.1.*

Proof. The reduced binary images are nested as closed subschemes: a binary form of degree b has a divisor of each degree $a \leq b$, and reduced closed subschemes are determined by their closed points. Their ideals are thus nested, so their sum is the largest ideal. The assertion about branches has already been proved as an identity of coordinate ideals, without taking radicals. $\square$

16. AN INCIDENCE ATLAS FOR ALL BINARY ROOT COLLISIONS

The squarefree theorem has a uniform replacement when any number of root clusters collide. This replacement specifies the full completed image, not merely a transverse family in it.

Theorem 16.1 (Simultaneous collision atlas). *Let $K \in Y_g$ be binary with exact common divisor $\prod_{i=1}^b \ell_i^{s_i}$ and $s = \sum_i s_i$. There are formal ambient coordinates consisting of the coefficients of monic polynomials P_i of degree s_i , polynomials R_{ij} of degree less than s_i ($1 \leq j \leq m$), and $N_0 - rs$ smooth parameters z . They are centred at $P_i = x_i^{s_i}$ and $R_{ij} = 0$. For every $\alpha = (a_i)$ with $0 \leq a_i \leq s_i$ and $\sum a_i = g$, let B_α be the regular power-series ring in the coefficients of monic A_i, B'_i of degrees $a_i, s_i - a_i$, polynomials U_{ij} of degree less than $s_i - a_i$, and the same z . The completed normalization and image are exactly*

$$T \longrightarrow \prod_{\alpha} B_{\alpha}, \quad P_i \mapsto A_i B'_i, \quad R_{ij} \mapsto A_i U_{ij}, \quad \widehat{\mathcal{O}}_{Y_g, K} = T / \bigcap_{\alpha} \ker(T \rightarrow B_{\alpha}). \quad (16.1)$$

Here $T = \mathbb{C}[[\text{coeff}(P_i), \text{coeff}(R_{ij}), z]]$. Degree-zero monic factors are one, and negative-degree remainder spaces are zero. The formula is independent, up to isomorphism of completed incidence diagrams, of the frame, the selected F_0 , and the local root coordinates. It covers every higher and simultaneous binary collision.

Proof. Choose $F_0 \in K$ having exact order s_i at every common root. The required nonvanishing of its residual value excludes finitely many hyperplanes in K , so a choice exists. Complete it to a frame. Weierstrass preparation at the distinct roots produces P_i ; division of the remaining frame members produces R_{ij} . The map

$$S_D/K \longrightarrow \bigoplus_i \mathbb{C}[x_i]/(x_i^{s_i})$$

is surjective: its kernel contains K , and the usual interpolation map from S_D is surjective. Thus the Grassmannian tangent space permits independent variation of all rs jets. Division and preparation change these jets by an invertible triangular matrix, whose diagonal contains the nonzero residual values of F_0 . Their differentials are independent. Formal inverse coordinates supply the remaining $N_0 - rs$ parameters; this integer is nonnegative since $r \leq D - s + 1$.

A marked divisor near a chosen normalization point splits uniquely among the coprime root clusters. In cluster i its monic representative has degree a_i and divides both P_i and all R_{ij} . Monic division is equivalent, over every local Artin base, to the unique equations $P_i = A_i B'_i$ and $R_{ij} = A_i U_{ij}$. Conversely these equations give the marked divisor and the residual subbundle. Thus the normalized incidence functor is represented by the displayed regular ring. It is finite over T : monic factor coefficients are integral over the coefficients of P_i , while the remaining factors and quotients are uniquely determined by division. Lemma 13.1 now gives the exact completed image and the intersection of kernels.

Changing a frame or a coordinate changes only a presentation of the same functor of a subbundle with a marked divisor. The mutually inverse changes of incidence data therefore induce isomorphisms of its complete representing rings and of the resulting finite-image diagrams. This also proves that the calculation describes the entire completed image. A point of that image over an Artin ring need not lift to its normalization; no such lifting assertion is used or implied. $\square$

17. THE SINGULAR FITTING IDEAL OF EVERY DOUBLE-ROOT MODEL

The atlas specializes for $s = 2, g = 1$ to the following ring, after completion of the square and suppression of smooth parameters:

$$A_m = R \oplus It \subset B = \mathbb{C}[[u_1, \dots, u_m, t]], \quad R = \mathbb{C}[[u_1, \dots, u_m, \Delta]], \quad I = (u_1, \dots, u_m), \quad \Delta = t^2.$$

Put $w_i = u_i t$. The defining equations are $u_i w_j - u_j w_i$ and $w_i w_j - \Delta u_i u_j$.

Theorem 17.1 (Closed singular-ideal formula). *For every $m \geq 1$, the intrinsic singular Fitting ideal is*

$$\text{Fitt}_{m+1} \Omega_{A_m/\mathbb{C}}^{\text{cont}} = (I^{m+1} + \Delta I^m) \oplus I^m t = I^{m-1} (I^2, (w_1, \dots, w_m), \Delta I). \quad (17.1)$$

The first equality is a decomposition as an R -submodule of B ; the right side is an ideal of A_m . It is not merely an ideal of Jacobian minors left unevaluated. For $m = 1$ it is $(u^2, w, \Delta u)$, and its radical is the conductor (u, w) for all m .

Proof. There are $2m + 1$ ambient variables and dimension $m + 1$, so the Fitting ideal consists of the m -minors of the Jacobian of the displayed relations. Evaluate this matrix in B using $w_i = u_i t$ and $\Delta = t^2$. A minor not using the Δ column has u -degree m . It vanishes at $t = 0$: the only remaining rows are the Koszul rows in the w columns, of rank at most $m - 1$. Hence its terms have a positive power of t , and belong to $\Delta I^m \oplus I^m t$. A minor using the Δ column has u -degree $m + 1$, and hence lies in $I^{m+1} \oplus I^{m+1} t$. This proves the first inclusion.

Use the m rows

$$u_1 w_j - u_j w_1 \quad (2 \leq j \leq m), \quad w_1^2 - \Delta u_1^2.$$

The columns $(w_1, w_2, \dots, w_m)$, $(u_1, w_2, \dots, w_m)$, and $(\Delta, w_2, \dots, w_m)$ give, up to sign and nonzero scalar, $u_1^m t$, Δu_1^m , and u_1^{m+1} . The ideal is invariant under simultaneous linear change of u and w . In characteristic zero the powers of linear forms span each symmetric power, so these three minors generate the three indicated spaces of all u -monomials. They generate the claimed R -module; it is an A_m -ideal since multiplication by $w_i = u_i t$ preserves it. This proves equality. The formula also immediately gives its radical, without discarding the nonreduced Fitting structure in the assertion itself. $\square$

18. A STRUCTURAL DICHOTOMY FOR PURE-POWER DIVISOR FIBRES

Let $E = \mathbb{C}a \oplus W$, $d = \dim W \geq 1$, and $g, k \geq 1$. The factorization fibre of a^{g+k} records monic factors of degrees g and k . Interchanging these factors identifies its two descriptions; we may and shall take $g \geq k$. Write

$$Q_i \in \text{Sym}^i W \quad (1 \leq i \leq k), \quad C_0 = 1, \quad C_j = - \sum_{i=1}^k Q_i C_{j-i}, \quad C_j = 0 \quad (j < 0).$$

Let $P = \text{Sym}(\bigoplus_{i=1}^k (\text{Sym}^i W)^*)$, with its generators of weight i , and let $\mathfrak{n}$ be its variable ideal. Put

$$F_{g,k}(W) = P/\mathcal{I}, \quad \mathcal{I} = (\text{coeff } C_{g+1}, \dots, \text{coeff } C_{g+k}). \quad (18.1)$$

It is an Artin local algebra. Indeed its complex points are exactly the factorizations of a^{g+k} , hence the single origin by unique factorization. The polynomial and completed presentations coincide. If $K = a^{g+k} L$ with $\gcd(L) = 1$, cancellation by a member of L not divisible by a , as in Lemma 13.2, identifies this with the full degree- g normalization fibre at K .

Theorem 18.1 (Minimal equations and complete-intersection classification). *The embedding dimension and minimal number of equations of $F_{g,k}(W)$ in a minimal regular local presentation are*

$$\text{edim } F_{g,k}(W) = \binom{d+k}{k} - 1, \quad (18.2)$$

$$\mu(\mathcal{I}) = \sum_{j=g+1}^{g+k} \binom{d+j-1}{j} = \binom{d+g+k}{d} - \binom{d+g}{d}. \quad (18.3)$$

More precisely the minimal relation module has the equivariant weighted decomposition

$$\mathcal{I}/\mathfrak{n}\mathcal{I} \simeq \bigoplus_{j=g+1}^{g+k} (\mathrm{Sym}^j W)^*. \quad (18.4)$$

Consequently this fibre is a complete intersection if and only if $d = 1$. For $d = 1$, it has length $\binom{g+k}{k}$, weighted Hilbert series $\prod_{i=1}^k (1 - z^{g+i})/(1 - z^i)$, and one-dimensional socle generated by Q_k^g , of weight gk . For $k = 1$ and arbitrary d ,

$$F_{g,1}(W) = \mathrm{Sym}(W^*)/\mathfrak{n}^{g+1}, \quad \mathrm{length} F_{g,1}(W) = \binom{d+g}{g}, \quad \mathrm{Soc} F_{g,1}(W) = \mathrm{Sym}^g(W^*). \quad (18.5)$$

In particular the last algebra is Gorenstein precisely when $d = 1$. No Gorenstein classification for $d > 1, k > 1$ is needed for the theorem.

Proof. Since $g \geq k$, each defining form has no linear term in the coefficient variables. Thus the presentation is minimal and (18.2) follows by summing $\dim \mathrm{Sym}^i W$. For $d = 1$, the regular-sequence argument in the proof of Theorem 14.1 applies to the k reciprocal relations. In particular none of them belongs to $\mathfrak{n}\mathcal{I}$. It gives the stated Hilbert series, length, and the classical rectangular socle class.

For general W , taking coefficients of C_j gives an equivariant map $(\mathrm{Sym}^j W)^* \rightarrow \mathcal{I}/\mathfrak{n}\mathcal{I}$. Its source is irreducible. This map is nonzero: choose one coordinate x of W , keep only the coefficients of x^i in every Q_i , and set every other coefficient to zero. The coefficient of x^j maps to the j th binary reciprocal relation. Were it in $\mathfrak{n}\mathcal{I}$ before specialization, its specialization would be in the binary $\mathfrak{n}\mathcal{I}$, contradicting the preceding minimality. Irreducibility now makes the entire coefficient map injective. The different j have different weights, so their images are independent. They generate the relation module by the chosen presentation, proving (18.4) and (18.3).

If $d \geq 2$, the sequence $\binom{d+j-1}{j}$ strictly increases in j . Comparing its terms at $j = g+i$ and $j = i$ shows that the minimal number of relations is strictly greater than the embedding dimension. A zero-dimensional complete intersection in a minimal regular local presentation has exactly that many minimal relations as variables; hence these fibres are not complete intersections. This proves both directions of the classification. Finally when $k = 1$ the sole form is $C_{g+1} = (-Q_1)^{g+1}$. Its coefficients are, with nonzero multinomial factors, every monomial of degree $g+1$ in the d variables. This gives (18.5), including the socle and length. $\square$

Corollary 18.2 (The ternary-pencil ramification fibre). *At the common limiting relation algebra with $e = 9$, $g = 6$ and $k = 2$, the normalization fibre has embedding dimension 44 and exactly*

$$\binom{14}{7} + \binom{15}{8} = 9867$$

minimal scalar equations. Their two weighted pieces have dimensions 3432 and 6435, in weights seven and eight. In particular this actual pencil-orbit boundary fibre is not a complete intersection.

Proof. Here $d = e - 1 = 8$. Apply (18.2) and (18.4). These are minimal equation counts, not the number of coefficient variables or the length of the fibre. $\square$

The binary factorization rings in this section are classical. The additional point of the multivariate argument is that polarization preserves every full irreducible coefficient space in the minimal relation module. Neither the Grassmannian identification nor a finite calculation in a few variables is being used as a claim of priority or as a substitute for that argument.

19. HOMOLOGICAL INVARIANTS OF THE DIVISOR FIBRES

We first extract all graded Betti numbers from the squarefree local model and then identify a determinant-apolar quotient of the multivariate fibres. A separate exact calculation treats an entire multivariate fibre, rather than only a quotient or its minimal relations. The grading conventions will be specified in each case.

Theorem 19.1 (Betti numbers of the squarefree block model). *Let $S = \mathbb{C}[u_{ij} : 1 \leq i \leq s, 1 \leq j \leq m]$ with its standard multigrading, let $0 \leq t < s$, and put $A = S/J_{t+1}$, with the block-support ideal of Theorem 15.1. For a nonempty squarefree multidegree F let $b(F)$ be the number of blocks it meets. Besides $\beta_{0,0} = 1$, the only nonzero multigraded Betti numbers are*

$$\beta_{i,F}^S(A) = \binom{b(F)-1}{t} \quad \text{if } b(F) > t \text{ and } i = |F| - t. \quad (19.1)$$

For $t = 0$ this is the Koszul resolution of $\mathbb{C}$. For all t , the singly graded numbers are

$$\beta_{i,i+t}^S(A) = \sum_{b=t+1}^{\min(s,i+t)} \binom{s}{b} \binom{b-1}{t} [z^{i+t}]((1+z)^m - 1)^b \quad (i \geq 1). \quad (19.2)$$

The ideal J_{t+1} has a $(t+1)$ -linear resolution. The projective dimension of A is $sm - t$. In the Cohen–Macaulay case $m = 1$, its type is $\binom{s-1}{t}$; it is Gorenstein exactly when $t = 0$ or $t = s - 1$. The same Betti numbers over the completed ambient regular ring apply to the local model after adjoining its smooth parameters.

Proof. The Stanley–Reisner complex consists of sets meeting at most t blocks. If $t \geq 1$, its induced subcomplex on F has, in each nonempty block, a simplex all of whose vertices have identical external links. Contract that simplex to one of its vertices: the map and the inclusion are contiguous, because adjoining the chosen vertex to any face does not change its block support. Applying this one block at a time gives a homotopy equivalence with the $(t-1)$ -skeleton of the simplex on $b(F)$ vertices. If $b(F) \leq t$ this simplex is contractible. Otherwise its reduced homology is concentrated in degree $t-1$, with dimension $\binom{b(F)-1}{t}$. The latter assertion follows by attaching its top faces successively, or from the augmented simplicial chain complex and the binomial identity for its Euler characteristic. Hochster’s multigraded formula now gives (19.1); see [26, Chapter 5]. If $t = 0$, every variable is in the ideal and the Koszul formula gives the same result directly.

For a support of size b , the coefficient in (19.2) counts choices of $i+t$ vertices meeting each selected block. Summing proves the formula. All ideal syzygies therefore have the indicated linear degrees. Taking F to be all sm vertices gives the nonzero last Betti number $\binom{s-1}{t}$ in homological degree $sm - t$; no larger degree occurs. When $m = 1$, Auslander–Buchsbaum gives depth t , equal to dimension, and the last Betti number is the Cohen–Macaulay type. Completion and adjoining independent regular parameters are flat operations preserving the minimal free resolution. These are classical squarefree-monomial methods. The assertion here is their full specialization to the normalization image model, not a claim of priority for Hochster’s formula or skeleton resolutions. $\square$

For the remaining results use the weighted reciprocal algebra $F_{g,k}(W)$ of (18.1), where $d = \dim W$ and $g \geq k \geq 2$. Set all Q_i except Q_2 equal to zero and, in addition, impose every coefficient of Q_2^2 . Denote the resulting algebra by

$$B_d = \text{Sym}((\text{Sym}^2 W)^*) / (\text{coeff } Q_2^2). \quad (19.3)$$

Its coefficient variables have reciprocal weight two. When referring to its ordinary polynomial grading, we instead give them degree one.

Theorem 19.2 (A determinant-apolar quotient of every higher fibre). *There is a $\mathrm{GL}(W)$ -equivariant graded surjection $F_{g,k}(W) \twoheadrightarrow B_d$ for every $g \geq k \geq 2$. After fixing the standard coefficient/differential pairing, B_d is the apolar algebra of the determinant of a generic symmetric $d \times d$ matrix. Its polynomial-degree Hilbert function is*

$$\dim(B_d)_p = \frac{1}{d+1} \binom{d+1}{p} \binom{d+1}{p+1} \quad (0 \leq p \leq d). \quad (19.4)$$

It is Gorenstein, its reciprocal-weight socle is $2d$, and

$$\mathrm{length} F_{g,k}(W) \geq \mathrm{length} B_d = \frac{1}{d+2} \binom{2d+2}{d+1}. \quad (19.5)$$

For $k = 2$ and $g = 2$ or 3 , the quotient by the coefficients of Q_1 alone is already B_d . In particular the actual ternary-pencil fibre $F_{6,2}(\mathbb{C}^8)$ has length at least 4862, independently of its embedding dimension 44 and its 9867 minimal equations.

Proof. Under the stated substitution, the reciprocal series becomes $(1 + Q_2 z^2)^{-1}$. Its odd coefficients vanish and its even coefficients are powers of Q_2 . Since $g + 1 \geq 3$, imposing $Q_2^2 = 0$ annihilates all defining coefficients. This proves the surjection, functorially for linear changes of W . When $k = 2$ and $g = 2$ or 3 , the first even defining coefficient is Q_2^2 and there are no further independent equations after $Q_1 = 0$.

Choose coordinates $x_1, \dots, x_d$ on W and write $Q_2 = \sum_i y_{ii} x_i^2 + \sum_{i < j} y_{ij} x_i x_j$. Let $X = (X_{ij})$ be symmetric, with independent entries for $i \leq j$, and interpret y_{ij} as $\partial/\partial X_{ij}$. Then $Q_2(x)$ acts as directional differentiation in the rank-one symmetric direction xx^t . The determinant of $X + zxx^t$ is linear in z , so every coefficient of $Q_2(x)^2$ annihilates $\det X$. These coefficients are, up to nonzero scalars, y_{ii}^2 , $y_{ii}y_{ij}$, $y_{ij}^2 + 2y_{ii}y_{jj}$, $y_{ii}y_{jl} + y_{ij}y_{il}$ for distinct i, j, l , and $y_{ij}y_{kl} + y_{ik}y_{jl} + y_{il}y_{jk}$ for four distinct indices. They are exactly the quadratic generators in [27, Proposition 3.1 and Theorem 3.10]. That theorem proves equality with the full apolar ideal, not merely containment. Its Corollary 2.7 gives (19.4) and the Catalan length. Alternatively, that Hilbert function counts the independent minors obtained by differentiating $\det X$. Apolarity to one homogeneous form gives a perfect pairing between complementary degrees and a one-dimensional socle in degree d . Replacing polynomial degree by twice that degree proves the weight statement. The length inequality follows from the surjection. For $d = 8$ the Catalan number is 4862.

The apolar Hilbert function and its Gorenstein property are classical results of the cited determinant theory. The new identification in this setting is the functorial quotient of the entire family of normalization fibres. A quotient's Gorenstein property is not being attributed to the original fibre. $\square$

Proposition 19.3 (The complete first multivariate two-factor fibre). *Let $A = F_{2,2}(\mathbb{C}^2)$ and write $Q_1 = ax + by$, $Q_2 = cx^2 + dxy + ey^2$. Then*

$$H_A^{\mathrm{wt}}(z) = 1 + 2z + 6z^2 + 6z^3 + 7z^4, \quad \mathrm{length} A = 22, \quad \dim \mathrm{Soc} A = 7, \quad (19.6)$$

$$H_{\mathrm{grm} A}(z) = 1 + 5z + 6z^2 + 5z^3 + 5z^4. \quad (19.7)$$

The weight grading gives a, b weight one and c, d, e weight two. In the minimal weighted free resolution over $S = \mathbb{C}[a, b, c, d, e]$, the shift polynomials $B_i(z) = \sum_j \beta_{i,j}^S(A) z^j$ are

$$\begin{aligned} B_0 &= 1, & B_1 &= 4z^3 + 5z^4, \\ B_2 &= 4z^5 + 12z^6 + 12z^7, & B_3 &= 21z^8 + 20z^9 + z^{10}, \\ B_4 &= 21z^{10} + 8z^{11}, & B_5 &= 7z^{12}. \end{aligned}$$

In particular the total Betti numbers are $(1, 9, 28, 42, 29, 7)$. This fibre is not Gorenstein. The assertion concerns this whole five-variable fibre, not the determinant-apolar quotient.

Proof. The nine initial equations are the coefficients of $-Q_1^3 + 2Q_1Q_2$ and $Q_1^4 - 3Q_1^2Q_2 + Q_2^2$. For graded reverse lexicographic order $e > d > c > b > a$, a Groebner basis is the following list (a scalar multiple does not change a member):

$$\begin{aligned} & (c, d, e)^3, \quad b(c, d, e)^2, \quad a(c, d, e)^2, \\ & b^2e - e^2, \quad b^2d - de, \quad b^2c - a^2e, \quad b^3 - 2be, \\ & 2abe - de, \quad 2a^2e + 2abd - 2ce - d^2, \quad 2abc - cd, \\ & 3ab^2 - 2ae - 2bd, \quad a^2d - cd, \quad a^2c - c^2, \\ & 3a^2b - 2ad - 2bc, \quad a^3 - 2ac. \end{aligned}$$

Here a monomial ideal in the list means all its minimal monomial generators. There are $10 + 6 + 6 + 12 = 34$ polynomials. Polynomial division gives both equality with the ideal of the nine equations and zero remainders for all their basis S-pairs. This is a finite exact arithmetic verification; the explicit input and all reduction and rank tests are supplied in the accompanying calculation. The standard monomials, grouped by weight, are

$$\begin{array}{c|l} 0 & 1 \\ 1 & a, b \\ 2 & a^2, ab, b^2, c, d, e \\ 3 & ac, bc, ad, bd, ae, be \\ 4 & c^2, cd, d^2, a^2e, ce, de, e^2. \end{array} \quad (19.8)$$

Reduction of products of these monomials gives $\dim A/\mathfrak{m} = 1$ and $\dim(\mathfrak{m}^i) = (21, 16, 10, 5, 0)$ for $1 \leq i \leq 5$. The common kernel of multiplication by a, b, c, d, e is exactly the last row of (19.8). This proves both distinct Hilbert series and the type.

For precision, the resolution calculation uses no numerical rank thresholds. In weight w and homological degree i , take basis pairs (J, m) with $J \subset \{a, b, c, d, e\}$ of size i , m a standard monomial, and $\text{wt}(J) + \text{wt}(m) = w$. The differential is

$$d(J, m) = \sum_{v \in J} (-1)^{\text{pos}(v, J) - 1} (J \setminus \{v\}, \text{NF}(vm)). \quad (19.9)$$

All entries are rational, explicitly determined by the displayed 34 polynomials. Exact row reduction and $\beta_{i,w} = \dim C_{i,w} - \text{rank } d_{i,w} - \text{rank } d_{i+1,w}$ give the six shift polynomials above. The calculation is fully specified by (19.8)–(19.9); the archived rank table makes each finite elimination independently repeatable. As additional checks, their alternating sum is $H_A^{\text{wt}}(z)(1-z)^2(1-z^2)^3$ and the last Betti space is the socle shifted by the total variable weight eight. These checks do not replace the exact rank computations. $\square$

20. QUADRATIC SECTIONS OF RECIPROCAL NORMALIZATION FIBRES

The reciprocal presentation has a distinguished linear coefficient section on which the entire algebra, rather than only its minimal relations, can be determined. Throughout this section a decomposition $E = \mathbb{C}a \oplus W$ is fixed. The section is equivariant for $\text{GL}(W)$; it is not asserted to be independent of an arbitrary change of that decomposition.

Put $d = \dim W$, $g \geq k \geq 2$, and $h = \lfloor g/2 \rfloor$. In the algebra $F_{g,k}(W)$ of (18.1), let $\mathfrak{l}$ be the ideal generated by the coefficients of all Q_i with $i \neq 2$. Write $Q = Q_2$ and

$$B_{d,h} = \text{Sym}((\text{Sym}^2 W)^*) / (\text{coeff } Q^{h+1}). \quad (20.1)$$

Here the ordinary degree of a coefficient of Q is one; its reciprocal weight is two. These two gradings will always be specified separately. A partition $\lambda \subseteq (h^d)$ has at most d rows, each of length at most h .

Theorem 20.1 (The entire quadratic coefficient section). *There is an isomorphism of $\mathrm{GL}(W)$ -equivariant graded algebras*

$$F_{g,k}(W)/\mathfrak{l} \simeq B_{d,h}. \quad (20.2)$$

Under the coefficient/differential pairing, $B_{d,h}$ is the apolar algebra of the h th power of the determinant of a generic symmetric $d \times d$ matrix. Its ordinary-degree decomposition is

$$(B_{d,h})_p \simeq \bigoplus_{\substack{\lambda \subseteq (h^d) \\ |\lambda|=p}} \mathbb{S}_{2\lambda}(W^*). \quad (20.3)$$

In particular it is Gorenstein of socle degree dh , with one-dimensional socle representation $(\det W^)^{2h}$. Its Hilbert function and length are*

$$H_{d,h}(p) = \sum_{\substack{\lambda \subseteq (h^d) \\ |\lambda|=p}} \prod_{1 \leq i < j \leq d} \frac{2\lambda_i - 2\lambda_j + j - i}{j - i}, \quad (20.4)$$

$$L_{d,h} = \prod_{i=1}^d \frac{h + d - i + 1}{d - i + 1} \prod_{1 \leq i < j \leq d} \frac{2h + 2d - i - j + 2}{2d - i - j + 2}. \quad (20.5)$$

The Hilbert function is symmetric. A degree-one element is strong Lefschetz precisely when its associated symmetric matrix has full rank. More generally, an element associated with a rank- r matrix has nilpotence index $hr + 1$ (index one for the zero element). Consequently

$$\text{length } F_{g,k}(W) \geq L_{d,h}, \quad \mathfrak{m}^{dh} \neq 0 \text{ in } F_{g,k}(W). \quad (20.6)$$

The Gorenstein and Lefschetz assertions concern the section $B_{d,h}$, not the full fibre $F_{g,k}(W)$.

Proof. After setting the indicated coefficients to zero, the reciprocal series is $(1 + Qz^2)^{-1}$. The first even integer in $\{g + 1, \dots, g + k\}$ is $2(h + 1)$; it belongs to this interval because $k \geq 2$. Every odd coefficient vanishes, and the even ones are signed powers of Q . Thus the specialized ideal is exactly $(\text{coeff } Q^{h+1})$: all its higher powers have coefficients in this ideal. This proves (20.2) as an equality of ideals, not only an equality of radicals.

For clarity about the pairing, write $Q(x) = \sum_i y_{ii} x_i^2 + \sum_{i < j} y_{ij} x_i x_j$ and let X have independent symmetric entries X_{ij} , $i \leq j$. Send y_{ij} to $\partial/\partial X_{ij}$. Then $Q(x)$ acts by differentiation in the rank-one direction xx^t . Since $\det(X + zxx^t)^h$ has degree at most h in z , the coefficients of $Q(x)^{h+1}$ annihilate $(\det X)^h$. The reverse containment is the symmetric-matrix case of [30, Remark 11, pp. 8–9]: that apolar ideal is generated by the full general-linear orbit of the $(h + 1)$ st power of a rank-one differentiation direction. Powers of linear forms span their symmetric power, so this orbit span is precisely the coefficient space just displayed. No assertion that an arbitrary quotient of an apolar algebra is Gorenstein is used.

We record explicitly how the classical representation formulas translate here. In [30, Example 9 and Proposition 10], the surviving highest weights are indexed by $k_1, \dots, k_d \geq 0$ with $\sum k_i \leq h$, and their degrees are $\sum ik_i$. Set $\lambda_i = \sum_{j \geq i} k_j$. These are exactly the partitions in (h^d) , of size $\sum ik_i$, and their polynomial representations are $\mathbb{S}_{2\lambda}(W^*)$. This proves (20.3); the ordinary Weyl dimension formula gives (20.4). The same apolar module is the finite-dimensional highest-weight $\mathfrak{sp}_{2d}$ -module of weight $h\omega_d$, with its Levi character shifted by $\det^h$; see [30, Proposition 16 and Lemma 13]. For the positive roots $e_i - e_j$, $e_i + e_j$, and $2e_i$, its Weyl dimension factors are respectively 1, $(2h + 2d - i - j + 2)/(2d - i - j + 2)$, and $(h + d - i + 1)/(d - i + 1)$. Their product is (20.5). Apolarity to one homogeneous polynomial of degree dh gives the perfect pairings and

the socle. Equivalently, the complementary partition $(h - \lambda_d, \dots, h - \lambda_1)$ gives the dual summand with the indicated determinant twist.

The strong Lefschetz assertion is the symmetric-matrix case of [30, Theorem 4 and Example 5]. There is a direct argument for the asserted nilpotence index. If $\ell_A = \sum_{i \leq j} A_{ij} y_{ij}$, then $\ell_A^p = 0$ in the apolar algebra exactly when $\partial_z^p \det(X + zA)^h = 0$ as a polynomial in X . Congruence reduces a rank- r symmetric matrix to $\text{diag}(1, \dots, 1, 0, \dots, 0)$. The last determinant power has degree exactly hr in z , as is seen by taking X diagonal with independent entries. This proves the index. Finally a surjective local algebra map surjects on every power of its maximal ideal. The section has nonzero degree dh , proving both assertions in (20.6). $\square$

Corollary 20.2 (Size and Loewy depth of the actual common boundary fibre). *In the notation of Theorem 42.1, put $d = n^2 - 1$ and $h = n(n - 1)/2$. The entire normalization fibre at K_∞ has length at least $L_{d,h}$ and its maximal ideal has nonzero dh th power. For the ternary pencil boundary,*

$$\text{length } F_{6,2}(\mathbb{C}^8) \geq 922268360, \quad \mathfrak{m}^{24} \neq 0. \quad (20.7)$$

Its quadratic coefficient section has exactly that length, ordinary socle degree 24, and reciprocal-weight socle 48.

Proof. The common boundary fibre is $F_{g,k_n}(\mathbb{C}^{n^2-1})$, with $g = n(n - 1) \geq k_n \geq 2$. Apply the theorem. For $n = 3$, substitute $(d, h) = (8, 3)$ into (20.5). This computes a whole closed section, not the length of the original fibre. The previously obtained 44 coefficient variables and 9867 minimal equations remain counts of different invariants. $\square$

21. SPECTRAL FITTING SCHEMES AS POWER-ZERO LOCI IN ONE RECIPROCAL SECTION

The algebra just calculated supplies a single geometric object in which all spectral Fitting schemes can be read simultaneously. This avoids constructing a separate binary form after each Fitting ideal has already been recovered.

Let V have dimension n and $R \subset \text{Sym}^2 V$ be any pencil. Fix $h \geq 1$ and put $\tilde{E} = \mathbb{C}a \oplus V^*$. Consider the divisor-incidence point

$$K_h = a^{2h+2} \tilde{E} \subset \text{Sym}^{2h+3} \tilde{E}. \quad (21.1)$$

The entire degree- $2h$ normalization fibre over K_h is $F_{2h,2}(V^*)$, because $\tilde{E}$ is gcd-free. Its section by the linear-factor coefficients is $B_{n,h}$ and $(B_{n,h})_1 = \text{Sym}^2 V$. Thus the given pencil defines a line $\Lambda = \mathbb{P}(R)$ in the projectivized degree-one space of this actual reciprocal section. This construction uses a new formal line $\mathbb{C}a$ and is functorial for isomorphisms of V .

For $0 \leq p \leq hn$ let $\mathcal{J}_p$ be the ideal sheaf on Λ locally generated by the coordinates of

$$\ell^p \in (B_{n,h})_p \otimes \mathcal{O}_\Lambda(p), \quad (21.2)$$

where ℓ is the universal pencil member. Multiplying a local representative of ℓ by a unit does not change this ideal. We set $\mathcal{J}_0 = \mathcal{O}_\Lambda$. Let $\mathcal{T}_R$ be the cokernel of the universal symmetric matrix $V^* \otimes \mathcal{O}_\Lambda(-1) \rightarrow V \otimes \mathcal{O}_\Lambda$. This definition includes singular pencils.

Lemma 21.1 (The local power-ideal formula). *Let $A(t)$ be a symmetric matrix over $\mathbb{C}[[t]]$ of rank r over $\mathbb{C}((t))$. Write its finite elementary exponents as $0 \leq e_1 \leq \dots \leq e_r$ and put $e_i = +\infty$ for $i > r$. For $1 \leq p \leq hn$, write $p = hq + s$, $0 \leq s < h$. The ideal generated by the coordinates of $\ell_{A(t)}^p$ in $(B_{n,h})_p \otimes \mathbb{C}[[t]]$ is*

$$\left(t^{h \sum_{i=1}^q e_i + s e_{q+1}} \right). \quad (21.3)$$

An empty sum is zero, the last term is omitted if $s = 0$, and $t^{+\infty}$ means the zero ideal.

Proof. A symmetric matrix over this discrete valuation ring is congruent to

$$\text{diag}(t^{e_1}, \dots, t^{e_r}, 0, \dots, 0).$$

Here is the required elementary reduction. Divide by the least valuation of an entry. A diagonal unit is a pivot; if only an off-diagonal entry is a unit, the value of the form on a sum of two basis vectors supplies a unit because 2 is invertible. Split off the resulting rank-one summand by elimination and continue. Every remaining unit has a square root in $\mathbb{C}[[t]]$, so the units can be removed from the diagonal. The finite exponents are the Smith exponents, as congruence is in particular equivalence.

Congruence acts on every graded piece of $B_{n,h}$ by an invertible matrix over $\mathbb{C}[[t]]$. It therefore preserves the ideal in question. The apolar embedding

$$(B_{n,h})_p \longrightarrow \mathbb{C}[X_{ij}]_{hn-p}, \quad D \longmapsto D(\det X)^h$$

is injective and splits as a map of complex vector spaces. After tensoring with $\mathbb{C}[[t]]$, the coordinates of a vector and of its image generate the same ideal. We may therefore calculate the coefficient ideal of $[z^p] \det(X + zA(t))^h$.

For diagonal $A(t)$, expand each of the h determinant factors in its chosen diagonal entries. Every term is divisible by $t^{\sum c_i e_i}$ with $0 \leq c_i \leq h$ and $\sum c_i = p$. The least such exponent is obtained by filling the smallest e_i to capacity h , giving (21.3). There is no cancellation at this least exponent: specialize X to $\text{diag}(x_1, \dots, x_n)$. Then

$$\det(X + zA(t))^h = \prod_i (x_i + zt^{e_i})^h$$

(with x_i^h for a zero diagonal entry of A). The monomial $\prod x_i^{h-c_i}$ has coefficient $\prod \binom{h}{c_i} t^{\sum c_i e_i}$, which is nonzero and has exactly the required valuation. If $p > hr$, every term is zero. This proves all the conventions, including the zero-ideal case. $\square$

Theorem 21.2 (A reciprocal multiplication model for every spectral Fitting ideal). *For every complex quadratic pencil, regular or singular, and for $0 \leq j \leq n$, there is an equality of coherent ideal sheaves on its intrinsic line*

$$\mathcal{J}_{h(n-j)} = \text{Fitt}_j(\mathcal{T}_R)^h. \quad (21.4)$$

In particular, when $h = 1$, the complete spectral Fitting schemes, including their multiplicities and any whole-line components, are exactly the power-zero schemes inside the one quadratic section of the normalization fibre at K_1 .

For a regular pencil and a spectral point x , set $b_{x,j} = \text{length } \mathcal{O}_{\Lambda,x} / \text{Fitt}_j(\mathcal{T}_R)_x$. The local length of the power-zero scheme is $hb_{x,j}$, and, for $0 \leq j < n$,

$$\lambda_{x,j+1} = \frac{1}{h} \left(\text{length}(\mathcal{O}_{\Lambda,x} / \mathcal{J}_{h(n-j),x}) - \text{length}(\mathcal{O}_{\Lambda,x} / \mathcal{J}_{h(n-j-1),x}) \right). \quad (21.5)$$

Thus its multiplication, together with the embedded pencil line, recovers the full Segre data and their common projective support. For $n \geq 3$, the unmarked inverse makes this an invariant of the finite failure algebra. For a singular pencil, the construction recovers its normal rank and all its spectral Fitting ideals; it does not assert that these ideals alone recover the singular minimal indices.

Proof. At a point of the smooth curve Λ , pass to the completed discrete valuation ring. The ideal of $(n-j)$ -minors of a matrix with the stated Smith exponents is $(t^{e_1 + \dots + e_{n-j}})$, with the zero-ideal convention. Apply Lemma 21.1 with $p = h(n-j)$. The equality of completed ideals descends by faithful flatness. This proves (21.4) at every stalk and hence globally. This proof does not replace an ideal by its radical.

For a regular pencil the positive Smith exponents at x , in reverse order, are $\lambda_{x,1} \geq \lambda_{x,2} \geq \dots$. Consequently $b_{x,j} = \sum_{i>j} \lambda_{x,i}$. Equation (21.4) multiplies these lengths by h , so taking successive differences proves (21.5). The powers that vanish at the generic point similarly determine the normal rank in the singular case. Frame changes act on Λ by projectivities and congruences act equivariantly on the reciprocal construction. Theorem 8.3 recovers precisely the needed pencil up to these changes, proving the intrinsic assertion. $\square$

Remark 21.3 (Families and the scope of the ideal identity). For a vector bundle and pencil subbundle over any complex base, the quadratic-section algebra is obtained by applying the fixed $\mathrm{GL}(V)$ -equivariant construction to local frames. Its graded pieces are vector bundles, and the coefficient ideal of each power map commutes with arbitrary base change: it is the image of the dual of a map between vector bundles. No flatness of its zero scheme is needed for that assertion. Formula (21.4) has been proved on each complex pencil line (and on a regular one-dimensional base by the same local argument). We do not infer an equality of those relative ideals on an arbitrary nonreduced parameter space from equality on its geometric fibres. Nor are the power-zero schemes asserted to be flat across a change in their length.

Corollary 21.4 (The native specialization order in reciprocal coordinates). *On the fixed-discriminant, fixed-corank locus $X_{\mathbf{m},\mathbf{c}}(A)$ of (L.1), a specialization from the labelled partition tuple λ to μ exists if and only if, at every labelled root and every j , the local length of the corresponding power-zero scheme does not decrease. Equivalently the special power-zero divisor contains the generic one, using the common labelled spectral line. The reduced closure of a full $O(A)$ -orbit within this locus is therefore specified by these length inequalities. Each inequality can be written as the vanishing of the requisite jets of the coordinates of the power map (21.2). Every strict transition is realized by a flat family of the original finite failure neighbourhoods and is first distinguished at order $n^2 + 2n - 4$.*

Proof. For fixed multiplicity m_x , dominance $\mu_x \preceq \lambda_x$ is equivalent to $\sum_{i>j} \mu_{x,i} \geq \sum_{i>j} \lambda_{x,i}$ for every j . Use (21.5) and Theorem L.2. In a local coordinate t at a labelled root, a coefficient ideal has colength at least b exactly when each of its generators is divisible by t^b ; this is the asserted finite jet condition. The closure order and the flat realization are the inherited full-orthogonal results, ultimately using Ohta's theorem. The new assertion is their expression inside one reciprocal multiplication algebra. No reducedness assertion is made for the scheme cut out by the jet equations before taking its reduction. $\square$

Example 21.5 (A rank profile with a genuinely different power-zero scheme). At a root of multiplicity four the partitions $(3, 1)$ and $(2, 2)$ have equal determinant and corank. In $B_{4,h}$ their sixth powers when $h = 2$ (in general their $3h$ th powers) cut out local schemes of lengths h and $2h$, respectively. Their $4h$ th powers have length $4h$ in both cases. Thus the nonreduced power-zero scheme, not just the set of non-Lefschetz elements, distinguishes them. The specialization $(3, 1) \rightsquigarrow (2, 2)$ increases the first of these lengths. Additional simple spectral blocks give the same distinction in all larger dimensions, with power indices $h(n - 1)$ and hn .

The representation theory and strong Lefschetz property used above are classical inputs, explicitly credited to Nagaoka–Wachi and, for $h = 1$, Shafiei. The geometric statements here are the exact section of the whole reciprocal family, its size and Loewy-depth consequences for actual first-relation boundary fibres, and the power-ideal identity on pencil lines. The last construction is an intrinsically induced reciprocal model; it is not a claim that the common extreme $\mathrm{GL}(E)$ boundary point remembers the starting pencil. Its embedded pencil line is essential.

The preceding relative caution concerns the valuation-ring proof. The following polynomial identity establishes the equality over arbitrary complex base algebras.

22. UNIVERSAL POWER IDEALS

The power-zero construction admits an equality of ideals before any restriction to a pencil or to a valuation ring. This is the point at which the reciprocal section acquires a global boundary geometry. Let V have dimension $n \geq 2$, put $M = \text{Sym}^2 V$, and use the algebra $B_{n,h}$ of Theorem 20.1 with $W = V^*$. Thus $(B_{n,h})_1 = M$. If A is a symmetric matrix with entries in a commutative $\mathbb{C}$ -algebra T , denote its element of $M \otimes T$ by ℓ_A , and set

$$J_p(A) = \text{im}((B_{n,h})_p^* \otimes T \longrightarrow T, \quad \varphi \longmapsto \varphi(\ell_A^p)). \quad (22.1)$$

Write $I_q(A)$ for the ideal of all q -minors, with $I_0(A) = T$. In this section powers of ideals are ordinary powers unless parentheses explicitly indicate symbolic powers.

Theorem 22.1 (Universal determinantal formula). *For every commutative $\mathbb{C}$ -algebra T , every symmetric $n \times n$ matrix A over T , and $0 \leq p \leq nh$, write $p = hq + s$ with $0 \leq s < h$. Then*

$$J_p(A) = I_q(A)^{h-s} I_{q+1}(A)^s. \quad (22.2)$$

When $s = 0$ the second factor is omitted. For $p > nh$, $J_p(A) = 0$. These are equalities of ideals, not of radicals or integral closures, and commute with every homomorphism of $\mathbb{C}$ -algebras. In particular, for every symmetric map $A : T^n \rightarrow T^n$,

$$J_{h(n-j)}(A) = \text{Fitt}_j(\text{coker } A)^h, \quad 0 \leq j \leq n. \quad (22.3)$$

The formulas also hold locally, and hence as ideal sheaves, for symmetric maps of vector bundles with a line-bundle twist on an arbitrary $\mathbb{C}$ -scheme.

We separate the classical determinantal input. For an arbitrary matrix X over a characteristic-zero polynomial ring, the balancing inclusion

$$I_u(X)I_v(X) \subseteq I_{u+1}(X)I_{v-1}(X), \quad 0 \leq u \leq v-2, \quad (22.4)$$

is the determinantal symmetrization lemma. See [31, Lemma 2.5 and Theorem 2.4], which attributes the characteristic-zero product theorem to De Concini–Eisenbud–Procesi. Because this is a polynomial ideal inclusion, it remains true after specialization to a symmetric matrix; no flatness of that specialization is needed. This classical lemma is not a claim of the present paper.

Lemma 22.2 (The bounded-row generating space). *Let $S = \text{Sym}(\text{Sym}^2 V^*)$ and let A be its universal symmetric matrix. For $p = hq + s$, the degree- p part generating $I_q(A)^{h-s} I_{q+1}(A)^s$ is*

$$\bigoplus_{\substack{|\lambda|=p, \ell(\lambda) \leq n \\ \lambda_1 \leq h}} \mathbb{S}_{2\lambda}(V^*) \subseteq S_p. \quad (22.5)$$

Proof. The symmetric Cauchy decomposition of S_p is multiplicity free, with summands $\mathbb{S}_{2\lambda}(V^*)$, $|\lambda| = p$; see [32, §2.4]. We may use the dual general linear group to regard these as polynomial representations. Every minor has torus weight at most two in any one coordinate: its row set and column set each use that coordinate at most once. A product of h minors therefore has no weight with a coordinate exceeding $2h$. Its invariant span cannot contain a highest weight 2λ with $\lambda_1 > h$. This proves one inclusion.

For the converse, let $c_1 \geq \dots \geq c_h \geq 0$ be the column lengths of a partition λ occurring on the right, padded with zeros. The product of the leading principal minors of orders c_i is a nonzero highest-weight vector of weight 2λ in S_p . It belongs initially to $\prod_{i=1}^h I_{c_i}(A)$. Whenever two column lengths differ by at least two, apply (22.4) to increase the smaller length by one and decrease the larger by one. The sum of the squares of the lengths strictly decreases, so the procedure terminates. At termination the lengths are $q+1$ repeated s times and q repeated $h-s$ times, since their sum is p . Thus that highest-weight vector belongs to $I_q^{h-s} I_{q+1}^s$. The latter ideal

is invariant; its degree- p part contains the full irreducible summand. Multiplicity freeness proves the equality. The case $q = 0$ includes zero-length minors, which are the constant 1. $\square$

Proof of Theorem 22.1. First work over S with its universal matrix. The coefficient map of ℓ_A^p is the dual of the multiplication quotient $\text{Sym}^p M \rightarrow (B_{n,h})_p$, up to nonzero multinomial constants. Consequently its degree- p coefficient span is exactly $(B_{n,h})_p^* \subseteq S_p$. The character formula (20.3) identifies that span with (22.5). Both ideals in (22.2) are generated in degree p , so the lemma proves their equality in S . This step uses the full coefficient space, not just evaluation on diagonal matrices or closed points.

For a matrix over T , substitute its entries in this polynomial identity. Taking the image of the fixed finite-dimensional coefficient map commutes with scalar extension, as does taking minors and products of ideals. This proves the assertion even when T is nonreduced, non-Noetherian, or not a domain. The Fitting formula follows from its minor presentation. Changing a vector-bundle frame applies an invertible congruence to A and an invertible linear transformation to each coefficient space; changing a local frame of the twisting line multiplies generators by a unit. Therefore the local ideal identities glue. Finally, $(B_{n,h})_p = 0$ for $p > nh$. $\square$

Example 22.3 (A nonreduced test base). Take $T = \mathbb{C}[\epsilon]/(\epsilon^3)$, $n = h = 2$, and $A = \text{diag}(\epsilon, 0)$. Then $J_1(A) = (\epsilon)$ and $J_2(A) = (\epsilon^2) \neq 0$, whereas $J_3(A) = J_4(A) = 0$. Every closed-point matrix is zero. Thus equality on geometric fibres would not establish the theorem. Nor does (22.3) license taking unique ideal roots: for example, in $\mathbb{C}[\epsilon]/(\epsilon^2)$ the distinct ideals (ϵ) and 0 have the same square. At $h = 1$ the power ideals are the Fitting ideals themselves over every base.

23. THE RECIPROCAL POWER GRAPH AND COMPLETE QUADRICS

Let $P = \mathbb{P}(M)$, let P^{nd} be the nonsingular-quadric open, and define $\mathcal{G}_h$ as the scheme-theoretic closure of the graph

$$P^{\text{nd}} \longrightarrow P \times \prod_{p=1}^{nh-1} \mathbb{P}((B_{n,h})_p), \quad [A] \longmapsto ([A], [\ell_A], \dots, [\ell_A^{nh-1}]). \quad (23.1)$$

The first factor records the base point. We use the reduced, integral graph closure, equivalently its scheme-theoretic image; no additional normalization is inserted in this definition. Let $\text{CQ}(V)$ be the classical space of complete quadrics: the graph closure of $[A] \mapsto ([\wedge^2 A], \dots, [\wedge^{n-1} A])$ over P . For $n = 2$ the empty product gives $\text{CQ}(V) = P$.

Theorem 23.1 (A compactification determined by multiplication). *For every $h \geq 1$ there is a canonical $\text{GL}(V)$ -equivariant isomorphism over P*

$$\mathcal{G}_h \simeq \text{CQ}(V). \quad (23.2)$$

In particular the graph is smooth and projective, is independent of h as a scheme over P , and has a simple-normal-crossing boundary. Write $\pi : \text{CQ}(V) \rightarrow P$, and let E_r be its boundary divisor of rank break r , $1 \leq r < n$; for $r < n - 1$ it is exceptional over rank at most r , and E_{n-1} is the strict transform of the determinant divisor. Then every power ideal has the exact total transform

$$J_p \mathcal{O}_{\text{CQ}(V)} = \mathcal{O}_{\text{CQ}(V)} \left(- \sum_{r=1}^{n-1} (p - hr)_+ E_r \right), \quad 0 \leq p \leq nh, \quad (23.3)$$

where $(a)_+ = \max(a, 0)$. Removing this common divisor from the universal p th power gives a nowhere-zero projective section on the whole compactification.

The smoothness, boundary and flag interpretation of complete quadrics are classical, due to the complete-form construction; see [33, 34]. The assertion here identifies that particular compactification, with all of its scheme structure, from the multiplication in one reciprocal

normalization-fibre section. It does not assert that complete quadrics is a newly constructed variety.

Proof. We first recall the graph calculation with its scheme structure. If rational maps from an integral scheme are given by generators of nonzero ideals $K_1, \dots, K_a$, their simultaneous graph closure is $\text{Bl}_{K_1 \dots K_a}$ of the base. Indeed, after the Segre embedding the coordinates are all products of one generator from each ideal. The closure of this graph is the Proj of the Rees algebra of that product: on a generator chart its coordinate ring is the subring of the function field obtained by adjoining the generator ratios. These are exactly the affine charts of both constructions. Tensoring an ideal by an invertible ideal, or replacing it by a positive power, does not change the blow-up, the latter by the Veronese description of Proj.

Apply this calculation on P . The sheaf $I_1 \mathcal{O}_P$ is the unit ideal, and $I_n \mathcal{O}_P$ is invertible. For each $2 \leq q \leq n-1$, the exponent of I_q in $\prod_{p=1}^{nh-1} J_p$ is

$$\sum_{s=0}^{h-1} (h-s) + \sum_{s=1}^{h-1} s = h^2.$$

Theorem 22.1 therefore gives, up to an invertible ideal,

$$\prod_{p=1}^{nh-1} J_p = \left(\prod_{q=2}^{n-1} I_q \right)^{h^2}.$$

The same graph calculation for the exterior-power maps identifies $\text{CQ}(V)$ with the blow-up of $\prod_{q=2}^{n-1} I_q$. This proves (23.2) as an isomorphism of schemes, not only after normalization. Both isomorphisms are the identity on the common dense open, hence are canonical and equivariant.

For completeness, the local calculation behind the boundary formula is as follows. The complete-quadric construction blows up the strict rank loci successively. On its Gaussian charts, after a congruence and projective normalization, its matrix has the form

$$U \text{diag}(1, \tau_1, \tau_1 \tau_2, \dots, \tau_1 \cdots \tau_{n-1}) U^t, \quad (23.4)$$

with U lower unitriangular and $E_r = (\tau_r = 0)$. One obtains these charts inductively by making the first pivot one, taking a Schur complement, blowing up the zero residual matrix, and repeating. At each step the next nonzero residual quadratic form has a nonisotropic vector, so congruence translates of these charts cover the construction. This also explains the smooth charts and the normal crossings stated in the classical construction.

Cauchy–Binet, applied to (23.4), shows that every q -minor is divisible by $\tau_1^{q-1} \tau_2^{q-2} \cdots \tau_{q-1}$. The leading principal q -minor is exactly that product. Thus, on these charts and therefore globally,

$$I_q \mathcal{O}_{\text{CQ}(V)} = \mathcal{O} \left(- \sum_{r < q} (q-r) E_r \right). \quad (23.5)$$

Using (22.2), its exponent along E_r is $(h-s)(q-r)_+ + s(q+1-r)_+ = (p-hr)_+$. This proves (23.3). The coefficient sections generate the transformed ideal; division by its local generator makes them generate the unit ideal, proving the last assertion. $\square$

Corollary 23.2 (Native boundary strata). *The power graph has one congruence orbit for every subset $I \subseteq \{1, \dots, n-1\}$. It is the stratum $E_I^\circ = (\bigcap_{r \in I} E_r) \setminus \bigcup_{r \notin I} E_r$, of codimension $|I|$, and its closure is $\bigcap_{r \in I} E_r$. Its points are flags with nondegenerate projective quadratic forms on successive quotients; the quotient dimensions are the successive gaps in $\{0\} \cup I \cup \{n\}$. The simultaneous normalized power values recover this complete-quadric datum. They are not the one common limit for the full cotangent-coordinate group.*

Proof. The flag description and boundary stratification are the classical complete-quadric description. The general linear group acts transitively on flags of fixed dimensions, and over $\mathbb{C}$ every nondegenerate quadratic form on a fixed-dimensional quotient is congruent to the identity. Independent diagonal scalings remove the projective scalars. The chart (23.4) realizes each subset of vanishing τ_r and permits further coordinates to vanish, giving the stated codimensions and closures. The graph is a closed subscheme of the product in (23.1); its projections therefore retain the point, including its flag and quotient forms. The group here is the native congruence group of V . $\square$

Remark 23.3 (Relative construction and base change). For a vector bundle $\mathcal{V}$ on any $\mathbb{C}$ -scheme S , the associated bundle with fibre $\text{CQ}(\mathbb{C}^n)$ is a relative smooth projective scheme over S . Equivariance glues the power maps and the boundary formula. Formation of this associated bundle and the polynomial ideal identity commute with arbitrary base change. This does not say that taking the graph closure of an arbitrary restricted family commutes with nonflat base change. The relative ambient resolution and the closure of an individual family's graph are different constructions.

24. BOUNDARY CONTACTS AND SPECTRAL SCHEMES

Let $R \subset M$ be a regular pencil and $\Lambda = \mathbb{P}(R)$. Its map to P lifts uniquely to a morphism $\tilde{\iota}_R : \Lambda \rightarrow \text{CQ}(V)$: it lifts on the nonsingular open, and properness extends it uniquely across the missing points of the smooth curve. Put $D_r(R) = \tilde{\iota}_R^* E_r$. These are effective divisors because the generic member is nonsingular.

Theorem 24.1 (Contact-divisor reconstruction of spectral data). *For every $h \geq 1$, every regular pencil, and $0 \leq p \leq nh$,*

$$J_p|_{\Lambda} = \mathcal{O}_{\Lambda} \left(- \sum_{r=1}^{n-1} (p - hr)_+ D_r(R) \right). \quad (24.1)$$

At a point $x \in \Lambda$, let $0 = e_1 \leq \dots \leq e_n$ be the symmetric Smith exponents of the local pencil matrix. Then

$$\text{mult}_x D_r(R) = e_{r+1} - e_r \quad (1 \leq r < n). \quad (24.2)$$

Consequently the supported contact divisors determine every elementary divisor, hence the full Segre symbol. If $v_p(x) = \text{length}(\mathcal{O}_{\Lambda, x}/J_p)$ and $v_0(x) = 0$, then

$$\text{mult}_x D_r(R) = \frac{v_{h(r+1)}(x) - 2v_{hr}(x) + v_{h(r-1)}(x)}{h}. \quad (24.3)$$

Proof. Pull back (23.3); equality of extended ideals under composition gives (24.1). The pencil matrix is nonzero at every point, since R is a genuine two-dimensional subspace; hence its entries generate the unit ideal locally and $e_1 = 0$. Over $\mathbb{C}[[t]]$ symmetric elimination gives a congruence to $\text{diag}(t^{e_1}, \dots, t^{e_n})$, absorbing unit factors by square roots. Its unique lift has, in (23.4), $\tau_r = t^{e_{r+1} - e_r}$ up to a unit. This proves (24.2); faithful flatness of completion gives the local divisor statement. Taking valuations in (24.1) and second differences at multiples of h proves (24.3). Recover e_i by summing the first $i - 1$ contact multiplicities. The positive exponents, with their points on Λ , are precisely the elementary-divisor partitions of the spectral cokernel. $\square$

Example 24.2 (A boundary point and its contact order are distinct data). For a symmetric germ with exponents $(0, 1, 3)$ and $h = 2$, the valuations of $J_1, \dots, J_6$ are $(0, 0, 1, 2, 5, 8)$. Its two boundary contacts are 1 and 2. The germs $\text{diag}(1, t, t^3)$ and $\text{diag}(1, t^2, t^5)$ reach the same full-flag complete-quadric point but have contacts $(1, 2)$ and $(2, 3)$. The compactification retains the flag; the arc with its divisorial contacts retains the elementary exponents. These assertions must not

be conflated. For singular pencils the universal ideal theorem still holds and detects all Fitting schemes and the normal rank; Fitting ideals alone do not determine Kronecker minimal indices.

25. A PROJECTIVE INCIDENCE COMPACTIFICATION FOR PENCILS

The complete-quadric graph concerns members of pencils. There is also a projective parameter space carrying both the original finite failure algebra and flat limits of their resolved member curves. This construction keeps the pencil and does not identify distinct pencils at a common full-coordinate-change limit.

Proposition 25.1 (Resolved pencil curves and the failure family). *Let $n \geq 3$, $G = \text{Gr}(2, M)$, and let G° be the open set of regular pencils whose lines avoid the rank-at-most- $(n - 2)$ locus in P . There is a normal projective $\text{GL}(V)$ -equivariant scheme $\widehat{G}$ with a projective birational morphism $\rho : \widehat{G} \rightarrow G$, an isomorphism over G° , and a flat family of subschemes $\mathcal{C} \subseteq \widehat{G} \times \text{CQ}(V)$ with the following properties. Over G° it is exactly the lifted pencil line. Every fibre over R lies scheme-theoretically in $\pi^{-1}(\mathbb{P}(R))$. All the resolved power maps restrict to $\mathcal{C}$, for every h . The sharp finite failure algebra on G pulls back to a finite locally free algebra on $\widehat{G}$, with its original multiplication.*

Proof. The rank-at-most- $(n - 2)$ locus has codimension three in P , so a general line avoids it; a general pencil is regular. Thus G° is nonempty and dense. Over that open, π is an isomorphism along every pencil line and these lifts form a flat family of embedded projective lines.

For $1 \leq q < n$, let H_q be the pullback to $\text{CQ}(V)$ of the hyperplane bundle for its q th exterior-power projection, taking $q = 1$ to mean P . The bundle $L = \bigotimes_{q=1}^{n-1} H_q$ is very ample, by the graph embedding and Segre embedding. On a lifted line over G° , $\deg H_q = q$, since its q -minors have no common zero. The Hilbert polynomial with respect to L is therefore $\binom{n}{2}m + 1$. The lifted family defines a morphism

$$G^\circ \longrightarrow \text{Hilb}^{\binom{n}{2}m+1}(\text{CQ}(V)).$$

Take the reduced closure of its graph in G times this projective Hilbert scheme, and then its normalization; this is $\widehat{G}$. Normalization is finite here, so the result is projective and normal. The graph is already closed over G° and that open is smooth, so ρ is an isomorphism there. Equivariance of the family and the uniqueness of normalization give the stated group action.

Pull back the universal Hilbert family to obtain $\mathcal{C}$; it is flat. To verify the asserted scheme-theoretic incidence, pull back the linear equations for the universal pencil line to $\mathcal{C}$. They vanish over the dense open G° . On an affine open of the integral base $\widehat{G}$, flatness makes the coordinate modules of $\mathcal{C}$ torsion free over the base. A section which becomes zero at the generic point is therefore zero. Applied locally to these linear equations, this proves the incidence on the whole family. The power maps are everywhere defined on $\text{CQ}(V)$ by Theorem 23.1, so their restrictions exist. Finally pull back the finite locally free algebra of Corollary 7.5 along ρ ; arbitrary base change preserves its rank and multiplication. $\square$

This is an incidence compactification over the Grassmannian, with its native congruence action, not a properness assertion for the quotient stack by that action. A limiting Hilbert fibre may have extra or nonreduced components; no classification of all such fibres is used. The construction is independent of h : its target, polarization and Hilbert polynomial were specified using the h -independent complete quadric graph. The unmarked inverse in the companion reconstruction paper recovers the underlying pencil, so the incidence construction is compatible with that inverse rather than a replacement of it.

26. SINGULARITIES OF THE UNIVERSAL POWER-ZERO SCHEMES

There is an explicit further consequence for all the universal ideals, which we record separately from the geometry of the full reciprocal normalization fibre. Multiplier ideals are taken on the affine smooth space M , and $I_{r+1}^{(a)}$ denotes symbolic power, with exponent zero interpreted as the unit ideal.

Corollary 26.1 (Multiplier ideals and threshold). *For $1 \leq p \leq nh$, put $w_r = (p - hr)_+$ and $c_r = \binom{n-r+1}{2}$ for $0 \leq r < n$. For every real $b \geq 0$,*

$$\mathcal{J}(M, b \cdot J_p) = \bigcap_{r=0}^{n-1} I_{r+1}^{(\max\{0, \lfloor bw_r \rfloor + 1 - c_r\})}, \quad (26.1)$$

$$\text{lct}_M(J_p) = \min_{\substack{0 \leq r < n \\ p > hr}} \frac{\binom{n-r+1}{2}}{p - hr}. \quad (26.2)$$

Proof. Blow up the affine origin and then the strict rank loci. This is the affine complete-quadric resolution, whose charts are (23.4) with one additional scalar coordinate. Let E_0 be the divisor over the origin. The same minor calculation now includes $r = 0$, and the total transform of J_p is $\sum_{r=0}^{n-1} w_r E_r$. The discrepancy along E_r is $c_r - 1$: over the generic rank- r point the earlier blow-ups are isomorphisms, and the smooth rank- r locus has codimension c_r ; for $r = n - 1$ this is the strict determinant divisor and the discrepancy is zero. All boundary divisors have normal crossings, so this is a log resolution.

The defining pushforward formula for the multiplier ideal requires $\text{ord}_{E_r}(f) \geq \lfloor bw_r \rfloor + 1 - c_r$. At the generic rank- r point this valuation is the order in the regular local ring with maximal ideal I_{r+1} , so the condition on polynomials is exactly membership in the indicated symbolic power. Intersecting the conditions proves the first formula; the first value at which one becomes nontrivial proves the second. Equivalently this is the specialization, via Theorem 22.1, of the classical symmetric-determinantal multiplier-ideal calculation [32, §2.4 and Theorem 4.8]. $\square$

The ideal formula, graph identification and contact reconstruction are statements about the entire universal coefficient space. They do not supply the conductor of every multivariate common-divisor image or the Hilbert function of every full $F_{g,k}(W)$. Those objects remain in the preserved incidence theory; the present results connect their exact quadratic sections to a native global boundary and its spectral schemes.

27. POWER COEFFICIENTS AND THEIR TWISTS

For a vector space V , put $M(V) = \text{Sym}^2 V$, with congruence action $A \mapsto gAg^t$, and define

$$B_h(V) = \text{Sym}(M(V)) / \langle (v^2)^{h+1} : v \in V \rangle. \quad (27.1)$$

The degree of $M(V)$ is one. The generating space of the ideal is the Cartan inclusion $\text{Sym}^{2h+2} V \hookrightarrow \text{Sym}^{h+1}(\text{Sym}^2 V)$, normalized by $v^{2h+2} \mapsto (v^2)^{h+1}$. This definition requires neither a volume form nor a complementary space. It agrees with $B_{n,h}$ in Theorem 20.1 when $W = V^*$.

Proposition 27.1 (The coefficient map and its twists). *Let $n = \dim V$ and $N = nh$. The quotient multiplication μ_p gives an equivariant injection*

$$\mu_p^* : B_h(V)_p^* \hookrightarrow \text{Sym}^p(M(V)^*), \quad (\mu_p^* \varphi)(A) = \varphi(A^p). \quad (27.2)$$

Its image is the full coefficient space defining J_p , without a choice of apolar coordinates. The socle and perfect pairings are

$$B_h(V)_N = (\det V)^{2h}, \quad B_h(V)_p^* \simeq B_h(V)_{N-p} \otimes (\det V^*)^{2h}. \quad (27.3)$$

For a line L , $B_h(V \otimes L)_p = B_h(V)_p \otimes L^{2p}$, compatibly with all products. On $\mathbb{P}(M(V))$, the p th power is a section of $B_h(V)_p \otimes \mathcal{O}(p)$, and its base ideal is the image of $B_h(V)_p^* \otimes \mathcal{O}(-p) \rightarrow \mathcal{O}$.

Proof. A homogeneous polynomial of degree p is identified with its symmetric p -linear polarization, normalized so that its value on A is its value on $(A, \dots, A)$. Dualizing the surjection μ_p gives exactly (27.2). Multinomial coefficients appearing in a monomial basis express this normalization and are nonzero; no dual representation is silently replaced by the original one. Theorem 20.1 identifies the quotient with determinant apolarity. The determinant of $A : V^* \rightarrow V$ takes values in $(\det V)^2$. Choosing a trivialization gives a scalar polynomial whose inverse-system line has character $(\det V^*)^{2h}$; its dual socle therefore has character $(\det V)^{2h}$. The Gorenstein pairing gives the second formula. Each generator in $M(V)$ has scalar weight two, proving the line-twist assertion directly from (27.1). The tautological line $\mathcal{O}(-1) \rightarrow M(V) \otimes \mathcal{O}$ has p th product $\mathcal{O}(-p) \rightarrow B_h(V)_p \otimes \mathcal{O}$, which proves the last assertion. $\square$

27.1. Graph schemes and classical inputs. The projective graph argument uses the following precise statement. It applies to the ambient space and also to the two-dimensional total base of the degeneration computed below.

Lemma 27.2 (Simultaneous graph with invertible twists). *Let X be an integral noetherian complex scheme and let $W_i \otimes L_i^{-1} \rightarrow \mathcal{O}_X$ be nonzero coherent systems of sections, with finite-dimensional W_i and invertible L_i . Write K_i for their image ideals, and let $U = X \setminus \bigcup_i V(K_i)$. The scheme-theoretic graph closure of $U \rightarrow \prod_i \mathbb{P}(W_i^*)$ is $\text{Bl}_{\prod_i K_i} X$ with its specified graph embedding. This statement identifies the scheme and the maps, not only their normalizations. Replacing the product ideal by an invertible twist or a positive power does not change its blow-up over X .*

Proof. The open U is dense because all K_i are nonzero and X is integral. After the Segre embedding the simultaneous map is given by all products of one section from each system. On a trivializing affine open, these products generate $K = \prod_i K_i$. The homogeneous coordinate algebra of the graph is the image of the polynomial algebra on these products in $\mathcal{O}_X[t]$, namely the Rees algebra $\bigoplus_{m \geq 0} K^m t^m$. It is a subalgebra of the function-field polynomial ring. Thus its Proj is exactly the schematic closure of the graph, with no extra component. The Segre equations hold on the dense graph and hence on this integral closure scheme, so the graph lands in the product, not merely in its ambient projective space. Local trivializations glue. An invertible ideal changes the degree- m piece by its m th tensor power and does not change relative Proj; a positive power replaces the Rees algebra by a Veronese subalgebra. $\square$

For the power system, $L_i = \mathcal{O}(p)$ and $W_i = B_h(V)_p^*$; for the exterior system of order q , $L_i = \mathcal{O}(q)$ and its sections are the q -minors. The determinant system is invertible on the integral ambient space. It may be omitted from the blow-up product, although its zero divisor must still be retained in contact formulas. Likewise the first-coordinate identity map has unit base ideal. These observations specify the projective twists in the proof of Theorem 23.1.

The precise classical scheme-level complete-quadric construction is [33, Theorem 6.3]; the graph of exterior-power maps and its successive symmetric-rank blow-ups are stated in [34, Remark 2.5 and Construction 2.6]. The relative rank version is recalled in [35, Definition 2.5, Remark 2.6 and Theorem 2.14]. Consequently smoothness, the flag description, and the orbit stratification in Corollary 23.2 are classical consequences of the graph identification, not new classifications.

There is also a precise invariant-ideal predecessor to the universal identity. Henriques–Varbaro [32, §2.4, Theorems 2.6–2.8] recall Abeasis’s classification for the symmetric coordinate ring, its products of determinantal ideals and their integral closures, and the passage between invariant generators and products indexed by diagrams. Our coefficient-space comparison uses this same multiplicity-free setting. Theorem 22.1 identifies a particular multiplication coefficient map with

the balanced determinantal product as an ordinary ideal; it is not an independent classification of invariant ideals. The exact coefficient identification in Proposition 27.1 explains which map is being specialized on nonreduced bases. We assert no exhaustive historical nonanticipation of the displayed identity in all invariant-ideal terminology. Similarly the multiplier calculation in Corollary 26.1 is the substitution of its exponents in the symmetric determinantal formula [32, Theorem 4.8], not an independent multiplier-ideal theorem.

28. THE INTRINSIC SOURCE AND ITS APOLAR ENVELOPE

The source of a pencil and the coefficient complement in a divisor normalization fibre are different spaces. We construct the native power geometry from the first relation of the failure algebra, without taking a coefficient section of that fibre. The construction produces a canonical projective power diagram and an honest algebra bundle on the intrinsic source projective space. The location of the algebra bundle is essential: forgetting a scalar lift does not give a linear representation of the projective general linear group.

28.1. A construction from the first relation. Let A be an unmarked, ungraded complex algebra in the sharp pencil image, and write $\mathfrak{m}$ for its maximal ideal. Set

$$E_A = \mathfrak{m}/\mathfrak{m}^2, \quad H_A = \ker\{\mathrm{Sym}^d E_A \longrightarrow \mathfrak{m}^d\}, \quad d = n^2 + 2n - 4. \quad (28.1)$$

Here $n^2 = \dim E_A$, so neither n nor d is extra marking. The reduced support of the homogeneous ideal generated by H_A is a determinant hypersurface in $\mathbb{P}(E_A^*)$. Its successive reduced singular loci give the rank-one Segre variety. The scheme of maximal rank-one linear spaces has two components. Cancel the determinant factor from H_A and select the component singled out by coefficient support ranks $(1, M)$ in Lemma 8.1. Denote the resulting source projective space by S_A ; with a pencil presentation it is $\mathbb{P}(V)$, not $\mathbb{P}(\mathbb{C}^{n^2-1})$.

There is a direct jet-theoretic presentation which does not begin with a linear source lift. Let L be an ample generator of $\mathrm{Pic}(S_A)$, and let $J^1 L$ denote its first principal-parts bundle. Put

$$\mathcal{E}_A = \mathcal{H}om(J^1 L, L), \quad \mathcal{B}_{A,h} = \mathrm{Sym}(\mathrm{Sym}^2 \mathcal{E}_A) / \langle (u^2)^{h+1} : u \text{ a local section of } \mathcal{E}_A \rangle. \quad (28.2)$$

Scalar automorphisms of L act trivially on $\mathcal{E}_A$, so this bundle is independent of the representative of the ample generator. For $p > 0$ its projective coefficient space can be constructed as $\mathbb{P}H^0(S_A, \mathcal{B}_{A,h,p} \otimes L^{-2p})$; changing L changes the linear identification only by a scalar. Multiplication supplies the projective power maps. Thus jets, a Cartan quotient, and projectivization give the construction from the intrinsic source.

Theorem 28.1 (Intrinsic projective power geometry). *On the groupoid of complex sharp pencil-failure algebras and their ungraded isomorphisms, the following construction is functorial. There is a source projective space S_A , a finite locally free graded algebra $\mathcal{B}_{A,h}$ on S_A for each $h \geq 1$, and projective spaces $P_{A,p}$ with a multiplication power diagram. In any source lift $S_A = \mathbb{P}(V)$ they are*

$$\mathcal{B}_{A,h} = \bigoplus_{p=0}^{nh} B_h(V)_p \otimes \mathcal{O}_{\mathbb{P}(V)}(2p), \quad P_{A,p} = \mathbb{P}(B_h(V)_p). \quad (28.3)$$

The nondegenerate power graph on $P_{A,1}$ is the native $\mathrm{CQ}(V)$, canonically as a projective diagram, and every rank- r graph is the relative complete-quadric space of Theorem 29.1. Over a general base the projective spaces mean the associated projective schemes, which need not admit a global linear lift. These constructions use only (28.1), determinant cancellation, the oriented ruling, and (27.1); no pencil coefficient line or reciprocal-fibre splitting is needed to construct the ambient graph. Extracting that line afterwards supplies the intrinsic pencil line $\Lambda_A \subset P_{A,1}$.

All source-lift choices cancel. Once an oriented projective-source bundle is given, the same associated constructions descend over any complex base, including a nonreduced base, and commute

with arbitrary base change. This last assertion concerns the associated objects; it does not assert that reduction or graph closure commutes with arbitrary base change on unrelated degenerating families.

Proof. The first-relation construction is preserved by every ungraded isomorphism: terms of degree at least two in a change of generators cannot change the degree- d relation modulo $\mathfrak{m}^{d+1}$. The determinant and its rank-one locus are therefore intrinsic. Lemma 3.1 identifies the two ruling components as schemes; coefficient-support asymmetry orients them. These are the steps of the local inverse before the final recovery of the pencil coefficient line. Consequently an isomorphism of algebras induces a unique projective isomorphism of the selected source spaces, and compositions give compositions. This establishes the source functor without first choosing a congruence representative of the pencil.

To verify the jet presentation, take $S_A = \mathbb{P}(V)$ and $L = \mathcal{O}(1)$. The evaluation map $H^0(L) \otimes \mathcal{O} \rightarrow J^1 L$ is an isomorphism: a linear form whose value and first derivatives vanish at a point is zero, and both bundles have rank n . Hence $\mathcal{E}_A = V \otimes \mathcal{O}(1)$, and the Cartan quotient in (28.2) is exactly (28.3). Its degree- p twist by L^{-2p} is constant, with global section space $B_h(V)_p$; this proves the stated projective coefficient construction. Relative principal parts commute with base change on the locally trivial smooth projective-space bundle. Replacing L by its tensor product with a line from the base cancels in $\mathcal{H}om(J^1 L, L)$.

Fix a source lift temporarily. A scalar c acts on $B_h(V)_p$ by c^{2p} and on $\mathcal{O}_{\mathbb{P}(V)}(2p)$ by c^{-2p} . Hence every summand in (28.3), together with its multiplication, is naturally $\mathrm{PGL}(V)$ -equivariant. If two linear lifts of a projective isomorphism differ by a scalar, their induced algebra-bundle maps are equal. This also proves the cocycle identity on three overlaps; in a family, replacing V by $V \otimes L$ cancels L^{2p} against L^{-2p} . Thus the algebra bundle descends independently of all lifts. For a projective-source bundle, take fppf local frames of its PGL_n -torsor; the scalar-trivial action just proved supplies the cocycle descent even when no global vector-bundle lift exists. The same scalar acts trivially on each $P_{A,p}$, and power maps commute with scalar changes and congruence. They therefore define a projective diagram on the source torsor, with the same cocycle compatibility. Its first projective space can alternatively be described as the parameter space of quadric hypersurfaces in the dual projective space $S_A^\vee$.

Apply Theorem 23.1 to identify the nondegenerate graph. The rank- r identification is proved independently in Theorem 29.1. Equivariance makes both identifications independent of the source lift. Finally the residual left coefficient line, by the multiplicity-one exterior-duality map in the local inverse, gives the Pluecker point of a two-dimensional subspace in $\mathrm{Sym}^2 V$. Scalar changes do not change its projective line. This is Λ_A . Associated bundles, their specified equations, and their multiplication commute with any base change; no claim about arbitrary restricted graph closures is required. $\square$

Remark 28.2 (Exact intrinsic status). The envelope is a functor constructed from multiplication, not a claimed subalgebra or preferred quotient of A . It is an algebra bundle on S_A , not an untwisted algebra canonically based at $\mathrm{Spec} \mathbb{C}$. This distinction records, rather than suppresses, the scalar ambiguity: for $n \geq 3$ the central element $\zeta I \in \mathrm{SL}(V)$ acts on $B_h(V)_1 = \mathrm{Sym}^2 V$ by ζ^2 , which need not be one although it acts trivially projectively. Thus the usual linear degree-one action does not descend to $\mathrm{PGL}(V)$. The twist in (28.3) removes exactly this obstruction. There is no selection of an n -plane inside the auxiliary complement of dimension $n^2 - 1$. The latter still defines its own, different coefficient-section algebra in Corollary 20.2.

29. POWER GRAPHS OF EVERY RANK

For $1 \leq r \leq n$, let $P^{(r)} \subset \mathbb{P}(\text{Sym}^2 V)$ be the rank-exactly- r locus. Define $\mathcal{G}_{r,h}$ as the scheme-theoretic closure of

$$P^{(r)} \longrightarrow \mathbb{P}(\text{Sym}^2 V) \times \prod_{p=1}^{hr} \mathbb{P}(B_h(V)_p), \quad [A] \longmapsto ([A], [A], \dots, [A^{hr}]). \quad (29.1)$$

Keeping the final power is essential when $r < n$. Write $\mathcal{U}$ for the tautological rank- r bundle on $\text{Gr}(r, V)$ and $\text{CQ}(\mathcal{U})$ for its relative complete-quadric space. For rank one this means $\mathbb{P}(\text{Sym}^2 \mathcal{U}) = \text{Gr}(1, V)$.

Theorem 29.1 (The rank-stratified power graph). *There is a canonical isomorphism*

$$\mathcal{G}_{r,h} \simeq \text{CQ}(\mathcal{U}) \longrightarrow \text{Gr}(r, V). \quad (29.2)$$

The final power in (29.1) recovers the image r -plane by the degree- $2h$ Pluecker embedding. In particular the graph is smooth and projective, independently of h . For its relative boundary divisors E_i , $1 \leq i < r$, its power base ideals have total transforms

$$J_p \mathcal{O}_{\mathcal{G}_{r,h}} = \mathcal{O}_{\mathcal{G}_{r,h}} \left(- \sum_{i=1}^{r-1} (p - hi)_+ E_i \right), \quad 0 \leq p \leq hr. \quad (29.3)$$

The projective power maps in this formula have their common divisors removed. All powers $p > hr$ vanish on the rank- r locus.

Proof. For $U \subset V$, the map $B_h(U) \rightarrow B_h(V)$ from the presentation (27.1) is injective: any linear projection $V \rightarrow U$ splitting the inclusion induces a retraction on the quotient algebras. The injection itself is independent of that projection. Locally this gives subbundles $B_h(\mathcal{U})_p \subset B_h(V)_p$. If A has image U and rank r , it is a nonsingular element of $\text{Sym}^2 U$, and A^{hr} generates the socle $B_h(U)_{hr} = (\det U)^{2h}$. As U varies these lines give the homogeneous embedding

$$\text{Gr}(r, V) \hookrightarrow \mathbb{P}(\mathbb{S}_{(2h)r} V) \hookrightarrow \mathbb{P}(B_h(V)_{hr}).$$

Indeed the line at the coordinate r -plane is a highest-weight line of weight $(2h)^r$; its orbit map is the Pluecker embedding followed by its degree- $2h$ linear system. In particular this is a closed immersion, not just a recovery of the image plane on closed points.

Over the recovered Grassmannian, the remaining graph is precisely the relative simultaneous power graph for $B_h(\mathcal{U})$. By Theorem 23.1 on each frame chart, it is $\text{CQ}(\mathcal{U})$. The relative graph embedding, the subbundle injections above, and the final-power Grassmannian embedding together give a closed immersion into the product in (29.1). Its dense open is the graph of $P^{(r)}$. Both its source and that graph closure are integral, so equality of their scheme-theoretic images proves (29.2) without a normalization step. Applying (23.3) in rank r proves the transform formula. Local split inclusion of the coefficient bundles ensures that ambient and relative power ideals agree. The vanishing assertion follows from the socle degree of $B_h(U)$. $\square$

The varieties of complete singular quadrics are classical: see [35, Definition 2.5, Remark 2.6 and Theorem 2.14]. The point of the theorem is the exact recovery of their image-plane projection and entire scheme structure from the power maps in one fixed algebra $B_h(V)$, simultaneously for every r . The classical orbit stratification is background, not an additional novelty claim.

30. CONTACTS AND MINIMAL INDICES OF ARBITRARY PENCILS

Let $R = \langle A_0, A_1 \rangle \subset \text{Sym}^2 V$ be a genuine pencil, $\Lambda = \mathbb{P}(R)$, and r its normal rank. Its tautological symmetric map is $A : V^* \otimes \mathcal{O}_\Lambda \rightarrow V \otimes \mathcal{O}_\Lambda(1)$. Write

$$K = \ker A, \quad U = K^\perp \subset V \otimes \mathcal{O}_\Lambda.$$

The kernel is saturated and hence a subbundle on the smooth curve; U is the saturation of the generic image. Thus A is a generically nonsingular section of $\mathrm{Sym}^2 U \otimes \mathcal{O}_\Lambda(1)$. The map on the rank- r open lifts uniquely to $\tilde{\Lambda} \rightarrow \mathcal{G}_{r,h}$ by properness. Its Grassmannian projection is the map defined by U , and its source is still Λ . Let D_i be its contact divisor with E_i .

Lemma 30.1 (Symmetric diagonalization with a kernel). *Over $\mathbb{C}[[t]]$, a symmetric matrix of rank r over the fraction field is congruent to $\mathrm{diag}(t^{e_1}, \dots, t^{e_r}, 0, \dots, 0)$, with $e_1 \leq \dots \leq e_r$. For the local matrix of a genuine pencil, $e_1 = 0$. The exponents are its nonzero Smith exponents.*

Proof. Choose a matrix entry of smallest valuation. If a diagonal entry has that valuation, it is a pivot. Otherwise an off-diagonal entry does, and the value of the quadratic form on $e_i + ce_j$ has the same valuation for some $c \in \mathbb{C}$: its leading coefficient is a nonzero polynomial in c , since 2 is invertible. Extend this primitive vector to a basis. The pivot divides every entry in its row, so symmetric row-and-column elimination clears the row and column over the ring. All remaining entries have valuation at least that of the pivot. Repeat until the remaining block is zero. Units have square roots in $\mathbb{C}[[t]]$ and can be absorbed into the basis. Invertible congruence preserves determinantal ideals, which identify the resulting exponents with Smith exponents. A genuine pencil has a nonzero matrix at every projective parameter, giving a unit entry and $e_1 = 0$. This proof supplies the local congruence step rather than relying on left-right Smith form alone; compare the classical pencil congruence treatment in [36]. $\square$

For $j \geq 0$ define the finite multiplication-syzygy map

$$C_j(R) : V^* \otimes \mathrm{Sym}^j R^* \longrightarrow V \otimes \mathrm{Sym}^{j+1} R^*, \quad q \longmapsto Aq, \quad \kappa_j = \dim \ker C_j(R). \quad (30.1)$$

Set $\kappa_{-1} = \kappa_{-2} = 0$. Under a basis of R this is the block Toeplitz map with rows $A_0 q_0, A_0 q_i + A_1 q_{i-1}$, and $A_1 q_j$; its definition is independent of that basis.

Theorem 30.2 (Complete power-and-syzygy data). *For every complex symmetric pencil, including every singular pencil, the following formulas recover its complete congruence data. If $\varepsilon_1, \dots, \varepsilon_c$ are its minimal indices, where $c = n - r$, then*

$$K \simeq \bigoplus_{a=1}^c \mathcal{O}_\Lambda(-\varepsilon_a), \quad \kappa_j = \sum_{a=1}^c (j - \varepsilon_a + 1)_+, \quad (30.2)$$

$$\#\{a : \varepsilon_a = j\} = \kappa_j - 2\kappa_{j-1} + \kappa_{j-2}. \quad (30.3)$$

It is enough to use $0 \leq j \leq \lfloor (n-1)/2 \rfloor$. At $x \in \Lambda$, let $0 = e_1(x) \leq \dots \leq e_r(x)$ be the nonzero local Smith exponents. With $v_0 = 0$ and $v_p(x) = \mathrm{length}(\mathcal{O}_{\Lambda,x}/J_p)$ for $p \leq hr$, one has

$$v_{hq+s}(x) = h \sum_{i=1}^q e_i(x) + s e_{q+1}(x), \quad (30.4)$$

$$e_i(x) = \frac{v_{hi}(x) - v_{h(i-1)}(x)}{h}, \quad \mathrm{mult}_x D_i = e_{i+1}(x) - e_i(x). \quad (30.5)$$

In the first formula $s = 0$ at $q = r$ and the last term is omitted. The positive exponents at each supported point, together with the minimal indices, determine the pencil up to congruence and projective reparametrization of Λ . The spectral datum includes the positions of the points, not only their local partitions.

There is also a global conservation law:

$$\sum_{i=1}^{r-1} (r-i) \deg D_i + 2 \deg(\Lambda \longrightarrow \mathrm{Gr}(r, V)) = r, \quad \deg(\Lambda \longrightarrow \mathrm{Gr}(r, V)) = \sum_a \varepsilon_a. \quad (30.6)$$

The Grassmannian degree uses its Pluecker polarization. The first summand is the dimension of the regular summand in the Kronecker congruence form. Every object and number in the theorem is invariant under the ungraded failure-algebra isomorphisms of Theorem 28.1.

Proof. We use the classical complex symmetric Kronecker congruence theorem [36]. A singular block of minimal index ε has size $2\varepsilon + 1$ and form

$$S_\varepsilon(s, t) = \begin{pmatrix} 0 & L_\varepsilon(s, t)^\mathbf{t} \\ L_\varepsilon(s, t) & 0 \end{pmatrix}, \quad L_\varepsilon = \begin{pmatrix} s & t & & 0 \\ & s & t & \\ & & \ddots & t \\ & & & t \end{pmatrix},$$

where L_ε has ε rows and $\varepsilon + 1$ columns; S_0 is a zero 1×1 block. Its kernel is generated by $((-t)^\varepsilon, s(-t)^{\varepsilon-1}, \dots, s^\varepsilon)$ in the first block. These components have no common projective zero, so it gives $\mathcal{O}(-\varepsilon)$ as a kernel subbundle. A regular block has zero kernel as a sheaf. Direct sums prove the first formula in (30.2). Taking sections after twist by $\mathcal{O}(j)$ identifies them with (30.1), proving the second formula. The second finite difference of $(j - \varepsilon + 1)_+$ is one exactly at $j = \varepsilon$ and zero otherwise. Each singular block uses $2\varepsilon + 1 \leq n$ dimensions, proving the stated finite bound. The canonical-form classification is used here as a classical input, not reproved or claimed as new.

Locally, symmetric diagonalization factors the matrix through the saturated r -plane U with an $r \times r$ symmetric block. Theorem 22.1, or its rank- r version via the split inclusion $B_h(U) \hookrightarrow B_h(V)$, yields (30.4). Finite differences recover the e_i . In the relative complete-quadric chart the successive ratios of diagonal entries have valuations $e_{i+1} - e_i$, proving the contact formula. The torsion of the pencil cokernel has precisely the positive Smith exponents. The regular blocks of the classical congruence form are specified by these elementary divisors, and its singular blocks by the minimal indices; over $\mathbb{C}$ there are no additional signature parameters. Retaining the supported points on Λ makes the statement compatible with arbitrary projective changes of the pencil basis.

The exact sequence $0 \rightarrow U \rightarrow V \otimes \mathcal{O} \rightarrow K^* \rightarrow 0$ gives $\deg U = -\sum \varepsilon_a$, so the Grassmannian degree is $\sum \varepsilon_a$. The determinant of the induced symmetric map $U^* \rightarrow U(1)$ is a nonzero section of $(\det U)^2 \otimes \mathcal{O}(r)$; its zero divisor has degree $r - 2\sum \varepsilon_a$. At each point its order is $\sum e_i = \sum_{i < r} (r - i) \text{mult}_x D_i$. This proves (30.6). The block-size identity $n = n_{\text{reg}} + \sum (2\varepsilon_a + 1)$ identifies the first summand with n_{reg} . Finally the intrinsic first relation extracts Λ_A on its source and preserves all the maps involved; changes of source lift or pencil basis give isomorphic kernel maps and equal ideals, hence the asserted invariance. $\square$

Example 30.3 (Power contacts alone miss a singular invariant). The size-six pencils $S_0 \oplus S_2$ and $S_1 \oplus S_1$ both have constant rank four and no spectral torsion. All their power ideals are the unit ideal for $p \leq 4h$ and zero for $p > 4h$, and every contact divisor is empty. Their kernel splittings are respectively $\mathcal{O} \oplus \mathcal{O}(-2)$ and $\mathcal{O}(-1)^2$. Thus their first three syzygy nullities are $(1, 2, 4)$ and $(0, 2, 4)$, distinguishing the two congruence types. Both have Grassmannian degree two, as required by (30.6); the full splitting, rather than just its degree, is necessary.

Proposition 30.4 (Finite closed tests in families). *For $j \geq 0$ and $0 \leq a \leq n(j + 1)$, on the pencil Grassmannian each condition $\kappa_j \geq a$ is the closed determinantal condition on the bundle map C_j given by minors of size $n(j + 1) - a + 1$. These conditions commute with arbitrary base change as schemes. Their fibrewise ranks need not be constant, so no unqualified assertion that the kernel bundles commute with every base change is made. Along a specialization over an integral curve, κ_j can only increase.*

Proof. Use the tautological rank-two subbundle in (30.1). Its source has rank $n(j + 1)$ and its target rank $n(j + 2)$. Nullity at least a is equivalent to rank at most $n(j + 1) - a$, giving the stated minors. The ideals are polynomial in the bundle map and therefore commute with arbitrary

substitution. Upper semicontinuity of nullity, or specialization of these minors, gives the last assertion. $\square$

31. A COMPUTED BOUNDARY OF THE PENCIL INCIDENCE SPACE

We give the incidence compactification a precise universal property and compute a family of its boundary fibres in every dimension. The new boundary is selected by one power base ideal, not by pulling an unrelated algebra back to a parameter space.

Proposition 31.1 (The normal main-component flattening property). *Let $G^\circ \subset G = \text{Gr}(2, \text{Sym}^2 V)$ and $\widehat{G} \rightarrow G$ be as in Proposition 25.1. Among normal integral schemes T with a dominant morphism $f : T \rightarrow G$ for which the closure of the lifted pencil family over $f^{-1}(G^\circ)$ is an embedded flat family in $\text{CQ}(V) \times T$ with Hilbert polynomial $\binom{n}{2}m + 1$, there is a unique morphism $T \rightarrow \widehat{G}$ over G , and that family is the pullback of its universal family. Consequently $\widehat{G}$ is the minimal normal dominant modification with this specified embedded-flat-closure property.*

Proof. The flat embedded family gives a morphism to the indicated Hilbert scheme. Together with f it agrees, over a dense open of T , with the graph defining the reduced graph closure. Since T is integral, its morphism factors through this closed graph scheme: each equation vanishing on the dense open vanishes on T . The induced morphism is dominant. A dominant morphism from a normal integral scheme to an integral scheme factors uniquely through the normalization: locally, an element integral over the target ring becomes integral over the source ring and lies in its fraction field, hence belongs to the normal source ring. Thus the morphism factors uniquely through $\widehat{G}$. The Hilbert-scheme representing property identifies the families. Conversely its pulled-back universal family is flat; flatness over an integral base makes it the schematic closure of its restriction to the dense open. Uniqueness follows also from separatedness. $\square$

The hypothesis of dominance is part of this property; this proposition does not assert that normalization commutes with arbitrary base change. The property is a standard consequence of the Hilbert functor and normalization. Its usefulness here is that the following calculation identifies a nontrivial fibre of the resulting modification.

Assume $n \geq 3$. Choose distinct nonzero numbers $a_3, \dots, a_n$ and put $D = \text{Spec } \mathbb{C}[[\tau]]$. On $\mathbb{P}_D^1$, consider the symmetric pencil

$$A_\tau(s, t) = \text{diag}(s, s + \tau t, t - a_3 s, \dots, t - a_n s). \quad (31.1)$$

For $n = 3$ there is just one number a_3 . Write $X = \mathbb{P}_D^1$ and let o be the point $s = \tau = 0$ in the special fibre. Its generic pencil has n distinct rank- $(n - 1)$ spectral points, so defines a point of G° .

Theorem 31.2 (A transverse corank-two tail and its power maps). *The closure of the generic lifted curve of (31.1) in $\text{CQ}(V) \times D$ is*

$$Z = \text{Bl}_o X. \quad (31.2)$$

It is flat over D , and its special fibre is the reduced nodal union $C \cup F$, where $C \simeq \mathbb{P}^1$ is the lifted special pencil and $F \simeq \mathbb{P}^1$ is an exceptional tail. There are no embedded components. If H_q is the exterior-power hyperplane bundle of order q , then

$$(\deg_C H_q, \deg_F H_q) = \begin{cases} (q, 0), & 1 \leq q \leq n - 2, \\ (n - 2, 1), & q = n - 1. \end{cases} \quad (31.3)$$

Thus the special fibre has the same Hilbert polynomial $\binom{n}{2}m + 1$ as the generic lifted line.

The tail lies in E_{n-2} , its two intersections with E_{n-1} are distinct and reduced, and its attachment to C is neither of those two points. For any $h \geq 1$, let L_p be the hyperplane bundle of the resolved p th power map. Then

$$\deg_F L_p = \begin{cases} 0, & p \leq h(n-2), \\ \min\{b, 2h-b\}, & p = h(n-2) + b, \quad 0 \leq b \leq 2h. \end{cases} \quad (31.4)$$

When this degree is positive, the restricted map on F is the complete Veronese map of that degree into its linear span. Most directly, in the local coordinate $z = s/t$ at o ,

$$J_{h(n-2)+1}(A_\tau) = (z, \tau). \quad (31.5)$$

Hence a single multiplication power ideal selects the blow-up centre and its exceptional component.

Proof. The rank is at least $n-2$ on X . In a neighbourhood of o the last $n-2$ diagonal entries are units; all other spectral points have rank $n-1$ and are pairwise distinct. Consequently the extended minor ideals I_q are the unit ideal for $q \leq n-2$, while

$$I_{n-1} = (z, z + \tau) = (z, \tau), \quad I_n = (uz(z + \tau))$$

near o , with u a unit. Away from o , I_{n-1} is a unit ideal; I_n is everywhere invertible. The simultaneous exterior-power graph on the integral surface X is therefore the blow-up of (z, τ) . This is an application of the graph-by-generators lemma, not a claim that arbitrary base change of a blow-up is a blow-up. Since the universal family of pencil lines embeds X into $\mathbb{P}(\text{Sym}^2 V) \times D$, this graph embeds into $\text{CQ}(V) \times D$ and is the required scheme-theoretic closure.

On the chart with $\tau = zv$, the special fibre has equation $zv = 0$; on the other chart, $z = \tau w$, it has equation $\tau = 0$. These charts prove flatness over the DVR, multiplicity one of both components, their transverse intersection, and absence of embedded components. The map on C is the unique lift of the special pencil. The q -minors of that pencil have degree q . They have no common factor for $q \leq n-2$, and precisely one factor s for $q = n-1$. This proves their degrees on C . On F , all lower maps are constant, and the $(n-1)$ st exterior map is linear. This gives (31.3). For $L = \bigotimes H_q$, the degrees on C and F add to $\binom{n}{2}$. The normalization sequence of $C \cup F$ subtracts one point, so its Euler characteristic is one and its Hilbert polynomial is as stated. Thus it is the Hilbert limit used by $\hat{G}$; equivalently the generic morphism from D extends to $\hat{G}$ by properness, with this universal fibre.

Use homogeneous coordinates $[\alpha : \beta] = [z : \tau]$ on F . After removing the exceptional factor, the residual quadratic form on the two-dimensional kernel is $\text{diag}(\alpha, \alpha + \beta)$. This is a line in $\text{CQ}(\mathbb{C}^2) = \mathbb{P}(\text{Sym}^2 \mathbb{C}^2)$, and gives the claimed image of F in the fibre of E_{n-2} . Its determinant $\alpha(\alpha + \beta)$ has two simple zeros $[0 : 1]$ and $[-1 : 1]$, which are its intersections with E_{n-1} . The main component attaches at $[1 : 0]$, where this residual form is nondegenerate.

For clarity, the whole list of extended power ideals near o is

$$J_p = \begin{cases} \mathcal{O}, & p \leq h(n-2), \\ (z, \tau)^b, & p = h(n-2) + b, \quad 0 \leq b \leq h, \\ (z, \tau)^{2h-b} [z(z + \tau)]^{b-h}, & p = h(n-2) + b, \quad h \leq b \leq 2h, \end{cases} \quad (31.6)$$

up to an invertible factor. These are actual ideals, by Theorem 22.1. In particular $b = 1$ gives (31.5) for every h .

It remains to check the linear system on the tail, not only its degree. On diagonal matrices the coefficient polynomials of a p th power span exactly the monomials of total degree p in the diagonal entries with each exponent at most h . For containment, use the torus-weight bound in Lemma 22.2; for the reverse inclusion, take the coefficients of the independent diagonal entries of

X in $\det(X + uA)^h$. Thus, for $p = h(n - 2) + b$, the leading normal coefficients use exponent h in each of the $n - 2$ invertible entries and span

$$\alpha^j(\alpha + \beta)^{b-j}, \quad \max\{0, b - h\} \leq j \leq \min\{b, h\}.$$

If $b \leq h$ these span every binary form of degree b . If $b > h$, remove the common factor $[\alpha(\alpha + \beta)]^{b-h}$; the remaining monomials span every binary form of degree $2h - b$. Lower powers have nonzero constant normal term and are constant on F . This proves the complete-linear-system assertion and (31.4). $\square$

Remark 31.3 (What the failure family contributes). Along (31.1), the original finite failure algebras form the finite locally free family already constructed in Corollary 7.5. Its pullback to F is constant; we do not count that pullback as a new interaction. Instead, its first relation canonically recovers the source and the pencil line, and the source-normalized envelope has the total-base ideal (31.5). Resolving that multiplication ideal creates F , and the other products prescribe all the linear systems (31.4) on it. A degeneration arc, not only its closed failure algebra, supplies this total-base ideal. Thus the calculation distinguishes information in the family from information in the special point and supplies an explicit modular Hilbert limit.

32. UNIVERSAL PROPERTY OF THE INTRINSIC ENVELOPE

The intrinsic envelope is a construction from the multiplication table, not an identification of an auxiliary coefficient section with the original algebra. Its universal property can be stated before choosing a linear source or recovering a representative of the pencil.

Proposition 32.1 (The source-normalized nilpotence envelope). *Let S_A and $\mathcal{E}_A = \mathcal{H}om(J^1 L, L)$ be as in Theorem 28.1. The algebra $\mathcal{B}_{A,h}$ represents, on S_A , graded commutative algebras C equipped with a linear map $\mathrm{Sym}^2 \mathcal{E}_A \rightarrow C_1$ for which*

$$(\text{image of } u^2)^{h+1} = 0$$

for every local section u , also after base change. More precisely, giving such a map is equivalent to giving a unique graded algebra map $\mathcal{B}_{A,h} \rightarrow C$. The degree-zero part maps by the structure map. This property, its representing algebra, and its projectivized power maps descend along the oriented source torsor. The case $h = 1$ requires no additional integer choice and gives the same native graph as every $h \geq 1$.

Proof. The symmetric algebra is free on its degree-one module. In characteristic zero the span of $(u^2)^{h+1}$ is exactly the image of

$$\mathrm{Sym}^{2h+2} \mathcal{E}_A \longrightarrow \mathrm{Sym}^{h+1}(\mathrm{Sym}^2 \mathcal{E}_A), \quad u^{2h+2} \longmapsto (u^2)^{h+1}.$$

Polarization verifies this assertion on every trivializing open; the required denominators are nonzero integers, hence units on every complex base. Factoring a free-algebra map through the ideal generated by this image is equivalent to the displayed identities. This proves existence and uniqueness and their compatibility with base change.

On $\mathbb{P}(V)$, evaluation of linear sections identifies $J^1 \mathcal{O}(1)$ with $V^* \otimes \mathcal{O}$. Thus $\mathcal{E}_A = V \otimes \mathcal{O}(1)$. A scalar on V acts oppositely on $\mathcal{O}(1)$, so it acts trivially on $\mathcal{E}_A$, the Cartan map, and the quotient. This proves descent without choosing a global $\mathcal{O}(1)$ on a nonsplit source bundle. The scalar action on each untwisted coefficient space is scalar, and hence also disappears on its projectivization. The graph assertions follow from Theorems 23.1 and 29.1. $\square$

This universal property distinguishes an algebra bundle constructed from the intrinsic source from a preferred linear subquotient of A . The former is proved; the latter is neither needed nor asserted. The source is extracted by the determinant and oriented-ruling steps of the first-relation inverse, before its final pencil-coefficient step.

33. ORDINARY COLLISION STRATA AND THEIR HILBERT FIBRES

We compute simultaneous collisions of arbitrary multiplicities, and allow higher-order perturbations of the total family. Besides describing its reduced components, we determine every resolved power system and exhibit positive-dimensional families in a fibre of the pencil incidence modification. The calculation uses the ordinary power ideals on the two-dimensional total base.

33.1. A local ideal calculation.

Lemma 33.1 (Distinct tangent factors). *Let $\ell_1, \dots, \ell_c$ be pairwise nonproportional linear forms in two variables. For $0 \leq j < c$, the products of j distinct ℓ_i span all binary forms of degree j . Consequently, if R is a regular local complex surface ring with maximal ideal $\mathfrak{n} = (z, \tau)$, and a symmetric $c \times c$ matrix S satisfies*

$$S = zC + \tau D + O(\mathfrak{n}^2), \quad C \text{ nonsingular}, \quad \det(\alpha C + \beta D) \text{ squarefree},$$

then $I_j(S) = \mathfrak{n}^j$ for $j < c$, while $I_c(S) = (\det S)$.

Proof. Choose $j + 1$ of the linear forms. Their products obtained by omitting one factor are linearly independent: evaluation at the zero of the omitted factor kills all other products and not that one. Their number is $j + 1$, the dimension of the required binary-form space. For $j = 0$ the assertion is immediate.

The endomorphism $C^{-1}D$ is self-adjoint for C and has distinct eigenvalues. Its eigenspaces are orthogonal and none is isotropic, since otherwise C would be degenerate. A constant congruence therefore makes $\alpha C + \beta D$ diagonal with distinct linear entries. Its j -minors span $\text{Sym}^j(\mathfrak{n}/\mathfrak{n}^2)$. The entries of S lie in $\mathfrak{n}$, so $I_j(S) \subset \mathfrak{n}^j$ and $I_j(S) + \mathfrak{n}^{j+1} = \mathfrak{n}^j$. Nakayama's lemma applied to $\mathfrak{n}^j/I_j(S)$ proves equality. The argument may be made after completion and descended by faithful flatness. The determinant assertion is its definition. $\square$

Lemma 33.2 (Block products and the first normal term). *For $V = U \oplus W$, multiplication gives an injection*

$$B_h(U) \otimes B_h(W) \longrightarrow B_h(V).$$

If $\dim U = r$, $Q \in \text{Sym}^2 U$ is nonsingular, and $S \in \mathfrak{n} \otimes \text{Sym}^2 W$ has a generically nonsingular linear term S_1 , the first nonzero normal term of $(Q + S)^p$, for $hr < p \leq h \dim V$, is

$$\binom{p}{hr} Q^{hr} S_1^{p-hr}.$$

For $p \leq hr$ its order-zero term Q^p is nonzero. These statements remain valid with a nonsingular formal unit block and after a formal congruence, with the corresponding invertible changes of coefficients.

Proof. The Cartan relations on each summand vanish in $B_h(V)$, so the algebra map is defined. Its source is Artin Gorenstein. Its socle is the product of the two socles, and that product is nonzero in $B_h(V)$: in determinant apolarity this is the nonzero mixed top derivative of $\det(X_U)^h \det(X_W)^h$, obtained by restricting to block diagonal matrices. A nonzero ideal in an Artin local Gorenstein algebra meets its socle: successive multiplication of a nonzero ideal by the maximal ideal ends in a nonzero subspace annihilated by that maximal ideal. The kernel is therefore zero.

Expand $(Q + S)^p$ in the injected tensor product. Powers of Q above hr vanish, whereas Q^{hr} is a nonzero socle generator. For $p > hr$ the term with the fewest factors of S consequently has precisely $p - hr$ such factors and is the displayed term. The unit block and a formal change of frame specialize to invertible changes on the associated graded coefficient space. This proves the last assertion as well. $\square$

33.2. The simultaneous collision theorem. Let $D_0 = \text{Spec } \mathbb{C}[[\tau]]$, let $\mathcal{R}$ be a rank-two pencil subbundle of $\text{Sym}^2 V \otimes \mathcal{O}_{D_0}$, and put $X = \mathbb{P}(\mathcal{R})$. Assume its generic pencil has n distinct spectral points. Its special pencil is assumed regular. Apart from simple spectral points, suppose it has distinct points $x_1, \dots, x_k$ of coranks c_a , where $2 \leq c_a \leq n - 1$. Write $r_a = n - c_a$.

At x_a , choose a parameter z_a on the special pencil and a unit $r_a \times r_a$ block. Its symmetric Schur complement is required to satisfy

$$S_a = z_a C_a + \tau D_a + O((z_a, \tau)^2), \quad C_a \text{ nonsingular}, \quad \det(\alpha C_a + \beta D_a) \text{ has } c_a \text{ simple zeros.} \quad (33.1)$$

We call this an ordinary transverse cluster. Intrinsically, $C_a, D_a \in \text{Sym}^2 W_a$, where $W_a = (\ker(A(x_a) : V^* \rightarrow V))^*$. Their pencil is the first derivative of the form restricted to its radical. Taking a Schur complement does not alter that first derivative. Replacing the parameter changes the basis of this residual pencil and preserves its marked member $[C_a]$. Thus the hypothesis does not depend on the chosen complement or frame.

Theorem 33.3 (Simultaneous ordinary collision fibres). *Under (33.1), the schematic closure of the generic lifted pencil in $\text{CQ}(V) \times D_0$ is*

$$Z = \text{Bl}_{\{x_1, \dots, x_k\}} X. \quad (33.2)$$

It is regular and flat over D_0 . Its special fibre is the reduced nodal tree

$$C \cup F_1 \cup \dots \cup F_k,$$

where C is the lift of the special pencil and each $F_a \simeq \mathbb{P}^1$ meets C in one point. There are no embedded components and the tails are pairwise disjoint. The tail F_a is the complete-quadric lift of the pointed residual pencil $([\alpha C_a + \beta D_a], [C_a])$ on W_a . It lies in E_{r_a} , meets E_{n-1} in c_a distinct reduced points, meets no other boundary divisor, and attaches at the nonsingular member $[C_a]$.

Let H_q be the hyperplane bundle of the resolved q th exterior map, $1 \leq q \leq n - 1$. Then

$$\deg_{F_a} H_q = (q - r_a)_+, \quad \deg_C H_q = q - \sum_a (q - r_a)_+. \quad (33.3)$$

For the product polarization $\bigotimes_{q=1}^{n-1} H_q$ the Hilbert polynomial is $\binom{n}{2}m + 1$. In particular $\deg F_a = \binom{c_a}{2}$ and $\deg C = \binom{n}{2} - \sum_a \binom{c_a}{2}$.

For every $h \geq 1$ and $1 \leq p \leq hn$, set

$$b_a = (p - hr_a)_+, \quad s_p = (p - h(n - 1))_+, \quad \delta_{a,p} = b_a - c_a s_p. \quad (33.4)$$

These integers are nonnegative. The resolved p th power restricts to the complete linear system of degree $\delta_{a,p}$ on F_a . It is constant if that degree is zero, and otherwise the complete Veronese map into its linear span. Near x_a , with $\mathbf{n}_a = (z_a, \tau)$ and $f_a = \det S_a$, the exact ideals are

$$J_p(A) = \mathbf{n}_a^{\delta_{a,p}} (f_a)^{s_p}, \quad J_{hr_a+1}(A) = \mathbf{n}_a. \quad (33.5)$$

Equalities are up to multiplication by a unit after a local frame is fixed, not up to radical or integral closure. Consequently the multiplication powers select the graph modification and every tail linear system. This description depends only on the first normal jet in (33.1); with the special pencil fixed, higher-order terms do not change its embedded special fibre or its power maps.

Proof. Near x_a the unit block and its Schur complement give $I_q(A) = \mathcal{O}$ for $q \leq r_a$ and $I_{r_a+j}(A) = I_j(S_a)$. Lemma 33.1 therefore gives

$$I_{r_a+j}(A) = \mathbf{n}_a^j \quad (1 \leq j < c_a), \quad I_n(A) = (u_a f_a), \quad u_a \in \mathcal{O}^\times.$$

At every other point the rank is at least $n - 1$, so all minor ideals of order at most $n - 1$ are units. The determinant ideal is invertible. The simultaneous exterior graph is the blow-up of the product of its base ideals by Lemma 27.2. Near x_a this product is $\mathbf{n}_a^{\binom{c_a}{2}}$. A positive power has

the same blow-up as the ideal itself, and the points are disjoint. These local identifications glue uniquely over X to (33.2). This computes the graph on the total surface; it is not an assertion that graph closure commutes with arbitrary base change.

The tautological family of genuine pencil lines embeds X into $\mathbb{P}(\mathrm{Sym}^2 V) \times D_0$. Hence the simultaneous graph embedding places Z in $\mathrm{CQ}(V) \times D_0$, with exactly the prescribed generic curve. The two charts $\tau = z_a v$ and $z_a = \tau w$ of each point blow-up show that Z is regular. In the first chart its special fibre is $z_a v = 0$. Both branches have multiplicity one and intersect transversally. In the second chart only the exceptional branch occurs. The parameter τ is a nonzerodivisor, so Z is flat over the DVR. The charts also rule out embedded components.

On F_a , divide the Schur block by the exceptional parameter. Its value is $\alpha C_a + \beta D_a$. All earlier unit-block data specialize to their values at x_a , independently of the direction. The complete-quadric chart thus identifies F_a with the lifted residual pencil in the fibre over $[A(x_a)]$. Its only residual rank drops have corank one and occur at the c_a simple determinant zeros. This proves the boundary assertions, including the attachment at $[\alpha : \beta] = [1 : 0]$. For $q = r_a + j < n$, the leading q -minors are the j -minors of this residual pencil. They have no common zero and have degree j . The lower maps are constant. On C , the common factor in the q -minors has order $(q - r_a)_+$ at x_a and no other zeros. This proves (33.3). The sum of the component degrees is $\binom{n}{2}$. The normalization sequence for a tree of $k + 1$ projective lines subtracts its k nodes, so its Euler characteristic is one. This proves the Hilbert polynomial and identifies the curve as the fibre in the Hilbert incidence space.

For the power ideals apply Theorem 22.1. If $p \leq hr_a$, the ideal is the unit ideal. Otherwise write $p = hr_a + hj + s$ with $0 \leq s < h$. For $j < c_a - 1$, the two minor ideals are $\mathfrak{n}_a^j$ and $\mathfrak{n}_a^{j+1}$, so the power ideal is $\mathfrak{n}_a^{hj+s}$. For $j = c_a - 1$ it is

$$\mathfrak{n}_a^{(c_a-1)(h-s)}(f_a)^s.$$

At $p = hn$ it is $(f_a)^h$. These cases are exactly (33.5). The value $p = hr_a + 1$ gives the point ideal for every h .

It remains to identify the complete coefficient space on F_a . By Lemma 33.2, its first normal coefficients, for $p > hr_a$, are exactly those of the b_a th power of the residual pencil, multiplied by a nonzero unit-block socle vector. In the polynomial ring $\mathbb{C}[\alpha, \beta]$, Lemma 33.1 gives $I_j = (\alpha, \beta)^j$ for $j < c_a$. The universal power identity gives the homogeneous ideal

$$(\alpha, \beta)^{\delta_{a,p}} \det(\alpha C_a + \beta D_a)^{sp}.$$

All its original power generators have degree b_a ; therefore their span is its entire degree- b_a part. Remove the common determinant factor. The remaining space is exactly $H^0(\mathbb{P}^1, \mathcal{O}(\delta_{a,p}))$, proving the assertion about the linear system. This argument determines the map as well as its degree. All surviving leading coefficients use only the first normal jet; the higher-order terms vanish on the exceptional divisor. $\square$

33.3. A modular family over a fixed pencil. Let R_0 be a regular pencil whose spectral elementary divisors all have exponent one, with repeated spectral multiplicities $c_1, \dots, c_k \geq 2$. Thus these are semisimple collisions, not nontrivial Jordan blocks. At x_a keep the residual space W_a and the nonsingular tangent member $[C_a]$. Put

$$U_a = \{\ell \text{ a line through } [C_a] \text{ in } \mathbb{P}(\mathrm{Sym}^2 W_a) : \ell \text{ meets the determinant in } c_a \text{ simple points}\}. \quad (33.6)$$

This is a nonempty open in $\mathbb{P}(\mathrm{Sym}^2 W_a / \langle C_a \rangle)$, of dimension $\binom{c_a+1}{2} - 2$.

Theorem 33.4 (Residual-pencil moduli in an incidence fibre). *The product $U = \prod_a U_a$ parametrizes an algebraic flat family of distinct embedded Hilbert limits over the fixed pencil R_0 . Each is the fixed main curve with the tails prescribed by its residual lines, and each is realized*

by an ordinary transverse one-parameter smoothing of R_0 . Hence, for the normal incidence modification $\rho : \widehat{G} \rightarrow G$, one has

$$\dim \rho^{-1}(R_0) \geq \sum_a \left(\binom{c_a + 1}{2} - 2 \right). \quad (33.7)$$

Distinctness is detected by the resolved multiplication power $J_{h(n-c_a)+1}$ on the a th tail. The statement describes a specified family in the fibre, not the entirety of the fibre or its normalization.

Proof. Since C_a is nonsingular, lines through $[C_a]$ are parametrized by the displayed projective quotient. The condition on determinant zeros is open and nonempty, as is seen by choosing D_a diagonal with distinct eigenvalues relative to C_a .

The complete-quadric morphism is an isomorphism over forms of rank at least $c_a - 1$. Each line in U_a stays in this open, and thus its universal line has an algebraic lift. The fibre of $\text{CQ}(V) \rightarrow \mathbb{P}(\text{Sym}^2 V)$ over $[A(x_a)]$ contains the residual $\text{CQ}(W_a)$: the nondegenerate first quotient form is fixed, and its radical carries the remaining complete form. This follows also by taking the unit block in the Schur charts used in the preceding proof. Thus the universal lifted lines give an algebraic family of embedded tails, each passing through the fixed attachment corresponding to $[C_a]$.

Adjoin the fixed main curve. The scheme-theoretic intersection of each tail family with that main curve is just this reduced section. Indeed the main curve has contact order one with E_{r_a} , since its local nonzero Smith exponents are 0 and 1, whereas the tail lies inside that divisor. The tails lie over different points, hence do not intersect each other. The union exact sequence

$$0 \longrightarrow \mathcal{O}_{C \cup \bigcup F_a} \longrightarrow \mathcal{O}_{C \times U} \oplus \bigoplus_a \mathcal{O}_{F_a} \longrightarrow \bigoplus_a \mathcal{O}_{\{\text{attachment}_a\} \times U} \longrightarrow 0$$

proves flatness over U : the two terms on the right are flat, so its kernel is flat. The degrees from (33.3) give the required fixed Hilbert polynomial. We therefore obtain a morphism U to the Hilbert scheme. The embedded fibre over $[A(x_a)]$ recovers its residual tail. Its first nonconstant exterior map, equivalently the power of index $h(n - c_a) + 1$, has degree one and recovers the line ℓ itself. Distinct parameters give distinct embedded curves.

To see that these Hilbert points belong to the graph closure defining $\widehat{G}$, choose a nonsingular member of R_0 . Semisimplicity makes its spectral eigenspaces orthogonal, so in a congruence frame the pencil is a direct sum of blocks $(z - x_a)C_a$ and simple blocks. Choose any representative D_a of each residual line and replace these blocks by $(z - x_a)C_a + \tau D_a$. Each cluster splits into c_a simple spectral points and distinct clusters stay separated. Theorem 33.3 identifies the Hilbert limit with the constructed curve. This realizes every parameter by a smoothing, independently on all the blocks.

Let $\overline{G}$ denote the reduced Hilbert graph before normalization. The morphism from the smooth, reduced scheme U factors through its fibre over R_0 : the defining equations vanish at every closed point by the smoothing construction, hence vanish on U . Its geometric fibres are zero-dimensional by the distinctness just proved; its image consequently has dimension $\dim U$. Normalization $\widehat{G} \rightarrow \overline{G}$ is finite and surjective, and the preimage of this image has the same dimension. This gives (33.7). We use this finite preimage, rather than asserting that a nondominant map from a normal scheme must lift to a normalization. Such an assertion would not be a valid normalization universal property. $\square$

Example 33.5 (Triple collision). For one semisimple triple collision, $c = 3$, the tail has exterior multidegree $(0, \dots, 0, 1, 2)$ and total degree three. At $h = 2$ its successive residual powers $b = 0, \dots, 6$ have degrees

$$0, 1, 2, 3, 4, 2, 0.$$

The residual lines through a fixed nonsingular ternary quadratic form form an open in $\mathbb{P}^4$, and give a four-dimensional family of distinct Hilbert limits over the same special pencil. Every member has three distinct spectral points on its tail and one nonspectral attachment. This is a family of embedded limits, not just a count of branches of a normalization fibre.

The special finite failure algebra remains A_{R_0} throughout this family of limits. What varies is the total-family first-relation geometry: its intrinsic envelope produces the maximal-ideal centre, its successive products give the tail systems, and their degree-one member recovers the residual line. Thus the family supplies information not contained in its closed algebra alone. The construction is equivariant under source congruence and under reparametrization of the pencil, but the dimension bound concerns the embedded incidence space, not a quotient by its stabilizer.

34. THE SIMPLE CORANK-TWO INCIDENCE MODIFICATION

The ordinary-collision calculation describes specified families of limits. We now identify the entire normalized Hilbert incidence space on a natural open of the pencil Grassmannian. The result holds in every dimension. It does not require a semisimple collision or a squarefree residual pencil.

Let $n \geq 3$, $M = \text{Sym}^2 V$, $P = \mathbb{P}(M)$, and $G = \text{Gr}(2, M)$. Write $Z_j \subset P$ for the symmetric rank-at-most- j scheme, and put $Z_0 = \emptyset$ when $n = 3$. Let $\mathcal{U} \subset G$ consist of regular pencils R such that their lines $L_R = \mathbb{P}(R)$ avoid Z_{n-3} and their scheme-theoretic intersections with Z_{n-2} are either empty or a single reduced point. Write $\mathcal{P}$ for the universal line over $\mathcal{U}$. We use projective spaces of lines, so that the exceptional fibre of a smooth blow-up is the projectivization of the normal space.

Lemma 34.1 (The incidence centre and its normal space). *The locus $\mathcal{U}$ is open and contains G° of Proposition 25.1. Its incidence scheme*

$$\mathcal{D} = \mathcal{P} \times_P Z_{n-2} \longrightarrow \mathcal{U}$$

is a closed immersion with smooth image Σ_2 of codimension two. At $R \in \Sigma_2$, let $p = L_R \cap Z_{n-2}$ and let $a_R \subset N_{Z_{n-2}/P, p}$ be the nonzero image of $T_p L_R$. There is a canonical isomorphism

$$N_{\Sigma_2/\mathcal{U}, R} \simeq N_{Z_{n-2}/P, p}/a_R. \quad (34.1)$$

Near (p, R) , after an étale change of base and coordinates, the ideal of $\mathcal{D}$ in $\mathcal{P}$ is

$$I_{n-1}(A) = (x, u, v), \quad I_{\Sigma_2} = (u, v), \quad (34.2)$$

where x is a relative line coordinate and u, v are part of a regular system of parameters on the base. The generators may be changed by an invertible matrix; the asserted equality is an equality of ideals.

Proof. On $P \setminus Z_{n-3}$ the scheme Z_{n-2} is smooth of codimension three. A nonsingular $(n-2)$ -block reduces its equations to the three entries of a symmetric 2×2 Schur complement. In particular these are regular local equations for its given determinantal scheme structure.

The universal line evaluation $\mathcal{P}_G \rightarrow P$ is smooth: over a point of P its fibre is the projective space of lines through that point. Hence its inverse image of the smooth rank locus is smooth and has dimension $\dim G - 2$. Lines contained in the rank locus, or meeting the lower rank locus, form closed exceptional sets after the usual proper incidence projection. On the complementary open the incidence projection is finite. The condition that its fibre have length at most one is open, by upper semicontinuity for a finite module. This proves the asserted openness. On this open the unit map $\mathcal{O}_{\mathcal{U}} \rightarrow q_* \mathcal{O}_{\mathcal{D}}$ is surjective: on each geometric fibre its target is either zero or a one-dimensional algebra with unit, and Nakayama's lemma applies to its coherent cokernel. Thus

q is a closed immersion. Its source is smooth by the smooth evaluation just noted; its codimension is two.

First-order motion of a line evaluates surjectively onto $T_p P/T_p L_R$: in the description $T_R G = \text{Hom}(R, M/R)$ this is restriction to the one-dimensional subspace representing p . To remain incident to the rank locus, this evaluation must vanish in $T_p P/(T_p Z_{n-2} + T_p L_R)$. A reduced intersection means that the normal image of $T_p L_R$ is nonzero. The kernel is precisely $T_R \Sigma_2$, since the incidence point may move along both the line and the rank locus. The quotient is therefore (34.1).

For the local assertion choose a rank-locus equation with nonzero relative derivative. Its zero hypersurface is etale over the base near (p, R) ; after an etale base change it gives a section. Use that equation as the coordinate x . Subtract multiples of x from the other two equations. Their restrictions to the section are base functions u, v , and the ideal becomes (x, u, v) . The normal-space calculation proves that du, dv are independent. This proves (34.2). The argument is the smooth-centre coordinate calculation, not an assertion that arbitrary graph closures commute with base change. $\square$

The open is nonempty along its centre. For example, a diagonal pencil with two equal linear entries and all other entries having distinct zeros away from their common zero belongs to Σ_2 . Although $a_R \neq 0$, its residual binary quadratic form may have rank one. This case is allowed throughout the following theorem.

Theorem 34.2 (The full simple-incidence Hilbert modification). *For the normalized Hilbert incidence modification $\rho : \hat{G} \rightarrow G$ of Proposition 25.1, there is a canonical isomorphism over $\mathcal{U}$*

$$\hat{G} \times_G \mathcal{U} \simeq B := \text{Bl}_{\Sigma_2} \mathcal{U}. \quad (34.3)$$

It is equivariant under source congruence. In particular this part of the incidence space is smooth. Its exceptional divisor $F = \mathbb{P}(N_{\Sigma_2/\mathcal{U}})$ is a $\mathbb{P}^1$ -bundle, and

$$K_B = \beta^* K_{\mathcal{U}} + F, \quad \mathcal{O}_B(F)|_{F_R} = \mathcal{O}_{\mathbb{P}^1}(-1), \quad (34.4)$$

where $\beta : B \rightarrow \mathcal{U}$ is the blow-up.

For every $R \in \Sigma_2$ the entire fibre of ρ is $F_R \simeq \mathbb{P}^1$. Its point $[\bar{b}] \in \mathbb{P}(N_{Z_{n-2}/P,p}/a_R)$ represents the reduced nodal curve

$$C_R \cup \ell_{[\bar{b}]}, \quad \ell_{[\bar{b}]} = \mathbb{P}(a_R + \mathbb{C}b) \subset \pi^{-1}(p) = \mathbb{P}(N_{Z_{n-2}/P,p}). \quad (34.5)$$

Here C_R is the strict transform of L_R and b is any lift of $\bar{b}$. The tail attaches to C_R at $[a_R]$. Every line through $[a_R]$ in this exceptional $\mathbb{P}^2$ occurs, and no further components, embedded points, or nonreduced fibres occur over this open. The two components have exterior multidegrees

$$\deg_{C_R} H_q = q - \mathbf{1}_{q=n-1}, \quad \deg_{\ell} H_q = \mathbf{1}_{q=n-1} \quad (1 \leq q < n). \quad (34.6)$$

Their common Hilbert polynomial is $\binom{n}{2}m + 1$ for $\bigotimes_{q=1}^{n-1} H_q$.

The universal curve over B is the simultaneous multiplication graph on $\mathcal{P} \times_{\mathcal{U}} B$. Near a node it is etale locally $xy = t$, where $t = 0$ defines F . Thus it is a flat family of nodal curves, including along directions for which the residual determinant has a repeated zero. It commutes with base change as this specified flat family. The blow-up is characterized by the universal property that $I_{\Sigma_2} \mathcal{O}_B$ is an invertible ideal; the corresponding factorization property holds for every $T \rightarrow \mathcal{U}$ on which that image ideal is an effective Cartier ideal.

Proof. On $P \setminus Z_{n-3}$, the complete-quadric morphism is exactly the blow-up of the smooth centre Z_{n-2} : all earlier rank blow-ups are isomorphisms there and the determinant divisor requires no blow-up. We use the scheme-level classical construction [34, Remark 2.5 and Construction 2.6],

originating in [33, Theorem 6.3]. Nothing in this step declares the space of complete quadrics itself new.

Pull the universal line back to B . In a base blow-up chart $u = t$, $v = tw$, the ideal (34.2) becomes (x, t) . Its blow-up has the two charts

$$\mathcal{O}_B[x, y]/(xy - t), \quad \mathcal{O}_B[z], \quad x = tz.$$

They glue by the usual ratio change. Their total spaces are smooth, and they are flat over B . For example $\mathcal{O}_B[x, y]/(xy - t)$ is free as an $\mathcal{O}_B$ -module with basis $1, x, x^2, \dots, y, y^2, \dots$; reduction by the monic relation $xy - t$ gives this basis. Its fibres at $t = 0$ have two reduced branches and one ordinary node. The other base chart has the same description. Outside the incidence section the pulled ideal is the unit ideal. Blowing up the globally defined pulled ideal therefore gives a projective flat family $\mathcal{C}_B$ with these local charts. All smaller minor ideals are units on this open. The universal power identity makes the product of the nontrivial power ideals an h^2 th power of I_{n-1} , up to an invertible determinant ideal. Thus this same blow-up is the simultaneous multiplication graph for every h , not only the exterior graph.

There is a closed immersion $\mathcal{C}_B \hookrightarrow \text{CQ}(V) \times B$. Indeed the base-changed Rees algebra of the rank-locus ideal surjects onto the Rees algebra of its image on the universal line; taking Proj gives a closed immersion into the base-changed complete-quadric graph. The universal line is already closed in $P \times B$. Composing these two closed immersions proves the assertion. Equivalently, use the simultaneous graph lemma on the integral total space. This argument does not identify a general base change of a blow-up with the blow-up of the image ideal without the indicated Rees quotient.

In normal coordinates at p , the exceptional map is $[x : t] \mapsto [xa_R + tb]$, with b representing the normal direction of the base motion. Formula (34.1) implies that a_R, b are independent. This is a line embedded in $\mathbb{P}(N_{Z_{n-2}/P, p})$, not a multiple cover. The strict-transform branch meets it at $[a_R]$. This proves (34.5), including the converse parametrization of all lines through that point. The local charts show the absence of embedded components and multiplicities.

The minor systems of orders $q < n - 1$ have no base point on L_R . The $(n - 1)$ -minors have one simple common zero, by (34.2). Their degrees on C_R are therefore q and $n - 2$, respectively. On the exceptional $\mathbb{P}^2$ only the last exterior system has positive degree, namely one. This proves (34.6). The two projective lines have Euler characteristic $2 - 1 = 1$, since their intersection is one reduced point. Their total degree is $\binom{n}{2}$.

The flat embedded family defines a morphism from B to the reduced Hilbert graph closure over $\mathcal{U}$. It is proper because B is proper over $\mathcal{U}$ and the Hilbert graph is separated over that base. Its image is exactly the graph closure: it is closed and contains the dense graph over $\mathcal{U} \setminus \Sigma_2 = G^\circ$, while B is integral. Over an incident pencil the displayed exceptional lines are pairwise distinct as embedded curves; over a nonincident pencil the fibre is a point. The morphism thus has finite geometric fibres and is finite by [Stacks Project, Tag 02LS]. It is birational, and B is normal because its centre is smooth. A finite birational map from a normal integral scheme identifies it with the normalization: on affine charts its ring is the integral closure in the common function field. Normalization commutes with restriction to the open $\mathcal{U}$. This proves (34.3), without assuming that an arbitrary nondominant normal test scheme lifts to a normalization.

All maps are specified by the universal incidence and agree on the dense nonincident locus. Their uniqueness proves canonicity and congruence equivariance. For completeness, in coordinates $v = uw$, $du \wedge dv = u du \wedge dw$, giving the discrepancy one in (34.4). The exceptional normal bundle is tautological of degree -1 . The universal property is the usual Rees-algebra factorization for an effective Cartier image ideal [Stacks Project, Tag 0806]; here the preceding argument identifies the actual Hilbert incidence modification with that universal object. Flat pullback preserves the

explicitly constructed nodal family. No assertion about unrestricted graph-closure base change is used. $\square$

34.1. Every power on every exceptional direction.

Proposition 34.3 (Complete exceptional power systems). *Fix $R \in \Sigma_2$ and $p = L_R \cap Z_{n-2}$. For $h \geq 1$ and $1 \leq j \leq hn$, define*

$$\delta_j = (j - h(n - 2))_+ - 2(j - h(n - 1))_+. \quad (34.7)$$

On the full exceptional plane $\pi^{-1}(p) \simeq \mathbb{P}^2$, the resolved j th multiplication map is the complete degree- δ_j Veronese map into its linear span, with degree zero interpreted as constant. Its restriction to every tail in (34.5) is therefore the complete degree- δ_j system on $\mathbb{P}^1$. In particular $j = h(n - 2) + 1$ detects the tail as an embedded line. These assertions include tangent lines to the residual determinant conic and attachments on that conic.

Proof. A unit $(n - 2)$ -block leaves a residual two-dimensional space W . Lemma 33.2 identifies the first nonzero normal coefficient at degree $j > h(n - 2)$ with the power of order $b = j - h(n - 2)$ in $B_h(W)$, multiplied by a nonzero unit-block socle. For $j \leq h(n - 2)$ the order-zero coefficient is constant on the exceptional fibre. In the three-variable coordinate ring of $\text{Sym}^2 W$, put $\mathbf{a} = I_1$ and $f = \det$. The universal ideal formula gives

$$J_b = \mathbf{a}^b \quad (b \leq h), \quad J_b = \mathbf{a}^{2h-b} f^{b-h} \quad (h \leq b \leq 2h).$$

The generators all have degree b . Their span is the entire degree b part of the displayed ideal. After removal of the common factor $f^{(b-h)_+}$ it is the full space of ternary forms of degree $b - 2(b - h)_+ = \delta_j$. Hence it defines the complete Veronese system on the exceptional plane. Restriction of all ternary forms to any line gives all binary forms of that degree, independently of its intersection multiplicities with the conic $f = 0$. This proves all assertions, including the endpoint $j = hn$ where the map is constant. $\square$

35. THE INCIDENCE CENTRE FROM FAILURE MULTIPLICATION

We specify how the preceding modification is recovered from a family of sharp failure algebras. This is a construction of its centre and Rees algebra, not merely a pullback of the existing finite algebra. For a framed source let $\Phi : G \rightarrow \mathcal{H}$ be the first-relation closed immersion of Theorem 7.3. Here $\mathcal{H}$ denotes its relation-space Grassmannian. We use $\Phi(\mathcal{U})$ with its induced scheme structure, not an arbitrary neighbourhood in $\mathcal{H}$.

Theorem 35.1 (Intrinsic incidence ideal and normal direction). *On the effective family of sharp pencil-failure algebras corresponding to $\mathcal{U}$, its first relation, oriented source, and multiplication diagram determine I_{Σ_2} and the graded Rees algebra*

$$\bigoplus_{m \geq 0} I_{\Sigma_2}^m. \quad (35.1)$$

Their Proj is the modification in Theorem 34.2. More explicitly, restrict the power ideal $J_{h(n-2)+1}$ to the intrinsic pencil line and let $q : \mathcal{D} \rightarrow \mathcal{U}$ be its zero scheme. Then

$$I_{\Sigma_2} = \ker\{\mathcal{O}_{\mathcal{U}} \longrightarrow q_* \mathcal{O}_{\mathcal{D}}\}. \quad (35.2)$$

The construction is independent of h . It is equivariant for ungraded failure-algebra isomorphisms and descends along an oriented projective source torsor.

An arc through $R \in \Sigma_2$ whose first normal derivative is nonzero has exceptional limit given by its line in $N_{\Sigma_2/\mathcal{U},R}$. Thus its first-order failure-algebra family determines the embedded Hilbert limit. A fixed closed algebra A_R alone determines the pencil and its full exceptional parameter

space, but not a preferred point of that space. For higher-contact arcs the leading nonzero normal ideal quotient replaces the first normal derivative.

Proof. The first-relation inverse recovers the oriented projective source and the intrinsic pencil, with its arbitrary-base closed immersion Φ . The source-normalized envelope in Theorem 28.1 supplies its power diagram. The universal power identity gives

$$J_{h(n-2)+1} = I_{n-2}^{h-1} I_{n-1} = I_{n-1}$$

on $P \setminus Z_{n-3}$, since I_{n-2} is the unit ideal there. For $h = 1$ the same formula is $J_{n-1} = I_{n-1}$. Lemma 34.1 identifies its zero scheme on the universal line with Σ_2 by a closed immersion. Taking the kernel of the unit map gives its exact scheme ideal, establishing (35.2). Multiplication of this ideal produces (35.1), whose Proj was identified with the normalized Hilbert graph in the preceding theorem. Because Φ is a closed immersion onto its scheme image, this also gives

$$\mathrm{Bl}_{\Phi(\Sigma_2)} \Phi(\mathcal{U}) \simeq \mathrm{Bl}_{\Sigma_2} \mathcal{U}.$$

This is an identity on the effective image, not an unjustified assertion about blow-ups of the whole ambient Grassmannian.

In the local coordinates of (34.2), an arc with $u(\tau) = \tau b_1 + O(\tau^2)$ and $v(\tau) = \tau b_2 + O(\tau^2)$, $(b_1, b_2) \neq (0, 0)$, lifts to the exceptional point $[b_1 : b_2]$. Under (34.1) this gives exactly the plane $a_R + \mathbb{C}b$ and hence the embedded tail. The arbitrary-base inverse identifies first-order pencil deformations with their effective first-relation deformations. Terms of degree at least two in an ungraded generator change do not change the sharp relation, so they do not change this normal direction. Higher-contact arcs are handled by dividing u, v by their common smallest power of τ before specializing. This calculation explains why no preferred tail can be assigned to the closed algebra without its normal direction.

Every operation is invariant under the congruence and source-lift changes already treated in the intrinsic-envelope theorem. Its scalar normalization and projectivization remove the linear-lift ambiguity; fppf descent therefore applies to the incidence ideal, its Rees algebra, and the resulting projective family. The universal Cartier factorization property, rather than an assertion of arbitrary blow-up base change, specifies their functorial meaning on test bases. $\square$

Example 35.2 (A complete fibre with a tangential residual limit). For $n = 3$, take the affine two-parameter family of genuine pencils

$$A_{u,v}(s, t) = \begin{pmatrix} s & 0 & 0 \\ 0 & t + us & vs \\ 0 & vs & t - us \end{pmatrix}.$$

Near the pencil $R_0 = \langle \mathrm{diag}(1, 0, 0), \mathrm{diag}(0, 1, 1) \rangle$, the incidence centre is $(u, v) = (0, 0)$. Indeed in the chart $s = 1$ its minor ideal is (t, u, v) , while near $s = 0$ the rank is two. The exceptional fibre is the full projective line $[u : v]$. The arc $(u, v) = \tau(\alpha, \beta)$ gives the residual tail

$$[x : y] \mapsto \begin{pmatrix} x + \alpha y & \beta y \\ \beta y & x - \alpha y \end{pmatrix}, \quad \det = x^2 - (\alpha^2 + \beta^2)y^2.$$

All directions $(\alpha, \beta) \neq (0, 0)$ occur. For $\alpha^2 + \beta^2 \neq 0$ the two residual determinant zeros are distinct. For $[\alpha : \beta] = [1 : i]$ or $[1 : -i]$ they coalesce into a double zero. The tail remains an embedded reduced line and the total curve remains nodal. These two points of the full incidence fibre are not ordinary transverse residual pencils, and so are not covered by the squarefree hypothesis of Theorem 33.3. Every power system at these points is still the complete system of Proposition 34.3. The example separates nodality of the Hilbert fibre from transversality of its boundary contact.

36. MULTIPLE REDUCED INCIDENCE AND ITS COMPLETE HILBERT BOUNDARY

We remove the restriction to a single incidence point. The independence of the incidence conditions is proved from the radicals of distinct members of a regular pencil; it is not added as a transversality hypothesis. The resulting modification has a global Fitting-ideal description even though the incidence points need not be globally labelled.

Put $M = \text{Sym}^2 V$, $P = \mathbb{P}(M)$ and $G = \text{Gr}(2, M)$, with $n = \dim V \geq 3$. Let $Z_j \subset P$ have its symmetric determinantal scheme structure. When $n = 3$ put $Z_0 = \emptyset$. Let G_{reg} be the open of regular pencils and let $X_G = \mathbb{P}(\mathcal{R})$ be the universal line. The image of $X_G \times_P Z_{n-3}$ is closed, since the projection from this incidence is proper. Remove it and call the resulting open $G_* \subset G_{\text{reg}}$. Every line over G_* meets Z_{n-2} in a finite scheme: a positive-dimensional intersection would contain the line and contradict regularity. Consequently

$$q_*^{\text{inc}} : D_* := X_{G_*} \times_P Z_{n-2} \longrightarrow G_*$$

is proper and quasi-finite, hence finite. Define

$$U_{\text{red}} = G_* \setminus q_*^{\text{inc}}(\text{Supp } \Omega_{D_*/G_*}), \quad D = D_* \times_{G_*} U_{\text{red}}, \quad q : D \rightarrow U_{\text{red}}. \quad (36.1)$$

This is open because q_*^{inc} is finite. Its geometric incidences are precisely the finite reduced intersections: for a finite algebra over $\mathbb{C}$, vanishing of relative differentials is equivalent to being a product of copies of $\mathbb{C}$. Unlike the single-incidence open, it allows every possible number of such points. Set

$$\mathcal{Q} = q_* \mathcal{O}_D, \quad \mathcal{I} = \text{Fitt}_0(\mathcal{Q}), \quad B_{\text{red}} = \text{Bl}_{\mathcal{I}} U_{\text{red}}. \quad (36.2)$$

The zeroth Fitting ideal of the zero module is the unit ideal.

36.1. The general incidence calculation.

Lemma 36.1 (Transverse incidence of a family of curves). *Let S, Y be smooth complex varieties, let $Z \subset Y$ be smooth of codimension $c \geq 2$, and let $X \rightarrow S$ be a smooth family of embedded curves in Y . Suppose $D = X \times_Y Z \rightarrow S$ is finite and unramified. Fix $s \in S$ with incidence points $p_1, \dots, p_k$. Write $a_i \subset N_{Z/Y, p_i}$ for the normal image of $T_{p_i} X_s$. Assume $a_i \neq 0$ and that the joint normal evaluation map*

$$T_s S \longrightarrow \bigoplus_{i=1}^k N_{Z/Y, p_i} / a_i$$

is surjective. Etale locally at s , the incidence splits into closed immersions $D_i \simeq \Sigma_i \hookrightarrow S$, with independent smooth centres of codimension $c - 1$. Near p_i there are coordinates in which

$$I_D = (x_i, u_{i1}, \dots, u_{i,c-1}), \quad I_{\Sigma_i} = (u_{i1}, \dots, u_{i,c-1}), \quad (36.3)$$

where all base parameters u_{ij} are independent. If $\mathcal{Q}_S = q_ \mathcal{O}_D$, then*

$$\text{Fitt}_0(\mathcal{Q}_S) = \prod_i I_{\Sigma_i} = \bigcap_i I_{\Sigma_i}.$$

The blow-up of this ideal is the fibre product of the individual blow-ups. It is smooth, with etale normal-crossing exceptional boundary and fibre $\prod_i \mathbb{P}^{c-2}$ at s .

Proof. Over the strict henselization at s , the finite algebra splits into factors indexed by its geometric fibre. These factors and their idempotents descend to an etale neighbourhood. For each factor the unit map from the base ring has cokernel whose fibre at s is zero: finite base change identifies that fibre with the corresponding one-dimensional residue algebra. Nakayama's lemma, followed by shrinking, makes the unit map surjective. Each factor is therefore a closed immersion. This argument uses the finite algebra, not just the set of its geometric points.

The map of the curve tangent into the normal space is nonzero, so one local equation of Z has nonzero relative derivative. Its zero set is etale over the base. After an etale base change

choose this as a section and use that equation as the relative coordinate x_i . Modulo x_i , the other equations become base functions u_{ij} . Their differentials give exactly the joint normal evaluation in the statement. Surjectivity makes them independent regular parameters, proving the normal form and smoothness of all intersections of the centres. In particular the length-one factors in this argument are reduced on complex geometric fibres; reducedness has not been inferred from an arbitrary higher fibre length.

Now $\mathcal{Q}_S = \bigoplus_i \mathcal{O}_S / I_{\Sigma_i}$. Fitting ideals of a direct sum multiply. In the displayed independent coordinates, intersection equals product: in a polynomial coordinate ring a monomial in every ideal uses a variable from each disjoint group, and the equality persists under étale flat extension. Equivalently one may verify it in the faithfully flat completion. The simultaneous blow-up is covered by charts in which, for each i , one u_{ij} is retained and the others are its multiples by new ratios. These charts are smooth and identify the fibre product with the blow-up of the product ideal. The latter equality also follows from the simultaneous Rees graph lemma. There are k independent exceptional parameters, and their zero sets have normal crossings. The fibre consists of independent choices of a line in each normal space of dimension $c - 1$, proving the last assertion. $\square$

The arrangement blow-up mechanism is classical. Li's wonderful-model Theorems 1.2 and 1.3 [Li(2009)] give the normal-crossing and product-ideal description for a building set. Here we have proved the independent-centre case explicitly. The pencil-specific assertion needed to apply it at every point of U_{red} is the following.

Lemma 36.2 (Independence at distinct spectral points). *For every $R \in U_{\text{red}}$, with incidence points $p_1, \dots, p_k$, the joint normal evaluation of Lemma 36.1 is surjective. In particular the local incidence centres have codimension two and meet transversely, without any additional genericity assumption on the other elementary divisors of R .*

Proof. Choose a nonsingular member Q_0 of the pencil and write its other members as $Q_1 - \lambda Q_0$. Put $T = Q_0^{-1}Q_1$ on V^* . The radicals $K_i = \ker(T - \lambda_i)$ at distinct spectral points lie in distinct generalized eigenspaces. Their sum is direct. Restriction therefore gives a surjection

$$\text{Sym}^2 V \longrightarrow \bigoplus_i \text{Sym}^2(K_i^*) :$$

extend a basis of the direct sum of the K_i to a basis of V^* and prescribe the indicated diagonal blocks of a symmetric form. Varying Q_1 with Q_0 fixed realizes every one of these restrictions. At a corank-two point the normal space to the rank locus is the three-dimensional space of forms on K_i , with its harmless local projective twist. The tangent of the pencil maps to the nonzero line a_i , because the incidence is reduced. Quotienting by a_i gives the joint evaluation in the statement. A change of basis of the pencil contributes only multiples of its tangent image in each summand, so the map factors through $T_R G = \text{Hom}(R, M/R)$. It remains surjective after this quotient. This proves the claim also when the tangent form on K_i has rank one rather than rank two. $\square$

36.2. Identification of the full Hilbert graph. Let $G^\circ \subset U_{\text{red}}$ be the open with no corank-two incidence. A pencil over it has a unique lifted embedded curve in $Q = \text{CQ}(V)$. Use the polarization $H = \bigotimes_{q=1}^{n-1} H_q$, where H_q is the resolved exterior hyperplane bundle, and put $P_H(l) = \binom{n}{2}l + 1$. Let Γ_{red} be the reduced closure of this graph in $U_{\text{red}} \times \text{Hilb}^{P_H}(Q)$, and let $\hat{\Gamma}_{\text{red}}$ be its normalization. Thus $\hat{\Gamma}_{\text{red}} = \hat{G} \times_G U_{\text{red}}$ with the conventions of Proposition 25.1. This equality uses localization of integral closure [Stacks Project, Section 29.55, Lemmas 29.55.3–4]; it is an open restriction, not an arbitrary base-change assertion.

Theorem 36.3 (The complete multiple-incidence modification). *There is a canonical congruence-equivariant isomorphism*

$$\widehat{\Gamma}_{\text{red}} \simeq B_{\text{red}} = \text{Bl}_{\text{Fitt}_0(q_* \mathcal{O}_D)} U_{\text{red}}. \quad (36.4)$$

The space is smooth, and its exceptional boundary has normal crossings etale locally. Over a pencil with k distinct reduced corank-two incidences the whole scheme fibre is $(\mathbb{P}^1)^k$. A point of that fibre specifies independently, at each p_i , a line through $[a_i]$ in

$$\pi^{-1}(p_i) = \mathbb{P}(N_{Z_{n-2}/P, p_i}) \simeq \mathbb{P}^2.$$

Its Hilbert curve is the strict transform C_R of the pencil line, with these k embedded tails attached at their respective points. It is reduced and nodal, with no embedded components. For each tail F_i and each $1 \leq q < n$,

$$\deg_{F_i} H_q = \mathbf{1}_{q=n-1}, \quad \deg_{C_R} H_q = q - k \mathbf{1}_{q=n-1}. \quad (36.5)$$

The incidence scheme satisfies $k \leq \lfloor n/2 \rfloor$.

On a labelled etale neighbourhood the exceptional divisors E_i meet like coordinate hyperplanes; every subset occurs locally, and

$$K_{B_{\text{red}}} = \beta^* K_{U_{\text{red}}} + \sum_i E_i.$$

No global labelling, or global simplicity of each irreducible boundary component, is asserted. The universal curve is a flat nodal family; near its i th node its equation is $x_i y_i = t_i$, with $t_i = 0$ the corresponding exceptional divisor. The base blow-up has its usual effective-Cartier image-ideal universal property. These statements classify all fibres over U_{red} , not only the fibres of selected smoothing arcs.

Proof. Apply the preceding two lemmas with $c = 3$. Locally write $I_i = (u_i, v_i)$. In the simultaneous blow-up chart put $u_i = t_i$, $v_i = t_i w_i$. Near the i th incidence the image of the ambient rank-locus ideal on the pulled-back universal line is (x_i, t_i) . Its blow-up has charts

$$\mathcal{O}_B[x_i, y_i]/(x_i y_i - t_i), \quad \mathcal{O}_B[z_i], \quad x_i = t_i z_i.$$

The incidence points have disjoint neighbourhoods on the curve, so these modifications do not interfere. Outside them the ideal is the unit ideal. The local hypersurfaces are flat over B : they are relative Cartier divisors in a smooth relative plane and their nonzero defining equations remain nonzerodivisors on every fibre. The fibrewise Cartier flatness criterion applies. These are the standard nodal local models, so all the stated fibre properties follow. Their total spaces are smooth in the displayed independent coordinates.

Here is the scheme-level embedding needed for the Hilbert argument. If $\mathcal{K}$ is the rank-locus ideal on the open ambient space and $f : X_B \rightarrow P$ is evaluation, there is a surjection of graded algebras

$$f^* \left(\bigoplus_{d \geq 0} \mathcal{K}^d \right) \twoheadrightarrow \bigoplus_{d \geq 0} (\mathcal{K} \mathcal{O}_{X_B})^d. \quad (36.6)$$

Its kernel removes precisely the relations annihilated on passing to the image ideal. The target is the image Rees algebra inside $\mathcal{O}_{X_B}[T]$; it is not an assumed tensor-product identification. Taking Proj gives a closed immersion into the base-changed ambient blow-up, which is Q over rank at least $n - 2$. The universal line is already embedded in $P \times B$. We obtain a flat closed family of curves in $Q \times B$. This agrees with the lifted pencil on the dense nonincident open.

In normal coordinates its exceptional map at p_i is $[x_i : t_i] \mapsto [x_i a_i + t_i b_i]$. The vectors a_i, b_i are independent, and their span determines an embedded line, whose homogeneous ideal is its linear annihilator. Distinct directions modulo a_i therefore give distinct closed subschemes of the exceptional plane, not merely different parametrizations of one curve. Tails at different p_i cannot

be confused. The induced proper map $B \rightarrow \Gamma_{\text{red}}$ has finite geometric fibres and is birational. It is consequently finite, and the smooth source is the normalization of the reduced integral graph. This proves (36.4), including its scheme fibres. Both maps restrict to the identity on the dense graph, giving canonicity and equivariance and gluing the etale calculations.

The smaller minor systems have no base point, while the last minor system has one simple base point at each p_i . Subtracting these base points and adding one unit of degree on each tail gives (36.5). The determinant has vanishing order at least two at each distinct corank-two point, so $2k \leq n$. The nodal tree has Euler characteristic one and total H -degree $\binom{n}{2}$. This proves the Hilbert polynomial used above. Finally the product blow-up charts give independent parameters t_i ; the Jacobian for each codimension-two blow-up contributes one E_i to the canonical divisor. These computations also prove the local intersection statement. $\square$

Corollary 36.4 (Multiplication systems at every node). *The ideal $\mathcal{I}$ in (36.2) is also obtained by taking the zeroth Fitting ideal of the pushforward of the zero scheme of $J_{h(n-2)+1}$ on the universal line, for every $h \geq 1$. At $h = 1$ this is the distinguished construction without an exponent choice. On every tail the resolved j th power is the complete system of degree*

$$\delta_j = (j - h(n - 2))_+ - 2(j - h(n - 1))_+.$$

The exceptional parameter space at a k -incidence pencil has Chow ring $\mathbb{Z}[\xi_1, \dots, \xi_k]/(\xi_1^2, \dots, \xi_k^2)$. In particular at a double incidence the full fibre is a smooth quadric surface $\mathbb{P}^1 \times \mathbb{P}^1$, not merely two independent one-parameter examples.

Proof. On this rank open the universal formula gives $J_{h(n-2)+1} = I_{n-1}$ because $I_{n-2} = \mathcal{O}$. Its zero scheme is exactly D . Finite pushforward and Fitting ideals give the specified centre. The power assertion is Proposition 34.3 at each disjoint tail. For clarity, a coefficient ideal generated in degree b has degree- b part equal to the linear span of its actual generators. In the residual binary ring this part is $f^{(b-h)+} \text{Sym}^{b-2(b-h)+}(\text{Sym}^2 W^*)$; division by the common factor therefore leaves all forms, not merely a subspace with the same base locus. The Chow ring is the projective bundle formula applied to the identified scheme fibre $(\mathbb{P}^1)^k$. $\square$

37. HIGHER CONTACT, THICK TAILS, AND SINGULAR MULTIPLICATION GRAPHS

Nonreduced incidence has a different Hilbert boundary from reduced multiple incidence. We give a two-parameter family of genuine symmetric pencils for every contact order $m \geq 2$, determine its entire normalized Hilbert graph, and compute the nonreduced fibres and the singularities of the universal curve. The result is a slice theorem, not an assertion that the whole ambient Hilbert modification commutes with restriction to that slice.

37.1. Symmetric linear pencils of every contact order. Let E_m be the reversal matrix, N_m the nilpotent matrix with ones on its superdiagonal, and put

$$L_m(x) = E_m(xI_m - N_m), \quad e = e_1.$$

This is symmetric. In dimension $n = 2m$, consider the affine family of pencils, with x the pencil parameter and $(u, v) \in S = \mathbb{A}^2$,

$$A_{u,v}(x) = \begin{pmatrix} L_m(x) + uee^t & vee^t \\ vee^t & L_m(x) - uee^t \end{pmatrix}. \quad (37.1)$$

The projective pencil is obtained by homogenizing every entry to degree one. Its leading matrix is $E_m \oplus E_m$, so infinity is nonsingular. For $n > 2m$ one may adjoin fixed scalar blocks $x - \lambda_j$ with distinct nonzero λ_j and restrict the parameter base to a neighbourhood where these additional simple roots stay disjoint from the cluster. All assertions below then hold near the same cluster; we state the global slice formula first for $n = 2m$.

Lemma 37.1 (Exact contact slice). *For the family (37.1), the complement of the two e_1 directions is a nonsingular block over $\mathbb{C}[x]$, of constant nonzero determinant. Its Schur complement is exactly*

$$\begin{pmatrix} x^m + u & v \\ v & x^m - u \end{pmatrix}. \quad (37.2)$$

Consequently, on the affine pencil chart,

$$I_{n-1}(A) = (x^m, u, v), \quad \det A = c_m(x^{2m} - u^2 - v^2), \quad c_m \in \mathbb{C}^*.$$

All pencils are regular and have rank at least $n - 2$ everywhere. Only the origin of S has a corank-two incidence; its intersection scheme has length m . Its two positive local Smith exponents are m, m . The finite incidence algebra pushed to S is $(\mathcal{O}_S/(u, v))^{\oplus m}$, so its zeroth Fitting ideal is $(u, v)^m$.

Proof. Delete row and column one from $L_m(x)$. The remaining matrix is anti-triangular with -1 on its antidiagonal, hence has constant nonzero determinant. Since $\det L_m(x) = \det(E_m)x^m$, its scalar Schur complement is x^m ; the ratio of the two constant determinants is one. This can also be read directly by eliminating along the superdiagonal. The two kernel directions receive exactly the displayed perturbation, proving (37.2). The unit-block minor identity then gives $I_{n-1} = I_1(\text{Schur complement}) = (x^m, u, v)$ in characteristic zero. All smaller required minors are units. The determinant and rank claims follow from the same block decomposition, including infinity. At $u = v = 0$ the residual matrix is $x^m I_2$, which proves both the intersection length and the Smith exponents. The incidence algebra is $\mathbb{C}[x, u, v]/(x^m, u, v)$; its $\mathbb{C}[u, v]$ -module basis is $1, x, \dots, x^{m-1}$ with (u, v) annihilating every basis vector. The direct-sum formula for Fitting ideals proves the last assertion. $\square$

Let Γ_m be the reduced closure of the lifted-pencil graph over $S \setminus \{0\}$ in $S \times \text{Hilb}^{P_H}(\text{CQ}(V))$, with $P_H(l) = \binom{n}{2}l + 1$, and let $\widehat{\Gamma}_m$ be its normalization. This definition is independent of whether normalization of the global pencil graph commutes with this particular slice; no such commutation is assumed.

Theorem 37.2 (The full contact-slice Hilbert modification). *For every $m \geq 2$ the slice graph is*

$$\widehat{\Gamma}_m \simeq \text{Bl}_0 \mathbb{A}^2 = \text{Bl}_{\text{Fitt}_0(q_* \mathcal{O}_D)} S. \quad (37.3)$$

Its whole exceptional fibre is $\mathbb{P}^1$; the Hilbert curves over that exceptional fibre are nonreduced. In a base chart $u = t$, $v = tw$, the total multiplication graph is the blow-up of (x^m, t) on the pulled-back pencil surface. With $R = \mathbb{C}[x, t, w]$, its Rees presentation and its two projective charts are

$$\begin{aligned} \mathcal{R}(x^m, t) &= R[U, V]/(tU - x^m V), \\ D_+(U) : \quad t &= x^m y, \quad D_+(V) : \quad x^m = tz. \end{aligned} \quad (37.4)$$

Here U represents $x^m T$ and V represents tT in $R[T]$. The charts glue by $y = z^{-1}$. The universal curve is flat over the base blow-up, is normal, and has a transverse A_{m-1} surface singularity along the section $x = t = z = 0$ of its exceptional divisor.

For every direction $[\alpha : \beta] \in \mathbb{P}^1$, its special Hilbert curve is

$$C_0 \cup T_m, \quad T_m \simeq \mathbb{P}_{\mathbb{C}[x]/(x^m)}^1, \quad C_0 \cap T_m \simeq \text{Spec } \mathbb{C}[x]/(x^m). \quad (37.5)$$

The intersection is at one section of T_m ; the support of T_m is the residual line through $[I_2]$ with second member $\begin{pmatrix} \alpha & \beta \\ \beta & -\alpha \end{pmatrix}$ in the exceptional plane. The fibre has no embedded associated points. Its cycle is $[C_0] + m[F]$, where $F = (T_m)_{\text{red}}$. Every direction occurs, and these embedded thickened curves are pairwise distinct. The multidegrees are

$$\deg_{C_0} H_q = q - m \mathbf{1}_{q=n-1}, \quad \deg_F H_q = \mathbf{1}_{q=n-1}.$$

For $\delta_j = (j - h(n - 2))_+ - 2(j - h(n - 1))_+$, the j th resolved multiplication system on T_m is, over $\mathbb{C}[x]/(x^m)$, the complete system $\mathcal{O}_{T_m}(\delta_j)$. In particular its reduced degree is δ_j and its degree as a cycle is $m\delta_j$.

Proof. All rank loci below corank two are avoided, so the relevant ambient complete-quadric map is the blow-up of the rank- $(n - 2)$ locus. Pull the exact ideal in Lemma 37.1 to $\text{Bl}_0 S$. In the stated chart it is (x^m, t) . These two elements are a regular sequence. The only homogeneous Rees relation is $tU - x^m V$: successive comparison of powers of T , using that x^m and t are relatively prime, expresses every relation as a multiple of this one. Thus the relation is saturated in the image subalgebra of $R[T]$, and the stated Proj charts follow. The other base chart is identical with the roles of u, v exchanged.

Both chart equations remain nonzerodivisors on every fibre of their smooth relative affine planes, so the fibrewise Cartier flatness criterion proves flatness. The first chart eliminates t and is smooth. The second is a hypersurface with singular locus $x = t = z = 0$, of codimension two in its total space. It is a domain and Cohen–Macaulay, so regularity in codimension one proves normality. Its transverse completed equation is $x^m = tz$, the A_{m-1} surface singularity. This describes a singular universal curve over a smooth parameter blow-up; the two smoothness assertions are not conflated.

At $t = 0$ the first chart has ring $\mathbb{C}[x, y]/(x^m y)$ and the second has ring $\mathbb{C}[x, z]/(x^m)$. In the first ring the ideals (y) and (x^m) intersect in $(x^m y)$. These pieces glue to the strict transform C_0 and to the thick tail whose two rings are $(\mathbb{C}[x]/(x^m))[y]$ and $(\mathbb{C}[x]/(x^m))[z]$, with $y = z^{-1}$ and x unchanged. This proves (37.5), including its length- m intersection. The local hypersurfaces of the fibre are Cohen–Macaulay of dimension one and have only their component primes as associated primes. There are no embedded points.

The Rees quotient (36.6) embeds this flat family into the fixed complete-quadric space times the base blow-up. Its generic curve is the prescribed lift. The support of each tail is the indicated residual line. Distinct $[\alpha : \beta]$ give distinct residual lines, hence distinct embedded closed subschemes. The resulting proper map to Γ_m is quasi-finite and birational, so the smooth base blow-up is its normalization. This proves the whole fibre statement in (37.3). The Fitting formula follows from $(u, v)^m$ and invariance of Proj under a positive Veronese power.

The last minor system loses a base divisor of length m on C_0 , while the smaller systems lose none. The graph line bundle restricts to $\mathcal{O}(1)$ on the residual projective line over $\mathbb{C}[x]/(x^m)$. It follows that $\chi(\mathcal{O}_{T_m} \otimes H^l) = m(l + 1)$ and the intersection has Euler characteristic m . Adding to the main component gives $\binom{n}{2}l + 1$, as required. More generally the unit-block congruence is invertible over $\mathbb{C}[x]$ and thus over $\mathbb{C}[x]/(x^m)$. The binary multiplication coefficient calculation of Corollary 36.4, extended to this Artin ring, spans every degree- δ_j binary form. Multiplication by the unit-block socle is a split injection on these coefficient modules: it has a unit coordinate in a trivializing frame. This proves the relative complete-system assertion, not just its reduced-fibre version. The degree- j coordinates originally have the twist $\mathcal{O}(j)$ on the pencil line; it is trivial on this local thickening. Removing their Cartier common factors produces the stated graph line bundle. $\square$

37.2. The singularity and the stable-map comparison.

Proposition 37.3 (Local deformation module of the multiplication graph). *At a singular point of a one-parameter direction in Theorem 37.2, let $A_m = \mathbb{C}[[x, t, z]]/(x^m - tz)$. The abstract local cotangent deformation modules are*

$$T^1(A_m) = \mathbb{C}[[x]]/(x^{m-1}), \quad T^i(A_m) = 0 \quad (i \geq 2). \quad (37.6)$$

The first equality has dimension $m - 1$. A formally versal family of hypersurface deformations is obtained by adding $\sum_{a=0}^{m-2} c_a x^a$ to $x^m - tz$. These are deformation modules of the total graph singularity, not of its fixed-base relative curve or of the raw failure algebra.

In a transverse surface the reduced tail is $\mathbb{Q}$ -Cartier with $F^2 = -1/m$. Its minimal resolution introduces the A_{m-1} chain; the total fibre has multiplicities m on the strict transform of F and $m - 1, m - 2, \dots, 1$ along the consecutive chain, and multiplicity one on C_0 .

Proof. The cotangent complex of a hypersurface is represented by the conormal free module mapping to the three ambient differentials. Dualizing gives cokernel by the partial derivatives $mx^{m-1}, -z, -t$, hence (37.6); there are no higher terms. Lifting an equation across a small extension has no obstruction, and the displayed monomials form a basis of the cokernel, giving the formal versality assertion. This is the classical hypersurface calculation applied to the exact Rees equation.

In the first Rees chart the exceptional Cartier ideal is generated by x^m , so its Cartier divisor is mF . Its associated ideal bundle restricts to $\mathcal{O}_F(1)$; thus $mF^2 = -1$. The standard cyclic quotient presentation can be checked directly:

$$x = ab, \quad t = a^m, \quad z = b^m, \quad A_m = \mathbb{C}[[a, b]]^{\mu_m}, \quad (a, b) \mapsto (\zeta a, \zeta^{-1} b).$$

Subdivide its two-dimensional cone by the primitive rays $((m - i)/m, i/m)$, $1 \leq i < m$, in the quotient lattice. Consecutive cones are smooth. The internal curves form the chain with self-intersection -2 , and the orders of $t = a^m$ are $m - i$. The end attached to the tail has multiplicity $m - 1$; the next ones decrease by one. The main component meets the other, already smooth, end of the tail, where $t = x^m y$ gives multiplicities $m, 1$. These charts prove the stated fibre description. $\square$

Theorem 37.4 (A thick Hilbert tail and a covered stable-map tail). *Fix a nonzero direction $(u, v) = \tau(\alpha, \beta)$ in the slice, with contact order m . After the base change $\tau = s^m$, the normalization of its total Hilbert graph is the blow-up of the smooth pencil surface at $(x, s) = (0, 0)$. Its special fibre is a reduced nodal union $C_0 \cup F'$. The induced map to complete quadrics is an embedding on C_0 and a degree- m map on F' onto F ,*

$$[X : S] \mapsto [X^m : S^m] \tag{37.7}$$

in the residual line coordinates. It is a stable map with tail deck group μ_m . It has the same cycle $[C_0] + m[F]$ as the Hilbert limit, but the latter has the nonreduced scheme structure (37.5), not the reduced image of this map.

Proof. The base-changed graph is the blow-up of (x^m, s^m) . This particular base change is flat, and the Rees presentation also verifies it directly. On the chart $x^m = s^m z$ adjoining $w = x/s$ gives the integral element $w^m = z$ and the normal chart $\mathbb{C}[s, w]$, with $x = sw$. On the other chart $s^m = x^m y$ adjoining $r = s/x$ gives $r^m = y$ and the normal chart $\mathbb{C}[x, r]$, with $s = xr$. These charts are finite and birational over the original charts and glue by $w = r^{-1}$. They are exactly the point blow-up charts. They also show a reduced nodal fibre after normalization.

The old graph coordinates $[x^m : s^m]$ restrict to (37.7) on the exceptional line of the point blow-up. Away from the contact point normalization does not alter the original lifted line. Both components are nonconstant; the main component is embedded and the nonconstant tail has precisely the finite deck group μ_m , which fixes the attachment point. Thus the map is stable. The image of F' is reduced while the flat Hilbert fibre has the explicit nilpotent x of order m , so the two objects differ scheme-theoretically even though pushforward cycles and degrees agree. This comparison is calculated here; it is not deduced from a general identification of Hilbert and stable-map compactifications. $\square$

For the reduced-incidence local model ($m = 1$), the two local descriptions coincide with the nodal model. For every $m \geq 2$, the length of the contact, the tail multiplicity, the transverse graph singularity A_{m-1} , and the stable tail covering degree record the same integer in four different objects. The assertions concern all orders in the specified symmetric Jordan slices. They do not classify arbitrary higher-corank contacts or the entire ambient Hilbert fibre over a nonreduced-incidence pencil.

38. EFFECTIVE BOUNDARY GROUPOIDS AND RELATIVE LIFTING OBSTRUCTIONS

This section places the intrinsic envelope and the boundary modification in a single category. It also distinguishes three deformation questions: deforming the effective failure object, lifting a prescribed base deformation to its boundary modification, and deforming a singularity of the universal curve. These questions have different answers.

38.1. The category and the envelope functor. Let $\mathcal{F}_{\text{pen}}^{\text{eff}}$ denote the fppf stack of Theorem E.2. Starting with the sharp pencil-failure families, its arrows are first divided by the tangent-identity nonlinear generator kernel and then by the ineffective right tensor-factor action, and the resulting groupoid is stackified. Its oriented source is specified by coefficient support, rather than by an arbitrary choice between the determinant rulings. The theorem gives, on every complex base,

$$\mathcal{F}_{\text{pen}}^{\text{eff}} \simeq [G/\text{PGL}(V)]. \quad (38.1)$$

The notation does not denote the stack of all Artin algebras of the same length. In particular deformations leaving the effective pencil image are not included in this equivalence.

Proposition 38.1 (The envelope as a morphism of stacks). *Let $\mathcal{A}_h$ be the stack whose objects over T are a projective-space bundle $S \rightarrow T$ of relative dimension $n - 1$ and a finite locally free graded algebra bundle on S locally isomorphic to $\bigoplus_{p=0}^{nh} B_h(V)_p \otimes \mathcal{O}(2p)$, with its projectivized power diagram. Isomorphisms preserve the grading, algebra structure, and diagram. The construction of Theorem 28.1 defines a morphism*

$$\mathcal{F}_{\text{pen}}^{\text{eff}} \longrightarrow \mathcal{A}_h.$$

It is compatible with arbitrary complex base change and does not require a global line bundle $\mathcal{O}_S(1)$. The case $h = 1$ is distinguished without an additional exponent choice. The envelope is a functorial algebra bundle on the recovered source, not a claimed subalgebra or quotient of the original finite algebra.

Proof. An object on the right side of (38.1) is a $\text{PGL}(V)$ -torsor with an equivariant map to G . On a lift of its source to $\mathbb{P}(V)$, the first-jet construction is $\mathcal{H}om(J^1\mathcal{O}(1), \mathcal{O}(1)) = V \otimes \mathcal{O}(1)$. A scalar λ acts on $B_h(V)_p$ with weight $2p$ and on $\mathcal{O}(2p)$ with weight $-2p$. It therefore acts trivially on their product, on its multiplication, and on the polarized Cartan relation maps. The algebra bundle is genuinely $\text{PGL}(V)$ -equivariant, not only equivariant up to a scalar. Associate it to the torsor and apply fppf descent. This defines both objects and arrows of the asserted morphism and gives its compatibility with pullback. This argument uses local $\mathcal{O}(1)$ only to exhibit a scalar-trivial representation; it never posits such a global line bundle on a Brauer-nonsplit projective-space bundle. The universal property of Proposition 32.1 verifies the same construction independently of the chosen trivializations. $\square$

38.2. A modular modification and its defining equations. Restrict (38.1) to U_{red} of (36.1). With B_{red} and its invariant Fitting ideal as in (36.2), put

$$\mathcal{B}_{\text{red}} = [B_{\text{red}}/\text{PGL}(V)], \quad \mathcal{F}_{\text{red}}^{\text{eff}} = [U_{\text{red}}/\text{PGL}(V)]. \quad (38.2)$$

Theorem 38.2 (The effective multiple-incidence boundary). *The map $\mathcal{B}_{\text{red}} \rightarrow \mathcal{F}_{\text{red}}^{\text{eff}}$ is representable, proper, birational, and carries the flat embedded curve family of Theorem 36.3. Both stacks are smooth. This is a modular modification over U_{red} , not an assertion that either stack is proper over $\mathbb{C}$ or has finite stabilizers. On an etale neighbourhood with labelled incidence centres $I_i = (u_i, v_i)$, the multi-Rees algebra has presentation*

$$\mathcal{R}(I_1, \dots, I_k) = R[U_1, V_1, \dots, U_k, V_k] / (v_i U_i - u_i V_i)_{i=1}^k, \quad (38.3)$$

where $\deg U_i = \deg V_i$ is the i th coordinate vector and $U_i = u_i T_i$, $V_i = v_i T_i$. There is no torsion kernel left in this presentation. Its diagonal subalgebra is

$$\bigoplus_{d \geq 0} (I_1 \cdots I_k)^d (T_1 \cdots T_k)^d, \quad (38.4)$$

so its diagonal Proj is the computed Hilbert modification. All these objects descend without a global labelling of the points.

Proof. For a test scheme $T \rightarrow [U_{\text{red}}/\text{PGL}(V)]$, take its defining torsor P_T . The pullback is the associated space $(P_T \times_{U_{\text{red}}} B_{\text{red}})/\text{PGL}(V)$. The projective equivariant morphism descends to a projective morphism over T . This proves representability and properness; birationality holds over the dense nonincident open. Smoothness of the two atlases and of the acting group proves smoothness of the stacks. The curve family descends by the same construction. No quotient by the stabilizer of a pencil has been mistaken for its fixed-frame scheme fibre.

For the algebra calculation begin in a polynomial ring in the independent parameters u_i, v_i and complementary smooth coordinates. For one pair, the kernel of $R[U_i, V_i] \rightarrow R[T_i]$ is generated by $v_i U_i - u_i V_i$: comparison of successive powers and cancellation using the regular sequence (u_i, v_i) proves the assertion. For several pairs the variables are disjoint. Tensoring the one-pair presentations over the ground field preserves their injections into the polynomial ring in the T_i , giving (38.3). The result persists after the etale flat extension supplied by Lemma 36.1; flat base change preserves both the coefficient ideals and the displayed kernel. This proves the absence of an additional saturation or torsion relation. The multidegree $(d, \dots, d)$ part is exactly $I_1^d \cdots I_k^d (T_1 \cdots T_k)^d$, which proves (38.4). It gives the actual defining equations before taking the ordinary-power diagonal, rather than treating the formation of ordinary powers as a second recovery theorem.

The ideal itself is $\text{Fitt}_0(q_* \mathcal{O}_D)$, where D is the zero scheme of the distinguished power J_{n-1} on the intrinsic pencil line. Finite base change identifies the pushforward module, and Fitting ideals commute with arbitrary base change. Its formation is invariant under the group action. The local multi-Rees presentations are labelled charts of this global ideal blow-up; permutation of the labels does not affect their diagonal. Descent therefore follows from the intrinsic Fitting construction, without claiming that graph closure or blow-up commutes with every base change. A pulled-back curve family is defined by its flat pullback; its existence does not require such a graph-closure assertion. $\square$

38.3. Relative lifting is not absolute obstruction. For precision a deformation of an effective object at R over a local Artin complex algebra A is an object of $\mathcal{F}_{\text{pen}}^{\text{eff}}(A)$ with an identification of its closed fibre with R . A source framing identifies it with a point of the completed pencil parameter scheme; changing the framing acts by $\text{PGL}(V)(A)$. A boundary deformation also specifies a lift to B_{red} of the closed boundary point.

Proposition 38.3 (Exact relative lifting equations). *Fix a boundary point with k incidences. In a labelled chart of the base blow-ups write $v_i = u_i w_i$. Let $A' \twoheadrightarrow A$ be a square-zero extension with kernel J . Fix a lift of the base object to A' , with coordinates u'_i, v'_i , and an existing lift to the*

boundary over A , with slopes w_i . For any lifts $\tilde{w}_i \in A'$, define

$$o_i = [v'_i - u'_i \tilde{w}_i] \in J/u'_i J. \quad (38.5)$$

These classes do not depend on the lifts $\tilde{w}_i$. The prescribed base deformation lifts in this chart if and only if all o_i vanish. When nonempty, its set of framed lifts is a torsor under $\bigoplus_i \ker(u'_i : J \rightarrow J)$. At a closed boundary point with slopes w_i^0 , a tangent vector $u_i = \epsilon a_i$, $v_i = \epsilon b_i$ lifts if and only if $b_i = w_i^0 a_i$. The kernel and cokernel of the differential of the fixed-frame blow-up map both have dimension k there.

The absolute effective and boundary stacks are smooth, with tangent complexes $[\mathfrak{p}\mathfrak{gl}(V) \rightarrow T_R U_{\text{red}}]$ and $[\mathfrak{p}\mathfrak{gl}(V) \rightarrow T_b B_{\text{red}}]$ in degrees $-1, 0$. Thus the relative conditions above do not assert an absolute obstruction to deforming an effective failure object. Nor do they identify all deformations of its raw unmarked Artin algebra.

Proof. The defining equation over A' is $v'_i = u'_i w'_i$. Its defect for a chosen lift lies in J because it vanishes over A . Replacing that lift by $\tilde{w}_i + \eta_i$, with $\eta_i \in J$, changes the defect by $-u'_i \eta_i$. This proves the well-defined obstruction class, its necessity and sufficiency, and the stated torsor of solutions. The independent coordinate charts introduce no further equations. For dual numbers at the closed point, multiplication of ϵa_i by a change $\epsilon \eta_i$ is zero, giving precisely the displayed tangent condition. In the chart $(u_i, w_i) \mapsto (u_i, u_i w_i)$ the differential has rank one for each normal parameter pair and kills its slope direction. Complementary coordinates are unchanged, so both kernel and cokernel have dimension k .

The tangent complexes are those of the smooth quotient atlases in (38.2), and Theorem E.2 identifies the effective failure deformation groupoid with the first of them over every Artin base. To check explicitly the nonlinear kernel used here, let the coefficient ring be $\mathbb{C}[\epsilon]/(\epsilon^2)$ and replace a generator x by $gx + q(x)$, with q of degree at least two. In a homogeneous relation of sharp degree d , every term containing a factor from q has degree at least $d + 1$ and vanishes in the sharp truncation. The surviving relation is exactly the linear action $\text{Sym}^d(g)$. This argument uses degrees in the generators, not powers of ϵ , and is valid for coefficients in any base algebra. It therefore separates the tangent-identity generator kernel from an actual infinitesimal change of the pencil over the base. Passing to the two prescribed rigidifications gives the claimed categorical interpretation of the coordinate calculation. $\square$

The ideal cutting out the incidence is scheme-theoretically intrinsic to the effective failure family *with its recovered pencil line*. That line is an essential output of the inverse; no bypass of it, and no canonical choice of a slope from a closed algebra, is asserted. The additional content is the complete multi-Rees presentation and relative lifting equations, and, on the nonreduced-contact slices, the singular graph and its computed cotangent module in Proposition 37.3. These calculations specify the interaction with boundary singularities rather than merely transporting a pre-existing finite algebra by pullback.

38.4. An accessible completeness input for singular pencils. The congruence classification used by Theorem 30.2 is recalled with explicit blocks in [De Teran–Dmytryshyn–Dopico(2018), Theorem 2.1]. In that statement a symmetric minimal-index block with index ε has dimension $2\varepsilon + 1$; hence no index in an $n \times n$ symmetric pencil exceeds $\lfloor (n - 1)/2 \rfloor$. The Toeplitz nullities in that finite range recover the multiplicities by the second-difference formula already proved in our theorem. The regular blocks are indexed by their projective spectral points and local exponent partitions. Accordingly “complete data” includes both the points and the partitions, modulo one common projective reparametrization of the pencil line. This is an accessible theorem-level classical completeness input, not a new symmetric Kronecker classification. It is distinct from the local congruence diagonalization proved in the paper.

39. DESCENT AND COMPLETE-LOCAL DETAILS

The following statements supply the local-to-global steps used in Sections 13–18. They also specify which assertions concern a completed image, a normalization fibre, or a family on an exact-divisor stratum.

Lemma 39.1 (Complete-local surjectivity and reduced completion). *In the cotangent-surjectivity assertion of Lemma 13.1, the rings are Noetherian complete local $\mathbb{C}$ -algebras with residue field $\mathbb{C}$, and the homomorphism is local, continuous, and the identity on that residue field. Under these hypotheses cotangent surjectivity implies ring surjectivity. A reduced local ring essentially of finite type over $\mathbb{C}$ has reduced completion.*

Proof. Choose preimages in the source maximal ideal of generators of the target cotangent space. Their degree- q products span every quotient $\mathfrak{m}_B^q/\mathfrak{m}_B^{q+1}$. Given an element of B , first lift its residue, then its error successively in degrees $q = 1, 2, \dots$. The correction in degree q lies in $\mathfrak{m}_B^q$. Their sum converges in R , and continuity and separatedness imply that its image is the desired element of B . This is not a statement about arbitrary noncomplete local homomorphisms.

Finite-type complex rings are excellent. Quasi-excellent local rings are Nagata and have geometrically reduced formal fibres; see [29, Tags 07QV and 0BJ0]. For a reduced local ring A , the injection into the product of its finitely many minimal-prime quotients remains injective after flat completion. Each quotient is a Nagata domain and its completion is reduced. Thus $\hat{A}$ is a subring of a product of reduced rings. This justifies the reduced completed-image step of the earlier lemma. $\square$

Lemma 39.2 (Coprime cluster products over Artin bases). *Let A be a local Artin $\mathbb{C}$ -algebra. If the reduction of a monic polynomial has a factorization into pairwise coprime monic factors, there is a unique factorization into monic lifts of the same degrees. In a factorization fibre, the functor of divisors with prescribed degrees in the distinct residual clusters is the product of the cluster divisor functors. Its coordinate algebra is therefore the tensor product of their Artin coordinate algebras.*

Proof. For two coprime factors p, q , the linear correction map

$$(\dot{p}, \dot{q}) \mapsto q\dot{p} + p\dot{q}, \quad \deg \dot{p} < \deg p, \quad \deg \dot{q} < \deg q,$$

is an isomorphism on coefficient spaces over the residue field. Its kernel is zero by coprimeness, and source and target have equal dimension. It is therefore an isomorphism over A as well. Across a square-zero extension of Artin rings it uniquely corrects a lifted product to the given polynomial. Induction on the nilpotence order and then on the number of clusters proves existence and uniqueness. Apply the same argument to both the marked divisor and its complement. The cluster product and decomposition constructions are mutually inverse on every Artin base; Yoneda gives the asserted product of finite schemes. Passing to inverse limits gives the corresponding assertion on complete local bases. $\square$

Lemma 39.3 (The local choice and descent in the binary stratum). *The base-change identification of Theorem 14.1 holds as an identification of incidence functors, not only on closed fibres. The required choice of a member of the residual space coprime to the common divisor can be made on a Zariski open cover of the exact-gcd stratum; the choices are auxiliary and the resulting identifications glue uniquely.*

Proof. Write a point of the stratum as (G, L) in $\mathbb{P}(S_s) \times X_{D-s,r}$. The vanishing of a member of L at one of the finitely many roots of G is a proper linear condition on that member. Over $\mathbb{C}$ a finite union of these proper hyperplanes does not cover L . Choose a linear combination avoiding them, lift its coefficients on a bundle-trivializing neighbourhood, and invert its resultant with

G . This gives the required Zariski neighbourhood. These neighbourhoods cover the stratum and may be pulled back by any base change.

On the marked-divisor incidence, pass locally to a monic coordinate for the proposed divisor f . The quotient by f is finite free. On every geometric fibre f divides G , so the chosen residual member is invertible in that quotient, by coprimeness. Its multiplication matrix has invertible determinant locally, hence it is invertible also over a nonreduced base. Thus $f \mid GL$ implies $f \mid G$ as an equation on that base, and the converse is immediate. Monic division forces the remaining subspace. The identification is independent of which coprime residual member established cancellation, so it glues on overlaps. No lifting of arbitrary image points to the normalization is asserted.

The finite factorization map used there is flat by miracle flatness: its target local rings are regular, its source local rings are Cohen–Macaulay, it is finite, and dimensions agree. These are exactly the hypotheses of [29, Tag 00R4]. This names the flatness input separately from the incidence-functor argument. $\square$

Lemma 39.4 (The coordinate equalizer and coherent conductor). *For the block ideals I_A of Theorem 15.1, the completed image ring is the equalizer of the branch rings with their restrictions to all pairwise intersections. This assertion uses the coordinate monomial structure. The completed conductor calculation there implies equality of the corresponding coherent ideal sheaves on the squarefree open.*

Proof. A monomial has a coefficient on each branch where its support survives. If it survives on two branches, it also survives on their coordinate intersection. Pairwise agreement therefore assigns it one well-defined coefficient. Collecting these coefficients gives one formal series modulo the intersection of the I_A ; conversely such a series has compatible restrictions. Formal convergence causes no issue, since there are finitely many monomials in each degree. The claim would be false for general configurations of ideals without this coordinate argument.

Write $A \subset B$ for the local finite normalization algebra. The conductor is $\text{Ann}_A(B/A)$, a coherent ideal. Flat completion preserves the exact finite-module sequence and its annihilator: compute the latter as the intersection of the kernels of multiplication on a finite generating set. The completed finite algebra is the product of the normal branch rings. Thus the conductor and the ideal of Y_{g+1} have equal completed stalks at every squarefree closed point. Faithful flatness gives equal stalks, and coherence on a finite-type complex scheme gives equality of the ideal sheaves. $\square$

Proposition 39.5 (Finiteness and uniqueness of the collision diagram). *Every branch map in Theorem 16.1 is finite, and the completed incidence diagram is determined, up to the unique isomorphism respecting its incidence data, by its marked-divisor functor. It is independent of the chosen frame and root coordinates.*

Proof. For each cluster $P_i = A_i B'_i$, the coefficients of the monic A_i are integral over the coefficients of P_i . This follows from the finite factorization morphism, or by expressing them as elementary symmetric functions of a selected subset of the roots in a finite splitting algebra. Division by a monic polynomial expresses every coefficient of B'_i polynomially in those of A_i and P_i . Likewise $R_{ij} = A_i U_{ij}$ expresses all quotient coefficients in terms of A_i and R_{ij} . Consequently the branch algebra is generated over the ambient coefficient algebra by finitely many integral coefficients of the A_i , so it is finite. Completing preserves this finite-module assertion.

The equations represent the functor of a subbundle with a marked divisor and its uniquely determined quotient. A change of frame or local coordinate gives inverse natural transformations of this same functor and of its forgetful map to the ambient Grassmannian. The complete representing rings and their maps are uniquely isomorphic by Yoneda. Their scheme-theoretic

images consequently agree as completed diagrams. This proves coordinate independence without identifying arbitrary Artin points of an image with lifted Artin points of its normalization. $\square$

Partition closures. For a fixed partition $(s_1, \dots, s_b)$, the proper map $(\mathbb{P}^1)^b \rightarrow \mathbb{P}(S_s)$ sending (ℓ_i) to $\prod \ell_i^{s_i}$ has closed image. Its open set of distinct ℓ_i is dense. At the boundary precisely some of these points coincide, and the associated parts add. Every such coincidence can be approached from distinct points. This proves exactly the merging closure order in the fixed-degree symmetric product; compare the coincident-root constructions of [19, 20]. It is not a closure classification for all analytic singularities of the completed collision atlas.

Equivariance of the singular Fitting calculation. The group used in Theorem 17.1 is $\mathrm{GL}_m(\mathbb{C})$, acting simultaneously on the columns u and w and fixing Δ . It preserves the presentation and therefore its intrinsic Fitting ideal. In the normalization the three displayed minors span, under this group, respectively $\mathrm{Sym}^m(\mathbb{C}^m)t$, $\Delta \mathrm{Sym}^m(\mathbb{C}^m)$, and $\mathrm{Sym}^{m+1}(\mathbb{C}^m)$, with the coordinate/dual convention determined by the action on u . To see spanning directly, their translates are nonzero scalar multiples of the corresponding powers of all linear forms; polarization in characteristic zero spans each symmetric power. Multiplication by R gives exactly the modules asserted in (17.1). This is stronger than invoking invariance without identifying the representations.

Specialization of minimal relations. For the surjective coefficient specialization $\pi : P \rightarrow P'$ to one coordinate in Theorem 18.1, one has $\pi(\mathfrak{n}) = \mathfrak{n}'$ and $\pi(\mathcal{I}) = \mathcal{I}'$. Consequently $\pi(\mathfrak{n}\mathcal{I}) = \mathfrak{n}'\mathcal{I}'$. A coefficient whose image is a minimal binary relation cannot lie in $\mathfrak{n}\mathcal{I}$. This explicitly justifies the nonzero irreducible coefficient map used in that proof. All reciprocal series and socle weights in these arguments use the weight of a Q_i coefficient equal to i . They do not claim ordinary maximal-ideal Hilbert functions; (19.7) gives one such function separately.

40. CONDUCTORS AND LOCAL MODELS OF DIVISOR COLLISIONS

We retain the notation of Section 12. All completed local rings below are completed at closed complex points. For a reduced local ring A with finite normalization B , its *conductor* is $\mathfrak{c} = \mathrm{Ann}_A(B/A)$, regarded also as an ideal of B . We use the intrinsic singular subscheme defined by $\mathrm{Fitt}_{\dim A} \Omega_{A/\mathbb{C}}$; on completions, differentials mean continuous differentials. The conductor and this Fitting ideal need not agree. Our first result determines both on a multivariate stratum. The second identifies the way two branches merge when a common binary root becomes double.

The formal-image and separated-germ steps below are isolated in Lemma 13.1. Lemma 13.2 gives the Artin cancellation and a direct positive lower bound for the split codimension. Theorem 16.1 supplies the full Weierstrass incidence functor in every collision order; Theorem 17.1 evaluates the singular ideal for all ranks.

40.1. Two coprime divisor choices.

Theorem 40.1 (The split-divisor stratum). *Suppose $h > g$, $2 \leq r \leq s_{h-g}$, and*

$$K = abL, \quad [a], [b] \in \mathbb{P}(S_g), \quad L \in \mathrm{Gr}(r, S_{h-g}), \quad \mathrm{gcd}(L) = 1.$$

Assume that a, b are coprime and that the only degree- g divisors of ab , up to scalar, are a and b . Put

$$\begin{aligned} t &= 2(s_g - 1) + r(s_{h-g} - r), \\ c &= r(s_h - s_{h-g}) - (s_g - 1), \quad d_g = t + c. \end{aligned}$$

Then $c \geq 1$, and there are formal coordinates in which

$$\widehat{\mathcal{O}}_{Y_g, K} = \mathbb{C}[[z_1, \dots, z_t, x_1, \dots, x_c, y_1, \dots, y_c]] / (x_i y_j : 1 \leq i, j \leq c). \quad (40.1)$$

Its normalization is $\mathbb{C}[[z, x]] \times \mathbb{C}[[z, y]]$. The conductor is

$$\mathfrak{c} = (x_1, \dots, x_c, y_1, \dots, y_c), \quad \widehat{\mathcal{O}}_{Y_g, K}/\mathfrak{c} = \mathbb{C}[[z]], \quad (40.2)$$

and the intrinsic singular subscheme has ideal $\mathfrak{c}^c$. The two smooth branches meet scheme-theoretically in a smooth t -dimensional subscheme. The local ring has multiplicity two and depth $t + 1$; it is Cohen–Macaulay exactly when $c = 1$.

Proof. We first determine the intersection as a scheme, not by comparing closed points or tangent dimensions. The normalization has precisely two points above K , namely (a, bL) and (b, aL) . At either point, the overlap in Lemma 12.3 is zero. The map from the completed local ring of the ambient Grassmannian to each completed source ring is consequently surjective: its cotangent map is surjective, and completeness and Nakayama’s lemma lift successive homogeneous terms. Thus each source completion is a closed smooth formal branch of the image.

Here is the required assertion over infinitesimal bases. Let A_0 be a local Artin $\mathbb{C}$ -algebra. A point on both formal branches carries unique marked lifts a_{A_0}, b_{A_0} near a, b . If homogeneous polynomials have coprime reductions, those reductions form a regular sequence in the polynomial ring over $\mathbb{C}$. Its graded Koszul complex is exact. In each degree the polynomial modules are finite free; lifting a splitting of the complex, or applying the local criterion for flatness successively, shows that its lift over A_0 is exact and that the quotient is A_0 -flat. In particular

$$(a_{A_0}) \cap (b_{A_0}) = (a_{A_0} b_{A_0}). \quad (40.3)$$

This also follows by noting that b_{A_0} is a nonzerodivisor modulo a_{A_0} and solving a relation between them. Applying (40.3) to the columns of the relation subbundle gives, uniquely,

$$K_{A_0} = a_{A_0} b_{A_0} L_{A_0}.$$

The residual module is a subbundle: multiplication by $a_{A_0} b_{A_0}$ is a split injection in degree D , and the quotient of the ambient free module by K_{A_0} is free. The kernel of the induced surjection onto the complementary quotient is therefore free. This proves that the formal intersection is represented by the product of the two divisor charts and the Grassmannian chart for L , with no nilpotents omitted.

The differential of this product parametrization is injective. Indeed, an infinitesimally constant product gives

$$(\dot{a}b + a\dot{b})l + ab\dot{l} \in abL \quad (l \in L).$$

Taking valuations at the factors of a , and using $\gcd(a, b) = \gcd(L) = 1$, forces $a \mid \dot{a}$; similarly $b \mid \dot{b}$. These are the two projective scalar directions, and then $\dot{L} = 0$. The intersection is thus a smooth formal closed subscheme of dimension t .

Let R_0 be the ambient completed local ring and I_a, I_b the ideals of these branches. Finiteness of the normalization, exactness of completion on finite modules, and the definition of the scheme-theoretic image give

$$\widehat{\mathcal{O}}_{Y_g, K} = R_0/(I_a \cap I_b) = (R_0/I_a) \times_{R_0/(I_a + I_b)} (R_0/I_b).$$

This is the elementary conductor-square construction; see also [21, Lemma 1.3 and §1.3.2]. All three quotient rings are regular. Choosing compatible regular parameters for the two surjections gives (40.1). The branch dimension is $d_g = s_g - 1 + r(s_h - r)$, so its codimension along the intersection is the asserted c . It is positive: if $c = 0$, both branch ideals would coincide locally with the intersection ideal, contradicting the two distinct normalization branches of a reduced local ring.

In the fibre-product description, an element belongs to the conductor exactly when both its restrictions to the intersection vanish. This gives (40.2). The Jacobian of the relations $x_i y_j$ has nonzero entries only among the x_i and y_j . The Fitting ideal in question is its ideal of c -minors.

Every such minor belongs to $\mathfrak{c}^c$. Conversely, choosing the c columns indexed by the y_j and one row (i_j, j) for each j gives every monomial $x_{i_1} \cdots x_{i_c}$. The symmetric choice gives every degree- c monomial in the y 's. Mixed monomials vanish in the ring. Thus the ideal is exactly $\mathfrak{c}^c$, including its nonreduced structure.

Finally use

$$0 \longrightarrow \widehat{\mathcal{O}}_{Y_g, K} \longrightarrow \mathbb{C}[[z, x]] \oplus \mathbb{C}[[z, y]] \longrightarrow \mathbb{C}[[z]] \longrightarrow 0,$$

where the last arrow is the difference of restrictions. The depth lemma gives depth $t + 1$; for $c = 1$ this equals the dimension. For $c > 1$ it is strictly smaller. Additivity of leading Hilbert coefficients gives multiplicity two, since the last module has smaller dimension than the two regular branches. $\square$

Remark 40.2. The hypotheses are nonempty in every number of variables: for $e = 2$ one may take $g = 1$, and for $e \geq 3$ one may take two distinct irreducible forms of degree g . The theorem treats an explicitly specified stratum, not all pairs of overlapping degree- g divisors. In particular, it does not substitute a binary factorization rule for multivariate divisibility. Its high-codimension branch intersections need not be normal-crossing divisors.

40.2. A double common root. For $m \geq 1$, put $R_m = \mathbb{C}[[u_1, \dots, u_m, \Delta]]$ and $I = (u_1, \dots, u_m) \subset R_m$. Define

$$A_m = R_m \oplus It \subset B_m = \mathbb{C}[[u_1, \dots, u_m, t]], \quad \Delta = t^2. \quad (40.4)$$

The right side of (40.4) is a subring, not a square-zero extension: $(It)^2 = \Delta I^2$. For $m = 1$ it is the ordinary pinch-point ring.

Theorem 40.3 (Collision of two binary divisor choices). *Let $e = 2$, $g = 1$, $D = h + 1$, and $2 \leq r \leq D - 1$. At a point $K_0 = \ell^2 L$ with $L \in \text{Gr}(r, S_{D-2})$ and $\gcd(L) = 1$, set*

$$m = r - 1, \quad q = r(D - 1 - r) + 1.$$

Then

$$\widehat{\mathcal{O}}_{Y_1, K_0} \simeq A_m[[z_1, \dots, z_q]]. \quad (40.5)$$

Writing $w_i = u_i t$, a complete presentation of A_m is

$$\mathbb{C}[[u_1, \dots, u_m, \Delta, w_1, \dots, w_m]] / (u_i w_j - u_j w_i, w_i w_j - \Delta u_i u_j), \quad (40.6)$$

where the first relations have $i < j$ and the second have $i \leq j$. Its normalization is B_m , and its conductor is

$$\mathfrak{c} = (u_1, \dots, u_m, w_1, \dots, w_m), \quad \mathfrak{c} B_m = I B_m.$$

The induced map on conductor quotients is

$$\mathbb{C}[[\Delta]] \longrightarrow \mathbb{C}[[t]], \quad \Delta \longmapsto t^2. \quad (40.7)$$

Thus the two reduced divisor lifts merge through a ramified double cover of the conductor. The completed local ring has dimension $q + m + 1$ and depth $q + 2$. It is Cohen–Macaulay if and only if $r = 2$. For $r = 2$, its intrinsic singular ideal, with the smooth variables suppressed, is $(w, \Delta u, u^2)$, whereas its conductor is (u, w) .

Proof. Choose an affine coordinate x with the root of ℓ at zero. Some member $F_0 \in K_0$ has multiplicity exactly two there, since $\gcd(L) = 1$. Extend it to a frame $F_0, \dots, F_{r-1}$. In the completed local ring of the ambient Grassmannian the formal Weierstrass theorem writes the universal F_0 as a unit times

$$P(x) = x^2 + ax + b.$$

Divide each F_i , $i > 0$, by this monic polynomial. Its remainder is $u_i x + v_i$. The $2r$ functions $a, b, u_1, v_1, \dots, u_m, v_m$ are part of a regular parameter system. To see this rather than assume versality, note that K_0 has zero two-jet at the root, while $S_D/K_0 \rightarrow \mathbb{C}[x]/(x^2)$ is surjective. The

Grassmannian tangent space $\text{Hom}(K_0, S_D/K_0)$ therefore allows independent variations of both jets in every frame column. Passing from these jets to the displayed remainders is an invertible triangular operation, with leading coefficient $F_0/x^2(0) \neq 0$. Their differentials are consequently independent.

Set

$$t = x + a/2, \quad \Delta = a^2/4 - b, \quad w_i = au_i/2 - v_i.$$

The incidence of a common root near zero is now exactly

$$\Delta = t^2, \quad w_i = u_i t.$$

There are no other normalization points above K_0 , because its exact common divisor is ℓ^2 . Consequently the image of this completed incidence is the completed image defining Y_1 . The ambient dimension is $N_0 = r(D + 1 - r)$. Of the $2r$ parameters above, a is free, as are the other $N_0 - 2r$ parameters. This accounts for $q = N_0 - 2r + 1$ smooth variables in (40.5).

It remains to justify elimination scheme-theoretically. The relations in (40.6) reduce any series to $p + \sum_i w_i p_i$ with $p, p_i \in R_m$. If its image in B_m is zero, the decomposition $B_m = R_m \oplus R_m t$ gives $p = 0$ and $\sum_i u_i p_i = 0$. The first syzygies of the regular sequence $u_1, \dots, u_m$ are generated by the Koszul syzygies. They are precisely the relations $u_i w_j - u_j w_i$. This proves the presentation without taking a radical. The same argument works first over polynomial rings and then by exact completion.

The extension $A_m \subset B_m$ is finite, since $t^2 = \Delta$, and birational, since $t = w_1/u_1$ in the fraction field. The regular ring B_m is thus the normalization. For $p + vt \in R_m \oplus It$, multiplication by t stays in A_m if and only if $p \in I$. As $1, t$ generate B_m , the conductor is $I \oplus It$, proving the quotient map (40.7). Equivalently,

$$A_m = B_m \times_{\mathbb{C}[[t]]} \mathbb{C}[[\Delta]], \quad (40.8)$$

where the first map sets $u = 0$ and the second sends Δ to t^2 . This is a specific conductor square, not an assumption that the normalization descends through its reduced fibres.

As an R_m -module, $A_m = R_m \oplus I$. For $m = 1$, the ideal I is free. For $m > 1$, the exact sequence $0 \rightarrow I \rightarrow R_m \rightarrow \mathbb{C}[[\Delta]] \rightarrow 0$ gives $\text{depth}_{R_m} I = 2$. Hence A_m has depth two for every $m \geq 1$. The maximal ideal of R_m generates an ideal with maximal radical in A_m , so this is also its local ring depth. Adding the q smooth parameters gives the assertion. When $m = 1$, the equation is $w^2 - \Delta u^2 = 0$; its partial derivatives generate $(w, \Delta u, u^2)$ in characteristic zero. $\square$

Corollary 40.4 (The split and ramified parts of the same model). *On the binary locus where the common divisor has degree exactly two, there are two strata. Over two distinct roots the formal local ring has the split form (40.1), with $t = q + 1$ and $c = r - 1$. Over a double root it has (40.5). The support of the singular locus in the latter model is the conductor, and the normalization fibre at a collision is $\text{Spec } \mathbb{C}[t]/(t^2)$. For arbitrary m , its singular subscheme is given without radicalization by the m -minors of the Jacobian of the displayed m^2 relations in (40.6).*

Proof. The assertion at distinct roots is Theorem 40.1. In the collision model the normalization is an isomorphism wherever some u_i is invertible. On $u = w = 0$, a nonzero value of Δ gives two distinct points after an étale square-root extension, while $\Delta = 0$ gives the stated length-two fibre. The exact normality criterion proves the assertion about singular support. There are $2m + 1$ nonsmooth coordinates and dimension $m + 1$; the cotangent presentation therefore gives the m -minors as claimed. This formula specifies the whole singular scheme, even where it is thicker than its conductor support. $\square$

Remark 40.5 (Descent to intrinsic first relations). On the maximal-lower-Hilbert-function stratum of Corollary 12.8, these statements compute the completed smooth-atlas singularities of the intrinsic first-relation algebra stack after tangent-identity rigidification. The conductor is equivariant,

and formation of its finite-module annihilator and of the Fitting ideals commutes with flat base change. Thus the calculations descend through the frame atlas. This does not identify these rings with completed local rings of a coarse orbit space, where stabilizers can introduce further singularities.

41. INTERSECTIONS OF DIVISOR-DEGREE IMAGES

Here the total relation degree D is fixed. Write $Y_a \subset \text{Gr}(r, S_D)$ for the image admitting a divisor of degree a ; empty Grassmannians are omitted. In several variables these images are not generally nested. Their reduced intersections nevertheless have a finite description by common parts of two marked divisors.

Proposition 41.1 (The common-part formula). *For $1 \leq a, b < D$ and $0 \leq j \leq \min(a, b)$, let M_j be the scheme-theoretic image of*

$$\begin{aligned} \mathbb{P}(S_j) \times \mathbb{P}(S_{a-j}) \times \mathbb{P}(S_{b-j}) \times \text{Gr}(r, S_{D-a-b+j}) &\longrightarrow \text{Gr}(r, S_D), \\ (c, u, v, L) &\longmapsto cuvL. \end{aligned} \quad (41.1)$$

A negative last degree or insufficient dimension gives the empty image, and $\mathbb{P}(S_0)$ is a point. Then

$$(Y_a \cap Y_b)_{\text{red}} = \left(\bigcup_j M_j \right)_{\text{red}}. \quad (41.2)$$

On the locus of two marked divisors f_a, f_b with exact common part of degree j , the corresponding factorization is $c = \gcd(f_a, f_b)$, $f_a = cu$, $f_b = cv$, with $\gcd(u, v) = 1$. In particular, at a relation $K = a_0^m L$ with a_0 a nonzero linear form and $\gcd(L) = 1$, the set of positive divisor degrees is exactly $\{1, \dots, m\}$. For the boundary point in Theorem 42.1, $m = D - q_n$.

Proof. Each map in (41.1) is defined on a projective scheme: multiplication by a nonzero form has constant rank, so the subspace construction is a morphism. Its image is closed and admits the two divisors cu and cv , proving one inclusion. Conversely, if K has divisors f_a, f_b , unique factorization gives their common part c of some degree j . Their least common multiple is cuv and divides every member of K . Division yields L of degree $D - a - b + j$, of the same rank. This proves equality on geometric points. Over $\mathbb{C}$, a reduced closed subscheme of a finite-type scheme is determined by its closed points, giving (41.2). The last assertion follows because every homogeneous divisor of a power of one linear form is a power of that form. $\square$

The reduction in (41.2) is essential to the statement of this global formula. It is not a prescription to reduce the individual Y_a , their normalization fibres, or their singular subschemes. The full intersection of the two formal branches on the coprime stratum was computed over Artin bases in Theorem 40.1; Theorem 40.3 computes the thickening when those branches collide. Thus the degree-poset description and the local scheme calculations have distinct, explicit roles.

42. A COMMON RAMIFIED BOUNDARY OF PENCIL FAILURE ORBITS

The fixed tensor presentation of a pencil has exact common divisor δ^{n-1} . Varying the pencil alone therefore stays in the exact-divisor locus. The intrinsic finite algebra permits a larger, linear coordinate-change action: changing the basis of its entire cotangent space. We now describe a boundary point produced by such changes. The generic members are genuine pencil failure algebras. The central member has a larger common divisor, and its normalization fibre is nonreduced. Thus the boundary of Section 12 occurs inside the orbit closures of the motivating algebras themselves.

Let $n = \dim V = \dim U \geq 3$, $E = V^* \otimes U$, and put

$$e = n^2, \quad N = \binom{n+1}{2}, \quad r = \binom{N}{2}, \quad g = n(n-1), \quad h = 3n-4, \quad D = g+h.$$

For a pencil R , write $K_R = \delta^{n-1} C_R \subset \text{Sym}^D E$ as in Lemma E.1. Define the projective orbit closure

$$Z_R = \overline{\text{GL}(E) \cdot K_R} \subset \text{Gr}(r, \text{Sym}^D E).$$

It is independent of a chosen basis of the cotangent space. Since Y_g is closed and invariant, $Z_R \subset Y_g$.

The group in Z_R is the full $\text{GL}(E)$, not the $\text{PGL}(V)$ congruence group of the pencil. Fixing the standard flag, scalar direction and complement specifies the displayed common point; no choice-independent distinguished moduli point is asserted.

Theorem 42.1 (A common point in full first-relation orbit closures). *Set*

$$q_n = \begin{cases} 3, & n = 3, \\ 4, & n \geq 4, \end{cases} \quad k_n = h - q_n.$$

There is an explicitly constructed point K_∞ belonging to Z_R for every quadratic pencil R , with

$$K_\infty = a^{g+k_n} L_n, \quad L_n \subset \text{Sym}^{q_n} E, \quad \text{rank } L_n = r, \quad \gcd(L_n) = 1, \quad (42.1)$$

where $a \in E$ is a nonzero linear form. The flag pencil $R_ = \langle x_1^2, x_1 x_2 \rangle$ admits a finite-flat family of first-relation algebras over $\mathbb{A}^1$, with special relation space K_∞ and every nonzero fibre isomorphic to its failure algebra. Together with Theorem G.1, this gives a chain of two explicit finite-flat specializations from the failure algebra of every R to the same boundary algebra.*

The fibre of ν_g over K_∞ has a single geometric point, marked by a^g , and is nonreduced. Its tangent dimension is

$$\binom{e + k_n - 1}{k_n} - 1. \quad (42.2)$$

In particular K_∞ lies in the singular locus of Y_g and outside the exact-degree- g locus. For $n = 3$ this tangent dimension is 44.

The construction requires the exact osculating order of one elementary flag embedding. We include the calculation to distinguish the theorem from a computationally suggested degeneration.

Lemma 42.2 (The flag coefficient filtration). *Write a matrix of independent linear coordinates as*

$$T = aI_n + B, \quad B_{11} = 0.$$

For $C_ = C_{R_*}$, the filtration by vanishing order at $T = I_n$ on the chart $T_{11} = 1$ has last nonzero graded piece in degree q_n . It has a nonzero degree-zero piece. Hence an adapted basis $c_1, \dots, c_r$ has orders $0 \leq d_j \leq q_n$, attaining both endpoints, and*

$$\lim_{\tau \rightarrow 0} \tau^{-d_j} c_j(aI_n + \tau B) = a^{h-d_j} p_j(B), \quad (42.3)$$

where the displayed initial forms are linearly independent.

Proof. The complementary-minor identity for $\text{Sym}^2 T$ identifies the vector of primitive coefficients, up to a fixed invertible coordinate change, with

$$\delta(T)^3 (v^2 \wedge vw), \quad v = T^{-1} e_1, \quad w = T^{-1} e_2. \quad (42.4)$$

Indeed $\det(\text{Sym}^2 T) = \delta^{n+1}$, the complementary minors have order two, and division of δI_{N-2} by δ^{n-1} leaves δ^3 . This identity is initially on $\delta \neq 0$; both sides are the same polynomial vector of degree h .

A standard affine flag chart has representatives

$$v = e_1 + \sum_{i=2}^n z_i e_i, \quad w = e_2 + \sum_{i=3}^n y_i e_i.$$

Every coordinate of $v^2 \wedge vw$ is a polynomial of degree at most four. The map from GL_n to this flag chart is smooth, and restricting to $T_{11} = 1$ remains smooth: the scalar subgroup is contained in each flag stabilizer and is transverse to this chart. Normalizing v, w only multiplies the whole wedge by a common unit. Smooth pullback preserves the order of a nonzero function, as does multiplication by a unit. The coordinates in (42.4) are linearly independent by Lemma 6.2 and the nonzero matrix-coefficient inclusion in (6.6). Thus no nonzero linear combination can have order exceeding four. For $n \geq 4$, the coordinate indexed by $e_3^2 \wedge e_3 e_4$ is

$$z_3^2(z_3 y_4 - z_4 y_3),$$

which is nonzero and has order exactly four.

For $n = 3$, put $z = z_2$, $w = z_3$, $t = y_3$. In the ordered basis $e_1^2, e_1 e_2, e_1 e_3, e_2^2, e_2 e_3, e_3^2$, the two factors are

$$(1, 2z, 2w, z^2, 2zw, w^2), \quad (0, 1, t, z, zt + w, wt).$$

Their fifteen wedge coordinates have the same linear span as

$$\begin{aligned} 1, z, w, t, z^2, zw, w^2, zt, wt, z^2 t, zwt, w^2 t, \\ z^2 w - z^3 t, \quad zw^2 - z^2 wt, \quad w^3 - zw^2 t. \end{aligned} \tag{42.5}$$

This is obtained by subtracting the two coordinates with a common quadratic term and rescaling by 2. Their lowest homogeneous terms are distinct monomials, all of degrees at most three, and some have degree three. Thus the third jet is injective on this coefficient space and the second is not. This proves $q_3 = 3$. The coordinate $e_1^2 \wedge e_1 e_2$ is 1 in every dimension, proving the order-zero assertion. Finally, homogeneity of degree h gives (42.3); an adapted basis is by definition a basis on each associated-graded piece, so its initial forms are independent. $\square$

Proof of Theorem 42.1. The linear substitution $T \mapsto aI_n + \tau B$ is invertible for $\tau \neq 0$. Divide the adapted coefficient c_j by its exact power τ^{d_j} after substitution. There are no negative powers: all its terms have B -degree at least d_j . The resulting columns span a rank- r subbundle $J_\tau \subset \mathrm{Sym}^h E \otimes \mathbb{C}[\tau]$. Independence at zero follows from the lemma, and away from zero from the invertible substitution. The ideal of its maximal minors in $\mathbb{C}[\tau]$ has no common closed zero, and is therefore the unit ideal. On each maximal-minor open a corresponding square submatrix is invertible and splits the image. These opens cover $\mathrm{Spec} \mathbb{C}[\tau]$, proving the subbundle assertion over the whole line, including all scheme points. Set

$$K_\tau = \det(aI_n + \tau B)^{n-1} J_\tau.$$

Multiplication by the nonzero homogeneous polynomial in every fibre again has constant rank. Consequently K_τ is a subbundle, and

$$\mathcal{A}_\tau = \mathrm{Sym} E \otimes \mathbb{C}[\tau] / (K_\tau, \mathfrak{m}^{D+1})$$

is finite locally free of rank $\binom{e+D}{D} - r$. For $\tau \neq 0$, its relation space is a linear transform of K_{R_*} .

At zero the marked factor becomes a^g , while $J_0 = \langle a^{h-d_j} p_j(B) \rangle$. It contains a nonzero multiple of a^h , so its gcd is a power of a . It also contains a polynomial with exact a -adic order $h - q_n$, and every column is divisible by that power. Therefore $\mathrm{gcd}(J_0) = a^{k_n}$, proving (42.1). Notice that $0 < k_n < g$ for every $n \geq 3$.

The closed-flag specialization of Theorem G.1 and its failure-algebra lift imply $K_{R_*} \in Z_R$. A closed invariant subset containing a point contains that point's orbit closure. Hence the just-constructed $K_\infty \in Z_{R_*}$ belongs to every Z_R . This proves the assertion about the two-stage

specialization without replacing it by an unproved assertion about one fixed one-parameter subgroup acting on every initial pencil.

The sole degree- g divisor of a^{g+k_n} is a^g . Its overlap with the residual gcd is a^{k_n} . Lemma 12.3 gives (42.2); the fibre is supported at one point and has positive tangent space. The normalization criterion then proves singularity. With $n = 3$, $e = 9$ and $k_n = 2$, the dimension is $\binom{10}{2} - 1 = 44$. $\square$

42.1. The complete normalization fibre at the limiting algebra. The last theorem is not restricted to the tangent space of its fibre. The full local algebra can be given uniformly.

Proposition 42.3 (Pure-power divisor fibre). *Let $E = \mathbb{C}a \oplus W$, $g \geq k \geq 1$, and $K = a^{g+k}L$ with $\gcd(L) = 1$. Introduce coefficient variables for $q_i \in \text{Sym}^i W$, $1 \leq i \leq k$, and define forms c_j recursively by*

$$c_0 = 1, \quad c_j = - \sum_{i=1}^k q_i c_{j-i}, \quad c_j = 0 \ (j < 0). \quad (42.6)$$

The entire fibre of ν_g over K , supported at its unique point, is the Artin algebra

$$\mathbb{C}[[\text{coefficients of } q_1, \dots, q_k]] / (\text{coefficients of } c_{g+1}, \dots, c_{g+k}). \quad (42.7)$$

Its tangent dimension is $s_k(E) - 1$.

Proof. Choose $p \in L$ not divisible by a . Over any local Artin base, a divisor f near a^g is monic in a , and the quotient by f is flat over the base. The reductions of f, p are coprime. The regular-sequence argument used in (40.3) makes p a nonzerodivisor modulo f . Thus $f \mid a^{g+k}p$ is equivalent to $f \mid a^{g+k}$, functorially on these bases. Once this condition holds, it holds for every member of K .

Polynomial division gives a unique monic residual factor $H = a^k + q_1 a^{k-1} + \dots + q_k$. The equation $fH = a^{g+k}$ determines the coefficients of f successively as $c_1, \dots, c_g$ in (42.6). The remaining k product coefficients vanish if and only if $c_{g+1}, \dots, c_{g+k}$ vanish. This follows successively by comparing the product with the reciprocal series $(1 + \sum q_i a^{-i})^{-1}$. Hence (42.7) represents the full infinitesimal fibre, without reduction. Finiteness of ν_g makes it Artin; there is no other geometric divisor, so completion loses no part of the fibre. Since $g \geq k$, these equations have no linear terms in the coefficient variables. The number of coefficient variables is $\sum_{i=1}^k \dim \text{Sym}^i W = \dim \text{Sym}^k E - 1$, as asserted. $\square$

Example 42.4 (The nine-generator limit). For $n = 3$, the construction can be written with four of the nine linear coordinates, denoted a, u, v, w . The other five coordinates remain generators of the algebra. Up to nonzero scalar changes of the relation basis, $K_\infty = a^8 L_3$, where

$$L_3 = \langle a^3, a^2u, a^2v, a^2w, au^2, auv, av^2, auw, avw, \\ u^2v, u^2w, uv^2, uvw, v^3, v^2w \rangle. \quad (42.8)$$

Here one may take $u = B_{21}$, $v = B_{31}$, $w = B_{32}$; the monomials are the lowest terms in (42.5). The relation space has dimension 15 and exact gcd a^8 . The selected divisor has degree $g = 6$, and its fibre has 44 coefficient variables: eight coefficients of a linear form Q_1 and thirty-six of a quadratic form Q_2 in W . Its defining ideal consists of all coefficients of the two forms

$$c_7 = -Q_1^7 + 6Q_1^5Q_2 - 10Q_1^3Q_2^2 + 4Q_1Q_2^3, \\ c_8 = Q_1^8 - 7Q_1^6Q_2 + 15Q_1^4Q_2^2 - 10Q_1^2Q_2^3 + Q_2^4.$$

This is a full presentation of the normalization fibre. Its embedding dimension 44 is not being identified with its length, nor is this fibre algebra being identified with the completed local ring of Y_g .

Remark 42.5 (Relation to pencil invariants). The action defining Z_R is the intrinsic $\mathrm{GL}(E)$ action on first relations, not merely the congruence action on V . The flag pencil is singular: every member has rank at most two. It is therefore outside the regular-pencil open set treated by [4, Theorem 1.1 and §5], where Segre symbols organize classification and degenerations. The theorem above does not purport to classify that poset or all orbits in Z_R . Its conclusion is instead a uniform common boundary and an exact ramification algebra for genuine finite failure algebras. The increase from g to $g + k_n$ in the intrinsic gcd is precisely what is invisible in a fixed-determinant pencil specialization.

APPENDIX A. DETAILED RECONSTRUCTION STATEMENTS AND COMPARISONS

A degeneracy locus usually forgets the map that defines it. We study an inverse problem in which neither the ambient embedding nor the normal directions are retained: can the abstract nonreduced failure scheme recover a quadratic pencil? The answer is affirmative for every complex pencil, including singular pencils. The first relation, rather than its reduced determinant support, carries this information. The inverse is also global: a thickening over a projective reduction recovers an entire varying pencil subbundle and its actual source bundle, up to one constant left transformation. No fibrewise choice of coordinates or projective line-bundle ambiguity is left. A single unmarked local algebra suffices for the constant-pencil problem; a Schubert line gives another geometric realization.

Let V be a complex vector space of dimension $n \geq 3$, and put

$$W = \mathrm{Sym}^2 V, \quad N = \binom{n+1}{2}, \quad R \in \mathrm{Gr}(2, W), \quad S_R = W/R, \quad p = N - 2.$$

The cube-zero algebra $A_R = \mathbb{C} \oplus V \oplus S_R$ has multiplication $\mathrm{Sym}^2 V \rightarrow S_R$, with $(V \oplus S_R)S_R = 0$. On $X_R = \mathrm{Gr}(n, V \oplus S_R)$, let $\mathcal{K}$ denote the tautological bundle. For any $m \geq 2$, set

$$\widehat{D}_R = V(\mathrm{Fitt}_0 \mathrm{coker}[\mathrm{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \rightarrow A_R \otimes \mathcal{O}]). \quad (\mathrm{A.1})$$

This is the maximal-minor scheme of multiplication, not merely its support. It is independent of $m \geq 2$. Its reduction is the Schubert divisor where $\mathcal{K} \rightarrow V$ drops rank. Write $B_R = \mathrm{Gr}(n, S_R)$, let $\mathfrak{b}_R$ be its ideal in $\widehat{D}_R$, and define

$$Z_R^{[k]} = V(\mathfrak{b}_R^{k+1}) \subset \widehat{D}_R, \quad d = n + 2p = n^2 + 2n - 4.$$

An isomorphism of these neighbourhoods means an abstract complex scheme isomorphism; it is supplied with no ambient or tensor marking.

Theorem A.1 (Sharp intrinsic reconstruction). *For all quadratic pencils $R, R' \subset \mathrm{Sym}^2 V$,*

$$Z_R^{[d]} \simeq Z_{R'}^{[d]} \iff R' = gR \text{ for some } g \in \mathrm{PGL}(V).$$

For every $0 \leq k < d$, the isomorphism class of $Z_R^{[k]}$ is independent of R . Thus d is the smallest uniform reconstruction order. The same statements hold after restriction, using the graph projection, to any Schubert line in B_R . The resulting abstract curve-supported neighbourhood is independent of the choice of that line.

The Grassmannian statement is Theorem 7.1. The curve-supported statement is Corollary C.2. Its more general form, Theorem C.1, applies over any smooth connected projective curve with a rank- n bundle having nontrivial projectivization. For example, $\mathcal{O}_C(-P) \oplus \mathcal{O}_C^{n-1}$ works on every such pointed curve. The curve is intrinsic as the reduction; a chosen embedding of the curve is not part of the isomorphism data.

A.1. A moving family, not only its individual fibres. The same finite object also reconstructs a pencil that varies over its projective reduction. Let B be a smooth connected projective complex variety, let $\mathcal{U}$ have rank n , and let $\mathcal{R}$ be a rank-two subbundle of $\mathrm{Sym}^2 V \otimes \mathcal{O}_B$ with locally free quotient. Use the local ideal $(\det T)I_p(\gamma \mathrm{Sym}^2 T)$ in $\mathrm{Hom}(\mathcal{U}, V)$ and truncate at order d . Theorem 8.2 states that two resulting abstract finite schemes are isomorphic exactly when their data are related by an isomorphism of reduced bases, one constant linear left map, and an actual isomorphism of the source bundles, carrying one pencil subbundle to the other. The projection and both tensor factors are forgotten in the input. The varying coefficient line is recovered as a subbundle, not by pointwise choices of projective coordinates.

This strengthens the constant-pencil curve theorem in two directions: the reduced base and source bundle may vary in the comparison, and the pencil need not be constant on that base. Example D.6 has a nonconstant unordered cross-ratio and six spectral collision points, all retained by the same finite scheme. Theorem D.4 then identifies the remaining freedom in an isomorphism. Its quotient is the group of the recovered linear data; its kernel acts trivially on the associated graded scheme and has a terminating commutator filtration. Explicit nonlinear shears show why that kernel cannot be omitted. These statements are consequences of the unmarked inverse, not additional claims of novelty for the classical Pluecker map or for spectral cross-ratios.

A.2. A single unmarked local algebra suffices. The strongest form of the inverse needs no positive-dimensional support. Choose an n -dimensional source space U , write $T = (t_{ij})$ for its universal map to V , and let $\mathfrak{m} = (t_{ij})$. Define

$$\mathfrak{A}_R^{[d]} = \mathbb{C}[t_{ij}] / ((\det T)I_p(\gamma_R \mathrm{Sym}^2 T) + \mathfrak{m}^{d+1}).$$

Theorem A.2 (Local algebraic inverse). *For every pair of complex quadratic pencils, an ungraded algebra isomorphism $\mathfrak{A}_R^{[d]} \simeq \mathfrak{A}_{R'}^{[d]}$ exists if and only if $R' = gR$ for some $g \in \mathrm{PGL}(V)$. The order $d = n^2 + 2n - 4$ is the smallest uniform order already for these single-point objects.*

The full statement and proof are Theorem 8.3. Proposition 2.1 separates the general truncation argument from the pencil-specific orbit rigidity: nonlinear coordinate changes form an explicitly described unipotent kernel, whereas the first relation space carries the reconstruction data. The extra local ingredient is that the residual coefficient module has factor-support ranks $(1, \binom{N}{2})$. Transposition reverses these unequal ranks and therefore cannot carry one pencil ideal to another. This orients the unordered matrix factors without a nontrivial projective bundle. Theorem 8.2 consequently removes that restriction from the moving-family theorem as well. This is not a claim that the reduced determinant cone alone remembers the pencil: the nonreduced first relation kernel and its residual coefficients are indispensable. Proposition 8.4 also identifies the entire tangent-identity kernel, including its affine-space parametrization and nonadditive group law.

A.3. Conductors, collision singularities, and actual failure orbits. Normalization alone does not describe the gluing of its branches. Theorems 40.1 and 40.3 determine two successive local models. On the coprime two-divisor stratum the completed image is

$$\mathbb{C}[[z, x_1, \dots, x_c, y_1, \dots, y_c]] / (x_i y_j).$$

Its conductor is (x, y) , its intrinsic singular ideal is $(x, y)^c$, and its depth is $\dim(z) + 1$. When two common binary roots collide, the model is instead

$$\mathbb{C}[[u, \Delta]] \oplus (u_1, \dots, u_m)t \subset \mathbb{C}[[u, t]], \quad \Delta = t^2,$$

up to a specified smooth factor. Its conductor has a ramified double cover in the normalization. These formulas determine the Cohen–Macaulay cases and distinguish the singular scheme from its reduced support. Proposition 41.1 organizes the reduced intersections of different divisor-degree images by the common part of two chosen divisors.

The boundary also has a direct role in the pencil inverse. Theorem 42.1 constructs a common point in every intrinsic first-relation orbit closure of a pencil failure algebra. A scalar-matrix contraction of the flag-pencil relation has gcd a^{D-q_n} , where $q_3 = 3$ and $q_n = 4$ for $n \geq 4$. The selected degree- $n(n-1)$ divisor is unique but ramified; its normalization fibre has tangent dimension $\binom{n^2+3n-5-q_n}{3n-4-q_n} - 1$. Proposition 42.3 gives its entire Artin algebra by reciprocal-polynomial equations. This specializes *actual* failure algebras through flat families; it does not merely place an unrelated common-factor example beside the inverse theorem. The exact order of the flag coefficient filtration, proved in Lemma 42.2, is the link between the pencil representation and the divisor collision.

A.4. Factorization loci and the geometric comparison. The normalization mechanism has classical precedents. Chipalkatti's coincident-root locus parametrizes one binary form with a prescribed root partition. His Proposition 5.1, Theorem 5.4, and Corollary 5.8 distinguish multiple normalization points from failure of tangent injectivity and describe the singular support [19]. Kurmann gives local incidence equations, proves their scheme-theoretic equality in Theorem 1.5, and expresses smoothness through splittings of partitions in Proposition 2.1 [20]. In particular, uniqueness plus tangent injectivity is not a new general principle of the present paper.

Here the datum is an r -plane of forms in e variables, not one binary form; the selected divisor has prescribed degree, not a root partition. The residual datum is a Grassmannian subspace, and for $e > 2$ divisor degrees need not fill an interval. These changes produce high-codimension conductor strata and non-Cohen–Macaulay branch gluings. Theorems 40.1 and 40.3 compute the image rings, not merely the incidence graph or singular support. Ferrand's conductor-square formalism [21] is the standard algebraic framework; the particular conductor ideals, singular Fitting ideals, and depths are calculated here for these Grassmannian strata.

There is a second, closer binary comparison. Hu–Lin–Shao [23, §2.2, Corollary 3.6, and Theorem 1.1] study tuples of binary forms, define common-factor strata using resultant homomorphisms, and resolve the map-space boundary by successive blow-ups. On the open set of linearly independent r -tuples, passing to their span is the frame quotient onto $\mathrm{Gr}(r, S_D)$; the common-divisor condition is invariant under this quotient. Thus our binary loci genuinely meet that theory rather than forming an unrelated example. We do not identify its resultant scheme structure with our image structure without a scheme comparison, nor claim a new general resolution. Our multivariate local rings and the pencil-orbit boundary are the specific additional conclusions.

Rigidification is used as infrastructure, not as an independent headline innovation. The appropriate primary general result is Abramovich–Olsson–Vistoli, Appendix A, Theorem A.1 [22], for compatible flat finitely presented normal inertia subgroups, without a centrality assumption. Our argument identifies the particular substitution subgroup and twisted right-factor subgroup and retains their actual group laws. The boundary statement is on the maximal-lower-Hilbert-function first-relation stratum throughout.

The comparison with regular-pencil Segre strata is made in Remark 42.5. The common closed flag orbit is not presented as a new classification of congruence orbits. The new pencil-side conclusion is its explicit ramified boundary under intrinsic cotangent changes, together with the complete normalization-fibre algebra. The separate documentary status of Ballico's 1993 paper, discussed below, remains unchanged: absent its full theorem text, no nonanticipation conclusion is drawn.

A.5. The intrinsic mechanism. In the graph neighbourhood of B_R , with tautological bundle $\mathcal{U}$ and graph map $T : \mathcal{U} \rightarrow V$, the ideal is

$$(\det T)I_p(\gamma_R \mathrm{Sym}^2 T).$$

Its generators have degree d . This degree calculation alone only explains why lower orders are constant. The inverse requires three further intrinsic operations. Multiplication in the finite scheme recovers the first relation kernel and hence the whole homogeneous cone. The rank-one Fano scheme of its reduced determinant cone then recovers the two tensor rulings. On the socle Grassmannian one may distinguish them by projective triviality, already on a Schubert line. In general the unequal coefficient-support ranks orient them without that restriction. Properness forces the induced map to $\mathrm{PGL}(V)$ to be constant. After choosing its linear lift, the actual normal-bundle map lies in the commutant of $\mathrm{End}(V)$, which recovers the right bundle without a line twist (Lemma 4.2). Finally, cancellation of the intrinsic determinant ideal recovers the coefficient module. For pencils its exterior representation is irreducible, and exterior duality identifies its single coefficient line with the Pluecker line of R .

Theorem 5.1 separates this geometric mechanism from the particular multiplication algebra. The curve theorem shows why the mechanism is not tied to the dimension of the original parameter Grassmannian. The geometric orientation used here has a local replacement: coefficient-support asymmetry. Theorem 8.3 uses it to reconstruct the pencil from one unmarked Artin local algebra.

Theorem 4.3 gives the broader principle. It reconstructs simultaneously varying subspaces in several Schur modules from the unmarked first-relation thickening, provided one coefficient support is nonzero and proper. The coefficient spaces may have arbitrary prescribed ranks; they need not be lines coming from quadratic pencils. Its proof isolates the actual factor-descent step from classical linear-preserver theory. Example 4.4 shows why the global descent is not equivalent to fibrewise reconstruction: two Zariski locally trivial families have the same Artin fibre and isomorphic nilradical graded vector bundles in every degree, but their abstract total schemes differ. The inverse detects the inequivalent degree-three coefficient maps. This is the mechanism behind the moving-pencil theorem, rather than an assertion of novelty for canonical grading or for the numerical value of the first relation degree.

A.6. An exact image criterion and a pencil-native global consequence. The coefficient inverse admits an intrinsic converse. Theorem B.1 starts with an unmarked first relation whose homogeneous radical is a determinant cone. After recovering its unordered rulings, invariance under one full factor group is necessary and sufficient for the residual relation to come from coefficient subspaces in the Cauchy modules. Both orderings satisfy this test precisely when every coefficient support is zero or full. Thus the strict-support condition is exactly the criterion for a unique admissible orientation in the recognized class. Theorem B.2 computes its entire affine stabilizer, including the transpose component and the universal nonlinear automorphism kernel. Corollary B.4 expresses the inverse as an equivalence of geometric orbit sets, with those ambiguities specified rather than suppressed.

A global consequence lies inside the quadratic-pencil problem itself. Fix a plane $H = \langle x, y \rangle \subset V$. For a degree- m map $f = [a : b] : \mathbb{P}^1 \rightarrow \mathbb{P}(H)$, take the pencil subbundle

$$\mathcal{R}_f = (ax + by)H.$$

Every fibre is equivalent to $\langle x^2, xy \rangle$. Nevertheless, Theorem I.1 proves

$$Y_f \simeq Y_{f'} \iff f' = \alpha \circ f \circ \sigma, \quad \alpha, \sigma \in \mathrm{PGL}_2(\mathbb{C}),$$

for their unmarked order- d failure thickenings with trivial source bundle. Their fibre algebras and every nilradical graded vector bundle depend only on n, m , not on f . Their abstract pencil, quotient and first-relation bundles are fixed as well. Thus the entire covering equivalence problem, not only one chosen pair, is retained in the global multiplication and is invisible to those additive data. The maps z^3 and $z^3 + z$ give an explicit separating pair of quadratic pencils (Corollary I.2).

Neither complete reducibility of Cauchy modules, linear preservers of a Segre cone, nor division by the common factor of a pencil is new in isolation. Their synthesis here has a precise output:

recognition of the image and its exact orientation ambiguity after all matrix markings are forgotten, followed by recovery of a whole covering map from a locally isotrivial pencil thickening. A projective factor map alone would not recover the actual source bundle; the intrinsic normal map and its commutant do so. A pointwise pencil inverse would not recover a covering map; one constant left transformation over the entire proper base does so. These distinctions specify the geometric content beyond the classical ingredients. The recognized coefficient class is not the class of all determinant-radical relations. Its geometric orbit statement alone does not describe nonreduced deformation groupoids; the relative comparison below uses a further scheme-theoretic inverse.

A.7. Intrinsic strata, infinitesimal parameters, and pencil moduli. The relative inverse begins with a geometric condition on the first relation, not invariance under a factor group. For a space K of homogeneous forms, the divisor-incidence functor defines a canonical scheme-theoretic image in the relation Grassmannian. Its exact-gcd open is the locally closed stratum of Lemma 11.1, now identified by its incidence universal property in Corollary 12.5. Multiplication by the universal common-divisor line is an isomorphism onto this open. Consequently the common factor is recovered even over a nonreduced parameter base. The condition is scheme-theoretic: merely asking all geometric fibres to have the specified common factor is insufficient. When that divisor is a power of a determinant, the remaining parameter is an arbitrary gcd-free residual subspace, modulo the full determinant-preserver group (Theorem 11.3). This is a smooth quotient stack, and neither of the two factor symmetries is required.

For pencils the exact common divisor is δ^{n-1} , and the primitive residual relation has degree $3n - 4$ (Lemma E.1). The Cauchy projection and the Pluecker equations recognize its image as a locally closed scheme, not just a set of complex points. The resulting relative theorem is

$$\mathcal{F}_{\text{pen}}^{\text{eff}} \simeq [\text{Gr}(2, \text{Sym}^2 V) / \text{PGL}(V)]. \quad (\text{A.2})$$

Here the superscript records two explicit operations on morphisms: first remove tangent-identity nonlinear substitutions, then remove the ineffective right matrix factor, and fppf stackify. Theorem E.2 proves this equivalence over arbitrary complex parameter schemes, retaining all singular pencils. It does not identify the unrigidified algebra stack with the pencil stack. Its tangent complex is $[\text{pgl}(V) \rightarrow \text{Hom}(R, \text{Sym}^2 V/R)]$, so the effective stack is smooth of dimension $n - 3$, with precisely the pencil stabilizer jumps. Rank-incidence and spectral Fitting strata retain their scheme structures and deformation equations by Corollary E.3. This is an exact-image and rigidification comparison; the displayed rank and spectral consequences are its functorial restrictions, not an independent classification of those loci. The two quotient operations have the universal properties of Theorem F.1 and Corollary F.2.

The ambient determinant-divisor stratum is substantially larger. Theorem H.1 computes its normal space to the pencil stratum in three parts: non-Pluecker lines, non-scalar right-factor directions, and other Cauchy summands. Theorem H.2 constructs through every pencil a projective Grassmannian of relations with the same determinant radical, containing a projective line whose other points admit neither factor symmetry. Thus the family inverse describes how the coefficient and pencil loci sit in, and can be left within, a larger intrinsic stratum. It does not falsely infer deformation stability from fibrewise recognition.

A.8. The normalization of a natural common-divisor boundary. There is a different geometric question beyond the pencil image: what happens when a divisor of a first relation is no longer uniquely recoverable in a family? Let E have dimension $e \geq 2$, set $S_j = \text{Sym}^j E$ and $s_j = \dim S_j$, and fix $g, h \geq 1$, $D = g + h$, and $2 \leq r \leq s_h$. Let $Y_g \subset \text{Gr}(r, S_D)$ be the scheme-theoretic image of the incidence of a line of degree- g divisors of a rank- r relation bundle.

This definition uses the vanishing of the universal quotient map; it does not first recover a pencil, impose a factor symmetry, or choose a reduced structure on a set.

Theorem A.3 (The divisorial boundary). *The multiplication morphism*

$$\mathbb{P}(S_g) \times \mathrm{Gr}(r, S_h) \longrightarrow Y_g, \quad ([f], J) \longmapsto fJ$$

is finite and is the normalization. The locus Y_g is integral, of dimension $s_g - 1 + r(s_h - r)$. At a geometric relation K it is normal exactly when $F_K = \gcd(K)$ has one degree- g divisor $[f]$ and $\gcd(f, F_K/f) = 1$. At the preimage $([f], K/f)$, if this last gcd has degree k , the fibre tangent dimension is $s_k - 1$.

The theorem is proved in Section 12. Lemma 12.3 supplies equations for the entire normalization fibre, not only its tangent space. For example, $K = \langle x^2y, x^2z \rangle$ has a unique linear divisor, but the normalization fibre is $\mathrm{Spec} \mathbb{C}[a, b]/(a, b)^2$, of length three. A different relation has two reduced divisor lifts; yet another has a gcd-degree jump at a normal point. The exact criterion distinguishes these behaviours. Corollary 12.8 carries this intrinsic boundary to the first-relation algebra stack. It is not a classification of all finite-flat smoothings or of all linear orbits in that boundary.

A direct consequence on the pencil side is proved separately in Section G. Every quadratic pencil admits an explicit congruence degeneration to $\langle x_1^2, x_1x_2 \rangle$; its orbit is the closed flag variety $\mathrm{Fl}(1, 2; V)$, and it is the unique closed orbit in the full pencil Grassmannian. The construction is uniform for regular and singular pencils and lifts to a finite-flat family of their failure algebras with exact common factor δ^{n-1} . This proves a particular universal orbit-closure adjacency by equations, rather than by copying a prescribed invariant locus through the effective-stack equivalence. Its elementary highest-weight mechanism is distinct from the normalization theorem, which deals with a larger boundary of arbitrary first relations.

A.9. Framed families and spectral consequences. Once tensor coordinates are retained, the parameter map is simpler. The first relation spaces give a closed immersion of $\mathrm{Gr}(2, W)$ into a Grassmannian, by Theorem 7.3. Its proof factors through the classical Pluecker embedding, tensoring a line with a fixed space, and a fixed linear inclusion. This is the framed encoding consequence of the coefficient calculation, not a substitute for the unmarked inverse. Theorem K.1 glues the actual universal order- d neighbourhood. It is finite locally free over the relative socle of rank $\binom{n^2+d}{d} - \binom{N}{2}$, and projective and flat over the pencil parameter scheme. It commutes with arbitrary complex base change with that socle retained. Over a nonreduced base, the specified relative socle ideal need not be the absolute nilradical. These marked statements alone do not identify unmarked moduli stacks or deformation groupoids. The common-divisor descent and ineffective-inertia analysis in Theorem E.2 supply that separate step.

For regular pencils, the spectral sheaf and its Fitting incidence schemes are read from the reconstructed pencil. We retain these consequences because they exhibit information genuinely lost by coarser invariants: fixed discriminants and fixed reduced rank loci can conceal different elementary divisors. In Theorem L.2, every classical dominance specialization is transported to a flat family of the finite neighbourhoods, with every strict transition detected at order d . The spectral orbit classification itself is classical, not a second independent Torelli theorem.

A.10. Prior work and the publication object. The Weierstrass–Segre classification is used in its root-labelled form; compare [4, Theorem 1.1 and Corollary 2.1]. The orthogonal nilpotent input is isolated in Lemma L.1, with precise references to [5, Proposition 1 and Theorem 1], an independent dimension calculation, and an explicit distinction between full orthogonal orbits and their special-orthogonal components.

Ballico’s relative multiplication construction [2, Theorem 0.2; §1, Remark 1.2 and equation (6)] is a substantive antecedent for studying quadratic-normality failure on finite linear sections of an

embedded surface. Here the defining algebra varies, and the inverse starts from an unmarked infinitesimal failure scheme. The opening page of the distinct earlier paper [1, p. 5] concerns higher-order embedding properties and restriction maps to finite subschemes. Its remaining theorem and proof text has not been inspected; that first page does not establish the absence of an inverse statement. No anticipation or nonanticipation conclusion relative to that article is asserted.

The Artin-local aspect is compared at theorem level with [7, 8] in Section 2, including the distinction between canonical grading of a nonhomogeneous local algebra and orbit rigidity of our already homogeneous first relation. The linear-preserver input is classical [9]; its precise hypotheses and a proof in the required case are included in Lemma 4.1. The algebraic-group quotient step is supplied by the homomorphism theorem [11, Theorem 5.39 and Remark 5.42]; its use in the exact stabilizer calculation distinguishes a scheme-theoretic quotient from a bare bijection of complex points.

The present article consists of intrinsic reconstruction, the canonical common-divisor boundary, relative rigidification, explicit pencil specialization, and spectral readout. The reciprocal-likelihood theorem and projective critical schemes are preserved in a separate applications manuscript. Its finite-flat critical divisor requires supplied split semisimple data and simple disjoint projective poles; its totally real cover requires a supplied real definite realization and separated data. None of those hypotheses enters the all-pencil theorem here. Independent web, exterior-operator, K3, polarized, and boundary results are preserved in an explicitly non-submitted research archive. They are not inputs to any proof in this article and are not part of its significance claim. Every proof required here is included here.

APPENDIX B. RECOGNITION AND THE EXACT AMBIGUITY OF COEFFICIENT SYSTEMS

The support inequality in the preceding section is not merely a sufficient test in the class under consideration. We characterize that class intrinsically and determine exactly when its orientation is unique. We then compute the full linear stabilizer, including the cases in which an orientation is lost. These are statements about unmarked first relations, not about a supplied matrix presentation.

Throughout this section $n \geq 2$, $q \geq 1$, and $D = n + q$. Let E have dimension n^2 , put $S = \text{Sym } E$, and let $0 \neq K \subset S_D$. The affine cone is in E^* . Suppose that the radical of the homogeneous ideal (K) is a reduced hypersurface linearly equivalent to the determinant hypersurface in square n -by- n matrices. We call this the *determinant-radical condition*. The ideal (K) is taken in the full symmetric algebra, not in its Artin truncation. Its reduced equation gives a line $\mathbb{C}\delta \subset S_n$, and unique factorization gives the intrinsic residual subspace

$$J = K/\delta \subset S_q. \quad (\text{B.1})$$

Changing the generator δ does not change this subspace. The reduced singular loci of the recovered cone determine its unordered tensor rulings. An ordering identifies $E = V^* \otimes U$, with $\dim U = \dim V = n$, up to separate factor changes. The subgroup induced by $\text{GL}(U)$ on the second factor is independent of those changes. An ordering is called *admissible* when J is invariant under this full subgroup. Admissibility is therefore an intrinsic condition on an ordered ruling, and does not ask for a chosen basis of either factor.

Theorem B.1 (Intrinsic recognition and orientation dichotomy). *Under the determinant-radical condition, an ordering is admissible if and only if its Cauchy decomposition has*

$$J = \bigoplus_{\lambda \vdash q, \ell(\lambda) \leq n} L_\lambda \otimes S_\lambda U, \quad L_\lambda \subset S_\lambda V^*. \quad (\text{B.2})$$

The coefficient spaces are unique and are recovered by projection and contraction. Suppose at least one ordering is admissible and put $r_\lambda = \dim L_\lambda$ and $m_\lambda = \dim \mathbb{S}_\lambda \mathbb{C}^n$. Then exactly one of the following alternatives occurs.

- (i) Some $0 < r_\lambda < m_\lambda$. Precisely one of the two orderings is admissible, and every linear isomorphism between such relation spaces preserves that ordering.
- (ii) Every r_λ is either zero or m_λ . Both orderings are admissible. The residual space is a sum of full Cauchy summands, and transposition preserves K after identifications of the two matrix factors.

Conversely, every nonzero system in (B.2) satisfies the determinant-radical condition after multiplication by δ . Thus the theorem characterizes the image of the coefficient construction, without assuming a coefficient presentation as part of the input.

Proof. The determinant polynomial is irreducible for $n \geq 2$. Every element of K belongs to the radical (δ) , so $K \subset \delta S_q$ and the cancellation in (B.1) is legitimate. The intrinsic recovery of the unordered factors follows from Lemma 4.1 and the rank-one rulings. Changing either factor frame conjugates its full general linear group into itself; scalar changes act as scalars on S_q . Consequently the definition of admissibility is independent of all these choices.

For an ordering, the Cauchy formula is

$$S_q = \bigoplus_{\lambda \vdash q, \ell(\lambda) \leq n} \mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U.$$

As representations of $\mathrm{GL}(U)$, the nonzero $\mathbb{S}_\lambda U$ are irreducible and pairwise nonisomorphic. Finite-dimensional rational representations of this group over $\mathbb{C}$ are completely reducible. Hence an invariant subspace splits across its isotypic components. In the λ -component, evaluation identifies that subspace with

$$\mathrm{Hom}_{\mathrm{GL}(U)}(\mathbb{S}_\lambda U, J) \otimes \mathbb{S}_\lambda U.$$

Schur's lemma embeds the displayed multiplicity space uniquely in $\mathbb{S}_\lambda V^*$; this is L_λ . This proves necessity in (B.2), including the absence of unrecorded cross-summand graphs. Sufficiency follows since the second factor is full. Projection onto a Cauchy summand followed by contraction with the dual of its second factor recovers L_λ .

If the reverse ordering is also admissible, J is invariant under both full factor groups. Their product acts irreducibly on each Cauchy summand, and these product representations are pairwise nonisomorphic. Complete reducibility now makes J a sum of whole summands. Equivalently, all r_λ are zero or full. Conversely a sum of whole summands is invariant under both groups and under the factor exchange. This proves the dichotomy. One may also see uniqueness directly from the contraction ranks (r_λ, m_λ) : a factor exchange reverses them, which is impossible at a nonzero proper support. Every linear isomorphism of relation spaces preserves the determinant cone; Lemma 4.1 leaves just the two factor alternatives. It must therefore preserve the unique admissible ordering in case (i).

Finally choose a nonzero L_λ . At an invertible matrix $T : U \rightarrow V$, the map $\mathbb{S}_\lambda T$ is invertible. A nonzero covector in L_λ consequently has a nonzero matrix coefficient at T . Thus the residual coefficients do not vanish simultaneously at any invertible matrix. All elements of δJ vanish on the determinant hypersurface. Their common zero set is exactly that hypersurface; the Nullstellensatz and its reduced irreducible equation give $\mathrm{rad}(\delta J) = (\delta)$, as asserted. $\square$

The complete reducibility and Cauchy statements here are classical representation theory; see [3]. Their role is to supply an intrinsic image criterion *after* the matrix factors have been recovered. For a relation space that fails the admissibility criterion, the theorem does not assert a one-sided coefficient description. In particular it is not a classification of arbitrary homogeneous ideals or arbitrary degeneracy schemes.

B.1. Stabilizers and the full transpose ambiguity. Use the normal action $T \mapsto gTh^{-1}$. Its action on polynomial functions is inverse pullback, so it gives an honest representation ρ on $E = V^* \otimes U$. The kernel of $\rho : \mathrm{GL}(V) \times \mathrm{GL}(U) \rightarrow \mathrm{GL}(E)$ is the diagonal scalar subgroup. For a recognized system define

$$G_L = \{g \in \mathrm{GL}(V) : (\mathbb{S}_\lambda g^{-1})^* L_\lambda = L_\lambda \text{ for every } \lambda\}, \quad H_K = \mathrm{Stab}_{\mathrm{GL}(E)}(K).$$

These are closed subgroup schemes, with the stabilizer scheme structures from their actions on Grassmannians.

Theorem B.2 (Exact coefficient stabilizer). *In case (i) of Theorem B.1,*

$$H_K \simeq (G_L \times \mathrm{GL}(U))/\mathbb{G}_m. \quad (\text{B.3})$$

In case (ii), after choosing bases for the two factors,

$$H_K \simeq ((\mathrm{GL}_n \times \mathrm{GL}_n)/\mathbb{G}_m) \rtimes C_2, \quad (\text{B.4})$$

where C_2 is generated by transposition. These are isomorphisms of affine algebraic groups over $\mathbb{C}$, and hence of their represented functors on commutative complex algebras.

For the ungraded algebra $A(K) = S/((K) + \mathfrak{m}^{D+1})$, they give the split exact sequence

$$1 \longrightarrow U_K \longrightarrow \mathrm{Aut}(A(K)) \longrightarrow H_K \longrightarrow 1. \quad (\text{B.5})$$

Here U_K is the universal higher-order substitution group from Proposition 2.2, of dimension $n^2(\mathrm{length} A(K) - 1 - n^2)$.

Proof. Every element of $H_K(\mathbb{C})$ preserves the radical of (K) , and therefore the determinant cone. In case (i), the unique admissible orientation excludes its transposed component. The remaining map is $\rho(g, h)$. Cancellation of the determinant line and projection onto each Cauchy summand show that it preserves J if and only if $g \in G_L$; the arbitrary right map acts on the full factor and introduces no additional condition. The kernel is the diagonal $\mathbb{G}_m$. In case (ii), every separate factor transformation and transposition preserves each active full summand. These exhaust the determinant preservers, so they exhaust $H_K(\mathbb{C})$. The transposition has order two and lies outside the preserving component for $n \geq 2$, giving the asserted semidirect product. Its action on the left-right quotient is conjugation by the matrix-transpose map; no particular formula for its action on chosen factor lifts is needed.

We justify the scheme statement as well. The homomorphism theorem for algebraic groups identifies the quotient by the kernel with the closed image [11, Theorem 5.39 and Remark 5.42]. Every finite-type group scheme over a characteristic-zero field is smooth [10, Lemma 39.8.2]. Thus the stabilizer and the image are reduced closed subgroup schemes. The equalities on complex points just proved imply equality of their defining radical ideals. This gives (B.3) and (B.4) as algebraic-group isomorphisms, not only as bijections on complex points. Finally Proposition 2.2 applies in degree D and gives (B.5) with the stated kernel, splitting and dimension. $\square$

For completeness, the omitted zero system has $K = 0$. Then $A(K)$ is the free truncated symmetric algebra, $H_K = \mathrm{GL}(E)$, and its linear automorphisms need not preserve any matrix factors. The determinant-radical condition deliberately excludes this third possibility. A full nonzero coefficient system and a zero system therefore lose information in different ways: the former retains unordered matrix geometry and has exactly a transpose ambiguity, while the latter retains no intrinsic matrix geometry at all.

B.2. The zero/full case over a projective reduction. The local transpose ambiguity also has an exact global form. Fix a nonempty set of active partitions and take the corresponding $\mathcal{L}_\lambda$ to be full, with all other coefficient bundles zero. Use the same construction as in (4.4), without its strict-support hypothesis.

Proposition B.3 (Global ambiguity for full coefficient systems). *Let Y_i be two such schemes, for the same n, q and active partitions, over smooth connected projective bases B_i with source bundles $\mathcal{U}_i$. They are isomorphic as abstract complex schemes if and only if there are an isomorphism $\sigma : B_1 \rightarrow B_2$ and an actual bundle isomorphism $\mathcal{U}_1 \simeq \sigma^* \mathcal{U}_2$. An isomorphism exchanging the two ruling families over a fixed σ exists precisely when, in addition, the common source bundle is projectively trivial. Equivalently,*

$$\mathcal{U}_1 \simeq \mathcal{M}^{\oplus n}, \quad \sigma^* \mathcal{U}_2 \simeq \mathcal{M}^{\oplus n}$$

for a line bundle $\mathcal{M}$ on B_1 . Thus even the full systems recover the actual source-bundle class; the extra ambiguity is an exchange, not an undetected line twist.

Proof. An abstract isomorphism recovers the reduction, the actual normal bundle map, the first relation, and the unordered determinant rulings by the proof of Theorem 4.3, which uses strict support only to orient the rulings. Over the connected base the choice to preserve or exchange the two components is constant. In the preserving case, projectivity makes the left projective map constant. A constant lift and Lemma 4.2 give an actual isomorphism of the source bundles.

In the exchanging case each source projective bundle is identified with the constant left projective bundle on the other side. It is therefore trivial. A vector bundle with a trivialized projectivization is a line twist of a constant vector space: locally lift the given projective trivialization to linear frames; on intersections their differences are scalar matrices, whose scalar cocycle is the required line bundle. After pullback by σ , write $\mathcal{U}_i^* = \mathcal{M}_i^* \otimes W_i$, with W_i constant of dimension n . The normal bundles become $\mathcal{M}_i^* \otimes (V_i \otimes W_i)$. Their actual isomorphism induces a morphism from B_1 to the projective-linear isomorphisms between two constant n^2 -spaces. This target is affine; the morphism is constant. Choose its constant linear lift. On removing it, the actual bundle map is scalar on each projective fibre. In local line frames its off-diagonal entries vanish and all diagonal entries agree. Reducedness makes these equalities hold as bundle maps. Their common entry is a nowhere vanishing section of $\mathcal{H}om(\mathcal{M}_1^*, \mathcal{M}_2^*)$, and hence $\mathcal{M}_1 \simeq \mathcal{M}_2$. This proves necessity of the common actual line twist, and thus of the source-bundle isomorphism.

Conversely any actual source-bundle isomorphism, together with a constant left identification, preserves all active full Cauchy summands, so it gives an isomorphism of the schemes. If the source bundles have the common line-twist presentation above, choose identifications of the constant factor spaces and transpose their tensor factors, leaving the line factor intact. This is a global actual normal map. It preserves the determinant line, every active full summand, and the truncation, and supplies an exchanging isomorphism. These constructions prove all the claims. $\square$

B.3. The intrinsic orbit space. Fix a rank vector $\mathbf{r}$ with at least one proper nonzero entry and form the product of Grassmannians

$$X_{\mathbf{r}} = \prod_{\lambda \vdash q, \ell(\lambda) \leq n} \text{Gr}(r_{\lambda}, \mathbb{S}_{\lambda} V^*).$$

Zero and full factors in this product are points. Let $\mathcal{Z}_{\mathbf{r}}$ denote the set of subspaces of $\text{Sym}^D \mathbb{C}^{n^2}$ satisfying the determinant-radical condition and the intrinsic admissibility test, with ranks $\mathbf{r}$ in the unique admissible orientation.

Corollary B.4 (Geometric orbit classification). *The coefficient construction induces bijections*

$$X_{\mathbf{r}}(\mathbb{C}) / \text{PGL}(V) \simeq \mathcal{Z}_{\mathbf{r}} / \text{GL}_{n^2}(\mathbb{C}) \simeq \{A(K) : K \in \mathcal{Z}_{\mathbf{r}}\} / \simeq. \quad (\text{B.6})$$

Their stabilizers, including the ineffective right factor and the nonlinear kernel of the last passage, are precisely those in Theorem B.2. For a nonempty zero/full rank pattern there is just one relation orbit for that pattern, with the stabilizer in (B.4).

Proof. The recognition theorem gives surjectivity of the first map. A linear isomorphism of two such relations is orientation-preserving; its left factor carries every L_λ to the corresponding coefficient space, and its right factor is unrestricted. Scalars act trivially on the Grassmannians. This proves injectivity and the first bijection. The first-relation orbit proposition gives the second. For a zero/full pattern each coefficient Grassmannian is a point, and the full-stabilizer theorem supplies the last claim. $\square$

Equation (B.6) is a statement about geometric orbit sets with the explicitly computed affine stabilizers. It does not identify nonreduced moduli stacks or assert that every infinitesimal deformation remains in this class. For smooth projective reduced bases, its uniquely oriented case combines with actual factor descent in Theorem 4.3. The following pencil-native application shows that this global step separates an entire covering problem that fibrewise classification cannot see.

APPENDIX C. RECONSTRUCTION ON A PROJECTIVE CURVE

The global part of the inverse argument has a more economical geometric realization. It uses a proper base to make the projective left frame constant, and a nontrivial right projective bundle to distinguish the rulings. Neither step requires the whole socle Grassmannian. We now replace that Grassmannian by a curve. In particular, a neighbourhood supported on one Schubert line already retains every pencil.

Let C be a smooth connected projective complex curve, and let $\mathcal{U}$ be a rank- n bundle on C , where $n \geq 3$. Assume that $\mathbb{P}_C(\mathcal{U}^*)$ is not projectively trivial. A sufficient numerical condition is

$$\deg \det \mathcal{U} \not\equiv 0 \pmod{n}. \quad (\text{C.1})$$

Indeed a projectively trivial rank- n bundle is a line-bundle twist of a trivial bundle, so its determinant degree is divisible by n . Fix V of dimension n , put $E = \text{Hom}(\mathcal{U}, V)$, and let $\mathfrak{a}$ be the zero-section ideal in $\text{Sym } E^*$. For a pencil $R \subset \text{Sym}^2 V$, the quotient $\gamma_R : \text{Sym}^2 V \rightarrow S_R$ defines the homogeneous ideal

$$\mathcal{I}_{R,C} = (\det T) I_p(\gamma_R \text{Sym}^2 T), \quad p = \binom{n+1}{2} - 2, \quad (\text{C.2})$$

where $T : \mathcal{U} \rightarrow V$ is the universal map on $\text{Tot } E$. The expression is independent of local frames: both determinant multiplication and the maximal-minor construction are bundle constructions. Define

$$Y_{R,C}^{[k]} = \text{Spec}_C(\text{Sym } E^* / (\mathcal{I}_{R,C} + \mathfrak{a}^{k+1})).$$

Only its abstract complex scheme is retained, not its projection to C , its embedding in $\text{Tot } E$, or either tensor factor.

Theorem C.1 (Curve-supported reconstruction). *For fixed $(C, \mathcal{U}, V)$ as above and $d = n^2 + 2n - 4$, every pair of quadratic pencils satisfies*

$$Y_{R,C}^{[d]} \simeq Y_{R',C}^{[d]} \iff R' = gR \text{ for some } g \in \text{PGL}(V). \quad (\text{C.3})$$

All lower-order neighbourhoods are independent of R . The order- d schemes are finite locally free over C of rank $\binom{n^2+d}{d} - \binom{N}{2}$, where $N = \binom{n+1}{2}$. No regularity condition is imposed on the pencil.

Proof. On any frame chart, the ideal in (C.2) is generated in degree d . Its degree- d piece K_R is a subbundle of rank $\binom{N}{2}$, by the fixed coefficient inclusion (7.8); its quotient is locally free. The quotient algebra has its full symmetric powers in degrees less than d , has $\text{Sym}^d E^* / K_R$ in degree d , and is zero above that degree. This proves the rank and the assertion about lower orders.

Its reduction is C , and its nilradical $\mathfrak{n}$ recovers $L = \mathfrak{n}/\mathfrak{n}^2$ and

$$K_R = \ker[\mathrm{Sym}^d L \longrightarrow \mathfrak{n}^d].$$

Thus the abstract finite-order scheme reconstructs the entire graded algebra $\mathrm{Sym} L/(K_R)$, by the same argument as Lemma 7.2. The ideal here means the homogeneous ideal generated by the recovered kernel, not its radical. No splitting of the reduction or supplied projection is used.

The reduction of this reconstructed cone is the determinant cone. Indeed its ideal has the determinant factor, and when T is invertible the quadratic quotient has full rank. Its rank-one Fano scheme has two ruling components, $\mathbb{P}(V) \times C$ and $\mathbb{P}_C(\mathcal{U}^*)$. The hypothesis distinguishes the first one intrinsically. An abstract isomorphism of neighbourhoods may induce a nonidentity automorphism of C ; pull back the second bundle along that automorphism. The oriented ruling map is nevertheless a morphism from C to PGL_n . It is constant, since C is connected projective and $H^0(C, \mathcal{O}_C) = \mathbb{C}$. Lemma 3.2 therefore supplies one constant projective left frame.

Cancel the intrinsic determinant ideal and apply the coefficient construction of Theorem 5.1. For pencils, Lemma 6.2 leaves just the single line $\mathbb{C}\beta_R$. Exterior duality (6.8) recovers R with respect to that same projective frame. This proves the forward implication in (C.3). Conversely a linear lift of g induces the bundle map on E and the quotient isomorphism on S_R , and carries all the displayed ideals and their truncations to the corresponding ideals for R' . This proves the converse. $\square$

Corollary C.2 (One Schubert line suffices). *Let $\ell \simeq \mathbb{P}^1$ be any Schubert line in $B_R = \mathrm{Gr}(n, S_R)$, and use the canonical graph projection $Z_R^{[d]} \rightarrow B_R$ to form its restriction $Z_{R,\ell}^{[d]}$. Its abstract isomorphism class is independent of the choice of ℓ , and*

$$Z_{R,\ell}^{[d]} \simeq Z_{R',\ell'}^{[d]} \iff R' = gR \text{ for some } g \in \mathrm{PGL}(V).$$

The smallest uniform reconstruction order for these curve-supported neighbourhoods is again d . Their reductions all have dimension one, instead of $n(p-n)$.

Proof. The tautological bundle on a Schubert line is $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}^{n-1}$. Pulling back the homogeneous ideal of the graph neighbourhood gives exactly (C.2) with that bundle. Its isomorphism type depends on this bundle and R , not on the chosen inclusion of the line in the socle. The determinant degree is -1 , so Theorem C.1 applies. Its lower-order schemes are all independent of the pencil, while $\langle x_1^2, x_1x_2 \rangle$ and $\langle x_1^2, x_2^2 \rangle$ are inequivalent. This proves the uniform minimality assertion. $\square$

This restriction is made in the construction; it is not a claim that an abstract neighbourhood canonically chooses a Schubert line. No line is supplied to an isomorphism in the corollary: the reduced curve is recovered as the reduction of its source. Nor is the corollary a statement about one unmarked Artinian normal slice. The projective curve and the nontrivial ruling remain essential to this argument. The same theorem applies, for example, to $\mathcal{U} = \mathcal{O}_C(-P) \oplus \mathcal{O}_C^{n-1}$ on every smooth projective connected curve with a point P . It isolates the properness and ruling obstruction used in reconstruction, independently of the original Grassmannian realization.

APPENDIX D. RECONSTRUCTION OF MOVING PENCILS AND THEIR ISOMORPHISMS

The inverse need not be applied separately to constant pencils. A single unmarked finite scheme can retain a pencil varying over its entire projective reduction. We prove this assertion, including the recovery of the normal bundle and the ambiguity in lifting the recovered data to an isomorphism. The coefficient lines in this section are subbundles; constancy of the pencil is not assumed.

D.1. The family and its first relation bundle. Fix $n \geq 3$, and write

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n + 2p = n^2 + 2n - 4, \quad M = \binom{N}{2}.$$

An admissible datum consists of a smooth connected projective complex variety B , a rank- n vector bundle $\mathcal{U}$ on B , a constant n -dimensional vector space V , and a rank-two locally direct-summand subbundle

$$\mathcal{R} \subset \mathrm{Sym}^2 V \otimes \mathcal{O}_B.$$

We require that $\mathbb{P}_B(\mathcal{U}^*)$ is not a trivial projective bundle. Put $\mathcal{S} = (\mathrm{Sym}^2 V \otimes \mathcal{O}_B)/\mathcal{R}$ and $\mathcal{E} = \mathrm{Hom}(\mathcal{U}, V \otimes \mathcal{O}_B)$. On its total space, let T be the universal map and let γ denote the quotient onto $\mathcal{S}$. Define the homogeneous ideal and its truncations by

$$\begin{aligned} \mathcal{I}_{\mathcal{R}} &= (\det T)I_p(\gamma \mathrm{Sym}^2 T), \\ \mathcal{A}_{\mathcal{R}}^{[k]} &= \mathrm{Sym}(\mathcal{E}^*)/(\mathcal{I}_{\mathcal{R}} + \mathfrak{a}^{k+1}), \quad Y_{\mathcal{R}}^{[k]} = \mathrm{Spec}_B \mathcal{A}_{\mathcal{R}}^{[k]}, \end{aligned} \tag{D.1}$$

where $\mathfrak{a}$ is the positive-degree ideal. Determinant-line changes of frame multiply local generators by units, so this is an ideal sheaf, independent of all frames. This construction is the pullback of the coefficient construction in Theorem K.1, with an arbitrary rank- n source bundle. It is not necessary to choose an embedding of B in a socle Grassmannian.

Lemma D.1 (A varying coefficient line). *The ideal $\mathcal{I}_{\mathcal{R}}$ has no component of degree less than d and is generated by its degree- d component $\mathcal{K}_{\mathcal{R}}$. This component is a subbundle of $\mathrm{Sym}^d \mathcal{E}^*$ of rank M . After cancellation of the determinant line, its degree- $2p$ coefficient module determines $\mathcal{R}$ as a subbundle, once the two tensor factors of $\mathcal{E}$ are supplied. In particular, $Y_{\mathcal{R}}^{[d]} \rightarrow B$ is finite locally free of rank*

$$\binom{n^2 + d}{d} - M.$$

Proof. Put $F(H) = \bigwedge^p \mathrm{Sym}^2 H$ for rank- n vector spaces or bundles H . This is an irreducible polynomial representation by Lemma 6.2; its dimension is M . Write $\mathcal{L}_{\mathcal{R}} = \bigwedge^p \mathcal{S}^*$, viewed, by $\bigwedge^p \gamma^*$, as a line subbundle of $F(V)^* \otimes \mathcal{O}_B$. The residual minors are precisely the image of

$$\mathcal{L}_{\mathcal{R}} \otimes F(\mathcal{U}) \longrightarrow \mathrm{Sym}^{2p}(V^* \otimes \mathcal{U}), \quad \beta \otimes \xi \longmapsto [T \mapsto \beta(F(T)\xi)]. \tag{D.2}$$

The full map $F(V)^* \otimes F(U) \rightarrow \mathrm{Sym}^{2p}(V^* \otimes U)$ is the nonzero inclusion of its unique Cauchy summand: evaluation at $T = 1$ is nonzero, as in (6.6). Restricting to a nonzero line on the left therefore has rank M in every fibre. The coefficients vary algebraically. The constant-rank criterion makes the image a subbundle; equivalently one may choose a nonvanishing maximal minor of this coefficient map locally on the Grassmannian of pencils and pull it back to B . Multiplication by the determinant, a nonzero polynomial in every fibre, again has constant rank M .

All defining generators have degree d , proving the assertion about the ideal. For the recovery assertion, the residual image, inside the indicated Cauchy summand, is $\mathcal{L}_{\mathcal{R}} \otimes F(\mathcal{U})$. Contraction with $F(\mathcal{U})^*$ has image exactly $\mathcal{L}_{\mathcal{R}}$ as a line subbundle of $F(V)^* \otimes \mathcal{O}_B$. This statement is an equality of images of bundle maps, not just equality at closed points. Exterior duality identifies it with the Pluecker line $\bigwedge^2 \mathcal{R}$, up to the fixed determinant line of $\mathrm{Sym}^2 V$. The Grassmannian Pluecker embedding consequently recovers the morphism $B \rightarrow \mathrm{Gr}(2, \mathrm{Sym}^2 V)$, and its pullback of the tautological bundle recovers $\mathcal{R}$. Finally, the truncated algebra is the sum of its free graded pieces below d and the locally free degree- d quotient by $\mathcal{K}_{\mathcal{R}}$. Its rank is the asserted binomial sum minus M . $\square$

D.2. The unmarked inverse.

Theorem D.2 (Reconstruction of a moving pencil). *Let $(B_i, \mathcal{U}_i, V_i, \mathcal{R}_i)$, $i = 1, 2$, be admissible data of the same rank n . Then*

$$Y_{\mathcal{R}_1}^{[d]} \simeq Y_{\mathcal{R}_2}^{[d]} \quad (\text{D.3})$$

as abstract complex schemes if and only if there exist an isomorphism $\sigma : B_1 \rightarrow B_2$, a constant linear isomorphism $g : V_1 \rightarrow V_2$, and a bundle isomorphism $h : \mathcal{U}_1 \rightarrow \sigma^ \mathcal{U}_2$ such that*

$$\sigma^* \mathcal{R}_2 = (\text{Sym}^2 g) \mathcal{R}_1. \quad (\text{D.4})$$

The reduced base, normal tensor orientation, isomorphism class of the actual source bundle, and the varying pencil are recovered together. Neither the projection onto the base nor the ambient bundle is supplied in (D.3). The constant left map cannot be replaced by a separately chosen change of coordinates in each fibre. A pair (g, h) defining the same normal-bundle map is unique up to simultaneous multiplication by a nonzero constant.

Proof. Let $\mathfrak{n}_i$ be the nilradical of $\mathcal{O}_{Y_{\mathcal{R}_i}^{[d]}}$. Since the positive-degree ideal is nilpotent and B_i is reduced, the reduction is exactly B_i , and this ideal is exactly $\mathfrak{n}_i$. There are no relations below degree d , so multiplication intrinsically recovers

$$\mathcal{E}_i^* = \mathfrak{n}_i / \mathfrak{n}_i^2, \quad \mathcal{K}_i = \ker \left[\text{Sym}^d(\mathfrak{n}_i / \mathfrak{n}_i^2) \longrightarrow \mathfrak{n}_i^d \right]. \quad (\text{D.5})$$

The map is well defined since $\mathfrak{n}_i^{d+1} = 0$. An abstract isomorphism gives σ , an isomorphism of these normal bundles, and the matching relation subbundles. In particular it gives an isomorphism of the untruncated homogeneous cones

$$\text{Spec}_{B_i}(\text{Sym } \mathcal{E}_i^* / (\mathcal{K}_i)).$$

This cone is reconstructed from the first relation module; it is not the associated graded algebra of the finite scheme, whose degrees above d vanish.

The reduction of this cone is the determinant cone. Indeed the residual quadratic map is surjective wherever T is invertible, whereas the determinant factor annihilates the ideal wherever T is singular. On a frame chart the determinant hypersurface is reduced over the smooth base, so the radical ideal is precisely the determinant ideal. Successive relative reduced singular loci give the lower matrix-rank cones. The rank-one cone has the two ruling parameter schemes

$$\mathbb{P}(V_i) \times B_i, \quad \mathbb{P}_{B_i}(\mathcal{U}_i^*),$$

by Lemma 3.1. Only the first is projectively trivial. Thus an isomorphism cannot interchange them. The induced map on the trivial ruling is a morphism $B_1 \rightarrow \text{Isom}(\mathbb{P}(V_1), \mathbb{P}(V_2))$. After choosing one identification this target is the affine variety PGL_n . Since B_1 is connected and projective, every such morphism is constant. Choose one constant linear lift g .

We check that no line-bundle ambiguity remains in the other factor. Locally trivialize both source bundles. The linear rank-one preserver with its left projective map fixed to $[g]$ is of the form

$$T \longmapsto gTh^{-1} \quad (\text{D.6})$$

by Lemma 3.2. Once this particular g is fixed, the local linear map h is unique: if two right maps give the same value for every T , they are equal. The local maps therefore agree on overlaps and glue to an actual bundle isomorphism $\mathcal{U}_1 \rightarrow \sigma^* \mathcal{U}_2$. Alternatively, after removing g the normal-bundle map commutes with $\text{End}(V_1)$; the commutant is the right-factor homomorphism bundle. Rescaling the constant lift g rescales h by the same constant. This proves both existence and the asserted uniqueness for a given normal-bundle map.

Cancel the intrinsic determinant ideal in the recovered homogeneous ideal. On a frame chart the determinant polynomial is a non-zero-divisor and $((\det T)\mathcal{J} : (\det T)) = \mathcal{J}$; the identities glue

without selecting a global determinant generator. Apply Lemma D.1 to the degree- $2p$ part. The right map h acts on the full factor $F(\mathcal{U})$, so contraction against its dual removes that change of frame. The remaining line subbundle transforms by the one constant left map g . Exterior duality yields exactly (D.4). This uses the subbundle version proved above, not the constant-coefficient hypothesis in Theorem 5.1.

Conversely σ, g, h satisfying (D.4) identify the quotient bundles $\mathcal{S}_i$, the quadratic maps, and their maximal-minor ideals. The determinant changes by a unit in local frames. The map (D.6) consequently identifies both the homogeneous ideals and every truncation, proving (D.3). $\square$

Corollary D.3 (Uniform sharpness over an admissible base). *For fixed $B, \mathcal{U}, V$, all schemes $Y_{\mathcal{R}}^{[k]}$ with $k < d$ are independent of $\mathcal{R}$. Order d is the smallest uniform order at which the moving pencil is recoverable in the sense of Theorem D.2.*

Proof. Below d the ideal $\mathcal{I}_{\mathcal{R}}$ contributes no relations, and the algebra is $\bigoplus_{j=0}^k \text{Sym}^j \mathcal{E}^*$. Constant pencils are among the admissible subbundles. The pencils $\langle x_1^2, x_1 x_2 \rangle$ and $\langle x_1^2, x_2^2 \rangle$ are inequivalent: all members of the first have a common linear factor and those of the second do not. A base automorphism does not change either constant pencil. The theorem separates their order- d schemes, whereas every lower truncation agrees. This proves uniform, not pairwise, minimality. $\square$

D.3. The part of an isomorphism that the inverse does not record. The recovered data classify isomorphism classes, but an isomorphism of finite schemes need not itself be linear in the nilpotent directions. The following statement identifies this distinction without replacing it by a claim about moduli stacks.

Fix an admissible datum, put $Y = Y_{\mathcal{R}}^{[d]}$, and let $\mathcal{L}_{\mathcal{R}}$ be the group of triples (σ, g, h) satisfying (D.4), modulo $(g, h) \sim (cg, ch)$ for $c \in \mathbb{C}^*$. Its group law is composition of their induced normal-bundle maps. Write $\text{Aut}_{\mathbb{C}}(Y)$ for the group of complex scheme automorphisms. For $j \geq 1$, define G_j to consist of the automorphisms inducing the identity on the reduction and satisfying

$$(\alpha^* - 1)(\mathfrak{n}^k) \subseteq \mathfrak{n}^{k+j} \quad \text{for every } k \geq 0, \quad \mathfrak{n}^0 = \mathcal{O}_Y. \quad (\text{D.7})$$

They act on sheaves on the same underlying topological space.

Theorem D.4 (Linear data and the nilpotent kernel). *There is an exact sequence of groups*

$$1 \longrightarrow G_1 \longrightarrow \text{Aut}_{\mathbb{C}}(Y) \longrightarrow \mathcal{L}_{\mathcal{R}} \longrightarrow 1. \quad (\text{D.8})$$

The displayed homogeneous presentation of Y supplies a splitting. Moreover $G_{d+1} = 1$, $[G_i, G_j] \subseteq G_{i+j}$, and there are natural injections

$$G_j/G_{j+1} \hookrightarrow H^0(B, \text{Der}_{\mathbb{C}}(\text{gr}_{\mathfrak{n}} \mathcal{O}_Y)_j) \quad (\text{D.9})$$

into additive groups of degree- j derivations. Thus the kernel is nilpotent and records precisely the automorphisms invisible on the associated graded scheme. No surjectivity onto the derivation spaces is asserted.

Proof. An automorphism induces a base map and a degree-one bundle map. The proof of Theorem D.2 identifies these with a unique class of triples in $\mathcal{L}_{\mathcal{R}}$. The graded algebra is generated in degree one over degree zero, with relation subbundle $\mathcal{K}_{\mathcal{R}}$. Hence its entire graded automorphism is already determined by that triple. The kernel is the identity on every graded piece, which is exactly condition (D.7) for $j = 1$. Conversely the homogeneous maps $T \mapsto gTh^{-1}$ give actual automorphisms of Y , respect composition, and induce the prescribed triples. This proves exactness and the splitting. A choice of the homogeneous presentation supplies that section; the assertion does not endow an arbitrary unmarked presentation with a preferred section.

For $\alpha \in G_j$, put $D_{\alpha} = \alpha^* - 1$. It raises filtration by j , and is therefore nilpotent as an additive operator. The inverse of $1 + D_{\alpha}$ is the finite geometric series. Composition of filtration-raising

operators shows that a commutator of elements of G_i, G_j raises filtration by at least $i + j$. Also $G_{d+1} = 1$ follows by taking $k = 0$, since $\mathfrak{n}^{d+1} = 0$.

The leading term of D_α defines maps $\mathfrak{n}^k/\mathfrak{n}^{k+1} \rightarrow \mathfrak{n}^{k+j}/\mathfrak{n}^{k+j+1}$. They are independent of lifts. The identity

$$D_\alpha(ab) = aD_\alpha(b) + D_\alpha(a)b + D_\alpha(a)D_\alpha(b)$$

reduces to the Leibniz identity in the indicated graded degree, since $2j \geq j + 1$. Thus the leading term is a sheaf derivation over $\mathbb{C}$, including its action on degree zero; it need not be an $\mathcal{O}_B$ -linear derivation. Leading terms add under composition, and their kernel is exactly G_{j+1} . They glue globally, proving (D.9) and all assertions. $\square$

Example D.5 (A nontrivial invisible automorphism). Let $B = \mathbb{P}^1$ and $\mathcal{U} = \mathcal{O}(-1) \oplus \mathcal{O}^{n-1}$. The bundle $\mathcal{E}^* = V^* \otimes \mathcal{U}$ has at least two trivial direct summands. Choose their degree-one generators x, y . The substitution

$$x \mapsto x + y^2$$

fixing y , the other summands and $\mathcal{O}_B$, defines a nonidentity automorphism of $Y_{\mathcal{R}}^{[d]}$ in G_1 , for every admissible $\mathcal{R}$. Indeed each relation has degree d ; its change under this substitution has degree at least $d + 1$ and vanishes in the truncation. The inverse is $x \mapsto x - y^2$. The automorphism is nonidentity because there are no degree-two relations. Therefore identifying all finite-scheme automorphisms with the stabilizer of the pencil would be false, even in this simple case. The inverse theorem and the exact sequence retain, rather than suppress, this higher-order freedom.

D.4. A nonconstant family across spectral collisions.

Example D.6. Take $n = 4$, $B = \mathbb{P}_{[s:t]}^1$, $\mathcal{U} = \mathcal{O}(-1) \oplus \mathcal{O}^3$, and $V = \mathbb{C}^4$. In diagonal quadratic coordinates set

$$A = \text{diag}(1, 1, 1, 1), \quad B_{s,t} = \text{diag}(0, s, t, 2s + 3t).$$

The map $\mathcal{O} \oplus \mathcal{O}(-1) \rightarrow \text{Sym}^2 V \otimes \mathcal{O}$ with columns $A, B_{s,t}$ has rank two everywhere. The first diagonal entry shows that any scalar relation involving A has zero coefficient on A ; the next two entries then force $s = t = 0$, which is impossible on $\mathbb{P}^1$. Its image $\mathcal{R}$ is therefore a locally direct-summand pencil bundle. Since it contains the nondegenerate form A , it is regular at every point, including all spectral collisions.

On the open set of four distinct eigenvalues, an ordered cross-ratio of $0, s, t, 2s + 3t$ is

$$\chi = \frac{t(s + 3t)}{(2s + 3t)(t - s)}.$$

The rational function $J = (\chi^2 - \chi + 1)^3/(\chi^2(1 - \chi)^2)$ is invariant under the six cross-ratio transformations arising from permutations of four points. It is nonconstant: at $[s : t] = [1 : 2]$ and $[1 : 3]$, the values of χ are $7/4$ and $15/11$, and the corresponding values of J differ. Thus the pencil family is not projectively isotrivial, even allowing a change of parameter within each pencil. The discriminant of $\det(zA - B_{s,t})$ is

$$4s^2t^2(2s + 3t)^2(s - t)^2(s + 3t)^2(s + t)^2. \quad (\text{D.10})$$

No factor is inverted in forming $Y_{\mathcal{R}}^{[20]}$. This finite scheme is locally free over its entire projective reduction of rank $\binom{36}{20} - 45$. By Theorem D.2, its unmarked isomorphism class recovers the whole morphism into the pencil Grassmannian, including the six collision points and the pencil there, up to one constant transformation of V and an automorphism of the base. This is a consequence about a genuinely moving family, not just the pointwise Pluecker encoding. The cross-ratio and discriminant calculations are classical elementary readouts; their role here is to exhibit a family to which the unmarked global inverse applies.

Remark D.7 (Scope of the family statement). The reduction in this section is smooth, projective and reduced. The source projective bundle is assumed nontrivial, which orients the two rulings. The theorem allows all fibre pencils, singular or regular; regularity is used only in the spectral description of the last example. Theorem K.1 separately allows arbitrary complex base change with its specified relative socle ideal. The present intrinsic inverse does not replace that ideal by an absolute nilradical over a nonreduced base. Nor does (D.8) assert an equivalence of unmarked moduli stacks or a description of every infinitesimal deformation. It is an exact classification of geometric isomorphisms, their graded data, and the residual automorphism kernel for the stated admissible objects.

APPENDIX E. THE EFFECTIVE MODULI STACK OF PENCIL FAILURE ALGEBRAS

We apply the common-divisor inverse to the actual first relations of quadratic pencils. Their common divisor has a higher multiplicity than the single determinant factor used to recover their support. Removing that multiplicity places every pencil, without a regularity hypothesis, in the smooth stratum of the preceding section.

Keep $W = \text{Sym}^2 V$, $N = \binom{n+1}{2}$, $p = N - 2$, and put

$$\begin{aligned} M &= \binom{N}{2}, & s &= n - 1, & h &= 3n - 4, \\ \mu &= (\underbrace{3, \dots, 3}_{n-2}, 2, 0), & A &= \mathbb{S}_\mu V^*, & F &= \mathbb{S}_\mu U. \end{aligned} \tag{E.1}$$

Both A and F have dimension M . The Cauchy inclusion $A \otimes F \subset S_h(E)$ will be fixed; its subspace and all its projections are equivariant, independent of a choice of bases.

Lemma E.1 (Exact common multiplicity). *For every $R \in \text{Gr}(2, W)$, including every singular pencil, the first relation space of its order- d failure algebra is*

$$K_R = \delta^{n-1} \tilde{J}_R, \quad \tilde{J}_R = L_R^0 \otimes F \subset S_h(E), \quad \gcd(\tilde{J}_R) = 1. \tag{E.2}$$

Here $L_R^0 \subset A$ is the Pluecker line of R , under the canonical projective identification $\mathbb{P}(A) = \mathbb{P}(\wedge^2 W)$. The common divisor of K_R is therefore exactly δ^{n-1} , up to scalar. Its degree remains $d = sn + h = n^2 + 2n - 4$.

Proof. Exterior duality and $\wedge^2 \text{Sym}^2 H = \mathbb{S}_{(3,1)} H$ give

$$\bigwedge^{N-2} \text{Sym}^2 H = \mathbb{S}_\lambda H, \quad \lambda = (\underbrace{n+1, \dots, n+1}_{n-2}, n, n-2) = \mu + (n-2)^n.$$

Thus this representation is $(\det H)^{n-2} \otimes \mathbb{S}_\mu H$. Applied to the universal matrix, its coefficient polynomials factor as δ^{n-2} times the matrix coefficients of $\mathbb{S}_\mu T$. The additional determinant in the failure ideal gives (E.2). On the left, exterior duality also yields

$$A \simeq \bigwedge^2 W \otimes (\det V)^{-3},$$

which identifies the residual line with the Pluecker line.

We prove a slightly stronger assertion: the matrix coefficient space $\beta \otimes \mathbb{S}_\mu U$ has greatest common divisor 1 for every nonzero $\beta \in \mathbb{S}_\mu V^*$. Its common divisor is taken to a scalar multiple of itself by the full right group. Its zero divisor is therefore invariant under right multiplication. A nonconstant invariant divisor cannot meet the open orbit of invertible matrices. Since the complement of that orbit is the irreducible determinant hypersurface, any such common divisor is a power of δ .

It remains to exclude δ . Choose a rank- $n - 1$ map T_0 . The representation $\mathbb{S}_\mu$ has length $n - 1$, so $\mathbb{S}_\mu T_0 \neq 0$. Choose a nonzero vector in its image. The span of its left $\mathrm{GL}(V)$ -orbit is the entire irreducible representation. Consequently some left translate gT_0 has a coefficient not annihilated by β . Not all residual coefficients vanish on the rank- $n - 1$ locus. They have no determinant factor, proving the assertion. The same argument at an invertible matrix shows that the coefficient vector is nonzero everywhere on the invertible locus. $\square$

Let $\mathcal{P} = \mathrm{Gr}(2, W)$. The morphism

$$\iota : \mathcal{P} \longrightarrow X_{h,M}(E), \quad R \longmapsto L_R^0 \otimes F \quad (\text{E.3})$$

is a closed immersion. Indeed, the Pluecker embedding is followed by the closed immersion $\mathbb{P}(A) \rightarrow \mathrm{Gr}(M, A \otimes F)$, $L \mapsto L \otimes F$, and then by a linear Grassmannian inclusion. The inverse to the middle map is contraction with F^* . This factorization proves the assertion on schemes, including nonreduced test schemes, rather than only on points. Lemma E.1 shows that its image lies in the open set $X_{h,M}(E)$.

Define the *pencil relation stratum* Z_{pen} inside $Z_{n-1, 3n-4, M}^{\det}$ by the following intrinsic scheme conditions. Recover the common-divisor line, its determinant root, and its unordered tensor rulings by Theorem 11.3. In an ordering, the primitive residual relation is required to lie in the indicated Cauchy summand, to have the form $L \otimes F$, and to satisfy the Pluecker equations on L under $A \simeq \bigwedge^2 W \otimes (\det V)^{-3}$. These are equations on subbundles; a test on the fibres alone is not their definition. Use both orderings and descend. The two ordered images are disjoint: their support ranks are respectively $(1, M)$ and $(M, 1)$, with $M > 1$. Each is closed in the residual Grassmannian. Thus this prescription defines a locally closed scheme, independent of the tensor frames. It is an intrinsic image criterion for a natural family of first relations, not a definition by an algebra isomorphism to a preselected pencil.

Theorem E.2 (Effective all-pencil moduli and deformations). *The stack of first relations in Z_{pen} is*

$$[Z_{\mathrm{pen}} / \mathrm{GL}(E)] \simeq [\mathcal{P} / G^\circ]. \quad (\text{E.4})$$

There is an exact sequence of groups

$$1 \longrightarrow \mathrm{GL}(U) \longrightarrow G^\circ \longrightarrow \mathrm{PGL}(V) \longrightarrow 1, \quad (\text{E.5})$$

and its kernel acts trivially on $\mathcal{P}$. Quotient the morphisms in the algebra stack first by the nonlinear tangent-identity kernel of Proposition 11.4, and then by this ineffective right group, and fppf stackify. The resulting effective pencil failure stack satisfies

$$\mathcal{F}_{\mathrm{pen}}^{\mathrm{eff}} \simeq [\mathrm{Gr}(2, \mathrm{Sym}^2 V) / \mathrm{PGL}(V)]. \quad (\text{E.6})$$

This equivalence is valid over all complex parameter schemes. It identifies automorphisms after the specified quotients, infinitesimal deformation groupoids, and families of degenerating pencils, including singular pencils.

The stack is smooth of dimension $n - 3$. At R , its tangent complex is

$$[\mathfrak{pgl}(V) \xrightarrow{d_R} \mathrm{Hom}(R, W/R)], \quad d_R(X)(r) = (X \otimes 1 + 1 \otimes X)r \bmod R. \quad (\text{E.7})$$

In particular, the dimension of its space of first-order isomorphism classes is $n - 3 + \dim \mathrm{Stab}_{\mathrm{PGL}(V)}(R)$. Within this stratum, every infinitesimal pencil deformation and its effective failure-algebra deformation lift together across a small extension of local Artinian complex algebras.

Proof. In the determinant frame of Theorem 11.3, the residual pencil locus is $\iota(\mathcal{P}) \amalg \tau\iota(\mathcal{P})$, where τ exchanges the two factors. The disjointness just proved and the equivariance of the Cauchy and Pluecker maps give, as schemes with group action,

$$\iota(\mathcal{P}) \amalg \tau\iota(\mathcal{P}) \simeq G \times^{G^\circ} \mathcal{P}.$$

Quotienting by G gives (E.4). All maps used here are morphisms and all inverse operations are contractions, common-divisor descent, or the inverse on the Pluecker image. Consequently this calculation remains valid on a nonreduced base. It is not an inference from equality of complex-point stabilizers.

The map in (E.5) sends the class of (g, h) to $[g]$. If $g = aI$, multiplication by the diagonal scalar a^{-1} writes its class uniquely as $(I, a^{-1}h)$. This proves the asserted kernel. The action on the Pluecker line is the projective action of g ; the full right factor has no effect on R . Every $\mathrm{PGL}(V)$ -valued isomorphism lifts fppf locally to G° , and two lifts differ by that kernel. The quotient of morphisms and stackification therefore gives $[\mathcal{P}/\mathrm{PGL}(V)]$. Combining this with Proposition 11.4 proves (E.6). No splitting of (E.5) over the base is required.

A Grassmannian chart writes a nearby pencil as the graph of a map $R \rightarrow W/R$. Infinitesimal coordinate changes act by d_R , giving (E.7). Grassmannian charts and $\mathrm{PGL}(V)$ are smooth, so the quotient is smooth and the graph coefficients lift over every small extension. A coordinate change can also be lifted in the smooth group, and its action determines a compatible lift of the corresponding object. Torsors over a local Artinian complex algebra can be trivialized, since smooth torsors lift a point of their algebraically closed residue field. This proves the lifting assertion for deformation groupoids. Finally,

$$\dim \mathcal{P} - \dim \mathrm{PGL}(V) = 2(N - 2) - (n^2 - 1) = n - 3.$$

Taking kernel and cokernel of d_R gives the stated first-order dimension, including stabilizer jumps at singular pencils. $\square$

Corollary E.3 (Native specialization loci). *Every $\mathrm{PGL}(V)$ -invariant locally closed subscheme $Q \subset \mathrm{Gr}(2, \mathrm{Sym}^2 V)$, with its specified scheme structure, has a corresponding intrinsic effective failure stratum isomorphic to $[Q/\mathrm{PGL}(V)]$. Its infinitesimal equations and its specializations are retained, not just its closed points. In particular, this applies to rank-incidence and spectral Fitting conditions on the universal pencil, including their intersections. A family in such a locus specializes to a pencil in its closure exactly when its effective failure family has the corresponding specialization in the associated closure stratum.*

Proof. Restrict the equivariant closed immersion (E.3) and its inverse to Q . The construction of Z_{pen} and its two quotient steps then restrict to give the stated isomorphism. Equations for a rank or a Fitting condition are equivariant equations in the universal pencil, so they are included in this argument with their scheme structures. Apply the same construction to the closure of Q to obtain the specialization assertion. $\square$

Remark E.4. Theorem E.2 does not say that the unrigidified algebra stack is the pencil stack. At a complex pencil its linear stabilizer remains $(G_R \times \mathrm{GL}(U))/\mathbb{G}_m$, and its full algebra automorphism group also has the nonlinear kernel in Proposition 2.2. Both are real automorphisms, not redundant notation. They are explicitly removed only in the effective comparison. Nor is the quotient asserted to be proper, although its framed Grassmannian is projective.

The parameter base in this theorem is distinct from the intrinsic projective reduction in Theorem 8.2. Here projective coordinate changes may vary over the parameter scheme. In that moving theorem, properness of the intrinsic reduction forces one constant left map and recovers an actual source bundle. Neither assertion is substituted for the other. The present equivalence upgrades the geometric orbit statement to a family statement without changing the hypotheses or conclusions of the moving inverse.

APPENDIX F. UNIVERSAL PROPERTIES OF THE TWO RIGIDIFICATIONS

The effective stacks used above have a universal characterization. This specifies what is forgotten from isomorphisms, as opposed to what is retained from objects. We use rigidification in its standard sense of removing a flat normal subgroup of inertia; see [18, Tag 04V2]. The argument below gives the universal property directly in the present situation, including the nonconstant subgroup obtained by descent.

Let $H_0 \subset \mathrm{Gr}(r, S_D(E))$ be a $\mathrm{GL}(E)$ -invariant locally closed subscheme. Let $\mathcal{A}_{H_0}$ be the stack of algebras satisfying the trace, filtration and Hilbert-function conditions of Proposition 11.4, whose intrinsic kernel is in H_0 . Write $\mathcal{R}_{H_0} = [H_0 / \mathrm{GL}(E)]$.

Theorem F.1 (First-relation rigidification). *Taking the first relation defines an fppf gerbe*

$$q : \mathcal{A}_{H_0} \longrightarrow \mathcal{R}_{H_0}.$$

Its relative inertia $\mathcal{U}$ consists exactly of the substitutions which induce the identity on $\mathcal{N}/\mathcal{N}^2$. It is a smooth affine normal subgroup of the inertia. In a homogeneous presentation its underlying scheme is $\mathrm{Hom}(\mathcal{E}, \mathcal{N}^2)$, with the substitution group law. For every stack $\mathcal{Z}$ in groupoids on complex schemes, precomposition with q identifies

$$\mathrm{Hom}(\mathcal{R}_{H_0}, \mathcal{Z}) \simeq \{F \in \mathrm{Hom}(\mathcal{A}_{H_0}, \mathcal{Z}) : F(\mathcal{U}) = 1\}. \quad (\mathrm{F}.1)$$

The right side is the full subgroupoid, including all natural isomorphisms, of functors killing the indicated relative inertia. The construction is compatible with base change on the relation stack.

Proof. A relation bundle has its homogeneous truncated algebra as a lift. Every algebra over that relation bundle is locally isomorphic to this lift. The first-relation proposition shows that an isomorphism of relation bundles lifts fppf locally, and any two lifts differ by a unique element of the indicated kernel. These assertions are exactly the gerbe and relative-inertia assertions. The kernel is intrinsic, so it is stable under conjugation. Its local affine-space description shows smoothness and flatness; no additive group law is being asserted.

For clarity, we prove the descent implicit in (F.1). Given F on the right and an object y of $\mathcal{R}_{H_0}$, choose local lifts a_i over an fppf cover. Locally choose comparison isomorphisms on overlaps above the identity of y . Two choices differ by $\mathcal{U}$, hence have the same image under F . On triple overlaps the failure of the chosen lifts to satisfy the cocycle condition is again in $\mathcal{U}$. Their images therefore give a genuine descent datum for $F(a_i)$, which is effective because $\mathcal{Z}$ is a stack. Refinements and different lift choices give canonically isomorphic descended objects. An arrow between two objects of $\mathcal{R}_{H_0}$ is treated in the same way, using local lifts of that arrow. This constructs the descended functor. Natural isomorphisms descend by the same argument, proving full faithfulness as well as essential surjectivity. Conversely every functor pulled back by q kills its relative inertia. All the constructions use covers and isomorphism sheaves, so commute with base change. $\square$

Corollary F.2 (The effective pencil stack is a successive rigidification). *On the pencil relation stratum the second morphism*

$$[\mathcal{P}/G^\circ] \longrightarrow [\mathcal{P}/\mathrm{PGL}(V)], \quad \mathcal{P} = \mathrm{Gr}(2, \mathrm{Sym}^2 V),$$

is the rigidification by the normal relative inertia induced from $\ker(G^\circ \rightarrow \mathrm{PGL}(V)) = \mathrm{GL}(U)$. Consequently Theorem E.2 has the universal property of first killing $\mathcal{U}$ and then this right-factor inertia. It does not kill the stabilizer of the pencil in $\mathrm{PGL}(V)$.

Proof. The right factor acts trivially on $\mathcal{P}$, and the quotient map of groups is fppf surjective. Objects and isomorphisms of the second quotient therefore lift locally to the first, with differences in that kernel. Apply the descent proof of Theorem F.1. Over an object represented by a G° -torsor the subgroup is its conjugation-associated $\mathrm{GL}(U)$ -bundle; it need not be a constant group scheme

on the base. This describes the actual subgroup stack and shows that no global splitting is assumed. Composing with the first rigidification gives the asserted successive universal property. All remaining pencil automorphisms survive in the target. $\square$

APPENDIX G. A COMMON CLOSED SPECIALIZATION OF ALL QUADRATIC PENCILS

We give an orbit-closure consequence directly on the pencil side. Unlike restricting an equivalence to a previously specified locus, the following argument constructs the specialization. It works for every pencil, including pencils with singular generic member. Its mechanism is elementary projective representation theory; the additional failure algebra statement uses the all-pencil, exact-multiplicity construction.

Theorem G.1 (Explicit common closed orbit). *Let $\dim V = n \geq 3$. The action of $\mathrm{PGL}(V)$ on $\mathcal{P} = \mathrm{Gr}(2, \mathrm{Sym}^2 V)$ has a unique closed orbit, namely*

$$\mathcal{C} = \{\ell H : \ell \subset H \subset V, \dim \ell = 1, \dim H = 2\}. \quad (\mathrm{G}.1)$$

It is the flag variety $\mathrm{Fl}(1, 2; V)$, of dimension $2n - 3$. More precisely, after a fixed change of coordinates every pencil has a degeneration along the one-parameter subgroup

$$\lambda(t) = \mathrm{diag}(1, t, t^2, \dots, t^2)$$

to $\langle x_1^2, x_1 x_2 \rangle$. The pencil family is a subbundle on the whole affine line.

Proof. Take independent forms q_0, q_1 . The rational function q_1/q_0 on $\mathbb{P}(V^*)$ is nonconstant. In characteristic zero its differential is not identically zero: on an affine coordinate chart the common kernel of the coordinate derivations of the rational function field is $\mathbb{C}$. Choose a point $[v]$ where $q_0(v) \neq 0$ and this differential is nonzero. Replace q_1 by $q_1 - (q_1(v)/q_0(v))q_0$. Then $q_1(v) = 0$, whereas its differential has a nonzero value on some vector w independent of v . Euler's identity shows that its value on the radial vector v is zero. Extending v, w to a basis of V^* gives dual coordinates in which

$$[x_1^2]q_0 = a \neq 0, \quad [x_1^2]q_1 = 0, \quad [x_1 x_2]q_1 = b \neq 0.$$

Here brackets denote a coefficient.

Apply $\lambda(t)$ to the variables. The forms

$$Q_0(t) = q_0(x_1, tx_2, t^2 x_3, \dots, t^2 x_n), \quad Q_1(t) = t^{-1} q_1(x_1, tx_2, t^2 x_3, \dots, t^2 x_n)$$

have polynomial coefficients: the only possible weight-zero term in q_1 was removed. Their limits are ax_1^2 and $bx_1 x_2$. For $t \neq 0$ they span a coordinate transform of R ; at zero they are independent. Thus their coefficient matrix has rank two everywhere, defines a subbundle, and supplies the asserted morphism from $\mathbb{A}^1$ to $\mathcal{P}$.

The orbit of $\langle x_1^2, x_1 x_2 \rangle$ is exactly $\mathcal{C}$. A pencil of the form ℓH recovers ℓ as its gcd and H by division. These are scheme-theoretic inverse operations on the exact-gcd-one-degree stratum by Corollary 12.5, applied to $g = h = 1, r = 2$. Under that inverse the condition $\ell \subset H$ is the usual closed flag incidence. The flag-to-pencil map is therefore a locally closed immersion; since the flag variety is projective it is a closed immersion. It identifies $\mathcal{C}$ with the flag variety, and the dimension is $(n - 1) + (n - 2)$. Every pencil orbit closure contains $\mathcal{C}$, by the constructed family and equivariance. Any closed orbit must consequently be $\mathcal{C}$, proving uniqueness. $\square$

Corollary G.2 (Flat closed specialization of failure algebras). *For every quadratic pencil R there is a finite locally free family over $\mathbb{A}^1$, of rank $\binom{n^2+d}{d} - \binom{N}{2}$, whose nonzero fibres are isomorphic to $\mathfrak{A}_R^{[d]}$ and whose zero fibre is $\mathfrak{A}_{\langle x_1^2, x_1 x_2 \rangle}^{[d]}$. The intrinsic first relations throughout the family have exact gcd δ^{n-1} . In a fixed determinant frame, the two ordered copies of the flag orbit form the*

unique closed G -orbit in the pencil relation locus. Equivalently the effective pencil failure stack has the common closed specialization described in Theorem G.1.

Proof. Let R_t be the polynomial subbundle just constructed. Its Pluecker line gives, under the fixed exterior-duality identification, a line bundle $L_t^0 \subset \mathbb{S}_\mu V^* \otimes \mathcal{O}_{\mathbb{A}^1}$. Set

$$\mathcal{K}_t = \delta^{n-1}(L_t^0 \otimes \mathbb{S}_\mu U), \quad \mathcal{A}_t = \text{Sym}(E \otimes \mathcal{O}_{\mathbb{A}^1})/((\mathcal{K}_t) + \mathfrak{m}^{d+1}).$$

The first is a rank- M subbundle. The second is finite locally free by its graded direct-sum presentation, with the claimed rank. Lemma E.1 gives the exact gcd at every fibre, including zero; in fact this family lies scheme-theoretically in the fixed-divisor exact stratum. Equivariance and the local inverse identify its nonzero fibres as asserted. In a determinant frame the pencil locus is $G \times^{G^\circ} \mathcal{P}$. The previous closed-orbit calculation therefore gives the assertion for G . This is a statement inside that locus; it does not assert closedness under arbitrary degenerations of the determinant form in the full relation Grassmannian. $\square$

The closed specialization is very coarse compared with the unmarked local inverse: many pairwise nonisomorphic finite algebras have this same limiting fibre. The full first relation, not its common limit, retains the original pencil. The explicit family also differs from transport of spectral rank loci: it crosses from arbitrary pencils to a singular common-factor pencil without a regularity hypothesis. Theorem 12.4, on the other hand, describes divisor collisions in the larger natural first-relation boundary, outside the exact pencil image. These are two distinct geometric statements, and neither is a classification of every congruence-orbit adjacency of quadratic pencils.

APPENDIX H. TRANSVERSE DEFORMATIONS WITH UNCHANGED DETERMINANT SUPPORT

The larger determinant-divisor stratum has deformations which are not pencil deformations and need not satisfy either factor-invariance condition. We now identify their tangent directions and construct complete families in which the homogeneous radical remains exactly the determinant ideal. Thus the intrinsic stratum theorem has content outside the coefficient class, not merely a different description of the same symmetry hypothesis.

Let $J = L \otimes F$ be the primitive relation of a pencil $R \in \text{Gr}(2, W)$. In degree $h = 3n - 4$, write the equivariant Cauchy decomposition as

$$S_h(E) = (A \otimes F) \oplus H_{\text{other}},$$

where A, F, L are as in (E.1). The summand H_{other} contains every partition different from μ ; this notation denotes its full vector space, not a quotient chosen separately at each pencil.

Theorem H.1 (Normal directions to the pencil stratum). *The normal space at J of the pencil locus in $X_{h,M}(E)$ has the following equivariant direct-sum description:*

$$\begin{aligned} N_J \simeq & \text{Hom}\left(\bigwedge^2 R, \bigwedge^2 (W/R)\right) \\ & \oplus (\text{Hom}(L, A/L) \otimes \mathfrak{sl}(F)) \\ & \oplus \text{Hom}(L \otimes F, H_{\text{other}}). \end{aligned} \tag{H.1}$$

The summands respectively record departure from the Pluecker locus, departure from a full right multiplicity factor inside the same Cauchy summand, and departure into other Cauchy summands. They are all transverse directions in the effective first-relation stack: the infinitesimal G° -orbit lies in the pencil tangent space, so no vector in N_J is removed by that orbit. Every such direction is the first-order term of a local algebraic family in the determinant-divisor stratum. This last assertion refers to the common-divisor stratum; preservation of the entire homogeneous radical for complete families is supplied below.

Proof. The tangent space of a Grassmannian and the Cauchy splitting give

$$\mathrm{Hom}(J, S_h(E)/J) = (\mathrm{Hom}(L, A/L) \otimes \mathrm{End}(F)) \oplus \mathrm{Hom}(L \otimes F, H_{\mathrm{other}}). \quad (\mathrm{H}.2)$$

Under $L \mapsto L \otimes F$, the tangent to $\mathbb{P}(A)$ is the first factor tensored with the identity of F . In characteristic zero the trace splits $\mathrm{End}(F) = \mathbb{C}I \oplus \mathfrak{sl}(F)$. It remains to identify the normal space of the Pluecker embedding. The natural map $\wedge^2 W \rightarrow \wedge^2(W/R)$ has kernel $R \wedge W$. At the line $\wedge^2 R$, the tangent image is $\mathrm{Hom}(\wedge^2 R, (R \wedge W)/\wedge^2 R)$, which is the image of $\mathrm{Hom}(R, W/R)$. Its quotient in the projective tangent space is $\mathrm{Hom}(\wedge^2 R, \wedge^2(W/R))$. The common determinant twist in A cancels in this Hom space. Substitution in (H.2) proves (H.1) as a normal-space isomorphism. No splitting of the tangent sequence of the Pluecker embedding is being asserted.

The left group preserves the Pluecker locus, and the right group fixes $L \otimes F$; hence the differential of its orbit is in the pencil tangent. Passing to the quotient tangent complexes in (11.4) does not change this normal quotient. Finally $X_{h,M}(E)$ is open in a Grassmannian. A tangent vector is integrated by a graph with coefficients linear in one parameter, after restricting to a neighbourhood of zero which stays in this open set. Multiplication by δ^{n-1} and Proposition 11.4 give the claimed family of finite flat algebras. $\square$

Theorem H.2 (Complete fixed-support families through every pencil). *Put $e = n^2$. For every pencil R there is an e -dimensional subspace $B \subset J = \tilde{J}_R$ such that $\mathrm{gcd}(B) = 1$ and the forms in B have no common zero on the invertible matrix locus. The projective Grassmannian*

$$\Sigma_B = \{J' \in \mathrm{Gr}(M, S_h(E)) : B \subset J'\} \simeq \mathrm{Gr}(M - e, S_h(E)/B) \quad (\mathrm{H}.3)$$

contains J , is contained in $X_{h,M}(E)$, and satisfies

$$\sqrt{(\delta^{n-1}J')} = (\delta) \quad \text{for every geometric } J' \in \Sigma_B. \quad (\mathrm{H}.4)$$

Its universal first relation gives a finite flat family of the prescribed Hilbert function. It contains a projective line through J whose every other point is invariant under neither full factor group. Consequently determinant-radical relations through each pencil admit positive-dimensional complete families outside the recognized coefficient class, and their intrinsic deformation problem is described by Theorem 11.3.

Proof. By Lemma E.1, the residual coefficient vector is nonzero on every invertible matrix. Evaluation gives a morphism

$$\mathrm{GL}_n/\mathrm{G}_m \longrightarrow \mathbb{P}(J^*).$$

Its projective image closure Y has dimension at most $e - 1$. Since $M > e$ for $n \geq 3$, a general projective linear subspace of codimension e in $\mathbb{P}(J^*)$ misses Y . For such a subspace $\mathbb{P}(B^\perp)$, the associated e -plane $B \subset J$ has no common zero on the invertible locus. To justify the generic avoidance directly, the incidence of pairs consisting of a point of Y and an e -plane annihilated by it has dimension strictly smaller than $\mathrm{Gr}(e, J)$. Its projection is closed because Y is projective.

The condition $\mathrm{gcd}(B) = 1$ is a second nonempty open condition on $\mathrm{Gr}(e, J)$. To see nonemptiness, take any nonzero $b_1 \in J$. It has only finitely many irreducible factors. Since $\mathrm{gcd}(J) = 1$, a general $b_2 \in J$ avoids divisibility by each such factor; then $\mathrm{gcd}(b_1, b_2) = 1$. Extend their span to an e -plane. Openness follows from Lemma 11.1. The two nonempty opens intersect, as $\mathrm{Gr}(e, J)$ is irreducible.

Every J' containing this B has greatest common divisor 1, and its forms cannot vanish simultaneously on an invertible matrix. The common zero set of $\delta^{n-1}J'$ is thus exactly $V(\delta)$. The Nullstellensatz over an algebraically closed field proves (H.4). This is a statement about geometric homogeneous radicals. The scheme-theoretic family used here is the universal subbundle $\delta^{n-1}\mathcal{J}'$, not an operation of taking radicals on a nonreduced parameter ring. Multiplication by this fixed

nonzero form gives a subbundle; its quotient in degree d and all the lower symmetric powers are locally free. The truncated algebras are therefore finite flat.

For the final assertion, choose $j_0 \in J \setminus B$, a complement C to $\mathbb{C}j_0$ in J containing B , and a nonzero decomposable vector

$$\eta \in \mathrm{Sym}^h V^* \otimes \mathrm{Sym}^h U \subset H_{\mathrm{other}}.$$

The inequality $M > e$ allows the choice of C . The family

$$J_{[a:b]} = C \oplus \mathbb{C}(aj_0 + b\eta), \quad [a:b] \in \mathbb{P}^1,$$

is a projective line in Σ_B through $J = J_{[1:0]}$. For $b \neq 0$ its projection to the (h) -Cauchy summand is exactly the line $\mathbb{C}\eta$. A decomposable line in $\mathrm{Sym}^h V^* \otimes \mathrm{Sym}^h U$ is invariant under neither full factor group: each of the two irreducible factors has dimension greater than one. Equivariance of the Cauchy projection therefore excludes both possible admissible orientations of $J_{[a:b]}$. The determinant-preserver theorem says that these are the only orientations after any change of the recovered matrix coordinates. This proves the assertion intrinsically, not only in the chosen matrix frame. $\square$

Remark H.3. The three normal modes and the complete fixed-support families separate two questions. The effective pencil stratum has exactly the deformation theory of quadratic pencils, including their singular boundary. The larger determinant-divisor stratum contains additional directions, explicitly described here, which cease to be pencil or coefficient relations. Retaining the determinant radical does not suppress these directions. Conversely, no classification of all ideals with a prescribed radical is needed to describe this larger locally closed stratum and its family inverse. These assertions enlarge the reconstruction problem without replacing the all-pencil theorem by a generic statement.

APPENDIX I. COVERING MAPS INSIDE ONE FIBRE ORBIT OF QUADRATIC PENCILS

We give a consequence native to the pencil construction. It embeds the equivalence problem for maps $\mathbb{P}^1 \rightarrow \mathbb{P}^1$ into unmarked finite failure schemes, while keeping the fibre algebra and every nilradical graded vector bundle fixed. Unlike a varying spectral invariant, the phenomenon occurs within a single orbit of singular pencils.

Fix $n \geq 3$, a two-dimensional subspace $H = \langle x, y \rangle \subset V$, and the trivial source bundle $\mathcal{U} = \mathcal{O}_{\mathbb{P}^1}^n$. Let $f = [a:b] : \mathbb{P}^1 \rightarrow \mathbb{P}(H)$ be a morphism of degree $m \geq 1$, where a, b are homogeneous of degree m with no common zero. Its tautological line subbundle is

$$\mathcal{L}_f = \mathcal{O}(-m) \xrightarrow{(a,b)} H \otimes \mathcal{O}_{\mathbb{P}^1}.$$

Multiplication in $\mathrm{Sym} V$ defines a rank-two subbundle

$$\mathcal{R}_f = \mathcal{L}_f H \subset \mathrm{Sym}^2 V \otimes \mathcal{O}_{\mathbb{P}^1}, \quad \mathcal{R}_f|_{[s:t]} = \langle (ax + by)x, (ax + by)y \rangle. \quad (\text{I.1})$$

Let $Y_f = Y_{\mathcal{R}_f}^{[d]}$ be its order- d failure thickening, with

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n^2 + 2n - 4, \quad M = \binom{N}{2}, \quad e = n^2.$$

All schemes below are compared without projection, grading, embedding, matrix factors, or coefficient markings.

Theorem I.1 (Unmarked realization of the covering equivalence problem). *For fixed $n \geq 3$ and $m \geq 1$, the schemes Y_f have the following properties.*

(i) *They are finite locally free over their reduction $\mathbb{P}^1$, of rank $\binom{e+d}{d} - M$, and are Zariski locally trivial with the same Artin algebra as fibre. Every fibre pencil is equivalent to $\langle x^2, xy \rangle$.*

(ii) Writing $\mathfrak{n}_f$ for their nilradicals, their graded vector bundles are independent of f :

$$\begin{aligned} \mathfrak{n}_f^j / \mathfrak{n}_f^{j+1} &\simeq \mathcal{O}^{\binom{e+j-1}{j}} \quad (1 \leq j < d), \\ \mathfrak{n}_f^d &\simeq \mathcal{O}^{\binom{e+d-1}{d}-3M} \oplus \mathcal{O}(m)^{\oplus 2M}. \end{aligned} \quad (\text{I.2})$$

Their abstract first-relation bundles are all $\mathcal{O}(-2m)^{\oplus M}$, and their pencil and quotient bundles are respectively $\mathcal{O}(-m)^2$ and $\mathcal{O}(2m) \oplus \mathcal{O}^{N-3}$.

(iii) For two degree- m maps,

$$Y_f \simeq Y_{f'} \iff f' = \alpha \circ f \circ \sigma \quad \text{for some } \alpha, \sigma \in \text{PGL}_2(\mathbb{C}). \quad (\text{I.3})$$

Thus the double-orbit set of degree- m morphisms $\mathbb{P}^1 \rightarrow \mathbb{P}^1$ is realized faithfully by unmarked quadratic-pencil failure schemes with all the additive data in (ii) fixed. This is a statement about isomorphism classes, not an equivalence of their automorphism groupoids.

Proof. The pencil and local triviality. Multiplication by a nonzero linear form is injective on H . Since (a, b) never vanishes, the map $\mathcal{O}(-m) \otimes H \rightarrow \text{Sym}^2 H \otimes \mathcal{O}$ has rank two everywhere. Its image is a subbundle with locally free quotient. In the basis x^2, xy, y^2 its two columns are

$$\begin{pmatrix} a & 0 \\ b & a \\ 0 & b \end{pmatrix}, \quad \text{with Pluecker coordinates } (a^2, ab, b^2). \quad (\text{I.4})$$

It follows that $\mathcal{R}_f \simeq \mathcal{O}(-m)^2$. The quotient of $\text{Sym}^2 H \otimes \mathcal{O}$ is a line bundle of degree $2m$. A constant complement to $\text{Sym}^2 H$ in $\text{Sym}^2 V$ gives $\mathcal{S}_f \simeq \mathcal{O}(2m) \oplus \mathcal{O}^{N-3}$. On an open set where $\mathcal{L}_f$ is trivialized, complete its nonvanishing generator to a frame of $H \otimes \mathcal{O}$. The inverse frame map takes $\mathcal{R}_f$ to $\langle x^2, xy \rangle$. Extend it by the identity on a fixed complement of H in V . With the source bundle trivial, it identifies the defining minors and the truncated algebra with a product on that open set. This proves local triviality, not merely equality of fibre isomorphism classes. Lemma D.1 gives finite local freeness and the asserted rank.

The entire graded bundle calculation. The first relations occur in degree d , so all lower positive graded pieces are the full constant symmetric powers of $E = V^* \otimes \mathbb{C}^n$. Put $F(V) = \bigwedge^p \text{Sym}^2 V$, of dimension M . The coefficient line $\mathcal{L}_{\mathcal{R}_f} = \bigwedge^p \mathcal{S}_f^*$ is $\mathcal{O}(-2m)$. Under exterior duality

$$F(V)^* \simeq \bigwedge^2 (\text{Sym}^2 V) \otimes \det(\text{Sym}^2 V)^*,$$

its inclusion is the Pluecker line of $\mathcal{R}_f$. The constant determinant twist does not affect the bundle calculation. Thus it lies in a fixed three-dimensional subspace of $F(V)^*$, with column (a^2, ab, b^2) . There is an exact sequence

$$0 \longrightarrow \mathcal{O}(-2m) \xrightarrow{(a^2, ab, b^2)^t} \mathcal{O}^3 \xrightarrow{\begin{pmatrix} -b & a & 0 \\ 0 & -b & a \end{pmatrix}} \mathcal{O}(m)^2 \longrightarrow 0. \quad (\text{I.5})$$

Indeed the composite is zero; the maximal minors of the second map are $b^2, -ab, a^2$, which have no common zero. It is therefore everywhere surjective. Its rank-one kernel is the image of the displayed nowhere vanishing column. Taking a constant complement proves

$$(F(V)^* \otimes \mathcal{O}) / \mathcal{L}_{\mathcal{R}_f} \simeq \mathcal{O}^{M-3} \oplus \mathcal{O}(m)^2. \quad (\text{I.6})$$

The fixed Cauchy inclusion, followed by multiplication by the fixed determinant, embeds $F(V)^* \otimes F(\mathbb{C}^n)$ as an M^2 -dimensional constant subspace of $\text{Sym}^d E$. The first relation bundle is precisely $\mathcal{L}_{\mathcal{R}_f} \otimes F(\mathbb{C}^n)$ in that subspace. Choose a constant complement to the whole subspace, not to its varying relation subbundle. The top graded quotient is therefore

$$\mathcal{O}^{\binom{e+d-1}{d}-M^2} \oplus \left((F(V)^* \otimes \mathcal{O}) / \mathcal{L}_{\mathcal{R}_f} \right)^{\oplus M}.$$

Equation (I.6) gives exactly (I.2). It also shows that the first relation bundle is $\mathcal{O}(-2m)^{\oplus M}$. All assertions in (ii) concern vector bundles; their multiplication maps have not been identified.

The global inverse is the covering equivalence. Suppose $Y_f \simeq Y_{f'}$. Their reductions give an automorphism of $\mathbb{P}^1$. By Theorem 8.2, or the actual descent theorem with the pencil coefficient line, the varying pencil subbundles are identified by that base map and a single constant $g \in \mathrm{GL}(V)$. For a nonzero $\ell \in H$, the common polynomial divisor of the two-dimensional space $\ell H \subset \mathrm{Sym}^2 V$ is precisely ℓ , up to scalar: x and y have no common nonconstant divisor in $\mathrm{Sym} V$. Dividing the pencil space by this common factor recovers H . These operations are equivariant under $\mathrm{GL}(V)$. Consequently an equality between $g(\ell H)$ and a pencil of the form $\ell' H$ forces $gH = H$ and $[g\ell] = [\ell']$. The constant restriction $g|_H$ therefore supplies a single $\alpha \in \mathrm{PGL}(H)$ for the whole map. Replacing the recovered base map by its inverse if necessary gives the right-hand side of (I.3).

Conversely choose linear lifts of α on H and extend by the identity on a complement in V . Together with the base map and the identity identification of the trivial source bundles, this gives equivalent pencil data. Their homogeneous normal maps induce an isomorphism of the finite schemes. This proves (iii) and the theorem. $\square$

Corollary I.2 (A pencil-native separating pair). *For every $n \geq 3$, the degree-three maps*

$$f_0 = [s^3 : t^3], \quad f_1 = [s^3 + st^2 : t^3]$$

give nonisomorphic unmarked order- d failure schemes of quadratic pencils, although they have the same Artin fibres, the same conormal and first-relation bundles, the same pencil and quotient bundles, and isomorphic nilradical graded vector bundles in every degree. For $n = 3$ their common finite rank is 167945, and their top graded bundle is

$$\mathcal{O}^{75537} \oplus \mathcal{O}(3)^{\oplus 30}.$$

Proof. The pairs of cubics have no common zero. In the affine coordinate $z = s/t$ the maps are z^3 and $z^3 + z$. The first has two ramification points, both of index three. The second has the two distinct zeros of $3z^2 + 1$, each of index two, and infinity of index three. Independent projective changes in source and target preserve this multiset of indices, so the two maps are inequivalent. The theorem proves all the geometric assertions. For $n = 3$, $e = 9$, $d = 11$, $M = 15$, whence $\binom{20}{11} - 15 = 167945$ and $\binom{19}{11} - 45 = 75537$. $\square$

This consequence uses the global inverse essentially. Pointwise local reconstruction returns the same pencil orbit at every point for every f ; it cannot recover the map f . Even supplementing it with all the abstract bundles displayed in the theorem leaves the covering equivalence problem unresolved. It is the multiplication in the total unmarked thickening, followed by descent to one constant left transformation, that recovers that problem. The double-orbit description is not an assertion that the scheme has the automorphism group of the cover: the unused complement of H , the right factor, and the higher-order kernel remain present, as recorded by the previous automorphism theorems.

APPENDIX J. SPECTRAL TORSION AND RANK-PRESERVING SPECIALIZATION

The finite-neighbourhood theorem recovers more than the discriminant or the corank of each member of a regular pencil. We identify its classical spectral invariant and then give a flat specialization on which the entire discriminant and every reduced rank locus remain constant, while the recovered invariant changes. The two specializations are separated at exactly the order of Theorem 7.1. The parameter space throughout is the usual Grassmannian of pencils; the failure map is the original multiplication map, not an added coefficient-evaluation block.

J.1. The spectral torsion sheaf. Let $R \subset \text{Sym}^2 V$ be a regular pencil, meaning that it contains a nonsingular quadratic form. On its parameter line $L_R = \mathbb{P}(R)$, the tautological quadratic form gives an exact sequence

$$0 \longrightarrow V^* \otimes \mathcal{O}_{L_R}(-1) \xrightarrow{q_R} V \otimes \mathcal{O}_{L_R} \longrightarrow \mathcal{T}_R \longrightarrow 0. \quad (\text{J.1})$$

The first arrow is injective because its determinant is nonzero at the generic point of the smooth integral curve L_R . Thus $\mathcal{T}_R$ is a torsion sheaf of length n , whose zeroth Fitting divisor is the full discriminant. The pair $(L_R, \mathcal{T}_R)$ is considered up to isomorphism of the parameter line. Its definition involves no choice of a basis of R . The relation with elementary divisors is classical; see [4, §2, Corollary 2.1 and equation (5)]. We record the argument to make precise which datum is recovered.

Lemma J.1 (Sheaves on the affine spectral line). *Let A, B be symmetric complex matrices with A invertible. On the affine line with coordinate z , the cokernel of $zA - B$ is the finite-dimensional $\mathbb{C}[z]$ -module on $\mathbb{C}^n$ for which multiplication by z is $C = A^{-1}B$. An isomorphism of such sheaves, in this fixed coordinate, is equivalent to similarity of the corresponding matrices.*

Proof. Multiplication on the target by A^{-1} identifies the presentation with $zI - C$. Reduction of a polynomial vector modulo these relations sends $\sum_j z^j v_j$ to $\sum_j C^j v_j$; constant vectors give the inverse identification. The equivalence between coherent sheaves on an affine scheme and finitely generated modules identifies a finite-length sheaf supported in this affine line with this module. The support is finite, so its underlying complex vector space is finite dimensional. A module isomorphism is a linear isomorphism G satisfying $GC = C'G$, which is exactly similarity. Extension by zero across a point outside the support loses no sheaf data. $\square$

Lemma J.2 (Polynomial square roots of invertible operators). *For every invertible complex matrix K there is $h \in \mathbb{C}[x]$ with $h(K)^2 = K$. If K is self-adjoint for a nondegenerate symmetric form and commutes with C , then $h(K)$ has both properties.*

Proof. Write the minimal polynomial as $\prod_\lambda (x - \lambda)^{a_\lambda}$, where $\lambda \neq 0$. In its local Artinian factor choose a square root $\mu_\lambda^2 = \lambda$, and put

$$h_\lambda(x) = \mu_\lambda \sum_{j=0}^{a_\lambda-1} \binom{1/2}{j} \left(\frac{x - \lambda}{\lambda} \right)^j.$$

The formal binomial identity, truncated modulo $(x - \lambda)^{a_\lambda}$, gives $h_\lambda(x)^2 = x$. The Chinese remainder theorem joins these polynomials to a single class modulo the minimal polynomial. Evaluation at K proves the assertion even when K is not semisimple. Polynomials in one self-adjoint operator remain self-adjoint, and every polynomial in K commutes with every operator commuting with K . $\square$

Theorem J.3 (Spectral reconstruction at the first relation). *Let $n \geq 3$, $d = n^2 + 2n - 4$, and let R, R' be regular pencils. The following are equivalent:*

- (1) *the unmarked schemes $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic;*
- (2) *there is an isomorphism $f : L_R \rightarrow L_{R'}$ with $\mathcal{T}_R \simeq f^* \mathcal{T}_{R'}$;*
- (3) *the pencils R, R' are projectively equivalent.*

In particular the finite scheme determines the full root-labelled spectral torsion sheaf, not only its Fitting divisor or its support. This is an equivalence of isomorphism classes of objects; it is not an assertion identifying their automorphism groups or moduli stacks.

Proof. The equivalence of the first and third conditions is Theorem 7.1. A congruence of pencils induces an isomorphism of (J.1), so the third implies the second.

For the converse choose a common affine coordinate whose point at infinity avoids the two spectral supports, using f to identify the lines. Write the corresponding symmetric pairs as (A, B) and (A', B') , with A, A' invertible, and set $C = A^{-1}B$, $C' = A'^{-1}B'$. Choose the presentation $zA - B$ for the affine coordinate on both lines. Lemma J.1 identifies the two sheaves with the modules of C, C' , so their isomorphism gives a similarity. After applying that change of coordinates, assume $C' = C$. Both A and A' are nondegenerate symmetric forms for which C is self-adjoint. Then $K = A^{-1}A'$ commutes with C and is self-adjoint for A . Lemma J.2 provides a polynomial h with $h(K)^2 = K$. Put $H = h(K)$. It commutes with C , is self-adjoint for A , and satisfies

$$H^t A H = A', \quad H^t B H = H^t A C H = A' C = B'.$$

Thus the pairs, and hence the pencils, are congruent. This also shows why the spectral sheaf alone suffices here; no additional pairing on that sheaf is required over $\mathbb{C}$. $\square$

For $x \in \text{Supp } \mathcal{T}_R$, put

$$\ell_a(x) = \text{length}(\mathcal{T}_{R,x}/\mathfrak{m}_x^a \mathcal{T}_{R,x}), \quad a \geq 0. \quad (\text{J.2})$$

If the elementary-divisor exponents at x are $e_1, \dots, e_b$, the structure theorem over the discrete valuation ring $\mathcal{O}_{L_R,x}$ gives

$$\ell_a(x) = \sum_{j=1}^b \min(a, e_j), \quad \ell_a(x) - \ell_{a-1}(x) = \#\{j : e_j \geq a\}. \quad (\text{J.3})$$

Thus these intrinsic local lengths recover every exponent, including repeated roots. The corank of the pencil at x gives only $\ell_1(x)$, and the multiplicity of the discriminant gives only the eventual value $\ell_a(x) = \sum_j e_j$. The intermediate values are genuinely additional information.

J.2. A specialization with constant discriminant and coranks. The following family keeps both ends of this length profile fixed. Fix $i \in \mathbb{C}$ with $i^2 = -1$, and set

$$N_0 = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad P_t = \frac{t}{2} N_0, \quad A_0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad C_t = \begin{pmatrix} P_t & I_2 \\ 0 & P_t \end{pmatrix}. \quad (\text{J.4})$$

Put

$$B_t = A_0 C_t = \begin{pmatrix} 0 & P_t \\ P_t & I_2 \end{pmatrix}, \quad R_t = \langle A_0, B_t \rangle \subset \text{Sym}^2 \mathbb{C}^4. \quad (\text{J.5})$$

Both matrices are symmetric and A_0 is invertible. They are independent for every t . For $n > 4$, append I_{n-4} to A_0 and append $\text{diag}(\mu_5, \dots, \mu_n)$ to B_t , where the μ_j are fixed distinct nonzero numbers. Denote the resulting matrices by $A^{(n)}, B_t^{(n)}$ and their pencil again by R_t .

Theorem J.4 (Separation inside fixed reduced rank data). *For every $n \geq 4$, the family R_t , $t \in \mathbb{A}^1$, has the following properties.*

- (1) *Its full discriminant, with multiplicities, is independent of t :*

$$\det(sA^{(n)} + uB_t^{(n)}) = s^4 \prod_{j=5}^n (s + \mu_j u). \quad (\text{J.6})$$

For every minor size, the reduced determinantal locus on the marked parameter line is also independent of t .

- (2) *At the root $s = 0$, the spectral module has exponents $(3, 1)$ for $t \neq 0$ and $(2, 2)$ for $t = 0$. Consequently $Z_{R_t}^{[d]} \not\cong Z_{R_0}^{[d]}$ for every $t \neq 0$, whereas all the punctured fibres are isomorphic.*

- (3) The schemes $Z_{R_t}^{[d]}$ form a flat projective family over $\mathbb{A}^1$, with fixed reduction. After one identification of the socle quotient bundles, every truncation of order $k < d$ is a constant family. Thus the first separating order remains exactly $d = n^2 + 2n - 4$ even under the constraints in part (1).

Proof. The identities $N_0^2 = 0$ and $\text{rank } N_0 = 1$ give

$$C_t^2 = \begin{pmatrix} 0 & tN_0 \\ 0 & 0 \end{pmatrix}, \quad C_t^3 = 0, \quad C_t = \begin{pmatrix} I_2 \\ P_t \end{pmatrix} (P_t \quad I_2). \quad (\text{J.7})$$

The right factor in the last expression is surjective and the left factor is injective. Hence $\text{rank } C_t = 2$ for every t , whereas $\text{rank } C_t^2 = 1$ precisely when $t \neq 0$. The nilpotent partitions are therefore $(3, 1)$ and $(2, 2)$. Also $\det A_0 = 1$ and $\det(sI_4 + uC_t) = s^4$, proving (J.6). At $s = 0$ the full pencil has corank two for all t ; at each remaining root it has corank one, and elsewhere corank zero. Equality of these functions on $\mathbb{P}^1(\mathbb{C})$ implies equality of the reduced closed determinantal subschemes for each minor size.

At the unique multiplicity-four root, formula (J.3) yields respectively

$$(\ell_1, \ell_2, \ell_3, \dots) = (2, 3, 4, \dots), \quad (2, 4, 4, \dots).$$

No automorphism of the parameter line can exchange this root with a simple one. Theorem J.3 separates the special fibre from the punctured fibres and identifies the latter with one another. This is a statement about geometric fibres; no choice of a simultaneous trivializing congruence on $\mathbb{A}^1 \setminus 0$ is needed.

For flatness we use the relative gluing of Theorem K.1; the following coordinates identify its pullback in the present family. First note that $t \mapsto R_t$ is a morphism to $\text{Gr}(2, \text{Sym}^2 V)$, not merely a collection of pencils. In symmetric matrix coordinates the two fixed linear functionals

$$a(Q) = Q_{13} + iQ_{14}, \quad b(Q) = Q_{33}$$

satisfy $a(A^{(n)}) = b(B_t^{(n)}) = 1$ and $a(B_t^{(n)}) = b(A^{(n)}) = 0$. Their common kernel is a fixed complement to every R_t . It trivializes $S_{R_t} = \text{Sym}^2 V / R_t$, and hence identifies the relative socle Grassmannian with $B \times \mathbb{A}^1$.

On a frame chart of its tautological bundle, the graph description (7.4) identifies the truncated algebra with

$$\bigoplus_{j=0}^{d-1} \text{Sym}^j E^* \oplus (\text{Sym}^d E^* / K_{R_t}). \quad (\text{J.8})$$

By Theorem 7.3, K_{R_t} is a subbundle of constant rank $\binom{N}{2}$, with locally free quotient. This statement commutes with pullback to the present base and with right changes of frame, so the local descriptions glue. Thus (J.8) is locally free on $B \times \mathbb{A}^1$, of rank $\binom{n^2+d}{d} - \binom{N}{2}$. The graph projection is a finite flat morphism to that base. Since B is projective, its composition with projection to $\mathbb{A}^1$ is flat and projective. Below degree d there is no relation, and the fixed complement identifies the entire lower truncated algebra, including multiplication, with a constant family. These arguments prove flatness of the finite neighbourhoods; they make no assertion about flatness of the unrestricted failure schemes. $\square$

The qualifier “reduced” in part (1) is necessary. At $s = 0$, in a local parameter $z = s/u$, the gcd of the three-row minors of the four-dimensional block is a unit times z for $t \neq 0$ and a unit times z^2 for $t = 0$. Both radicals are (z) . Thus the family retains exactly the higher Fitting information discarded by reduction; it does not contradict the classical classification by the full determinantal ideals.

J.3. An independently visible change of reciprocal geometry. For a regular pencil let its reciprocal curve be the projective closure of the inverses of its nonsingular members. These curves and their strata are studied independently of multiplication-failure schemes in [4, §3].

Corollary J.5 (A conic specializes to a line). *For the four-dimensional family (J.5), the reciprocal curve is a smooth plane conic when $t \neq 0$ and a projective line when $t = 0$. The unmarked twentieth neighbourhood distinguishes these two cases, while all earlier neighbourhoods, the full discriminant, and all reduced rank loci agree.*

Proof. For $s \neq 0$, multiplication on the right by the fixed invertible matrix A_0^{-1} identifies the projective inverse with

$$(sI_4 + uC_t)^{-1} = s^{-1}I_4 - us^{-2}C_t + u^2s^{-3}C_t^2.$$

If $t \neq 0$, the matrices I_4, C_t, C_t^2 are linearly independent: their span has dimension equal to the degree three of the minimal polynomial. The closure is therefore the quadratic Veronese image $[s : u] \mapsto [s^2 : -su : u^2]$, a smooth conic in its spanning plane. At $t = 0$ the last term vanishes, and I_4, C_0 are independent, giving a line. The remaining assertions follow from Theorem J.4, with $d = 20$. The degree drop does not assert that the reciprocal curves themselves form a flat family; the flat family proved above consists of the finite failure neighbourhoods. $\square$

APPENDIX K. THE UNIVERSAL FINITE NEIGHBOURHOOD AND SPECTRAL STRATA

The finite-order construction is a family of actual neighbourhoods in multiplication-failure schemes. This fact is needed when passing from individual pencils to degenerations and infinitesimal parameter spaces. We establish it before discussing the spectral incidence schemes.

Put $P = \text{Gr}(2, \text{Sym}^2 V)$, and write $\mathcal{R}$ and $\mathcal{S} = (\text{Sym}^2 V \otimes \mathcal{O}_P)/\mathcal{R}$ for its universal relation and quotient bundles. Let

$$\pi : \mathcal{B} = \text{Gr}_P(n, \mathcal{S}) \longrightarrow P, \quad \mathcal{E} = \text{Hom}(\mathcal{U}, V \otimes \mathcal{O}_{\mathcal{B}}),$$

where $\mathcal{U}$ is tautological of rank n . As before, $N = n(n+1)/2$, $p = N - 2$, $d = n + 2p$, and $M = \binom{N}{2}$.

Theorem K.1 (Universal neighbourhood and arbitrary base change). *The universal multiplication-failure scheme has a canonical order- d neighbourhood $\mathcal{Z}^{[d]}$ of $\mathcal{B}$. There is a rank- M vector subbundle $\mathcal{K} \subset \text{Sym}^d \mathcal{E}^*$ such that*

$$\mathcal{Z}^{[d]} = \text{Spec}_{\mathcal{B}} \left(\text{Sym} \mathcal{E}^* / (\mathcal{K}, \text{Sym}^{d+1} \mathcal{E}^*) \right). \quad (\text{K.1})$$

This is finite locally free over $\mathcal{B}$, of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}, \quad (\text{K.2})$$

and is projective and flat over P . Its construction, its ideal-adic filtration, and its identification with the actual multiplication-failure neighbourhood commute with every morphism of complex schemes $T \rightarrow P$, including nonreduced ones. For $k < d$, its order- k truncation is $\text{Spec}_{\mathcal{B}} \bigoplus_{j=0}^k \text{Sym}^j \mathcal{E}^$, with truncated multiplication. After trivializing $\mathcal{S}$ on the parameter base, these lower-order families are constant.*

Proof. The locus in $\text{Gr}_P(n, V \oplus \mathcal{S})$ on which projection to $\mathcal{S}$ is injective is canonically $\text{Tot } \mathcal{E}$. It is an open neighbourhood of $\mathcal{B}$, not a choice of formal coordinates. On a frame chart of $\mathcal{U}$, let $T : \mathcal{U} \rightarrow V$ be the universal graph map and let $\gamma : \text{Sym}^2 V \rightarrow \mathcal{S}$ be the quotient. The same block elimination that proves (7.4) works over its coordinate ring: the ideal of the multiplication-failure scheme is

$$(\det T)I_p(\gamma \text{Sym}^2 T).$$

Every generator is homogeneous of degree d .

Here is also a coordinate-free description of its first relation bundle. The coefficient map, followed by determinant multiplication, is a morphism

$$\det V^* \otimes \det \mathcal{U} \otimes \det \mathcal{S}^* \otimes \bigwedge^p \mathrm{Sym}^2 \mathcal{U} \longrightarrow \mathrm{Sym}^d \mathcal{E}^*. \quad (\text{K.3})$$

The inclusion used in this expression is $\det \mathcal{S}^* \hookrightarrow (\bigwedge^p \mathrm{Sym}^2 V)^*$. The determinant factors evaluate $\det T$, and the remaining factors evaluate $\bigwedge^p \mathrm{Sym}^2 T$ followed by the quotient. Thus the image is exactly the span of the displayed maximal minors, not a separately inserted coefficient module.

On a frame chart this map is the fixed injective linear coefficient map ν of Theorem 7.3, restricted to a line subbundle tensored with the fixed space $\bigwedge^p \mathrm{Sym}^2 \mathbb{C}^n$. Its image is a subbundle of rank M : locally split the line in the source, and split the fixed complex linear inclusion ν . Both splittings remain splittings over any complex algebra. Denote its image by $\mathcal{K}$. A right change of frame of $\mathcal{U}$ changes the graph coordinates and this coefficient map contragrediently. The determinant character in (K.3) is exactly the one arising from $\det T$. Hence these local ideals and their degree- d subbundles agree on overlaps. In particular, they glue as ideals of the given graph neighbourhood.

Let $\mathfrak{b}$ be the ideal of the zero section inside the multiplication-failure scheme. In graph coordinates its preimage is the positive-degree ideal. Quotienting by $\mathfrak{b}^{d+1}$ therefore gives precisely (K.1). This identifies the gluing with the actual ideal-adic neighbourhood, not merely with algebras having the correct geometric fibres. Its underlying module is

$$\bigoplus_{j=0}^{d-1} \mathrm{Sym}^j \mathcal{E}^* \oplus (\mathrm{Sym}^d \mathcal{E}^* / \mathcal{K}).$$

All terms are locally free. Counting monomials in n^2 variables gives (K.2). The morphism $\mathcal{B} \rightarrow P$ is smooth and projective, so the asserted flatness and projectivity follow. Finally the graph description, the split coefficient inclusion, and the homogeneous quotient all commute with base change. Below degree d the ideal contributes no relation. This proves the final assertion as well. $\square$

Remark K.2. Over a nonreduced base, the theorem concerns the specified relative socle subscheme $\mathcal{B}_T$ and its ideal. It does not identify that ideal with the absolute nilradical of $\mathcal{Z}_T^{[d]}$, which can also contain nilpotents from the base. On geometric fibres $\mathcal{B}_t$ is exactly the reduction. Nor is this theorem an equivalence of unmarked deformation groupoids: extra automorphisms of truncated algebras are not being identified with pencil congruences.

K.1. Spectral incidence schemes from the first relation space. On the regular-pencil locus, pass locally to ordered generators (A, B) with A invertible, and put $C = A^{-1}B$. For a marked point with affine coordinate z , set $X = C - zI$. The next elementary presentation works over an arbitrary coefficient ring; no diagonalization or division by eigenvalue differences is involved.

Lemma K.3 (Truncated spectral presentation). *Let D be a commutative ring, $X \in \mathrm{Mat}_n(D)$, and $a \geq 1$. There is a natural isomorphism*

$$\mathrm{coker}(wI - X) \otimes_{D[w]} D[w]/(w^a) \simeq \mathrm{coker}(X^a : D^n \rightarrow D^n). \quad (\text{K.4})$$

Equivalently, the a -square block presentation with diagonal blocks X and subdiagonal blocks $-I$ is equivalent, by invertible row and column operations over D , to $I_{n(a-1)} \oplus X^a$.

Proof. Map the class of $\sum_{j=0}^{a-1} w^j v_j$ to the class of $\sum_{j=0}^{a-1} X^j v_j$ modulo $X^a D^n$. The relation $wv = Xv$ and the relation $w^a v = 0$ show that this map is well-defined. Sending a constant vector to its class is inverse. Eliminating successively the blocks with coefficient $-I$ gives the

stated presentation equivalence, using only unit pivots. This also proves naturality under ring homomorphisms. $\square$

Theorem K.4 (Scheme-theoretic spectral readout). *In the fixed tensor presentation of Theorem 7.3, the first relation subspace determines, on the regular-pencil locus, every closed spectral incidence scheme*

$$\mathfrak{S}_{a,h} = V(I_{n-h+1}((C - zI)^a)), \quad 1 \leq a, h \leq n. \quad (\text{K.5})$$

This determination commutes with arbitrary complex base change. At a geometric point, membership is equivalent to $\text{length}(\mathcal{T}_z/\mathfrak{m}_z^a \mathcal{T}_z) \geq h$. Consequently the schemes of specified spectral length profiles, and not only their reduced supports, can be read from the order- d relation subspace.

Proof. The closed immersion Φ of Theorem 7.3 has an inverse on its scheme-theoretic image. Thus a family of relation subspaces in that image recovers its morphism to P , over any base, and hence its universal pencil. In the displayed chart its spectral sheaf has presentation $(z+w)I - C = wI - X$. Lemma K.3 gives the presentation X^a for restriction to the order- a fat point centred at z . Its $(h-1)$ -st Fitting ideal is exactly the ideal in (K.5); Fitting ideals of finitely presented modules commute with base change. On a field the cokernel has dimension at least h exactly when these minors vanish.

These descriptions are independent of the chosen frames. A change of basis in V conjugates X ; a change of affine coordinate replaces the parameter ideal at the marked point by a unit multiple locally. Alternatively, restriction of the spectral sheaf to the fat diagonal on its parameter line defines the same finite module without a coordinate. Thus the Fitting ideals glue. Rank equality is imposed by the corresponding rank inequality together with the open complement of the next inequality. Intersecting these locally closed conditions gives each specified length profile. Formula (J.3) then identifies its geometric fibres with the prescribed elementary-divisor partition. $\square$

K.2. Realizing the complete Segre stratification. Let P_σ denote the reduced closure in P of a regular Segre stratum, and let $J_\sigma = \Phi(P_\sigma)$. Because Φ is a closed immersion, it gives isomorphisms

$$P_\sigma \simeq J_\sigma, \quad P_{\sigma_1} \cap_P \cdots \cap_P P_{\sigma_s} \simeq J_{\sigma_1} \cap_{\Phi(P)} \cdots \cap_{\Phi(P)} J_{\sigma_s}, \quad (\text{K.6})$$

where the intersections carry their scheme structures. In particular their local rings and formal completions are preserved. Pulling back Theorem K.1 realizes these parameter schemes by flat projective families of the actual finite failure neighbourhoods. The classical inclusions among Segre-stratum closures [4, Theorem 5.1] are therefore all realized, rather than just the one degeneration of Theorem J.4.

There are two distinct assertions here. The Segre closure order is classical, and transport of closed subschemes through a closed immersion is formal. The geometric input proved here is the base-change-compatible realization by actual multiplication-failure neighbourhoods and the recovery of their spectral Fitting schemes. The ambient parameter space in (K.6) is the image of Φ , not a moduli space of every possible unmarked deformation of a finite scheme.

APPENDIX L. FIXED-DISCRIMINANT STRATA AND ALL ADMISSIBLE SPECIALIZATIONS

The universal family of Section K and the example of Section J lead to a classification of the entire fixed-discriminant, fixed-reduced-rank locus. The spectral orbit order is classical. Every allowed specialization is realized here by a flat family of the original finite multiplication-failure neighbourhoods, and the same first relation detects every strict transition.

Fix distinct $\alpha_1, \dots, \alpha_s \in \mathbb{C}$, positive m_i with $\sum m_i = n \geq 3$, and $1 \leq c_i \leq m_i$. Fix a nonsingular symmetric A , and let $\mathfrak{p}_A$ be its self-adjoint endomorphisms. Consider the reduced locally closed

variety

$$X_{m,c}(A) = \left\{ C \in \mathfrak{p}_A : \det(zI - C) = \prod_i (z - \alpha_i)^{m_i}, \quad \dim \ker(C - \alpha_i I) = c_i \right\}. \quad (\text{L.1})$$

Exclude $s = 1, c_1 = n$, for which C is scalar and $\langle A, AC \rangle$ is not a pencil. Every other datum is realized. Put $R_C = \langle A, AC \rangle$, $d = n^2 + 2n - 4$, and $Z_C = Z_{R_C}^{[d]}$. The full binary determinant is $\det(A) \prod_i (u + \alpha_i v)^{m_i}$. The coranks and labelled roots specify all reduced determinantal loci on the line.

Write $\mathcal{P}(m, c)$ for partitions into exactly c positive parts. For equal-size partitions, $\mu \preceq \lambda$ means $\sum_{j \leq a} \mu_j \leq \sum_{j \leq a} \lambda_j$ for every a ; write λ' for the conjugate partition.

L.1. The classical orthogonal input. We isolate the orbit theorem used below, with its precise group and source. Let (H, A) be a nondegenerate complex symmetric space of dimension m . The acting group is the full orthogonal group $O(A)$, and the operators are A -self-adjoint, not elements of the skew-adjoint Lie algebra $\mathfrak{o}(A)$.

Lemma L.1 (Self-adjoint nilpotent orbits). *The full $O(A)$ -orbits of self-adjoint nilpotent operators are indexed by all partitions λ of m . If $\mathcal{O}_\lambda$ denotes the orbit, then*

$$\dim \mathcal{O}_\lambda = \frac{1}{2} \left(m^2 - \sum_a (\lambda'_a)^2 \right), \quad \mathcal{O}_\mu \subset \overline{\mathcal{O}_\lambda} \iff \mu \preceq \lambda.$$

An $O(A)$ -orbit is one $SO(A)$ -orbit when λ has an odd part. When every part is even, it is the disjoint union of two $SO(A)$ -orbits, interchanged by an isometry of determinant -1 . The closure statement above is for the full orthogonal orbit and does not identify the closures of its two components.

Proof. The classification and closure assertions are [5, Proposition 1, p. 444; Theorem 1, p. 447], respectively, with $\varepsilon = 1$ and $G(V)$ as defined in [5, §1.1, p. 443]. Remark 1 on p. 444 specifies that all partitions occur; the order in §1.4 on p. 446 is precisely the row-dominance order used here. The proof of Theorem 1 occupies pp. 447–451. These references concern the full group, not its identity component.

For the dimension formula there is a short independent calculation. The trace pairing on $\text{End } H$ is nondegenerate on both its self-adjoint and skew-adjoint summands. For a self-adjoint nilpotent N , the commutator with N exchanges the summands, and the two resulting maps are adjoint up to sign for that pairing. They have equal rank. Consequently $\dim \text{GL}(H)N = 2 \dim O(A)N$, also the identity in [5, §2.4, Remark 8(i), p. 456]. The general linear centralizer has dimension $\sum_{i,j} \min(\lambda_i, \lambda_j) = \sum_a (\lambda'_a)^2$, as follows by writing homomorphisms between the cyclic $\mathbb{C}[t]$ -modules of the Jordan decomposition. This proves the stated dimension.

We give the component argument explicitly. The good bases of [5, Lemma 1, p. 444] decompose the symmetric space orthogonally into cyclic Jordan blocks, each with a reversed-identity Gram matrix. If one block has odd size, its negative identity, extended by the identity on the other blocks, centralizes N and has determinant -1 . The centralizer then meets both components of $O(A)$, so the orbit is a single $SO(A)$ -orbit.

For the converse, write

$$E_k = \ker N^k / (\ker N^{k-1} + N \ker N^{k+1}).$$

The form induced by $A(u, N^{k-1}v)$ on E_k is nondegenerate and symmetric; this is immediate on the same cyclic blocks. A commuting isometry induces $g_k \in O(E_k)$. A filtration of the Jordan module by its cyclic lengths and powers of t gives

$$\det g = \prod_k (\det g_k)^k.$$

Indeed the multiplicity-space action of a length- k block occurs k times on the associated graded; the strictly triangular terms of a module automorphism do not change its determinant. If all lengths are even, this product is one. The centralizer lies in $SO(A)$, and the index-two quotient of $O(A)$ therefore gives exactly two $SO(A)$ -orbits. Since $SO(A)$ is connected, these are the connected components of the full orbit. This does not preclude their closures from meeting. $\square$

Theorem L.2 (Classification and specialization at the first relation). *The following statements hold.*

- (1) *The $O(A)$ -orbits in $X_{\mathbf{m},c}(A)$ are indexed by $\lambda \in \prod_i \mathcal{P}(m_i, c_i)$, with dimensions*

$$\dim \mathcal{O}_\lambda = \frac{1}{2} \left(n^2 - \sum_i \sum_a (\lambda'_{i,a})^2 \right). \quad (\text{L.2})$$

Unmarked isomorphism classes of Z_C are these tuples modulo the permutation group induced on the roots by projective transformations preserving their multiplicities and coranks.

- (2) *A specialization from λ to μ inside (L.1) exists if and only if*

$$\mu_i \preceq \lambda_i \quad \text{for every } i. \quad (\text{L.3})$$

Every such pair is realized over a smooth pointed algebraic curve, after shrinking at the special point, with A fixed and with constant full determinant and reduced rank loci.

- (3) *Every such pencil family induces a flat projective family of Z_C . After shrinking the curve, every truncation of order $k < d$ is constant. If at least one dominance inequality is strict, the generic and special Z_C are nonisomorphic; their first distinguishing order is d .*
- (4) *The recovered spectral module at α_i has length profile*

$$L_{i,a} = \sum_j \min(a, \lambda_{i,j}). \quad (\text{L.4})$$

Every $L_{i,a}$ can only increase in the stated specialization. All these inequalities together are equivalent to (L.3), giving an intrinsic numerical criterion for every admissible transition.

Proof. We use Lemma L.1. Its full-orthogonal classification and dominance order have exactly the group and self-adjoint convention needed here. The comparison for Grassmannian Segre strata is [4, Theorem 5.1].

The primary spaces $V_i = \ker(C - \alpha_i I)^{m_i}$ are an orthogonal direct sum: the Chinese remainder idempotents are polynomials in the self-adjoint C , hence self-adjoint projections. Each restricted form is nondegenerate, and $C|_{V_i} = \alpha_i I + N_i$, with N_i self-adjoint nilpotent. The prescribed corank is its number of Jordan blocks. The classical theorem gives the indexing. Conversely, a Jordan block is self-adjoint for the reversed-identity Gram matrix; an isometry with the chosen nondegenerate form realizes each partition.

The space of orthogonal decompositions has dimension $(n^2 - \sum_i m_i^2)/2$. Adding the nilpotent orbit dimensions proves (L.2). These are dimensions in the fixed- A matrix space, not in an unmarked moduli stack. Full orthogonal orbits need not be connected; no irreducibility assertion is needed. Theorem J.3 identifies the unmarked neighbourhood with the pencil, and its spectral module determines the partition at each root. A projective change of the line can therefore do exactly the stated root permutations. For one or two roots the projective stabilizer may be infinite; only its finite permutation image is used.

For necessity, the same polynomial idempotents decompose a family with fixed characteristic polynomial into subbundles of ranks m_i . The rank of N_i^a cannot increase at specialization. Since $m_i - \text{rank } N_i^a = \sum_j \min(a, \lambda_{i,j})$, this gives the length inequalities. Summing the first a columns of conjugate Young diagrams shows that they are equivalent to dominance: conjugation reverses the dominance order.

For sufficiency, fix an orthogonal decomposition of (V, A) . The closure criterion puts a chosen representative of μ_i in the closure of the orbit for λ_i . Thus the special tuple lies in the closure of the product of the generic orbits. Choose an irreducible component of that closure containing the point and meeting the generic orbit product. Successive general hyperplane sections through the point, avoiding the closed complement, give an integral algebraic curve with generic point in that open orbit product. Normalize it and choose a point over the prescribed point. A normal complex curve is smooth. Remove the other points outside the generic orbit product. A constant curve suffices in the equality case. The resulting block operator $\bigoplus_i(\alpha_i I + N_i(t))$ is self-adjoint for the same A . Its determinant is unchanged, and its coranks agree generically and specially, hence throughout the shrunk curve. This gives the required specialization. It does not assert constancy of the full higher Fitting ideals.

For part (3), pull back Theorem K.1 along the curve's morphism to $P = \text{Gr}(2, \text{Sym}^2 V)$. That theorem identifies the family with the actual ideal-adic neighbourhoods and proves flatness and projectivity under arbitrary base change. No inference from fibrewise isomorphisms is needed. An affine Grassmannian chart around the special pencil supplies a fixed complement to nearby R_C , identifying S_{R_C} , B , and E . There is no relation below degree d , so all lower neighbourhoods become constant. For a strict dominance transition, $\sum_{i,a}(\lambda'_{i,a})^2$ strictly increases; this sum is invariant under permutations of the decorated roots. The generic and special tuples therefore cannot be related by such a permutation. Part (1) separates their unmarked degree- d neighbourhoods. Flatness is asserted only for these finite neighbourhoods, not unrestricted failure schemes. $\square$

Corollary L.3 (Exactly when the coarse data lose a labelled class). *The number of root-labelled congruence classes with the prescribed data is $\prod_i \#\mathcal{P}(m_i, c_i)$. It is one precisely when, for every i , either $c_i = 1$ or $m_i - c_i \leq 1$. The first possible ambiguity with fixed full determinant and all reduced rank loci occurs in dimension four, at $(m, c) = (4, 2)$, with partitions $(3, 1)$ and $(2, 2)$.*

Proof. Subtracting one from every part identifies $\mathcal{P}(m, c)$ with partitions of $m - c$ into at most c parts. There is one if $c = 1$ or $m - c \leq 1$. Otherwise $(m - c)$ and $(m - c - 1, 1)$ are distinct admissible partitions. The theorem gives the count and the minimal example. $\square$

L.2. The first relation as a standard graded Betti space. The finite invariant has a usual Hilbert-scheme interpretation. The first Betti space means the generator space in the first free module of a minimal graded resolution of the quotient, not the space of relations among generators of its defining ideal. The following statement retains the marked quotient presentation.

Proposition L.4 (A graded Hilbert locus and its first Betti space). *In the fixed coordinates of Theorem 7.3, let $S = \text{Sym } H^*$, $N = \binom{n+1}{2}$, $M = \binom{N}{2}$, and $Q_R = S/((K_R) + S_{\geq d+1})$. Then*

$$\text{Tor}_1^S(Q_R, \mathbb{C})_d = K_R. \quad (\text{L.5})$$

This identity commutes with arbitrary complex base change. The functor of graded quotients with Hilbert function

$$h_j = \begin{cases} \dim S_j, & 0 \leq j < d, \\ \dim S_d - M, & j = d, \\ 0, & j > d \end{cases}$$

is represented by $\text{Gr}(M, S_d)$. The pencil locus is a closed copy of $\text{Gr}(2, \text{Sym}^2 V)$. Its degree- d first Betti space recovers every scheme-theoretic pencil parameter, including parameters over nonreduced bases.

Proof. Put $J = (K_R) + S_{\geq d+1}$ and $S_+ = \bigoplus_{j>0} S_j$. The augmented sequence $0 \rightarrow J \rightarrow S \rightarrow Q_R \rightarrow 0$ gives $\text{Tor}_1^S(Q_R, \mathbb{C}) = J/S_+J$, since its map to S/S_+ is zero. Its degree- d part is K_R . The

same argument over a base algebra applies to the degree- d subbundle with locally free quotient and commutes with pullback. A graded quotient with the indicated locally free components has zero ideal below degree d , a rank- M subbundle in degree d , and the full ideal in higher degrees. Conversely, every such subbundle defines an ideal, as its positive-degree products lie above degree d . This is the Grassmannian functor. The closed immersion and its inverse on the image are Theorem 7.3. This is not an inference from closed-point Torelli to unmarked deformation-groupoid equivalence. $\square$

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AI assistance was used in developing and drafting the arguments and finite verification scripts. All structural proofs are supplied for independent mathematical scrutiny; computational receipts are not formal proof certificates. No AI system is listed as an author.

For the present revision, GPT-6 Astra Pro assisted specifically with the incidence normalization and overlap criterion in Section 12, the universal-property proof in Section F, the explicit specialization in Section G, and the associated exact tests and exposition. These arguments require independent human verification. No statement here represents a completed journal review or a formal proof-assistant certification.

Additional disclosure for the present revision. GPT-6 Astra Pro assisted specifically with the conductor and collision models, the scalar-matrix degeneration, the flag-jet calculation, the reciprocal-polynomial fibre presentation, literature comparisons, and exact computational checks. The proofs above state the complete mathematical arguments; finite computations are consistency checks, not universal proof certificates. No journal decision or historical priority determination is inferred from the build and source audits.

ASSISTANCE IN THE PRESENT REVISION

An AI assistant helped formulate and write the binary conductor stratification, the simultaneous incidence atlas, the evaluated singular Fitting ideal, and the minimal-relation-module argument for pure-power fibres. It also prepared the source-preserving reorganization and finite exact consistency checks. The written proofs, rather than those finite checks, are the mathematical basis of the statements and are submitted to independent scrutiny.

Revision 154. AI assistance was used for the polarized moduli compatibility, spectral divisor-fibre construction, homological calculations and proof expansions. The fixed five-variable Betti table uses exact computer-assisted rational elimination with its complete finite inputs recorded. The general theorems rely on the written arguments and the explicitly attributed classical results, not on regression counts or source-preservation receipts.

Revision 155. AI assistance was used to formulate and draft the exact quadratic coefficient-section theorem and the reciprocal power-ideal description of spectral Fitting schemes, to check their finite examples, and to prepare the companion-paper organization. Classical apolar and orthogonal-orbit results are explicitly attributed. The general claims rely on the written proofs; finite exact tests and source preservation are not formal proof certificates.

Revision 157. AI assistance was used in deriving and drafting the universal power-ideal formula, its complete-quadric graph interpretation, the contact-divisor and resolved-pencil statements, and their exact finite checks. Classical determinantal, apolar and complete-quadric results are attributed at their points of use. The proofs, rather than the computations, support the general statements. No formal proof certification or journal endorsement is claimed.

Scope of the new intrinsic construction. The source-normalized envelope is an algebra bundle with an intrinsic projective power diagram. It is not a claimed preferred subquotient of the sharp failure algebra. Singular-pencil classification is read through the classical complex symmetric

Kronecker form. The explicit Hilbert tail is a boundary calculation for the stated transverse family, not a classification of every fibre of the incidence compactification. Exact computations accompanying the source check finite instances; they do not certify the general proofs.

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